The Volatility Surface as a Field of Probability Laws

A working draft

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Chapter 1

Introduction

[This chapter is written last. It will state the problem — finitely many noisy option prices, from which a surface of probability laws must be published — and the thesis that organizes every chapter: the separation of what follows from probability, what the market determines, and what the modeler must choose and disclose.]

Part I

Models

Chapter 2

The Log-Quantile-Density Model of the Volatility Smile

A volatility smile is a probability law expressed through the Black formula. This chapter constructs that law directly. The log-forward return is written as a monotone transport of a logistic variable, and the logarithm of the transport speed is expanded in Legendre polynomials of the percentile rank. Monotonicity, positive density, and unit mass are then automatic; a scalar shift imposes the martingale condition, and the sole remaining requirement is that the right endpoint speed be smaller than one. The endpoint speeds give the exact moment strip and the asymptotic Lee slopes. A single tail integral, the upper-share ledger, yields option prices, the density, the analytic calibration Jacobian, the variance-swap strike, and a complete test for calendar order. The construction separates what follows from probability, what is learned from quoted options, and what must be supplied by modeling convention. Theory, computation, and market examples are developed from the same representation.

2.1 A smile is a probability law

A listed option market gives finitely many prices. A smile model must turn them into a complete curve: a price and an implied volatility at every strike, including strikes that were not quoted. The completed curve will be differentiated for digitals and densities, integrated for variance contracts, and compared with curves at other expiries. It is therefore not enough to draw a smooth line through the quotes.

The obstruction is elementary. The second strike derivative of a call price is a probability density. A curve that fits every quote can nevertheless make this derivative negative between two quotes. It then assigns a negative price to a butterfly and a negative mass to an interval of terminal prices. No numerical routine has malfunctioned; the curve was allowed to leave the space of probability laws.

The useful question is thus not "which curve fits well?" but "in which coordinates is every fitted curve already a law?" The answer developed here uses quantiles. In rank coordinates, unit mass is automatic and positive density reduces to monotonicity. Taking the logarithm of the quantile speed removes even that constraint. The resulting log-quantile-density (LQD) family provides unconstrained coordinates for a broad class of mean-one laws with exponential log-return tails.

2.1.1 Normalized market

Fix one expiry T. Let F be its forward and D its discount factor. Under the T-forward measure define

$$X = \log \frac{S_T}{F}, \qquad Y = e^X = \frac{S_T}{F}, \qquad \mathbb{E}[Y] = 1. \tag{2.1}$$

The last equality is the martingale condition. Write $k = \log(K/F)$ for log-moneyness. Normalized call and put prices are

$$c(k) = \mathbb{E}[(Y - e^k)^+], \qquad P(k) = \mathbb{E}[(e^k - Y)^+], \qquad c(k) - P(k) = 1 - e^k. \tag{2.2}$$

A dollar price is DF times its normalized price.

Let τ denote variance time: calendar time, possibly adjusted for the desk's event-time convention. Total implied variance is $w(k) = \sigma(k)^2\tau$. The normalized Black call is

$$B(k, w) = \Phi(d_+) - e^k\Phi(d_-), \qquad d_\pm = -\frac{k}{\sqrt{w}} \pm \frac{\sqrt{w}}{2}, \tag{2.3}$$

and $w(k)$ is the root of $B(k, w(k)) = c(k)$ [4]. Rates and carry have now disappeared from the single-slice problem.

2.1.2 The static constraint set

Use normalized strike $y = e^k$. If Y has density f_Y, differentiation of the payoff gives

$$\frac{\partial c}{\partial y} = -\mathbb{P}(Y > y), \qquad \frac{\partial^2 c}{\partial y^2} = f_Y(y) \geq 0. \tag{2.4}$$

This is the Breeden–Litzenberger identity [6]. It identifies the geometry of the call curve with the law of Y: decreasing slope is a digital probability and convexity is positive density.

The identity can be read using traded portfolios. For a strike spacing δ, the symmetric butterfly

$$c(y - \delta) - 2c(y) + c(y + \delta)$$

is the price of a long call below, a long call above, and two short calls at the center. Dividing by δ^2 and shrinking the spacing recovers $f_Y(y)$. A negative second difference is therefore not merely undesirable curvature; it is a negative price for a nonnegative terminal payoff. The probability statement and the static-arbitrage statement are the same.

Definition 2.1 (Arbitrage-free slice)**.** An arbitrage-free slice is a call curve of the form (2.2) for a positive random variable Y with $\mathbb{E}[Y] = 1$. When Y has a density, this is equivalent to a decreasing convex function of strike y satisfying

$$(1 - y)^+ \leq c(y) \leq 1, \qquad -1 \leq c'(y) \leq 0, \qquad c(y) \longrightarrow 0 \quad (y \to \infty),$$

with $c'(y) \to -1$ as $y \downarrow 0$ when there is no mass at zero.

Each boundary condition has a probabilistic source. The upper bound follows from $(Y - y)^+ \leq Y$ and $\mathbb{E}[Y] = 1$. The lower bound is Jensen's inequality, $(\mathbb{E}[Y] - y)^+ \leq \mathbb{E}[(Y - y)^+]$. The slope is a survival probability and must lie between -1 and zero. Finally, calls vanish at infinite strike because the integrable random variable Y carries no mean mass arbitrarily far in its tail. A valid call curve is thus exactly a compact encoding of a mean-one law.

Convexity belongs to y, not to k. Indeed,

$$\frac{\partial^2 c}{\partial k^2} = e^k\{e^k f_Y(e^k) - \mathbb{P}(Y > e^k)\},$$

which is negative far in the left wing for any ordinary density. A convexity test performed in log-strike will therefore reject correct call curves.

The same constraint becomes less transparent in volatility coordinates. Differentiating $c(k) = B(k, w(k))$ twice yields

$$f_X(k) = g_{\mathrm{D}}(k)\frac{\varphi(d_-(k, w(k)))}{\sqrt{w(k)}}, \quad g_{\mathrm{D}}(k) = \left(1 - \frac{kw'}{2w}\right)^2 - \frac{(w')^2}{4}\left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w''}{2}. \tag{2.5}$$

Thus a volatility curve is butterfly-free precisely when $g_{\mathrm{D}}(k) \geq 0$ for all k [13]. The condition is global, nonlinear, and involves two derivatives. Direct volatility models must enforce it; the representation below makes it an identity.

This explains why good residuals alone are weak evidence. Quotes constrain only finitely many values of w. The function g_{D} also depends on the slope and curvature between them, and may become negative in a narrow unquoted interval. Increasing the strike grid reduces but does not remove the logical problem: the optimizer still moves in a space containing invalid curves. The quantile construction reverses the order of work. It first chooses a law and only then asks which smile the Black formula assigns to it.

Two checks fix the normalization in (2.5). If the smile is flat, $w' = w'' = 0$, so $g_{\mathrm{D}} \equiv 1$ and the remaining factor is exactly the Black density of X. At the money, the identities derived later in section 2.5 give $g_{\mathrm{D}}(0) = \sqrt{w_0} f_X(0)/\varphi(\sqrt{w_0}/2)$. Thus $g_{\mathrm{D}} \geq 0$ is not a penalty formula: it is positive density written in the volatility chart.

2.1.3 Notation ledger

The construction uses three horizontal axes. Strike y is the axis on which calls are convex; log-moneyness $k = \log y$ is the market axis; rank u and log-odds z are the model axes. For a rank $u \in (0,1)$, the *odds* are $u/(1-u)$ and the *log-odds* are

$$z = \log \frac{u}{1-u}.$$

The function taking rank to log-odds is called the *logit*: $\operatorname{logit}(u) = \log\{u/(1-u)\}$. It maps $(0,1)$ bijectively onto $\mathbb{R}$; its inverse is the logistic function $\Lambda(z) = 1/(1+e^{-z})$. Thus u and z record the same percentile on two different scales. The recurring notation is collected below.

$X,\ Y = e^X$	log and gross forward return	$u,\ z = \operatorname{logit} u$	rank and log-odds
$k,\ y = e^k$	log and normalized strike	$\Lambda,\ \rho$	logistic CDF and density
$c,\ P$	normalized call and put	$Q,\ q = Q'$	quantile and quantile density
$w = \sigma^2\tau$	total implied variance	g	smooth log-speed in rank
$x,\ m$	transport and martingale shift	G	upper-share ledger
L, R, a_n	polynomial coordinates	λ_-, λ_+	endpoint speeds
u_k, z_k	strike rank and log-odds	u_*, f_*, Δ	ATM rank, density, digital gap
β_L, β_R	Lee wing slopes	$N, Z_{\max}$	basis order and grid half-width

Table 2.1: Notation used throughout the chapter. A prime denotes an ordinary derivative of a one-variable function; ∂ denotes a partial derivative. The time τ is always variance time.

2.1.4 What the chapter separates

Three questions recur, and they should not be confused. *Validity* asks whether a parameter vector defines a probability law. *Information* asks which features of that law are determined by the quoted strip. *Convention* selects the remainder: basis order, regularization, tail coordinates, and the scope of calendar control. The LQD coordinates settle the first question structurally. The later sections measure the second and state the third explicitly.

The development is progressive. Section 2.2 constructs the coordinates. Sections 2.3 and 2.4 establish their probabilistic content. Section 2.5 derives prices and local smile quantities from one integral. Section 2.6 turns the representation into a fitter, section 2.7 into a computation, and section 2.8 into a surface. Section 2.9 then shows what the construction says on market and synthetic data. Proofs and implementation details are collected in the appendices.

2.2 From ranks to LQD coordinates

2.2.1 The speed of a quantile

Let $Q(u)$ be the quantile function of X, with rank $u \in (0,1)$. If U is uniform, then $Q(U)$ has the desired law. Hence probability one is built into the coordinate u, and the only remaining requirement is that Q increase.

Rank is a universal clock for distributions. The value $Q(0.01)$ is the first-percentile return whether the law is symmetric, skewed, or multimodal. Changing the law changes the returns assigned to the clock, not the clock itself. This is the main simplification: normalization of probability is handled once by the uniform variable rather than reimposed on every candidate density.

Write $q(u) = Q'(u)$ for the quantile density. Differentiating $F_X(Q(u)) = u$ gives

$$f_X(Q(u)) = \frac{1}{q(u)}. \tag{2.6}$$

The function q is the speed with which return space is traversed as rank increases. Slow motion packs probability into a narrow return interval; fast motion thins it. Since Q is increasing exactly when $q > 0$, an arbitrary real function may be used for $\log q$.

An unbounded distribution forces q to diverge at both endpoints. Here a *tail* means the probability lying beyond a remote threshold. We say that X has exponential tails, with positive scales λ_- and λ_+, when

$$F_X(x) \sim C_- e^{x/\lambda_-} \quad (x \to -\infty), \qquad 1 - F_X(x) \sim C_+ e^{-x/\lambda_+} \quad (x \to +\infty), \tag{2.7}$$

for constants $C_-, C_+ > 0$. Their logarithms therefore decay linearly in $|x|$. By contrast, a Gaussian tail has log-probability proportional to $-x^2$: it decays quadratically on the log-probability scale and hence faster than every exponential tail. For the exponential class, the leading endpoint singularities of q are $1/u$ and $1/(1-u)$. Remove them explicitly:

$$\log q(u) = -\log u - \log(1-u) + g(u). \tag{2.8}$$

The remainder g is bounded for the exponential-tail class considered here.

The subtraction is not arbitrary. The left relation in (2.7) implies $Q(u) \sim \lambda_- \log u$ as $u \downarrow 0$ and $q(u) \sim \lambda_-/u$. The right endpoint is analogous. More precisely, if $X \sim N(0, s^2)$, then

$$F_X(x) \sim \frac{s}{|x|\sqrt{2\pi}} e^{-x^2/(2s^2)} \qquad (x \to -\infty),$$

with the symmetric formula on the right. Its quantile density is $q(u) = s/\varphi(\Phi^{-1}(u))$ and, by the normal tail equivalent,

$$q(u) \sim \frac{s}{u\sqrt{2\log(1/u)}} \qquad (u \downarrow 0).$$

Its leading $1/u$ singularity is removed, but the remainder still tends to $-\infty$. Exponential tails therefore lie inside the LQD chart, whereas Gaussian tails lie on a different boundary.

Now set

$$z = \operatorname{logit} u = \log \frac{u}{1-u}, \qquad u = \Lambda(z) = \frac{1}{1+e^{-z}}, \qquad \rho(z) = \Lambda(z)\Lambda(-z). \tag{2.9}$$

The function Λ is the logistic CDF and $\rho = \Lambda'$ its density. Because $\mathrm{d}u/\mathrm{d}z = u(1-u)$, the singularities in (2.8) cancel:

$$\frac{\mathrm{d}Q}{\mathrm{d}z} = q(u)u(1-u) = e^{g(u)}. \tag{2.10}$$

Thus the entire distribution is generated by integrating a positive, ordinary function on the log-odds line.

The coordinate also places every percentile at a finite location. The median is $z = 0$; the tenth and ninetieth percentiles are $z = \mp \log 9$; moving one unit in z multiplies the odds $u/(1-u)$ by e. Meanwhile $\rho(z) \sim e^{-|z|}$, so rank integrals become ordinary integrals on $\mathbb{R}$ with exponentially decaying weight. This is why the same coordinate is convenient both analytically and numerically.

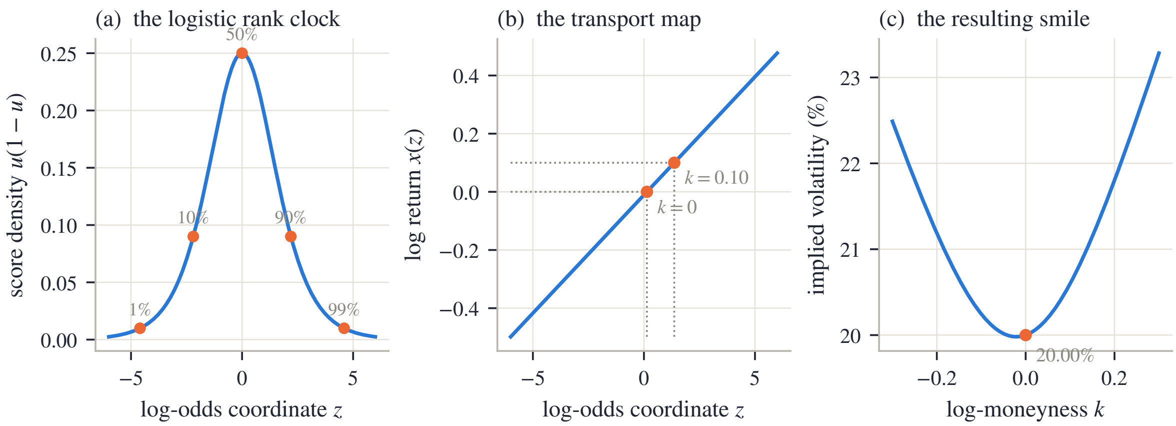

Figure 2.1: The transport construction for constant speed. Left: the logistic score and five percentile marks. Center: an affine map from log-odds to log-return, with two strike roots. Right: the implied-volatility smile priced from that law. The first two panels define the distribution; the last merely expresses it in market coordinates.

2.2.2 The LQD slice

Choose g to be a low-degree polynomial in rank. The shifted Legendre basis is convenient because it is orthogonal under the uniform rank measure. The constant and linear terms are written as an endpoint chord, leaving higher modes to describe the body.

Definition 2.2 (The LQD slice)**.** Fix $N \geq 2$ and $\theta = (L, R, a_2, \ldots, a_N) \in \mathbb{R}^{N+1}$. Let Z have CDF Λ and density ρ. The LQD slice is the law of

$$\boxed{X = x(Z), \qquad x(z) = m + \int_0^z e^{g(\Lambda(t))}\,\mathrm{d}t, \qquad g(u) = (1-u)L + uR + \sum_{n=2}^{N} a_n P_n(1-2u),} \tag{2.11}$$

where P_n is the Legendre polynomial of degree n and m is chosen so that $\mathbb{E}[e^X] = 1$.

For every finite coefficient vector, $x'(z) = e^{g(\Lambda(z))} > 0$. Consequently

$$F_X(x(z)) = \Lambda(z), \qquad f_X(x(z)) = \frac{\rho(z)}{x'(z)} = \Lambda(z)\Lambda(-z)e^{-g(\Lambda(z))} > 0. \tag{2.12}$$

The quantile is $Q(u) = x(\operatorname{logit} u)$, and (2.10) recovers

$$q(u) = \frac{e^{g(u)}}{u(1-u)}. \tag{2.13}$$

Equations (2.11) and (2.13) are the transport and log-quantile-density descriptions of the same law.

Since a polynomial g is bounded on $[0,1]$, its exponential is bounded above and away from zero. Therefore $x(z)$ tends to $-\infty$ and $+\infty$ with z and is a diffeomorphism of the line. The construction cannot fold, stop, or reverse. Positivity of the density in (2.12) is thus a geometric fact about the coordinate map, not a pointwise inequality checked after the fit.

The two endpoint values are

$$g(0) = L + \sum_{n\geq 2} a_n, \qquad g(1) = R + \sum_{n\geq 2} (-1)^n a_n, \tag{2.14}$$

and their exponentials

$$\lambda_- = e^{g(0)}, \qquad \lambda_+ = e^{g(1)} \tag{2.15}$$

are the endpoint speeds. If $\bar{x}(z) = \int_0^z e^{g(\Lambda(t))}\mathrm{d}t$, then Lipschitz continuity of g and exponential convergence of Λ to its endpoints imply

$$\bar{x}(z) = b_- + \lambda_- z + O(e^{z}) \quad (z \to -\infty), \qquad \bar{x}(z) = b_+ + \lambda_+ z + O(e^{-z}) \quad (z \to +\infty). \tag{2.16}$$

The endpoint speeds will therefore govern both tails.

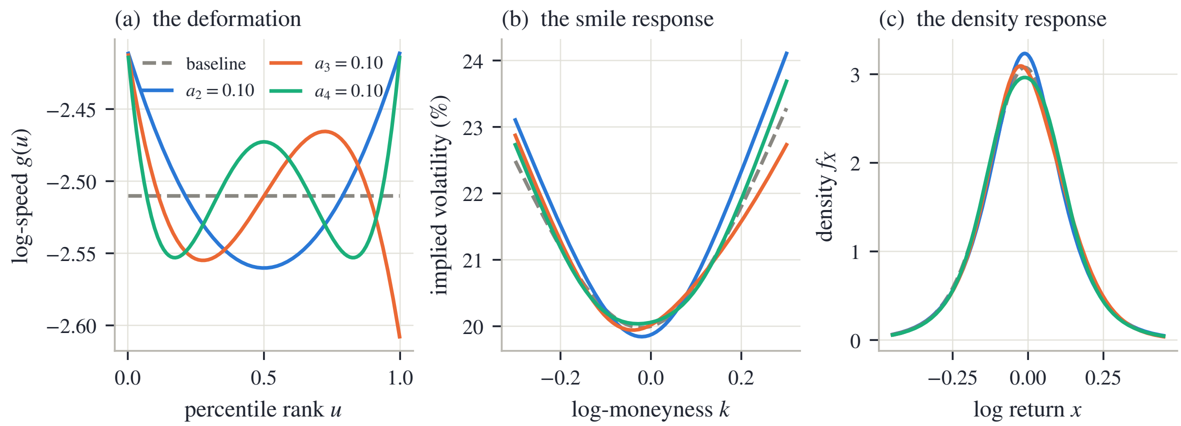

Figure 2.2: Three Legendre perturbations of the same reference law. Left: their action on log-speed g. Center: the response of implied volatility. Right: the response of the density. An even mode moves both endpoints in the same direction; an odd mode moves them in opposite directions. In all cases the density remains positive because the transport speed remains an exponential.

The modes have a direct interpretation. Raising g stretches returns near the affected ranks and therefore lowers density there by (2.6). Even modes act symmetrically; odd modes transfer shape from one side to the other; higher modes work on finer rank scales. The raw coefficients are not pure body controls, since (2.14) couples each of them to one or both tails. The endpoint chart introduced in section 2.6 removes that coupling without changing the family.

Legendre polynomials are a coordinate choice, not part of the probability argument. Rank integrals use uniform weight on $[0,1]$, under which Legendre modes are orthogonal. Their columns remain distinguishable in least-squares normal equations as order rises. Monomials span the same space but inherit Hilbert-matrix conditioning. Chebyshev modes would also be reasonable; the endpoint chord would remain useful because it names the two tail controls.

2.2.3 Martingale normalization and the finite-forward wall

Only a translation remains. Let

$$I = \int_{-\infty}^{\infty} e^{\bar{x}(z)} \rho(z)\, \mathrm{d}z.$$

Proposition 2.3 (Martingale normalization)**.** *If $I < \infty$, then $m = -\log I$ is the unique shift for which $\mathbb{E}[e^X] = 1$. Moreover, $I < \infty$ if and only if $\lambda_+ < 1$.*

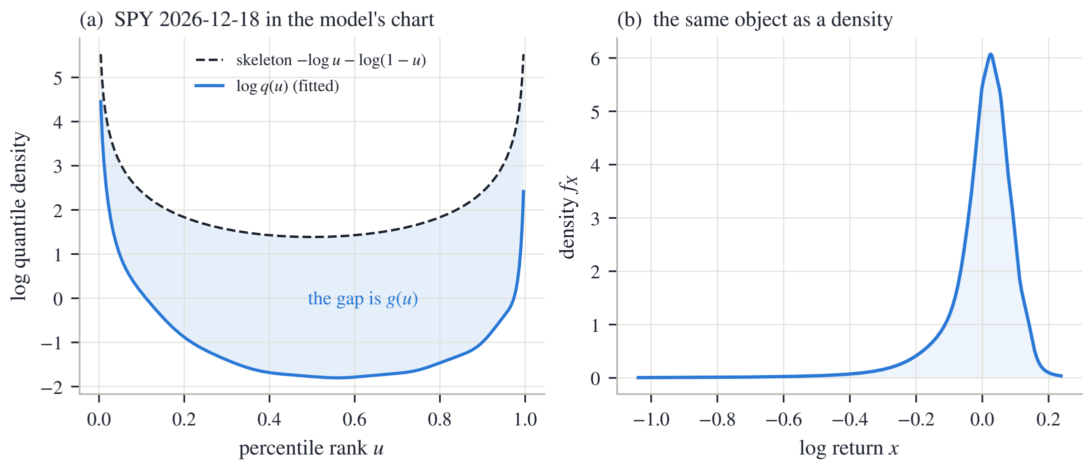

Figure 2.3: A fitted SPY December 2026 slice in the model's coordinates. Left: the universal boundary skeleton and the fitted $\log q$; their difference is g. Right: the density produced by the same transport. The quoted strikes occupy the central ranks; the continuation to the endpoints is extrapolation.

Proof. Since $X = m + \bar{x}(Z)$, $\mathbb{E}[e^X] = e^m I$, giving the unique shift. At the right endpoint, (2.16) and $\rho(z) \sim e^{-z}$ make the integrand asymptotic to a positive constant times $e^{(\lambda_+ - 1)z}$; it is integrable exactly when $\lambda_+ < 1$. At the left endpoint it behaves as $e^{(1+\lambda_-)z}$, which is integrable for every $\lambda_- > 0$. $\square$

The set of admissible raw parameters is therefore

$$\Theta_N = \{\theta \in \mathbb{R}^{N+1} : \lambda_+ < 1\}. \tag{2.17}$$

It is open and has a single boundary. A logistic coordinate for λ_+ will make that boundary unreachable.

2.2.4 The exactly solvable slice

Set $L = R = \log s$, $a_n = 0$, with $0 < s < 1$. Then $x(z) = m + sz$ and X is logistic with scale s. Its moment-generating function is the beta integral

$$\mathbb{E}[e^{tZ}] = \int_0^1 \left(\frac{u}{1-u}\right)^t \mathrm{d}u = \mathrm{B}(1+t, 1-t) = \Gamma(1+t)\Gamma(1-t) = \frac{\pi t}{\sin(\pi t)}, \qquad |t| < 1. \tag{2.18}$$

Hence

$$m = -\log \frac{\pi s}{\sin(\pi s)}, \qquad \mathrm{Var}(X) = \frac{\pi^2 s^2}{3}.$$

Matching a target ATM total variance w_0 gives the cold start

$$s = \frac{\sqrt{3w_0}}{\pi}. \tag{2.19}$$

This seed matches variance, not ATM price. As $s \downarrow 0$, $m = -\pi^2 s^2/6 + O(s^4)$. Moreover,

$$\mathbb{E}[Z^+] = \int_0^\infty z\rho(z)\,\mathrm{d}z = \sum_{n \geq 1} \frac{(-1)^{n-1}}{n} = \log 2,$$

where the logistic density is expanded as a geometric series on the positive half-line. Thus the LQD ATM call is $s \log 2 + O(s^2)$. Matching it to the small-variance Black price gives

$$\frac{w_{\text{implied}}}{w_0} \longrightarrow \frac{6(\log 2)^2}{\pi} \approx 0.918.$$

The seed is therefore about eight percent low in ATM variance, a known bias removed by calibration.

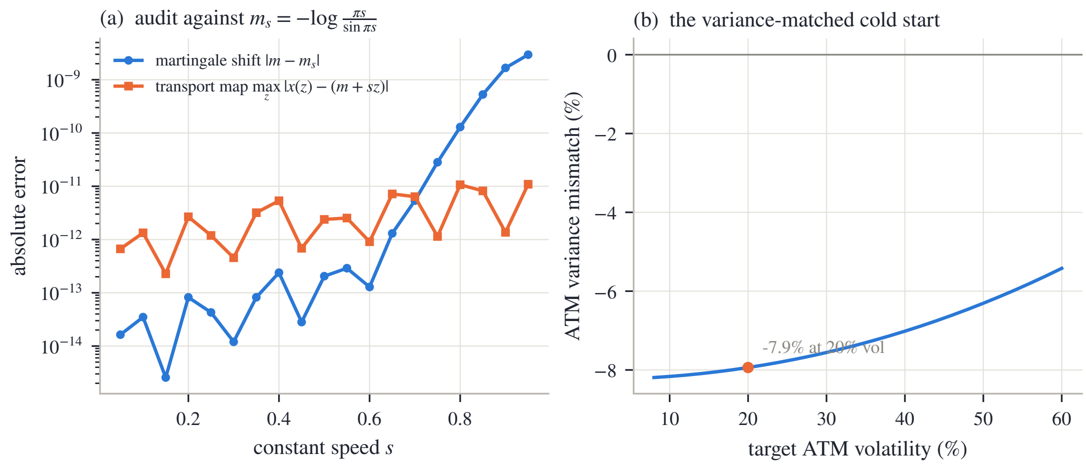

Figure 2.4: The solvable slice as an end-to-end numerical test. Left: errors in the computed transport and martingale shift against the closed forms from (2.18); the normalization becomes harder as s approaches the wall. Right: the variance-matched cold start and its derived small-s bias.

2.3 Validity and scope

The construction has two distinct claims. First, every admissible parameter vector defines an arbitrage-free slice. Second, increasing polynomial order fills out a specified class of laws. The first is exact at every order; the second determines the scope of the coordinates.

2.3.1 One-expiry validity

Proposition 2.4 (One-expiry no-arbitrage)**.** *Let $\theta \in \Theta_N$ and price calls from the LQD slice* (2.11)*. Then:*

(i) X has a strictly positive continuous density on $\mathbb{R}$, and $Y = e^X$ has a strictly positive density on $(0, \infty)$ with $\mathbb{E}[Y] = 1$;

(ii) as a function of strike y, the call is strictly decreasing and strictly convex, satisfies $(1 - y)^+ \leq c(y) \leq 1$, has $c'(y) = -\mathbb{P}(Y > y) \in (-1, 0)$, and tends to zero as $y \to \infty$;

(iii) every nondegenerate finite-width butterfly has positive price, and put–call parity holds identically;

(iv) as a function of log-strike k, the call is concave sufficiently far in the money.

Proof. Equation (2.12) gives a positive continuous density; the shift of proposition 2.3 gives unit mean. Rank integration gives

$$c(y) = \int_0^1 (e^{Q(u)} - y)^+ \mathrm{d}u.$$

Each integrand is decreasing and convex in y, and differentiation yields $c'(y) = -\mathbb{P}(Y > y)$. Positivity of the density makes both monotonicity and convexity strict. Jensen gives $(1 - y)^+ \leq c(y)$, while $c(y) \leq \mathbb{E}[Y] = 1$ and dominated convergence gives the right boundary. Butterfly positivity and parity follow immediately. Finally,

$$\frac{\partial^2 c}{\partial k^2} = e^k \{f_X(k) - \mathbb{P}(Y > e^k)\},$$

which is negative as $k \to -\infty$ because $f_X(k) \to 0$ and the probability tends to one. $\square$

The proposition is stronger than a dense-grid test. It covers every strike, including strikes between grid nodes and beyond the displayed smile, because the call is an expectation under a positive law. Calibration may move the coefficient vector arbitrarily inside Θ_N without approaching a hidden butterfly boundary. The numerical implementation must still preserve this continuous geometry; that separate question is treated by the certificates of section 2.7.2.

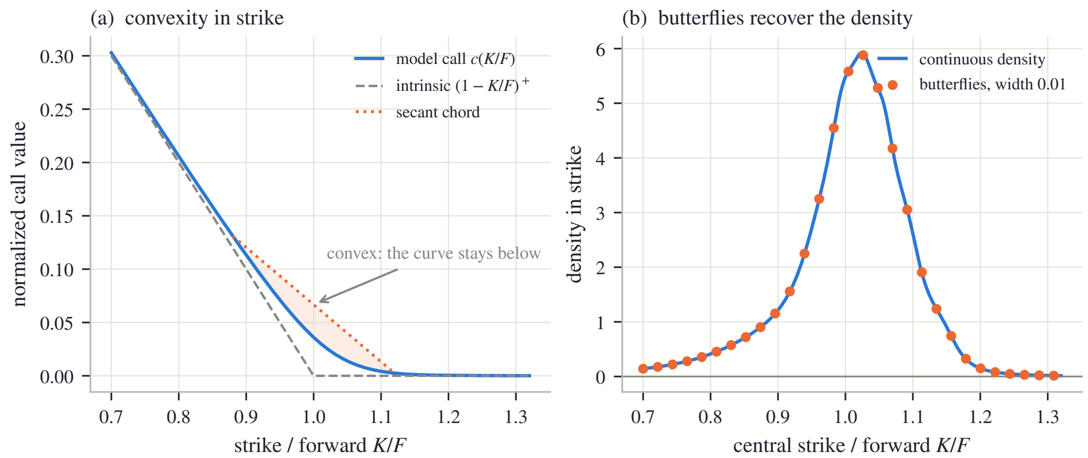

Figure 2.5: Validity on the fitted SPY December 2026 slice. Left: the call curve is convex in normalized strike and lies below its secant chord. Right: finite butterflies, divided by squared half-width, converge to the density. The geometry is produced by the law, not imposed during calibration.

2.3.2 Why the wall cannot be removed

Proposition 2.5 (Finite-forward wall)**.** *Within the LQD family the martingale normalizer exists if and only if $\lambda_+ < 1$. More generally, if a law satisfies $\mathbb{P}(X > x) \geq \kappa e^{-x}$ for some $\kappa > 0$ and all sufficiently large x, then $\mathbb{E}[e^X] = \infty$.*

Proof. The first statement is proposition 2.3. For the second, at any large x_0,

$$\mathbb{E}[e^X] \geq \int_{e^{x_0}}^{\infty} \mathbb{P}(X > \log t)\,\mathrm{d}t \geq \kappa \int_{e^{x_0}}^{\infty} \frac{\mathrm{d}t}{t} = \infty.$$

□

The boundary is therefore probabilistic, not parametric. A right log-return tail with scale at least one cannot support a finite forward. There is no left counterpart because e^X is bounded on the negative half-line.

2.3.3 Coverage of the coordinates

The natural target class can be described without reference to polynomials. For a differentiable quantile Q^* define

$$g^*(u) = \log Q^{*\prime}(u) + \log u + \log(1-u).$$

Continuity of g^* on $[0, 1]$ is exactly the condition that the law have a positive continuous density and exponential tails with finite nonzero endpoint scales.

Proposition 2.6 (Universality, with exclusions)**.** *Let X^* be a mean-one law for which g^* extends continuously to $[0, 1]$, with $e^{g^*(1)} < 1$. Then:*

(i) There exist admissible LQD slices with $g_N \to g^*$ uniformly. Their quantiles converge uniformly on compact subsets of $(0, 1)$, their call curves converge uniformly over all strikes, and their gross-return laws converge in Wasserstein-p distance for every $1 \leq p < e^{-g^*(1)}$.

(ii) *No finite-order LQD slice has an atom, a zero-density interval, bounded support, mass at zero, or Gaussian-or-faster tails. These exclusions persist under uniformly convergent log-speeds. More precisely, let admissible g_N converge uniformly to g_∞. If $g_\infty(1) < 0$, the limiting law remains in the class of part (i). If $g_\infty(1) = 0$, weak convergence can fail because mass escapes to $-\infty$; if a weak limit nevertheless exists, it has full support and a strictly positive continuous density, but sits at the finite-forward boundary. Its moments of order $r > 1$ diverge, and its mean may be smaller than one.*

Here the one-dimensional Wasserstein metric has the concrete quantile form

$$W_p(\mu, \nu)^p = \int_0^1 |Q_\mu(u) - Q_\nu(u)|^p \, \mathrm{d}u.$$

The proof is in appendix 2.A. Its approximation step is Weierstrass: polynomials approximate the continuous g^* uniformly. Since $g^*(1) < 0$, sufficiently accurate approximants remain inside the open admissible set. The tail margin supplies a common integrable envelope for normalizers, calls, and moments.

Uniform convergence of g is the correct strength for this result. It controls the transport speed at every rank, including both endpoints. Local approximation on the quoted ranks alone would say nothing about normalization or tail moments. Once a strict margin $g^*(1) < 0$ is fixed, the same exponential envelope controls all approximating normalizers and supplies uniform integrability of gross returns. This is what upgrades pointwise shape approximation to uniform convergence of call prices.

The exclusions follow from the same geometry. A positive continuous transport speed makes the quantile strictly increasing and unbounded. An atom would require a quantile jump; a gap, a flat interval; bounded support, a finite endpoint; and a Gaussian tail, $g(u) \to -\infty$ at an endpoint. None is available to a finite polynomial, nor to a uniform limit of bounded log-speeds.

Multimodality is not excluded. By (2.6), density modes are created where the transport slows and are separated where it accelerates. A positive speed can vary many times without ever becoming zero. Thus the family may represent a smooth double-hump density, but not a literal gap of zero density between the humps. Basis order determines how sharply the speed can vary, as the synthetic example in section 2.9 demonstrates.

Thus the LQD family is not a coordinate system for every probability law. It is a coordinate system for a flexible exponential-tail class. That class is large enough to approximate smooth unimodal and multimodal smile laws, while its boundary and exclusions remain explicit.

2.4 Tails, moments, and wings

The endpoint speeds introduced in (2.15) have an exact probabilistic meaning. This section follows the chain

$$\text{endpoint speed} \longrightarrow \text{tail scale} \longrightarrow \text{finite moments} \longrightarrow \text{asymptotic smile slope.}$$

2.4.1 Endpoint speeds are tail scales

Proposition 2.7 (Exponential tails). *For every admissible LQD slice there are constants $K_\pm > 0$ such that*

$$\mathbb{P}(X > x) \sim K_+ e^{-x/\lambda_+} \quad (x \to +\infty), \qquad \mathbb{P}(X < x) \sim K_- e^{x/\lambda_-} \quad (x \to -\infty). \tag{2.20}$$

Equivalently, $\mathbb{P}(Y > y) \sim K_+ y^{-1/\lambda_+}$ and $\mathbb{P}(Y < y) \sim K_- y^{1/\lambda_-}$.

Proof. On the right, (2.16) gives $x(z) = m + b_+ + \lambda_+ z + O(e^{-z})$. Its inverse satisfies

$$z(x) = \frac{x - m - b_+}{\lambda_+} + O(e^{-z(x)}).$$

Since $1 - \Lambda(z) \sim e^{-z}$,

$$\mathbb{P}(X > x) = 1 - \Lambda(z(x)) \sim e^{(m+b_+)/\lambda_+} e^{-x/\lambda_+}.$$

The left tail is identical after reversing signs. □

The gross return therefore has power tails, while the log-return has exponential tails. The two endpoint coefficients control extrapolation in a directly readable unit: log-return distance per e-fold reduction in tail probability.

2.4.2 The moment strip

Every expectation may be written either in rank or log-odds:

$$\mathbb{E}[h(X)] = \int_0^1 h(Q(u))\,\mathrm{d}u = \int_{-\infty}^{\infty} h(x(z))\rho(z)\,\mathrm{d}z. \tag{2.21}$$

Proposition 2.8 (Moment strip)**.** *For an admissible LQD slice,*

$$\mathbb{E}[e^{rX}] < \infty \quad \Longleftrightarrow \quad -\frac{1}{\lambda_-} < r < \frac{1}{\lambda_+}. \tag{2.22}$$

Consequently the last finite positive moment of Y beyond the first and the last finite inverse moment are

$$r_+^* = \frac{1}{\lambda_+} - 1, \qquad r_-^* = \frac{1}{\lambda_-}.$$

Proof. In (2.21), the right-tail integrand for $h(x) = e^{rx}$ is asymptotic to $e^{(r\lambda_+ - 1)z}$; the left-tail integrand is asymptotic to $e^{(r\lambda_- + 1)z}$. Integrability gives exactly (2.22), with divergence at both boundary values. □

The martingale condition is the point $r = 1$. It lies inside the strip exactly when $\lambda_+ < 1$, so normalization, the finite-forward wall, and the first moment are three forms of the same fact.

The asymmetry of the strip is financially important. A large λ_- may eliminate low-order inverse moments of Y without threatening the forward; negative log-returns contribute at most one to Y. A comparable right-tail scale would destroy the first positive moment and hence the forward itself. For the long NVDA example, $\lambda_- > 1$, so even $\mathbb{E}[Y^{-1}]$ diverges, while $\mathbb{E}[Y] = 1$ remains exact.

2.4.3 Closed Lee slopes

Lee's moment formula [20] relates the critical moments to the asymptotic slopes of total implied variance:

$$\beta_R = \limsup_{k \to +\infty} \frac{w(k)}{k} = \Psi(r_+^*), \qquad \beta_L = \limsup_{k \to -\infty} \frac{w(k)}{|k|} = \Psi(r_-^*), \tag{2.23}$$

where

$$\Psi(r) = 2 - 4(\sqrt{r^2 + r} - r), \qquad r \geq 0.$$

Proposition 2.9 (Speed-to-slope maps)**.** *The LQD wing slopes are*

$$\beta_R = \frac{2\lambda_+}{(1 + \sqrt{1 - \lambda_+})^2}, \qquad \beta_L = \frac{2\lambda_-}{(1 + \sqrt{1 + \lambda_-})^2}. \tag{2.24}$$

Both maps increase with their endpoint speed. For small speeds, $\beta_\pm \sim \lambda_\pm/2$; as $\lambda_+ \uparrow 1$, $\beta_R \uparrow 2$. For LQD slices the limsups in (2.23) are ordinary limits.

Proof. Substitute $r_+^* = 1/\lambda_+ - 1$ and $r_-^* = 1/\lambda_-$ into Ψ and rationalize. For example, with $s = \sqrt{1-\lambda_+}$,

$$\Psi(1/\lambda_+ - 1) = 2 - \frac{4s}{1+s} = \frac{2\lambda_+}{(1+s)^2}.$$

The left formula is obtained with $t = \sqrt{1+\lambda_-}$. Monotonicity and limits follow from the final expressions. By proposition 2.7, the return tails are regularly varying; the Benaim–Friz refinement of Lee's formula therefore gives convergence rather than only a limsup [2]. □

For completeness, the left algebra is

$$\Psi(1/\lambda_-) = 2 - \frac{4(\sqrt{1+\lambda_-} - 1)}{\lambda_-} = \frac{2\lambda_-}{(1+\sqrt{1+\lambda_-})^2}.$$

The formulas expose two useful scales. When λ is small, the wing slope is about half the endpoint speed. On the right, the map reaches Lee's model-free ceiling 2 exactly when the first moment reaches its boundary. The tail coordinate and the no-arbitrage wing bound meet at the same point.

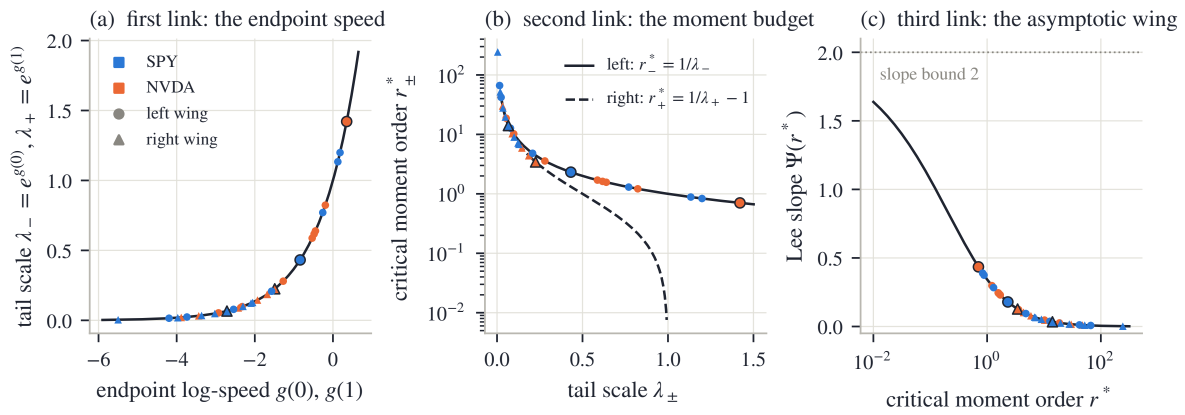

Figure 2.6: The tail chain. Left: endpoint speed determines the last finite moment; the right map reaches the finite-forward wall at $\lambda_+ = 1$. Center: Lee's moment-to-slope function. Right: the closed compositions in (2.24). Markers show the fitted SPY December 2026 and NVDA December 2027 slices.

The marked market slices occupy only a small part of the admissible range. For SPY December 2026 the endpoint speeds are $\lambda_- = 0.432$ and $\lambda_+ = 0.066$, giving slopes $\beta_L = 0.179$ and $\beta_R = 0.034$. The long NVDA slice has $\lambda_- = 1.421$ and $\lambda_+ = 0.225$, with slopes 0.435 and 0.128. The heavier single-name tails appear as larger endpoint speeds before the smile is ever plotted.

2.4.4 The limit and the traded wing

The numbers $\beta_\pm$ describe limits, not quoted strikes. If $w(k) = \beta|k| + b + o(1)$, then $w(k)/|k| = \beta + b/|k| + o(1/|k|)$ and may approach its limit from either side. The sign of the intercept b is not fixed.

No finite option strip identifies an asymptotic decay rate by itself. In a fit, $\lambda_\pm$ are selected jointly by the outer quotes, basis order, regularization, and endpoint coordinates. The model makes this choice low-dimensional and interpretable, but it does not turn extrapolation into observation.

The finite-strike discrepancy is large enough to matter. On the SPY slice in fig. 2.7, the effective right slope is 1.69 times its limit at ten delta and 1.05 times at one delta. On the left, the corresponding ratios are 0.70 and 0.70, both below one. A limiting slope can therefore overstate one traded wing and understate the other on the same fitted law.

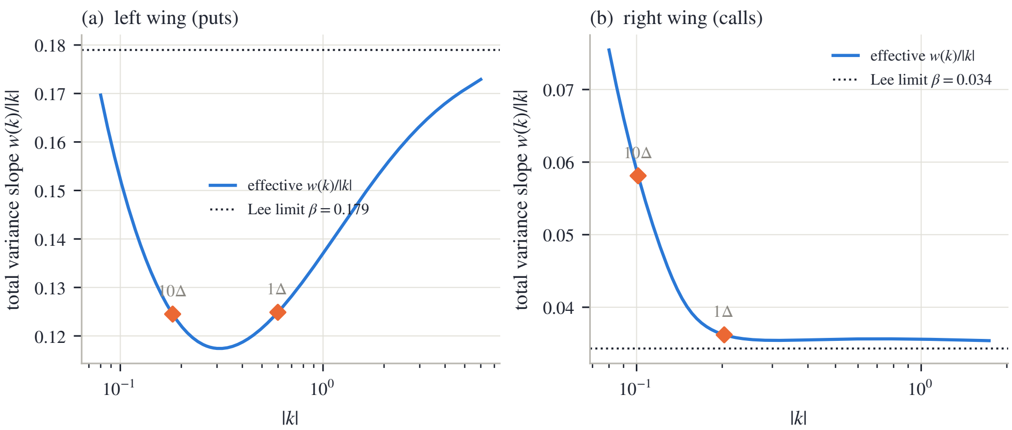

Figure 2.7: Effective variance slopes for the fitted SPY December 2026 slice. The left wing lies below its Lee limit at the marked deltas, while the right wing lies above. The asymptotic slopes are boundary conditions for extrapolation; they are not substitutes for prices at finite strikes.

2.5 The ledger calculus

Once the transport x is known, a single tail integral contains nearly every quantity needed by the fitter.

2.5.1 The upper-share ledger

Definition 2.10 (Upper-share ledger)**.** For an admissible slice define

$$G(z) = \int_z^\infty e^{x(t)}\rho(t)\,\mathrm{d}t = \mathbb{E}[Y\mathbf{1}\{Z > z\}]. \tag{2.25}$$

The name is literal: $G(z)$ is the fraction of the unit forward carried by outcomes above log-odds level z. It satisfies

$$G(-\infty) = 1, \qquad G(+\infty) = 0, \qquad G'(z) = -e^{x(z)}\rho(z) < 0. \tag{2.26}$$

For strike k, let z_k be the unique root and u_k its rank:

$$x(z_k) = k, \qquad u_k = \Lambda(z_k) = F_X(k). \tag{2.27}$$

Proposition 2.11 (Ledger pricing)**.** *For every admissible slice,*

$$c(k) = G(z_k) - e^k(1-u_k), \tag{2.28}$$

$$P(k) = e^k u_k - (1 - G(z_k)), \tag{2.29}$$

$$c'(k) = -e^k(1-u_k), \tag{2.30}$$

$$c''(k) = -e^k(1-u_k) + e^k\frac{\rho(z_k)}{x'(z_k)}, \tag{2.31}$$

$$f_X(k) = e^{-k}\{c''(k) - c'(k)\} = \frac{\rho(z_k)}{x'(z_k)}. \tag{2.32}$$

Proof. Because x increases, exercise is the event $Z > z_k$. Splitting asset and cash legs gives

$$c(k) = \mathbb{E}[Y\mathbf{1}\{Z > z_k\}] - e^k\mathbb{P}(Z > z_k) = G(z_k) - e^k(1-u_k),$$

and the put follows on the complementary event. Differentiating the call, the moving-root term is

$$\{G'(z_k) + e^k\rho(z_k)\}\frac{\mathrm{d}z_k}{\mathrm{d}k} = 0$$

because $G'(z_k) = -e^{x(z_k)}\rho(z_k) = -e^k\rho(z_k)$. This yields (2.30). Since $\mathrm{d}z_k/\mathrm{d}k = 1/x'(z_k)$, a second derivative gives (2.31); subtracting (2.30) gives (2.32). □

Pricing is thus the difference between the share delivered on exercise and the strike paid on the same event. The digital probability is $1 - u_k$, and the density is one division of two arrays already present in the transport. No price differentiation is required in the implementation.

The same rank change of variables prices any integrable terminal payoff:

$$\mathbb{E}[H(Y)] = \int_{-\infty}^{\infty} H(e^{x(z)})\rho(z)\,\mathrm{d}z. \tag{2.33}$$

The ledger is the specialization in which $H(y) = y\mathbf{1}\{y > e^{x(z_0)}\}$ and the threshold z_0 is allowed to vary. Calls are unusually cheap because their payoff splits into this asset tail and an ordinary probability tail. Digitals retain only the probability tail; power and log contracts become moments of the transport. The whole payoff calculus therefore remains on one fixed rank measure.

The constant-speed slice gives a concrete first calculation. Choose its scale so that the six-month ATM volatility is exactly twenty percent. The solution is $s = 0.081250$ and the closed-form shift is $m = -0.010883$. For the call with $k = 0.10$, the strike rank is $u_k = 79.65\%$. The upper-share entry is $G(z_k) = 0.247221$, while the cash entry is $e^k(1-u_k) = 0.224876$. Hence

$$c(0.10) = 0.247221 - 0.224876 = 0.022345,$$

corresponding to 20.60% implied volatility. The symmetric law produces a smile because its exponential return tails carry more remote-strike value than the lognormal matched at the money.

The fitted law also gives the log-contract variance directly:

$$w_{\mathrm{vs}} = -2\mathbb{E}[\log Y] = -2\int_{-\infty}^{\infty} x(z)\rho(z)\,\mathrm{d}z. \tag{2.34}$$

This is the direct-law form of the familiar strip replication: the classical construction synthesizes $-\log Y$ from calls and puts over all strikes [7, 22], while here the law is already in hand and the same payoff is integrated against its rank measure. Under the usual continuous-path, continuous-monitoring identity this is the fair variance-swap strike; jumps and discrete sampling take their standard separate corrections.

2.5.2 The ATM microscope

At the money, the smile is determined locally by three quantities from the law. Let z_* solve $x(z_*) = 0$, and write

$$c_0 = G(z_*) - (1 - u_*), \qquad u_* = \Lambda(z_*), \qquad f_* = \frac{\rho(z_*)}{x'(z_*)}. \tag{2.35}$$

The ATM Black variance is

$$w_0 = 4\left[\Phi^{-1}\left(\frac{1+c_0}{2}\right)\right]^2, \qquad d = \frac{\sqrt{w_0}}{2},$$

and define the digital mismatch

$$\Delta = u_* - \Phi(d). \tag{2.36}$$

Proposition 2.12 (Exact ATM handles)**.** *The total-variance smile $B(k, w(k)) = c(k)$ satisfies*

$$w(0) = w_0, \qquad\qquad w'(0) = \frac{2\sqrt{w_0}}{\varphi(d)}\Delta, \tag{2.37}$$

$$w''(0) = \frac{2\sqrt{w_0}}{\varphi(d)} f_* - 2 + \left(\frac{1}{2w_0} + \frac{1}{8}\right) w'(0)^2. \tag{2.38}$$

In volatility units, $\sigma(k) = \sqrt{w(k)/\tau}$,

$$\sigma(0) = \sqrt{w_0/\tau}, \qquad \sigma'(0) = \frac{\Delta}{\varphi(d)\sqrt{\tau}}, \tag{2.39}$$

$$\sigma''(0) = \frac{1}{\sqrt{\tau}} \left[\frac{f_*}{\varphi(d)} - \frac{1}{\sqrt{w_0}} + \frac{\sqrt{w_0}}{4} \left(\frac{\Delta}{\varphi(d)} \right)^2 \right]. \tag{2.40}$$

The full differentiation is given in section 2.A.2. The first derivative contains the main idea. At $(k, w) = (0, w_0)$,

$$\partial_k B = -(1 - \Phi(d)), \qquad \partial_w B = \frac{\varphi(d)}{2\sqrt{w_0}}, \qquad c'(0) = -(1 - u_*).$$

Implicit differentiation therefore gives

$$w'(0) = \frac{c'(0) - \partial_k B}{\partial_w B} = \frac{2\sqrt{w_0}}{\varphi(d)} \{u_* - \Phi(d)\}. \tag{2.41}$$

The model digital is $1 - u_*$; the matched flat-Black digital is $1 - \Phi(d)$. Hence Δ is Black digital minus model digital. A negative equity skew is exactly a model digital richer than the flat-Black digital. Level, digital, and density determine level, skew, and curvature.

Three sign checks make the interpretation concrete. For the matched Black law, $X \sim \mathcal{N}(-w_0/2, w_0)$, so $u_* = \mathbb{P}(X \le 0) = \Phi(d)$ and the skew vanishes. If $\sigma'(0) < 0$, then $u_* < \Phi(d)$ and the model digital exceeds the Black digital. Conversely, an upward skew is exactly a cheaper model digital. The comparison is with $\Phi(d)$, not with one half: the martingale shift may place the forward rank on either side of the median even when the skew is small.

The curvature formula adds one new local fact: f_*. Holding ATM price and digital fixed, a larger density at the forward raises $w''(0)$. This matches the geometry of a peaked distribution: more mass close to the forward makes nearby options cheap relative to options a little farther away, and the smile bends upward more rapidly. The other terms in (2.38) are the Black-coordinate correction and a quadratic skew contribution. Price level, first CDF value, and first density value form a hierarchy for the three smile handles.

The identities also run in reverse. A target ATM level fixes c_0; a target skew fixes u_* through (2.37); and a target curvature fixes f_* through (2.38). These are local constraints on the law rather than finite differences of a plotted smile.

2.5.3 Envelope cancellation

Calibration requires derivatives with respect to every parameter. The call depends on a parameter both through G and through the moving root z_k. The same cancellation used for (2.30) removes the latter.

Theorem 2.13 (Envelope cancellation)**.** *For every coordinate θ_j,*

$$\frac{\partial c(k;\theta)}{\partial \theta_j} = \left.\frac{\partial G(z;\theta)}{\partial \theta_j}\right|_{z = z_k(\theta)}. \tag{2.42}$$

Proof. Differentiate (2.28):

$$\frac{\partial c}{\partial \theta_j} = \left.\frac{\partial G}{\partial \theta_j}\right|_{z_k} + \{G'(z_k) + e^k \rho(z_k)\} \frac{\partial z_k}{\partial \theta_j}.$$

The bracket vanishes by (2.26) and $x(z_k) = k$. □

This is an envelope theorem: the exercise boundary is the point where the payoff integrand vanishes, so its displacement has no first-order value.

Because g is affine in the coefficients, all columns are obtained by two cumulative passes. With $\bar{x} = x - m$ and $I = \int e^{\bar{x}}\rho$, let $b_j(u) = \partial g(u)/\partial\theta_j$. Then

$$\frac{\partial \bar{x}(z)}{\partial \theta_j} = \int_0^z e^{g(\Lambda(t))} b_j(\Lambda(t))\,\mathrm{d}t, \tag{2.43}$$

$$\frac{\partial m}{\partial \theta_j} = -\frac{1}{I}\int_{-\infty}^{\infty} e^{\bar{x}(z)} \frac{\partial \bar{x}(z)}{\partial \theta_j}\rho(z)\,\mathrm{d}z, \tag{2.44}$$

$$\frac{\partial G(z)}{\partial \theta_j} = \int_z^{\infty} e^{x(t)} \left(\frac{\partial m}{\partial \theta_j} + \frac{\partial \bar{x}(t)}{\partial \theta_j}\right)\rho(t)\,\mathrm{d}t. \tag{2.45}$$

The basis columns are fixed functions,

$$b_L(u) = 1 - u, \qquad b_R(u) = u, \qquad b_{a_n}(u) = P_n(1 - 2u).$$

The first pass differentiates the transport speed and integrates it. The normalization dot product removes the component that would change $\mathbb{E}[e^X]$. The reverse pass differentiates the upper-share tail. Every stage follows the direction of the corresponding primal build, so the derivative is obtained by array operations rather than a separate root-and-price routine for each coefficient.

The variance-swap row is obtained from the same sensitivity of x:

$$\frac{\partial w_{\mathrm{vs}}}{\partial \theta_j} = -2\int_{-\infty}^{\infty} \left(\frac{\partial m}{\partial \theta_j} + \frac{\partial \bar{x}(z)}{\partial \theta_j}\right)\rho(z)\,\mathrm{d}z. \tag{2.46}$$

Both analytic and finite-difference Jacobians scale as $O(Pn_{\mathrm{grid}})$ for $P = N + 1$ parameters, but the analytic route avoids $P + 1$ complete rebuilds. In the reference implementation it reaches the same optimum, with costs agreeing to 1.7×10^{-10} relative and parameters to 3.7×10^{-7}, in 1.41–1.76× less time. The worst relative disagreement with an independent central difference is 3.7×10^{-6}.

The theorem is exact for the continuous curves. Numerically, x and G are interpolated separately, so between nodes the price interpolant is not algebraically identical to the integral of the quantile interpolant. The analytic derivative inherits this fourth-order interpolation error. The finite-difference comparison in fig. 2.8 measures the discrete discrepancy instead of assuming the continuous cancellation transfers without loss.

It is worth counting the shared work. One build stores (x, x', G, G') on a fixed grid. A price is one root and two ledger entries; a density is one division; the log-contract variance is one weighted sum; the Jacobian reads ledger sensitivities at the same roots; and calendar order, in section 2.8, will compare two such arrays. These are not five approximations to the law; they are five queries of the same law.

2.6 Calibration

Validity has already been settled by the coordinates. Calibration may therefore concentrate on information: selecting, among admissible laws, one that represents the quoted strip without inventing unsupported detail.

2.6.1 A one-expiry objective

Suppose a slice contains quotes (k_i, σ_i) with $w_i = \sigma_i^2\tau$. The reference fit minimizes

$$\sum_i \left(\frac{c(k_i;\theta) - B(k_i, w_i)}{\partial_\sigma B(k_i, w_i) + \eta}\right)^2 + \alpha \sum_{n \geq 4} n^{2\nu} a_n^2. \tag{2.47}$$

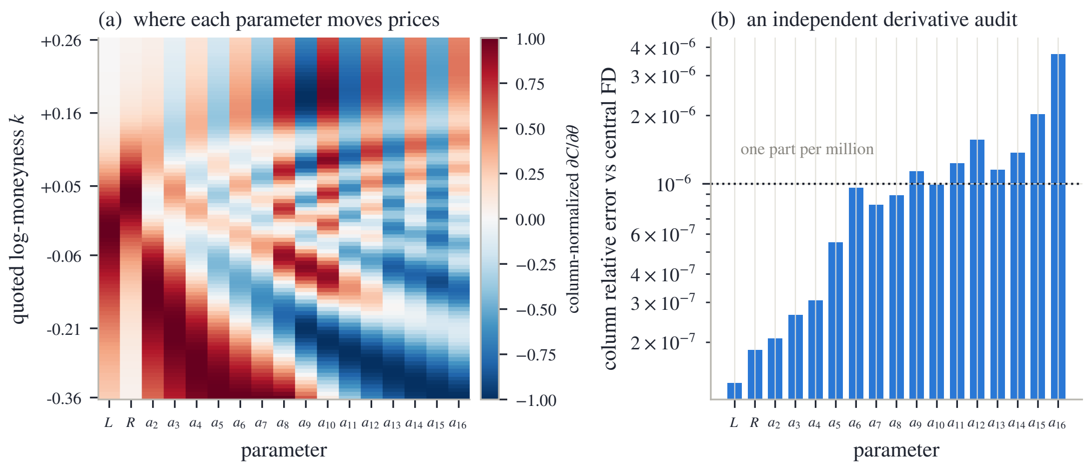

Figure 2.8: The analytic Jacobian on the fitted SPY December 2026 slice. Left: the normalized price-sensitivity pattern of every parameter across strikes. Right: columnwise disagreement with an independent central finite difference. The comparison tests the entire sensitivity pass by a separate route.

Each element of (2.47) answers one practical problem. The residual is computed in price, so implied-volatility inversion stays outside the optimization loop, where it cannot fail on an exploratory trial point. Division by the Black vega $\partial_\sigma B = \varphi(d_+)\sqrt{\tau}$ makes a central residual a volatility error — the unit in which quotes are judged — while the floor η lets the metric degrade to a scaled price error where vega vanishes, instead of assigning unbounded weight to a negligible option. The $n^{2\nu}$ ridge makes fine modes progressively more expensive but leaves a_2 and a_3, the main curvature and asymmetry modes, unpenalized.

The reference objective also contains a smooth row $\log(1 + e^{50(\lambda_+ - 0.90)})$. The admissibility chart already prevents the wall from being reached; this row is instead an explicit prior against a nearly exhausted right-tail moment budget.

A quoted option is an interval, not necessarily a point. Three targets are useful. A *mid* fit uses (2.47) directly. A *band* fit uses the outside-band hinge together with a weak mid anchor. If $[p_i^{\mathrm{lo}}, p_i^{\mathrm{hi}}]$ is the price band and ϕ_i the vega normalizer, its two residuals are

$$\frac{(c_i - p_i^{\mathrm{hi}})^+ + (p_i^{\mathrm{lo}} - c_i)^+}{\phi_i}, \qquad \sqrt{0.05}\,\frac{c_i - p_i^{\mathrm{mid}}}{\phi_i}. \tag{2.48}$$

The squared hinge is continuously differentiable at a band edge, and the weak mid row — its weight deliberately small — selects one point when a wide band leaves a flat valley of equally admissible laws. A *haircut* fit first shrinks the volatility band by h:

$$\sigma_i^{\mathrm{lo}} = \min(\sigma_i^{\mathrm{bid}} + h, \sigma_i^{\mathrm{mid}}), \qquad \sigma_i^{\mathrm{hi}} = \max(\sigma_i^{\mathrm{mid}}, \sigma_i^{\mathrm{ask}} - h). \tag{2.49}$$

If the original spread is narrower than $2h$, the band collapses to the mid instead of changing orientation. This occurs for every SPY quote in the frozen sample; the wider NVDA spreads retain live bands.

The two underliers illustrate both regimes. The SPY December node has median spread 4 vol bp, far inside the default $2h = 100$ vol-bp width, and none of its haircut bands survives. Its figures therefore show raw bid–ask intervals around a mid fit. On the featured NVDA nodes, 47% and 52% of the bands remain live. The same rule becomes a mid target on a liquid index and a genuine interval target on the wider single-name book.

The informational division can be read by location. Between liquid quotes a small change in the law moves observed prices and is strongly penalized: the strip determines the call curve there, and with it the ATM level, skew, curvature, and low-order moments. Near the edge of the strip, data and regularization share control; beyond the outer quote, the endpoint speeds and the ridge supply most of the continuation. The curve itself is smooth — what changes gradually is the source of information, a transition made visible by plotting quote locations with the density as in fig. 2.3. Tail coordinates, ridge, and basis order should be reported with a fit because they make these choices.

2.6.2 Order and starting values

At order N the model has $N+1$ parameters. The reference implementation uses the effective order

$$N_{\text{eff}} = \max\left(\min\left(N, \left\lfloor \frac{n_q}{2} \right\rfloor - 1\right), \min(N, 6)\right), \tag{2.50}$$

where n_q is the quote count. The cap keeps the parameter count well below the number of observations; the floor leaves thin strips enough shape to form a usable smile. The rule is an identification convention, not a validity condition. Changing it changes inter-quote density detail while every resulting slice remains arbitrage-free.

Away from the floor, (2.50) requires roughly two quotes per body degree of freedom. This is not a theorem about optimal smoothing; it is a guard against inferring one oscillatory direction from each noisy observation. On the shortest NVDA slice, 17 quotes give $N_{\text{eff}} = 7$. The fit remains visibly less certain there rather than borrowing high-order detail from the basis.

Cold starts use the constant-speed value (2.19). In a maturity sweep, the previous expiry supplies a warm start. A multi-start check on the featured SPY and NVDA nodes used 10 randomized starts per node and found 1 basin, with worst fit-quality spread 0.00 vol bp. This supports the starting rule on those slices; event-dominated books may still require a broader search.

2.6.3 Coordinates for optimization

Three charts describe the same polynomial family.

1. The *raw chart* is $(L, R, a_2, \ldots, a_N)$. It is simple, but (2.14) couples body modes to both tails.

2. The *endpoint chart* is $(\log\lambda_-, \log\lambda_+, a_2, \ldots, a_N)$. Solving (2.14) for (L, R) is linear. Equivalently, each body basis function is replaced by an endpoint-vanishing combination. Body moves then hold both tail scales fixed.

3. The *logistic chart*, the reference default, writes $\lambda_+ = \operatorname{expit}\xi$ with $\xi \in \mathbb{R}$. Every finite coordinate vector is admissible, and

$$\frac{\mathrm{d}\log\lambda_+}{\mathrm{d}\xi} = 1 - \lambda_+. \tag{2.51}$$

   Steps shrink smoothly as the wall is approached.

The endpoint-vanishing body functions are explicit:

$$\widetilde{P}_n(u) = P_n(1-2u) - (1-u) - (-1)^n u, \qquad n \geq 2. \tag{2.52}$$

They satisfy $\widetilde{P}_n(0) = \widetilde{P}_n(1) = 0$. A tail policy can therefore set (λ_-, λ_+) while calibration adjusts the body in a subspace that cannot move either endpoint.

The objective is identical in all three charts. Fits in the frozen sample agree across charts to 2.2×10^{-7} in the recovered raw parameters. In floating point, $\operatorname{expit}\xi$ eventually rounds to one, so the implementation also enforces $\lambda_+ \leq 1 - 10^{-6}$.

2.6.4 Calibration sequence

The complete one-slice calculation is short enough to state as an algorithm.

1. Convert strikes and prices to (k_i, c_i) using the node's forward and discount factor. Map bid and ask volatilities through Black once to construct price bands and compute the quote vegas used for normalization.

2. Choose the effective order from (2.50) and select mid, band, or haircut residuals. This fixes the data part of the objective before optimization begins.

3. Construct a cold start from the ATM variance using (2.19), or map the previous expiry into the endpoint chart for a warm start. In either case, convert to logistic coordinates so all finite trial vectors are admissible.

4. At each trust-region step, build (x, x', G, G') on the fixed grid of section 2.7, price every quote by (2.28), and obtain the whole residual Jacobian from (2.43)–(2.45). Failed exponent-budget trials return a large residual rather than an undefined price.

5. After convergence, rebuild the accepted parameters on the dense grid. Evaluate ATM handles, tail moments, Lee slopes, the log-contract variance, and the numerical certificates (section 2.7.2) from this one final object.

6. If the slice belongs to a maturity sequence, add calendar rows against the accepted nearer slice and repeat. Report any residual common-support gap in normalized price units.

This order matters. Market conventions are fixed before the solve; probability validity holds during the solve; diagnostics are computed after the solve from the same law that produced the quoted prices. No separate post-processing smoother is inserted between calibration and publication.

2.7 Numerical realization

The theorems of the preceding sections concern the continuous objects x and G; the fitter computes with discrete realizations of them. One grid build produces every array used above, analytic corrections handle what lies beyond the grid, and two certificates verify that discretization has preserved what the theory guarantees.

2.7.1 One grid build

All integrals use a symmetric log-odds grid $[-Z_{\max}, Z_{\max}]$, with $Z_{\max} = 40$. Optimization uses 2001 nodes; the accepted slice is rebuilt on 8001 nodes. A build performs four operations: evaluate g, integrate e^g cumulatively to form $\bar{x}$, compute the martingale shift, and integrate $e^x\rho$ backward to form G. Exact nodal derivatives $x' = e^g$ and $G' = -e^x\rho$ accompany the values. Cubic Hermite interpolation then gives fourth-order off-grid evaluation and a Newton-polished strike root.

The missing logistic tails are integrated analytically. To leading order,

$$\int_{Z_{\max}}^{\infty} e^{\bar{x}}\rho\,\mathrm{d}z \simeq \frac{e^{\bar{x}(Z_{\max})-Z_{\max}}}{1-\lambda_+}, \qquad \int_{-\infty}^{-Z_{\max}} e^{\bar{x}}\rho\,\mathrm{d}z \simeq \frac{e^{\bar{x}(-Z_{\max})-Z_{\max}}}{1+\lambda_-}. \tag{2.53}$$

For a strike beyond the right return grid, continue the root along the tail asymptote,

$$z_R(k) = Z_{\max} + \frac{k - x(Z_{\max})}{\lambda_+}.$$

Both the ledger and cash leg are then single exponentials. Their difference is

$$c(k) \simeq e^{k - z_R(k)} \frac{\lambda_+}{1 - \lambda_+}. \tag{2.54}$$

The left-wing continuation follows from parity.

To derive the factor in (2.54), restart the right-tail correction at $z_R(k)$, where $x = k$. The asset leg is approximately $e^{k - z_R(k)}/(1 - \lambda_+)$ and the cash leg is $e^{k - z_R(k)}$. Subtraction leaves $\lambda_+/(1 - \lambda_+)$. The continuation is therefore the tail form of the same asset-minus-cash identity (2.28), not an independent wing model.

Three floating-point rules are essential. First, compute $\rho(z) = \operatorname{expit}(z) \operatorname{expit}(-z)$ rather than $u(1-u)$, since u rounds to one for large positive z. Compute the cash leg as $\exp\{k - \log(1 + e^{z_k})\}$ using a stable log-sum-exp. Second, reject a trial if an interior value of g or $\bar{x}$ exceeds the exponent budget, even when the endpoints are admissible. Third, evaluate the Lee function at large r in the rationalized form

$$\Psi(r) = 2 - \frac{4}{\sqrt{1 + 1/r} + 1}, \tag{2.55}$$

which avoids subtracting nearly equal large numbers. The full build and sensitivity pseudocode appear in appendix 2.B.

2.7.2 Certificates

The continuous theorems concern x and G; numerical prices use their interpolants. Two inexpensive certificates connect the two levels. On each Hermite segment, the Fritsch–Carlson inequalities require both endpoint derivatives to lie between zero and three times the secant slope [10]. Applying them to x and $-G$ certifies monotone interpolation; segments below round-off are declared flat to a stated tolerance. Across the traded strike span, the analytic derivatives of $w(k)$ are also inserted in (2.5) on a dense grid and required to satisfy $g_{\mathrm{D}} \geq -10^{-4}$.

An independent randomized battery prices 27 admissible slices across several orders and stress classes at sub-grid strikes. Its worst errors in call bounds, butterflies, digitals, and agreement with a four-times-finer grid are respectively 1.1×10^{-11}, 8.2×10^{-13}, 1.5×10^{-12}, and 2.3×10^{-9} in normalized price. The exact constant-speed test of fig. 2.4 independently checks transport, normalization, tail correction, and root evaluation. These numerical tests do not create validity; they verify that the discrete realization has preserved it to the reported tolerance.

The two-grid rebuild supplies a final resolution check. Parameters fitted on 2001 nodes and refitted on 8001 nodes agree to 1.4×10^{-6}. The accepted vector is rebuilt once on the dense grid, so prices, densities, ATM handles, and calendar ledgers all read the same final discrete object.

These diagnostics answer different questions and should not be collapsed into one flag. Quote residuals measure agreement with the strip; effective order and ridge describe the asserted inter-quote structure; endpoint speeds describe the extrapolation and its moment budget; monotonicity and belly margins certify that discretization preserved validity; calendar gaps describe compatibility across expiries. A fit can be excellent on one axis and weak on another.

2.8 Calendar order

Butterfly freedom is a property of one marginal law. A surface also requires later expiries to dominate earlier ones in convex order. Under deterministic rates and carry, this comparison has a simple form in normalized units.

Lemma 2.14 (Calendar ordering at equal log-moneyness)**.** *Let $F_{0,t}$ be the deterministic forward curve and suppose $M_t = S_t/F_{0,t}$ is a positive martingale. For $T_1 < T_2$ and every k, the normalized calls satisfy*

$$c_1(k) \leq c_2(k). \tag{2.56}$$

Proof. Since $M_{T_i} = S_{T_i}/F_i = Y_i$, conditional Jensen gives

$$\begin{aligned} c_2(k) &= \mathbb{E}[(M_{T_2} - e^k)^+] \\ &\geq \mathbb{E}[(\mathbb{E}[M_{T_2} \mid \mathcal{F}_{T_1}] - e^k)^+] = \mathbb{E}[(M_{T_1} - e^k)^+] = c_1(k). \end{aligned}$$

□

Equal k means strikes $K_i = F_i e^k$, which are generally different dollar strikes. The comparison is a forward-scaled calendar relation, not the usual listed equal-strike spread. In probability language, the laws of Y_i have equal means and increase in convex order.

The scaling is essential. A violation at k is monetized by shorting $1/(D_1F_1)$ calls at $K_1 = F_1e^k$ and buying $1/(D_2F_2)$ calls at $K_2 = F_2e^k$. After normalization both payoffs are convex functions of the common martingale M_t. Without deterministic carry, one must first specify the numeraire and the objects being compared; equal log-moneyness alone is not a model-free calendar relation.

2.8.1 Calls and ledgers are conjugate

Lemma 2.15 (Ledger–call conjugacy). *For any admissible slice,*

$$\begin{aligned} c(k) &= \max_{z\in[-\infty,\infty]} \{G(z) - e^k(1 - \Lambda(z))\}, \\ G(z) &= \min_{k\in[-\infty,\infty]} \{c(k) + e^k(1 - \Lambda(z))\}. \end{aligned} \tag{2.57}$$

The optimizers are $z = z_k$ and $k = x(z)$, respectively.

Proof. For fixed k,

$$G(z) - e^k(1 - \Lambda(z)) = \int_z^\infty (e^{x(t)} - e^k)\rho(t)\,\mathrm{d}t.$$

The integrand is negative before z_k and positive after it. Starting the integral at z_k therefore maximizes it and gives the call price. The first identity implies $c(k) + e^k(1 - \Lambda(z)) \geq G(z)$ for every k, with equality at $k = x(z)$, proving the second. □

Geometrically, the first identity scans all possible exercise ranks. Before z_k, the integral includes outcomes where the payoff is negative; after z_k, it discards outcomes where the payoff is positive. The strike root is the unique point of tangency between call and ledger descriptions. The second identity reconstructs the ledger from the full call curve, so no information is lost in changing representation.

Theorem 2.16 (Calendar order is ledger order). *For two admissible mean-one slices,*

$$c_1(k) \leq c_2(k) \textit{ for all } k \quad \Longleftrightarrow \quad G_1(z) \leq G_2(z) \textit{ for all } z. \tag{2.58}$$

Proof. If $G_1 \leq G_2$, the maximum representation in (2.57) gives $c_1 \leq c_2$. If $c_1 \leq c_2$, the minimum representation gives $G_1 \leq G_2$. Both implications are pointwise. □

Thus the continuum of calendar inequalities is equivalent to ordering two arrays already produced by the slice builds. This is also the classical convex-order criterion: by the results of Strassen and Kellerer, a family of equal-mean laws increasing in convex order can be realized as the marginals of a martingale [19, 27].

This is an existence statement. Ordered ledgers make the marginal laws compatible with some martingale dynamics; they neither select that dynamics nor determine forward smiles. The calendar problem treated here is exactly the static marginal problem.

2.8.2 A support-confined control

The theorem is an exact diagnostic. A finite calibration requires a chosen enforcement rule. When fitting a farther expiry after a nearer one, the reference objective adds

$$\sqrt{w_{\mathrm{cal}}}\,\zeta_m\{c_{\mathrm{near}}(k_m) - c_{\mathrm{far}}(k_m)\}^+ \tag{2.59}$$

on a grid of strikes in the intersection of their quoted spans. The weights ζ_m taper to zero with a $\cos^2$ profile over the outer fifteen percent of that span. The number of nodes is $\max(49, 4N+1)$.

The rows are imposed in price space, not ledger space. A ledger value at one rank integrates the entire upper tail, so constraining it couples the quoted region of one expiry to the unquoted continuation of another: the ledger is ideal for diagnosis precisely because it sees the whole law, and unsuitable as a constraint for the same reason. In a reproduced stress comparison, a full-grid ledger floor moved the worst quote error from 10.6 to 1095.0 vol bp; fixed-strike call rows, each a specified payoff in the common quoted region, localize the control. The weight is finite, and any remaining violation is reported as a price gap and as a fraction of the local bid–ask width. Calendar order is thus controlled on common quoted support; outside it, the ledger comparison remains a diagnostic of the extrapolations.

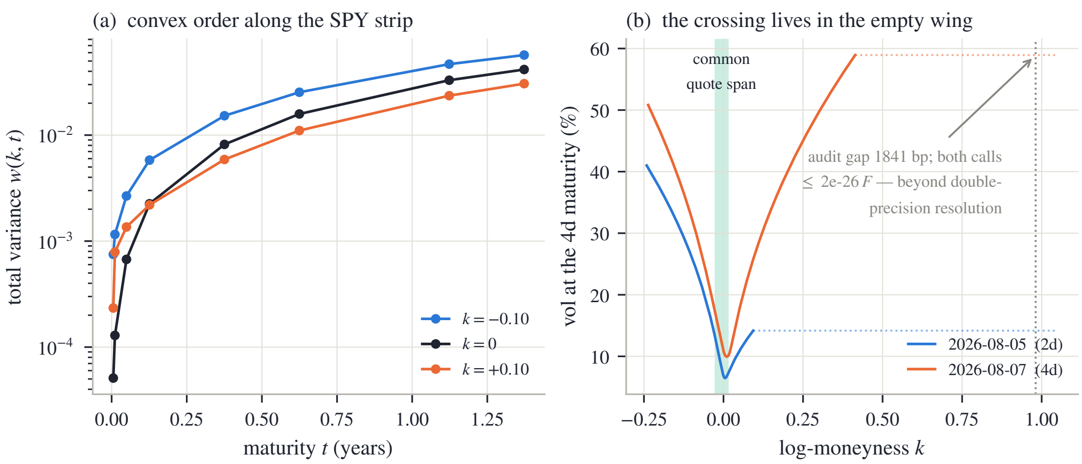

Figure 2.9: Calendar order in the frozen SPY sample. Left: total variance is ordered through maturity at representative log-moneyness values. Right: the two shortest expiries are ordered on their common quoted span, while a volatility-space crossing appears only in a numerically empty far wing. The price-space gap there is at round-off scale.

For the two shortest SPY expiries, a full-grid volatility audit flags a 1840.6 vol-bp crossing at $k = +0.981$, far outside the common quote span $[-0.028, +0.016]$. The corresponding calls are 5.4×10^{-107} and 1.6×10^{-26} of forward. A direct price audit finds a worst violation of only 2.2×10^{-15}. The crossing is a disagreement between extrapolations in a region where Black inversion is ill-conditioned, not an economically available calendar trade.

2.9 Examples and the model contract

All market examples use one frozen after-hours snapshot of SPY and NVDA listed options from the 2026-08-03 US session. Each underlier has eight expiries, for 16 fitted nodes in total. The common recipe is order $N = 16$ subject to (2.50), the logistic chart, the haircut target (2.49), ridge $\alpha = 10^{-6}$, and the calendar rows (2.59). No node is tuned separately. Across the sample the median quote error is 5.9 vol bp and the worst is 24.6 vol bp.

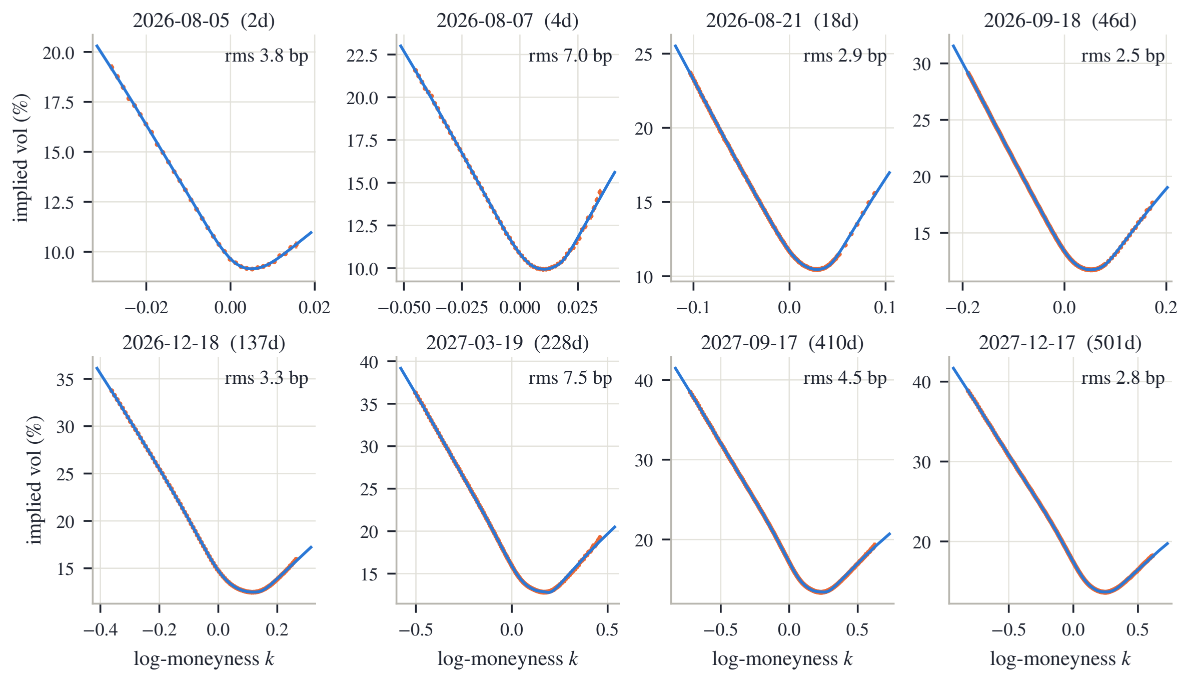

Figure 2.10: Eight SPY expiries fitted with one recipe. Each panel shows raw bid–ask quotes, mids, the fitted LQD slice, and rms error. The short end is steeper and more localized; the long end broadens smoothly. Beyond the last quote, each curve is determined by the fitted exponential-tail convention.

2.9.1 SPY: a term structure and one slice

The gallery separates recurring market shape from the completion chosen by the model. Through the quoted region the slices follow the term structure of level and skew; outside it the two endpoint speeds continue the curve. The boundary is most visible at the short end, where quoted spans are narrow in log-moneyness: the sharp local skew is supported by quotes, while the remote upward turn is the exponential-tail completion. At longer maturities the span widens and a larger fraction of the displayed smile is directly informed by prices. The median error across the eight slices is 3.6 vol bp; the worst is 7.5 vol bp.

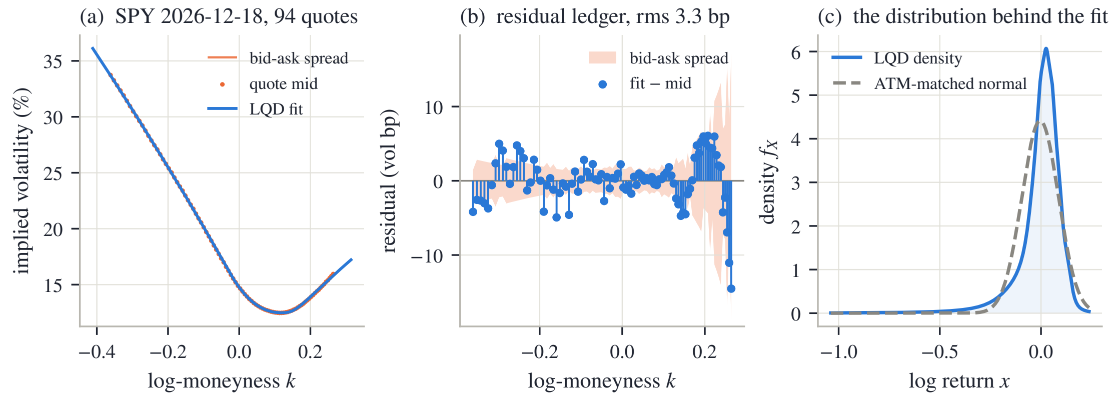

Figure 2.11: The SPY December 2026 slice. Left: quotes and fitted volatility. Center: signed residuals in vol bp. Right: the fitted density against an ATM-matched normal. The density view makes the negative skew visible as a transfer of mass toward the left tail.

The December 2026 slice contains 94 quotes and fits them to 3.3 vol bp rms. Its exact ATM quantities from proposition 2.12 are volatility 14.76%, skew −0.430, and curvature 4.40. The forward rank is 41.31%%, below the matched Black rank; hence $\Delta < 0$, in agreement with the negative skew. The log-contract variance corresponds to 19.84% annualized volatility.

The three panels answer different questions. The first tests the nonlinear map from law to implied volatility. The second asks whether residuals retain systematic strike structure. The third removes the Black coordinate and shows the fitted object itself: a left-heavy density. The ATM-matched normal has the same central price level but not the same digital or tail allocation, and therefore cannot reproduce the skew.

The ledger also makes an individual option transparent. For the listed December 800 call, $k = 0.0409$ and the strike rank is 65.27%%. The upper-share and cash entries are 0.378366 and 0.361834, so

$$c(0.0409) = 0.378366 - 0.361834 = 0.016532.$$

This is \$12.70 per share and corresponds to 13.37% implied volatility.

2.9.2 NVDA and the order guard

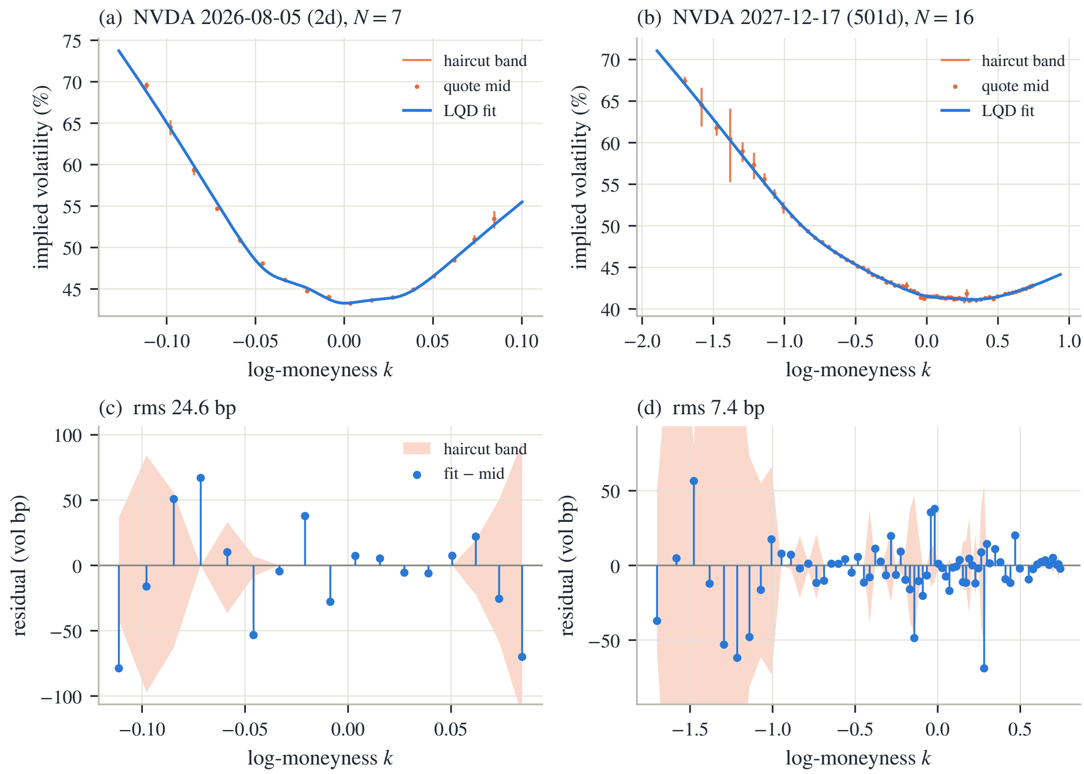

Figure 2.12: Two NVDA slices. Left: the shortest expiry, with 17 quotes and effective order $N_{\text{eff}} = 7$. Right: the December 2027 expiry at full order. Wider NVDA spreads leave live haircut bands and permit visibly larger residual variation than SPY.

The shortest node fits at 24.6 vol bp rms; the long node, with 69 quotes, at 7.4 vol bp. The latter has heavier fitted tails than SPY. In particular its left endpoint speed exceeds one, which is admissible: only the right endpoint is constrained by the finite-forward wall. The consequence is a last finite inverse-moment order 0.70 below one.

The shortest node is also the clearest example of the order guard. A thin strip cannot distinguish sixteen smooth modes, even though an optimizer can assign values to them. Reducing N_{eff} converts that lack of information into visible residual uncertainty instead of invisible oscillation between quotes. The long node has enough observations to use the full order and separate body curvature from tail continuation.

2.9.3 What basis order asserts

The synthetic benchmark is a martingale-normalized mixture of two lognormal components. It has a genuinely bimodal density but only gentle shoulder structure in implied volatility.

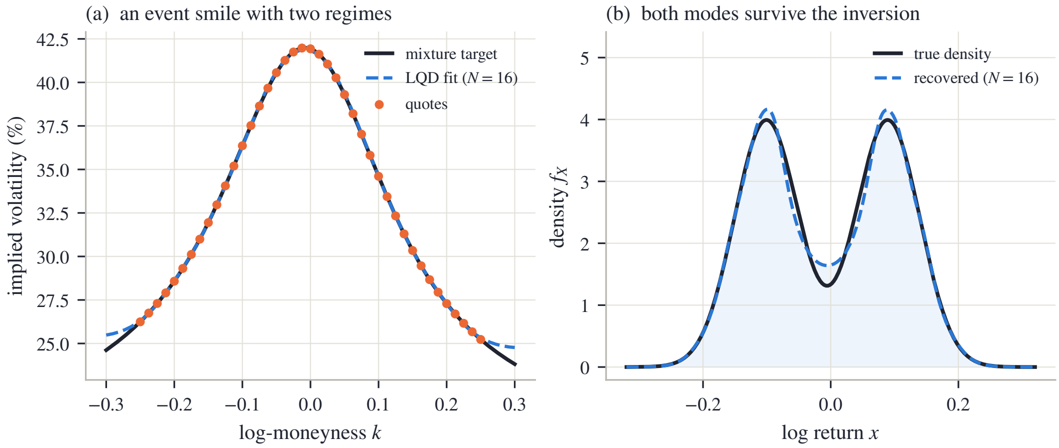

Figure 2.13: The double-hump benchmark at $N = 16$. Left: the fitted smile matches the synthetic quotes to 4.8 vol bp rms. Right: the fitted density recovers both modes, the trough, and the asymmetry of the target.

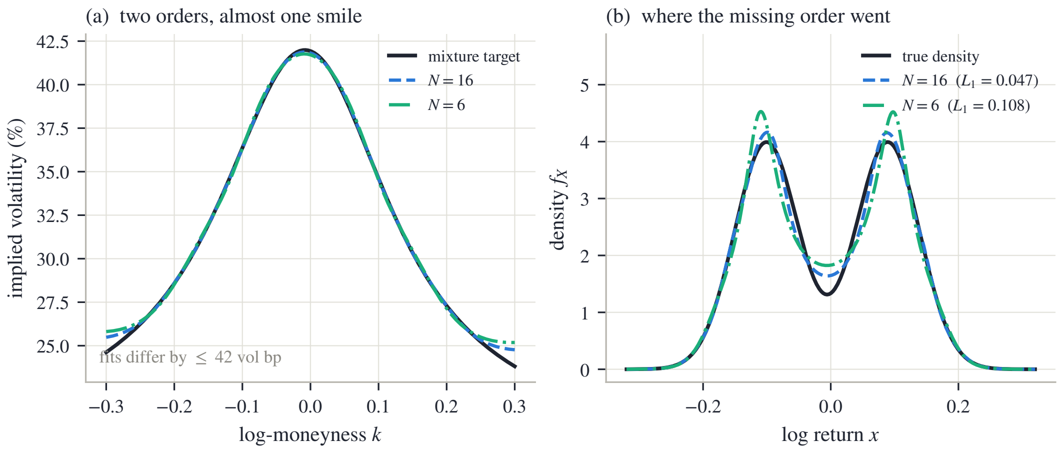

Figure 2.14: The same target at orders 6 and 16. Left: the implied-volatility curves nearly coincide over the quoted region. Right: the densities do not; the lower-order fit displaces the modes and fills part of the trough. Basis order controls how much inter-quote structure the model is allowed to assert.

At $N = 16$ the fitted mode locations are -0.099 and 0.088, against true locations -0.101 and 0.089. At $N = 6$ both modes remain, but their shape is distorted. The L^1 density error rises from 0.047 to 0.108, even though the two implied-volatility curves differ by at most 41.8 vol bp on the comparison grid. Option quotes can be nearly indifferent between laws that price density-sensitive structures differently.

The lesson is not that high order is always preferable. At $N = 16$ the synthetic target is known to contain fine structure, so extra modes recover a feature known to be real. In market data the truth is not supplied. High order then grants the model permission to assert finer inter-quote density shape. Digitals and narrow butterflies are sensitive to that permission; broad vanilla prices may not be. Basis order should be chosen for the downstream instruments, not solely for the in-sample volatility residual.

2.9.4 Position among existing models

The construction joins several classical ideas. Breeden–Litzenberger identify option prices with a density [6]; Lee and Benaim–Friz connect moments with wing slopes [2, 20]; quantile modeling supplies the rank coordinate [14, 23]; and Petersen–Müller use the same log-quantile-density transform to place density-valued data in an unconstrained function space [24]. The metalog family is a related direct quantile expansion [18].

SVI and SSVI work in volatility coordinates and characterize admissible parameter regions [11, 13]; arbitrage-free smoothing works directly with constrained prices [9]. LQD instead makes the probability law primary. Its contribution is the combined architecture: structural one-slice validity, readable tail coordinates, and one ledger for pricing, differentiation, density, variance, and calendar order.

2.9.5 The contract

Two things remain: how to read a fitted slice, and what precisely is claimed.

A fitted slice should be read in the same order as the theory. First inspect quote residuals over the retained span. Then read the three ATM handles as local level, digital, and density information. Next inspect $g(u)$ and the density over the ranks occupied by quotes. Only after that should one read $\lambda_{\pm}$, the moment strip, and Lee slopes, since these describe the extrapolated endpoints. Finally compare the ledger with neighboring expiries. This sequence prevents a tail convention or a calendar diagnostic from being mistaken for a directly observed quote.

Structural	**Numerically controlled**	**Outside the claim**
Positive density, unit mean, and butterfly-free calls for every admissible slice	Interpolation monotonicity, traded-range density, and randomized price audits	Atoms, mass at zero, density gaps, bounded support, and Gaussian-or-faster tails
Exact exponential tail scales, moment strip, and Lee limits	Tail quadrature, exponent budgets, grid convergence, and finite-strike wing prices	Tail parameters as observations, or asymptotic slopes as finite-strike quotes
Exact ATM handles, analytic Jacobian, and ledger characterization of calendar order	Quote fit, basis-order control, multi-start checks, and calendar order on common quoted support	Dynamics, jump decomposition, realized-variance corrections, and calendar structure outside the controlled support

Table 2.2: The model contract: theorem, numerical control, and excluded scope.

The central distinction is now visible in one place. Probability supplies validity. Quotes supply information over a finite strip. Basis, ridge, tails, and calendar scope supply convention elsewhere. The value of the LQD representation is not that convention disappears, but that it can no longer be mistaken for a theorem or for market information.

2.A Proofs deferred from the main text

2.A.1 Derivation of the volatility-space density factor

Let $c(k) = B(k, w(k))$. For general (k, w), the identity $e^k\varphi(d_-) = \varphi(d_+)$ gives

$$\partial_k B = -e^k\Phi(d_-), \qquad \partial_w B = \frac{\varphi(d_+)}{2\sqrt{w}}.$$

The second partials required by the chain rule are

$$\partial_{kk} B = -e^k\Phi(d_-) + \frac{\varphi(d_+)}{\sqrt{w}},$$

$$\partial_{kw}B = \frac{d_+\varphi(d_+)}{2w},$$
$$\partial_{ww}B = -\frac{d_+\varphi(d_+)}{2\sqrt{w}}\,\partial_w d_+ - \frac{\varphi(d_+)}{4w^{3/2}}, \qquad \partial_w d_+ = \frac{k}{2w^{3/2}} + \frac{1}{4\sqrt{w}}.$$

Differentiate $c(k)$ twice and subtract the first derivative:

$$c'' - c' = (\partial_{kk}B - \partial_k B) + (2\partial_{kw}B - \partial_w B)w' + \partial_{ww}B(w')^2 + \partial_w B w''.$$

Factor $\varphi(d_+)/\sqrt{w}$. Since $d_+ = -k/\sqrt{w} + \sqrt{w}/2$,

$$2\partial_{kw}B - \partial_w B = -\frac{k\varphi(d_+)}{w^{3/2}},$$

and a direct multiplication gives

$$d_+\partial_w d_+ = -\frac{k^2}{2w^2} + \frac{1}{8}.$$

Therefore

$$\begin{aligned} c'' - c' &= \frac{\varphi(d_+)}{\sqrt{w}}\left[1 - \frac{kw'}{w} + \left(\frac{k^2}{4w^2} - \frac{1}{4w} - \frac{1}{16}\right)(w')^2 + \frac{w''}{2}\right] \\ &= \frac{\varphi(d_+)}{\sqrt{w}}\left[\left(1 - \frac{kw'}{2w}\right)^2 - \frac{(w')^2}{4}\left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w''}{2}\right]. \end{aligned}$$

Finally, $f_X(k) = e^{-k}(c'' - c')$ and $e^{-k}\varphi(d_+) = \varphi(d_-)$, which proves (2.5).

2.A.2 Proof of the ATM formulas

The smile is defined by $B(k, w(k)) = c(k)$. Put $w_0 = w(0)$ and $d = \sqrt{w_0}/2$. The identity $e^k\varphi(d_-) = \varphi(d_+)$ follows from $d_+^2 - d_-^2 = -2k$ and gives

$$\partial_k B = -e^k\Phi(d_-), \qquad \partial_w B = \frac{\varphi(d_+)}{2\sqrt{w}}.$$

Differentiating once more and evaluating at $(0, w_0)$ gives

$$\begin{aligned} \partial_k B &= -(1 - \Phi(d)), & \partial_w B &= \frac{\varphi(d)}{2\sqrt{w_0}}, \\ \partial_{kk}B &= -(1 - \Phi(d)) + \frac{\varphi(d)}{\sqrt{w_0}}, & \partial_{kw}B &= \frac{\varphi(d)}{4\sqrt{w_0}}, \\ \partial_{ww}B &= -\frac{\varphi(d)}{4w_0^{3/2}} - \frac{\varphi(d)}{16\sqrt{w_0}}. \end{aligned} \tag{2.60}$$

By (2.30)–(2.32),

$$c'(0) = -(1 - u_*), \qquad c''(0) = f_* - (1 - u_*).$$

One derivative of the defining identity gives $\partial_k B + \partial_w B\, w'(0) = c'(0)$, hence

$$w'(0) = \frac{-(1 - u_*) + (1 - \Phi(d))}{\varphi(d)/(2\sqrt{w_0})} = \frac{2\sqrt{w_0}}{\varphi(d)}\Delta.$$

Two derivatives give

$$\partial_{kk}B + 2\partial_{kw}B\, w' + \partial_{ww}B\,(w')^2 + \partial_w B\, w'' = c''.$$

Solving at the money and using (2.60),

$$\begin{aligned} w''(0) &= \frac{2\sqrt{w_0}}{\varphi(d)}\Bigg[f_* + \Delta - \frac{\varphi(d)}{\sqrt{w_0}} - \frac{\varphi(d)}{2\sqrt{w_0}}w'(0) \\ &\qquad + \left(\frac{\varphi(d)}{4w_0^{3/2}} + \frac{\varphi(d)}{16\sqrt{w_0}}\right)w'(0)^2\Bigg] \\ &= \frac{2\sqrt{w_0}}{\varphi(d)}f_* - 2 + \left(\frac{1}{2w_0} + \frac{1}{8}\right)w'(0)^2. \end{aligned}$$

The terms linear in Δ cancel because $w'(0) = 2\sqrt{w_0}\Delta/\varphi(d)$.

Finally, $\sigma = \sqrt{w/\tau}$ gives

$$\sigma' = \frac{w'}{2\sqrt{\tau w}}, \qquad \sigma'' = \frac{w''}{2\sqrt{\tau w}} - \frac{(w')^2}{4\sqrt{\tau}\,w^{3/2}}.$$

Substitution yields (2.39)–(2.40).

2.A.3 Proof of universality and the closure dichotomy

Let g^* satisfy proposition 2.6, write $\lambda_+^* = e^{g^*(1)} < 1$, and fix $\bar\lambda \in (\lambda_+^*, 1)$.

Polynomial approximation. By Weierstrass there are polynomials p_N with $\varepsilon_N = \|p_N - g^*\|_\infty \to 0$. Every polynomial of degree at most N is represented by the endpoint chord and $P_2, \dots, P_N$. For all large N,

$$p_N(1) \le g^*(1) + \varepsilon_N < \log\bar\lambda < 0,$$

so the approximating slice is admissible.

Transport convergence and a common envelope. Let $\bar x_N$ be the corresponding unshifted transport. On every finite z-interval,

$$|\bar x_N(z) - \bar x^*(z)| \le (e^{\varepsilon_N} - 1)e^{\|g^*\|_\infty}|z|,$$

and therefore $\bar x_N \to \bar x^*$ locally uniformly. Continuity of g^* at the right endpoint gives z_0 such that, for all large N, $e^{p_N(\Lambda(z))} \le \bar\lambda$ when $z \ge z_0$. Hence a constant C_1, independent of N, satisfies

$$e^{\bar x_N(z)} \le C_1 e^{\bar\lambda z} \quad (z \ge 0), \qquad e^{\bar x_N(z)} \le C_1 \quad (z \le 0). \tag{2.61}$$

Normalizers, shifts, and quantiles. The functions $e^{\bar x_N}\rho$ converge pointwise and, by (2.61), are dominated by integrable multiples of $e^{(\bar\lambda - 1)z}$ on the right and e^z on the left. Dominated convergence gives $I_N \to I^*$, so $m_N = -\log I_N \to m^*$. Thus $x_N \to x^*$ locally uniformly and $Q_N \to Q^*$ uniformly on compact subsets of $(0, 1)$.

Uniform convergence of calls. Couple all laws with one uniform rank and let $Y_N = e^{Q_N(U)}$, $Y^* = e^{Q^*(U)}$. Since positive-part payoffs are one-Lipschitz,

$$\sup_k |c_N(k) - c^*(k)| \le \mathbb{E}|Y_N - Y^*|.$$

The integrand converges pointwise. The envelope above and boundedness of m_N give a constant C_2 such that $e^{Q_N(u)} \le C_2\{1 + (1-u)^{-\bar\lambda}\}$. This is integrable, so the calls converge uniformly.

Wasserstein convergence. The quantile coupling is optimal on the line:

$$W_p(Y_N, Y^*)^p = \int_0^1 |e^{Q_N(u)} - e^{Q^*(u)}|^p \,\mathrm{d}u.$$

For $1 \le p < 1/\lambda_+^*$ choose $\bar{\lambda} \in (\lambda_+^*, 1/p)$. The integrand is dominated by a multiple of $1+(1-u)^{-p\bar{\lambda}}$, proving the claim by dominated convergence.

Finite-order exclusions. At finite order, x'_N is positive and continuous. The quantile is a strictly increasing homeomorphism from $(0,1)$ onto $\mathbb{R}$; it can neither jump, remain flat, nor terminate. This excludes atoms, density gaps, bounded support, and mass at zero. A Gaussian-or-faster tail would require $g(u) \to -\infty$ at an endpoint, which no polynomial can do.

Uniform closure away from the wall. Suppose admissible g_N converge uniformly to g_∞. Necessarily $g_\infty(1) \le 0$. If $g_\infty(1) < 0$, the preceding dominated-convergence argument applies with g_∞ as target. The limit retains a positive continuous density and exponential tails.

The boundary branch. If $g_\infty(1) = 0$, weak convergence may fail. For $g_N \equiv \log(1 - 1/N)$, the solvable slice gives

$$I_N = \frac{\pi(1-1/N)}{\sin(\pi/N)} \longrightarrow \infty, \qquad m_N = -\log I_N \longrightarrow -\infty.$$

Every fixed-rank quantile tends to $-\infty$, so mass escapes.

Assume instead that the laws converge weakly. The shifts are bounded above because every normalizer has a positive central contribution. They are bounded below because $m_N \to -\infty$ along a subsequence would contradict tightness. Every convergent subsequence of shifts then yields

$$Q_\infty(u) = m_\infty + \int_0^{\operatorname{logit} u} e^{g_\infty(\Lambda(t))}\,\mathrm{d}t.$$

Weak convergence identifies this quantile uniquely, so the full shift sequence converges. The limiting speed is continuous and bounded away from zero; hence the limit has full support and a strictly positive continuous density, and all geometric exclusions remain.

The right endpoint speed is one. For each $\delta > 0$ it is eventually at least $1-\delta$, so the argument of proposition 2.8 gives $\mathbb{E}[e^{rX}] = \infty$ for every $r > 1$. Fatou also gives $\mathbb{E}[e^X] \le \liminf_N \mathbb{E}[e^{X_N}] = 1$, with equality not guaranteed. This is the boundary law in proposition 2.6.

2.B Numerical implementation

This appendix records the load-bearing parts of a reference implementation. The pseudocode follows the notation of the chapter; tail-sensitivity terms and error handling are described after the listings.

2.B.1 Slice build

Listing 2.1: Build the transport and ledger on one fixed grid.

```
def build(theta, grid):
    L, R, a = theta.L, theta.R, theta.a
    g = (1 - grid.u)*L + grid.u*R + a @ grid.Pmat
    if not finite(g) or max(g) > 700:
        raise InteriorOverflow

    dx = exp(g)                          # x'(z)
```

```
    xbar = cumulative_simpson(dx, anchor=grid.mid)
    if max(xbar) > 700:
        raise InteriorOverflow

    lam_m, lam_p = exp(g[0]), exp(g[-1])
    if lam_p >= 1 - 1e-6:
        raise InadmissibleTail

    mass = exp(xbar) * grid.rho
    I = integrate(mass) \
        + exp(xbar[-1] - grid.zmax)/(1 - lam_p) \
        + exp(xbar[0]  - grid.zmax)/(1 + lam_m)
    m = -log(I)
    x = m + xbar

    dG = -exp(x) * grid.rho              # G'(z)
    G = reverse_integrate(-dG) \
        + exp(x[-1] - grid.zmax)/(1 - lam_p)
    return Slice(x=x, dx=dx, G=G, dG=dG,
                 m=m, lam=(lam_m, lam_p), theta=theta)
```

The cached density must be computed as `expit(z)*expit(-z)`. Forming `u*(1-u)` loses the right tail when `u` rounds to one. Both x and G carry their exact nodal derivatives, so a cubic Hermite representation needs no global spline solve.

The median is the natural anchor for the cumulative transport. It fixes $\bar{x}(0) = 0$ and lets a single forward cumulative sum build the positive half-line while the same stencil, reversed, builds the negative half-line. The additive freedom is then carried entirely by m. Simpson quadrature is fourth order on the uniform grid; the Hermite interpolant has the same local order because both values and exact nodal derivatives are available.

The grid does not truncate the probability law. From (2.16), on the right

$$e^{\bar{x}(z)}\rho(z) = e^{b_+ + (\lambda_+ - 1)z}\{1 + O(e^{-z})\}.$$

Integrating its leading term from $Z_{\max}$ gives the first correction in (2.53); the relative error is $O(e^{-Z_{\max}})$. The left correction follows from $e^{\bar{x}(z)}\rho(z) \sim e^{b_- + (\lambda_- + 1)z}$. The factor $1 - \lambda_+$ in the right denominator is also a numerical warning: tail mass becomes increasingly sensitive as the finite-forward wall is approached.

2.B.2 Call evaluation

Listing 2.2: Evaluate a normalized call.

```
def call_price(slc, k, grid):
    if k > slc.x[-1]:
        z = grid.zmax + (k - slc.x[-1]) / slc.lam[1]
        return exp(k - z) * slc.lam[1] / (1 - slc.lam[1])
    if k < slc.x[0]:
        return 1 - exp(k) + put_asymptote_left(slc, k, grid)

    z = hermite_root(slc.x, slc.dx, k)
    share = hermite_eval(slc.G, slc.dG, z)
    cash = exp(k - logaddexp(0, z))       # e^k(1-Lambda(z))
    return share - cash
```

The log-space cash expression is essential. The algebraically simpler `exp(k)*(1-expit(z))` becomes zero after rank saturation and can create a spurious upward step in the far call wing.

The two branches beyond the return grid are part of the same law. On the right, (2.54) continues the exact power tail. On the left, put asymptotics are computed first and call parity is applied. This avoids subtracting two quantities close to one when a deep in-the-money call is evaluated directly.

2.B.3 Sensitivity pass

Listing 2.3: All raw-coordinate Jacobian columns on one build.

```
def sensitivities(slc, grid):
    # rows: dg/dL, dg/dR, dg/da_2, ..., dg/da_N
    B = vstack([1-grid.u, grid.u, grid.Pmat])
    dxbar = cumulative_simpson(slc.dx * B,
                               anchor=grid.mid, axis=1)
    weight = exp(slc.x) * grid.rho
    dm = -integrate(weight * dxbar, axis=1)
    dx = dm[:, None] + dxbar

    integrand = weight[None, :] * dx
    dG = reverse_integrate(integrand, axis=1)
    dG_slope = -integrand
    dvs = -2 * integrate(dx * grid.rho, axis=1)
    return dG, dG_slope, dvs

def price_jacobian_row(slc, sens, k):
    z = hermite_root(slc.x, slc.dx, k)
    return hermite_eval(sens.dG, sens.dG_slope, z)
```

The derivatives of the two tail corrections in (2.53) must also be included. They use the endpoint sensitivities

$$\partial\lambda_-/\partial\theta = \lambda_-(1,0,1,1,\ldots), \qquad \partial\lambda_+/\partial\theta = \lambda_+(0,1,1,-1,\ldots).$$

Derivatives in the endpoint or logistic chart follow by multiplication with the corresponding chart Jacobian.

The analytic pass performs one cumulative integral for each basis column, one normalization dot product, and one reverse cumulative integral. Its work is $O(Pn_{\text{grid}})$ and its storage may be either $O(Pn_{\text{grid}})$, when all columns are retained, or $O(n_{\text{grid}})$ per streamed column. Finite differences have the same formal order but repeat normalization, ledger construction, and every strike root $P+1$ times. Envelope cancellation removes the roots from the analytic derivative and is therefore responsible for most of the measured constant-factor gain.

2.B.4 Interpolation certificate

Listing 2.4: Fritsch–Carlson margin for a uniform-grid Hermite interpolant.

```
def monotone_margin(y, d, dz, flat_tol=1e-9):
    sec = diff(y) / dz
    d0, d1 = d[:-1], d[1:]
    active = ~((abs(sec) < flat_tol) &
               (abs(d0) < flat_tol) &
               (abs(d1) < flat_tol))
    margin = minimum.reduce([d0, d1,
                             3*sec-d0, 3*sec-d1])
    return min(margin[active])
```

An increasing interpolant is certified when the returned margin is positive. Apply the same routine to $-G$. Segments excluded by `flat_tol` are recorded as flat to tolerance rather than strictly monotone.

For a cubic Hermite segment, let s be its secant slope and d_0, d_1 its endpoint derivatives. The sufficient Fritsch–Carlson box is

$$0 \le d_0 \le 3s, \qquad 0 \le d_1 \le 3s.$$

The routine returns the minimum distance to the four faces of this box. A positive number is therefore not merely a pass flag but a margin. For smooth functions sampled on a fine grid, $d_i/s = 1 + O(h^2)$, so the certificate should clear the boundary comfortably. A small margin is a useful indication of under-resolution even before monotonicity fails.

Monotonicity of x and G is not by itself a convexity certificate for the interpolated call. The separate belly test uses the exact factorization (2.5): analytic w, w', w'' are evaluated at 801 points across the retained strike span and g_{D} must exceed -10^{-4}. Since a flat smile has $g_{\mathrm{D}} = 1$, the tolerance permits a negative density component no larger than 10^{-4} of the local Black density factor. The threshold is therefore dimensionless and interpretable.

2.B.5 Floating-point failure catalogue

Four failures deserve explicit tests.

1. *Rank saturation.* For z near 37, double precision rounds $\operatorname{expit} z$ to one. Any formula that subsequently forms $1 - u$ loses the right-tail probability. Use $\operatorname{expit}(-z)$ directly and keep the cash leg in log space.
2. *Interior overflow.* Admissible endpoints do not bound a polynomial between them. Large Legendre coefficients can make g exceed the exponent range in the body while $g(0)$ and $g(1)$ remain moderate. The build must inspect the full grid and reject the trial before exponentiation.
3. *Near-wall cancellation.* Normalization and right-tail prices contain $1/(1-\lambda_+)$. The logistic chart slows parameter steps near the wall, while the hard guard leaves a finite numerical margin.
4. *Lee cancellation.* In the textbook expression for $\Psi(r)$, $\sqrt{r^2 + r} - r$ subtracts nearly equal quantities for large r. Use the rationalized form (2.55), with the limiting value zero handled explicitly as $r \to \infty$.

Cached, parameter-independent grid arrays should be read-only. An accidental in-place modification of ranks, logistic weights, or basis values otherwise corrupts every later slice without necessarily producing an exception.

2.B.6 Validation stack

The solvable slice tests the most code with the least ambiguity. Across $s \in [0.05, 0.95]$, compare the computed transport with $m + sz$, the shift with $-\log(\pi s/\sin \pi s)$, and the martingale integral with one. The observed worst errors are 1.1×10^{-11}, 3.0×10^{-9}, and 3.3×10^{-16}. The shift error grows toward the wall because the normalizing integrand decays at rate $1 - s$.

The randomized battery serves a different purpose. It samples several basis orders, ordinary slices, near-wall slices, and slices with large interior modes. Strikes are placed between grid nodes, and convexity is checked in normalized strike rather than log-strike. A four-times-finer independent build provides a reference price. Together with the interpolation and belly certificates, this separates algebraic validity, interpolation validity, and grid accuracy.

2.B.7 Reference defaults and timing

The exact timings depend on hardware and caches. The portable facts are the $O(Pn_{\mathrm{grid}})$ analytic complexity and the removal of repeated builds. The reported medians and interquartile ranges come from a single run that alternates analytic and finite-difference solves after warm-up. Alternation reduces drift from cache state and background load; medians avoid a best-of-n claim. A timing table should always travel with its grid size, quote count, order, solver tolerances, and machine context.

Quantity	Default	Quantity	Default
Basis order N	16 (cap 24)	Order guard	(2.50)
Grid	$[-40, 40]$	Nodes	8001 (fit: 2001)
Wall guard	$\lambda_+ \leq 1 - 10^{-6}$	Exponent budget	700
Ridge (α, ν)	$(10^{-6}, 1)$, $n \geq 4$	Vega floor η	10^{-4}
Tail barrier	center 0.90, slope 50	Haircut h	0.005 vol
Solver	trust-region LSQ	Tolerances	10^{-10}
Chart	logistic	Calendar nodes	$\max(49, 4N + 1)$
Belly check	801 points	Cold start	$\sqrt{3w_0}/\pi$

Table 2.3: Reference implementation defaults.

Configuration	Order	Median	IQR	vs. finite difference
Analytic Jacobian	6	15.9 ms	0.5 ms	1.41×
Finite difference	6	22.5 ms	2.7 ms	—
Analytic Jacobian	12	26.0 ms	2.1 ms	1.68×
Finite difference	12	43.8 ms	3.2 ms	—
Analytic Jacobian	16	28.9 ms	0.5 ms	1.76×
Finite difference	16	50.9 ms	3.9 ms	—

Table 2.4: Single-slice calibration times from one interleaved run. The last column is the median finite-difference time divided by the median analytic time for the same configuration.

For a new implementation, the shortest reliable porting sequence is:

1. reproduce the constant-speed normalizer and transport;
2. implement the stable cash leg and tail corrections;
3. keep (x, x', G, G') as one immutable slice object;
4. add the envelope sensitivity pass;
5. run monotonicity, Durrleman, and randomized sub-grid checks on every release.

2.C Data and reproducibility

Every empirical number and figure is generated from one frozen JSON snapshot, captured after the 2026-08-03 US session from a delayed consolidated US listed-options feed. The file contains SPY and NVDA chains for eight expiries each, with spots 757.67 and 206.80, together with forwards, discount factors, prepared quotes, fitted parameters, and display curves. Maturities run from 2 calendar days to 1.3726 years.

The snapshot is self-contained. For each node it records the normalized chain, forward and discount provenance, retained quote set, fitted raw parameters, and diagnostic curves. Reproduction therefore requires neither credentials nor a market-data connection. The capture timestamp, reference date, and session are part of the manifest, so a later book cannot silently replace the one used for the chapter.

The manifest records the common fitting recipe: order 16 under (2.50), logistic chart, haircut $h = 0.005$, equal quote weights, ridge $\alpha = 10^{-6}$ with $\nu = 1$, and calendar rows active. All 16 nodes were fitted; none was tuned individually. The shortest NVDA node has the largest residual, consistent with its sparse and steep quote strip. SPY bands collapse to mids under the haircut rule; NVDA retains interval targets. These states are stored per quote rather than reconstructed from rounded display values.

The figure generator reads only this snapshot and fixed synthetic specifications. Real nodes are rebuilt from the frozen coefficients rather than refitted, and the rebuilt curves agree with the stored curves to 0.000 vol bp in the worst case. The script emits both the 14 figure PDFs and the LaTeX macro file supplying every empirical value in the chapter. Randomized audits use fixed seeds. Timing values follow a fixed interleaved protocol but vary with the machine.

The synthetic double-hump target is a fixed two-component lognormal mixture, martingale-normalized and sampled on a realistic strike ladder. The constant-speed benchmark requires no data and is evaluated directly from (2.18). Regeneration consists of running the figure generator and then compiling the present manuscript. Refetching the market feed would create a different dataset and is not part of reproduction.

One family-level diagnostic is retained with the data. The two shortest SPY expiries cross in implied-volatility space only far outside their common quote span, at call prices near numerical zero. Their direct normalized call prices remain ordered to 2.2×10^{-15} on the audited grid. Keeping both measurements makes the support-confinement example of section 2.8 reproducible without interpreting the picture by eye.

Chapter 3

Families in the Volatility Chart: SVI-JW and Superposition

Chapter 2 built a probability law first and read the smile from it. This chapter does what the industry does instead: it writes the smile down directly, as a formula for total implied variance. The gain is legibility. The SVI formula draws one expiry's smile with five parameters, and its jump-wing version speaks the desk's own vocabulary of level, skew, and wings. The price is that a formula for a curve is not automatically a probability law, so validity must now be checked rather than inherited. The chapter develops SVI slowly: the geometry of the curve, why its wing slopes are capped strictly inside Lee's bound of two, why passing the cheap checks does not yet make a slice a law, how the trader coordinates work and exactly where they fail, and how to calibrate so that the cheap conditions hold at every step. It then builds the Multi-Core Sigmoid, which adds local detail to the body of a smile without touching its wings, and closes by fitting both families and Chapter 2's model to the same frozen quotes, so that each family's strengths and blind spots show in the same three numbers.

3.1 Working in the volatility chart

Chapter 2 built a smile in two steps. First choose a probability law for the return; then read prices, and hence implied volatilities, off that law. The order of the steps did real work. Because every curve the model could produce came from a genuine law, validity never needed checking. The cost was indirection: the model's coordinates (L, R, a_n) — the endpoint speeds and polynomial coefficients of its rank-space construction — describe how probability is transported, and none of them is a number a trading desk quotes.

This chapter works in the opposite order, which is also the industry's order. Two definitions from Chapter 2 are used on every page that follows, so we restate them once. The *log-moneyness* of a strike K against the forward F is $k = \log(K/F)$: the money is $k = 0$, and negative k is the put side. The *total implied variance* at that strike is $w(k) = \sigma(k)^2\tau$, the squared implied volatility times the variance time τ. One expiry's smile is then a single function $w(k)$, and the direct route is to write that function down as an explicit formula with a few parameters. Working this way — on the picture whose axes are strike and variance, rather than behind it on the law — is what we call working *in the volatility chart.*

The direct route has a real cost, and it is exactly the one section 2.1 identified when it introduced the function g_{D}. Most functions $w(k)$ are not the smile of any probability law. A model that writes $w(k)$ freely therefore moves through a space that contains invalid curves, and validity — which Chapter 2 got for free — must now be policed. Part of the policing can be done by cheap inequalities that the optimizer respects while it runs. Part of it cannot, and needs a check on the finished curve. Learning which part is which is most of this chapter.

Why pay that cost? Because the direct route serves two audiences at once. An optimizer wants a formula that evaluates in one line and differentiates in closed form. A desk wants parameters it can talk about: where the at-the-money volatility sits, how steep the skew is, how heavy each wing is. The stochastic-volatility-inspired (SVI) family of Gatheral [11] answers the optimizer with a tilted hyperbola,

$$\boxed{w(k) = a_S + b_S\Big\{\rho_S(k - m_S) + \sqrt{(k - m_S)^2 + s_S^2}\Big\},} \tag{3.1}$$

five parameters under the structural conditions $b_S > 0$, $|\rho_S| < 1$, $s_S > 0$. (The subscript S is there to keep the book's notation unambiguous: bare ρ is Chapter 2's logistic density and bare m its martingale shift, so the SVI parameters carry their own mark.) The desk is answered by a second coordinate system on the same curve — the jump-wing handles of section 3.5, five numbers chosen to match the desk's questions. The two coordinate systems describe the same family, and theorem 3.5 will say exactly when one converts to the other.

Before any geometry, we need to be precise about what "valid" means here, because different validity conditions will be supplied by different mechanisms. Five graded conditions recur through the chapter; table 3.1 collects them once, and the sections that follow fill the table in. The first three are inequalities on the parameters, cheap enough to build into the fit itself. The fourth is the full butterfly condition $g_{\mathrm{D}} \geq 0$ of (2.5) — recall that $g_{\mathrm{D}}(k)$ is, up to a positive factor, the probability density the smile implies at strike k, so a valid smile needs $g_{\mathrm{D}} \geq 0$ everywhere. No cheap inequality implies it: section 3.4 exhibits a slice that passes every screen and still fails it. The fifth condition couples different expiries and is not new to this chapter: it is the calendar coupling of section 2.8, which applies to these families unchanged.

The table names two symbols and two mechanisms that are still ahead of us, so read it as a map of the chapter rather than a summary of things known: β_L and β_R are the smile's two wing slopes, defined in section 3.2; a *hinge penalty* is a cheap fence built into the fit, explained in section 3.4; and the *structural chart* is a choice of optimizer coordinates, built in section 3.6. We will return to this table as each row is earned.

Condition	Mathematical statement	Supplied by
Structural	$b_S > 0$, $\lvert\rho_S\rvert < 1$, $s_S > 0$	the optimization chart (section 3.6)
Positive floor	$\min_k w(k) \geq 0$	a hinge penalty; automatic under the structural chart
Wing-admissible	$\max(\beta_L, \beta_R) \leq \beta_{\max} < 2$	a hinge penalty; automatic under the structural chart
Butterfly-free	$g_{\mathrm{D}}(k) \geq 0$ for all k	*not* implied by the rows above; checked on a dense grid (section 3.4)
Calendar-ordered	$w_{\tau_1}(k) \leq w_{\tau_2}(k)$, $\tau_1 < \tau_2$	the coupling of section 2.8, applied model-agnostically

Table 3.1: Five grades of validity for a slice written directly in the volatility chart. In Chapter 2 the first four held by construction. Here each must be attributed to the mechanism that supplies it — and the fourth row is the one no cheap mechanism reaches.

The chapter now proceeds in order. Section 3.2 works out the geometry of the hyperbola (3.1): its vertex, its wings, and a rigidity that will matter later. Section 3.3 explains why the wing slopes must be capped, and why the cap sits strictly inside Lee's bound. Section 3.4 shows what the caps do *not* protect — the middle of the smile — and states the chapter's division of labor between fences and checks. Section 3.5 develops the trader coordinates and proves exactly where they can be inverted. Section 3.6 turns to calibration: what the least-squares objective integrates against, and how to choose coordinates so the cheap conditions hold at every step. Section 3.7 then builds the Multi-Core Sigmoid, which adds capacity in the body of the smile without touching the wings. Section 3.8 closes by fitting LQD, SVI, and MCS to the same frozen quotes under one protocol.

3.2 One hyperbola, five parameters

We have a formula, (3.1), and a claim that its five parameters are legible. This section earns the claim: every geometric feature of the curve — where it bottoms, how it curves there, how steep each wing is — comes out in closed form.

Start by computing, because the formula is friendlier than it looks. Take

$$a_S = 0.01, \qquad b_S = 0.10, \qquad \rho_S = -0.4, \qquad m_S = 0, \qquad s_S = 0.2,$$

with a quarter year of variance time, $\tau = 0.25$. At the money, $k = 0$, the square root is $\sqrt{0^2 + 0.2^2} = 0.2$, so

$$w(0) = 0.01 + 0.10\,(-0.4 \cdot 0 + 0.2) = 0.030,$$

and the implied volatility is $\sqrt{0.030/0.25} = \sqrt{0.120} \approx 34.6\%$. Further out on the call side, at $k = 0.3$: the square root is $\sqrt{0.09 + 0.04} \approx 0.361$, so $w(0.3) = 0.01 + 0.10\,(-0.12 + 0.361) \approx 0.0341$, an implied volatility of about 36.9%. And at $k = -0.3$: $w(-0.3) = 0.01 + 0.10\,(0.12 + 0.361) \approx 0.0581$, about 48.2%. The left side came out much higher than the right. That is the equity put skew, and it entered through the sign of ρ_S: a negative tilt makes the put side steep and the call side shallow. Everything else in this section is this computation done once and for all, in general.

For the general statement, three shorthands keep the algebra readable:

$$x_S = k - m_S, \qquad r_S = \sqrt{x_S^2 + s_S^2}, \qquad q_S = \sqrt{1 - \rho_S^2} \tag{3.2}$$

(the subscripts keep Chapter 2's transport x and the moment order r of (2.23) untouched).

Proposition 3.1 (Geometry of one slice)**.** *Under $b_S > 0$, $|\rho_S| < 1$, $s_S > 0$, the slice (3.1) is strictly convex with a single minimum:*

$$w'(k) = b_S\Big(\rho_S + \frac{x_S}{r_S}\Big), \qquad w''(k) = \frac{b_S\, s_S^2}{r_S^3} > 0, \tag{3.3}$$

$$k^* = m_S - \frac{s_S \rho_S}{q_S}, \qquad w^* = w(k^*) = a_S + b_S\, s_S\, q_S. \tag{3.4}$$

Its far field is two straight rays:

$$w(k) = \beta_L |k| + O(1)\ (k \to -\infty), \qquad w(k) = \beta_R\, k + O(1)\ (k \to +\infty), \tag{3.5}$$

with

$$\beta_L = b_S(1 - \rho_S), \qquad \beta_R = b_S(1 + \rho_S).$$

The curvature at the vertex is $\kappa^* = w''(k^*) = b_S\, q_S^3 / s_S$.

Proof. Differentiate (3.1) once and twice in k, using $r_S' = x_S/r_S$:

$$w' = b_S\Big(\rho_S + \frac{x_S}{r_S}\Big), \qquad w'' = b_S\,\frac{r_S - x_S \cdot (x_S/r_S)}{r_S^2} = b_S\,\frac{r_S^2 - x_S^2}{r_S^3} = \frac{b_S\, s_S^2}{r_S^3},$$

which is (3.3); it is positive because $b_S, s_S > 0$, so the curve is strictly convex and any stationary point is the unique global minimum. To find that point, set $w' = 0$, i.e. $x_S/r_S = -\rho_S$. Square both sides and use $r_S^2 = x_S^2 + s_S^2$:

$$x_S^2 = \rho_S^2\,(x_S^2 + s_S^2) \quad\Longrightarrow\quad x_S^2\,(1 - \rho_S^2) = \rho_S^2 s_S^2 \quad\Longrightarrow\quad |x_S| = \frac{s_S|\rho_S|}{q_S}.$$

Since $r_S > 0$, the equation $x_S/r_S = -\rho_S$ forces x_S to have the sign opposite to ρ_S, so $x_S = -s_S\rho_S/q_S$ and hence $k^* = m_S - s_S\rho_S/q_S$. At this point

$$r_S = \sqrt{\frac{s_S^2\rho_S^2}{q_S^2} + s_S^2} = s_S\,\frac{\sqrt{\rho_S^2 + q_S^2}}{q_S} = \frac{s_S}{q_S},$$

because $\rho_S^2 + q_S^2 = 1$. Substituting into (3.1),

$$w^* = a_S + b_S\Big(-\frac{s_S\rho_S^2}{q_S} + \frac{s_S}{q_S}\Big) = a_S + b_S\,s_S\,\frac{1-\rho_S^2}{q_S} = a_S + b_S\,s_S\,q_S,$$

which is (3.4), and plugging $r_S = s_S/q_S$ into (3.3) gives $\kappa^* = b_S s_S^2\,(q_S/s_S)^3 = b_S q_S^3/s_S$. Finally, in either wing $|x_S| \to \infty$ and $r_S = |x_S|\sqrt{1 + s_S^2/x_S^2} = |x_S| + O(1/|x_S|)$; substituting $r_S \approx |x_S|$ into (3.1) leaves $w \approx a_S + b_S(\rho_S x_S + |x_S|)$, whose slope in k is $b_S(\rho_S + 1) = \beta_R$ on the right and whose slope in $|k|$ is $b_S(1-\rho_S) = \beta_L$ on the left — which is (3.5). □

In words: the curve is a rounded hinge. Far from the vertex it is two straight rays, of slope β_L on the left and β_R on the right. The tilt ρ_S divides a fixed total steepness $\beta_L + \beta_R = 2b_S$ between the two sides — our worked slice had $\beta_L = 0.10 \times 1.4 = 0.14$ and $\beta_R = 0.06$, steep left, shallow right. Near the vertex, s_S decides how sharply one ray turns into the other, m_S slides the turn left or right, and a_S lifts the whole curve. (For the worked slice, (3.4) puts the vertex at $k^* = 0 - 0.2 \cdot (-0.4)/\sqrt{0.84} \approx 0.087$, with $w^* = 0.01 + 0.10 \cdot 0.2 \cdot 0.917 \approx 0.028$ — slightly right of the money, slightly below $w(0)$, as a left-steep curve requires.)

Figure 3.1 sweeps the three shape parameters one at a time around a common slice, and each panel isolates one of the readings above. In panel (a), only ρ_S varies: watch the left and right rays trade steepness while their total stays fixed — the vertex drifts, but the outer geometry only tilts. In panel (b), only s_S varies: the rays do not move at all, and the curves differ only in how tightly they round the corner between them. In panel (c), only b_S varies: both rays open or close together, like scissors. Every member, in every panel, is one convex curve with one minimum.

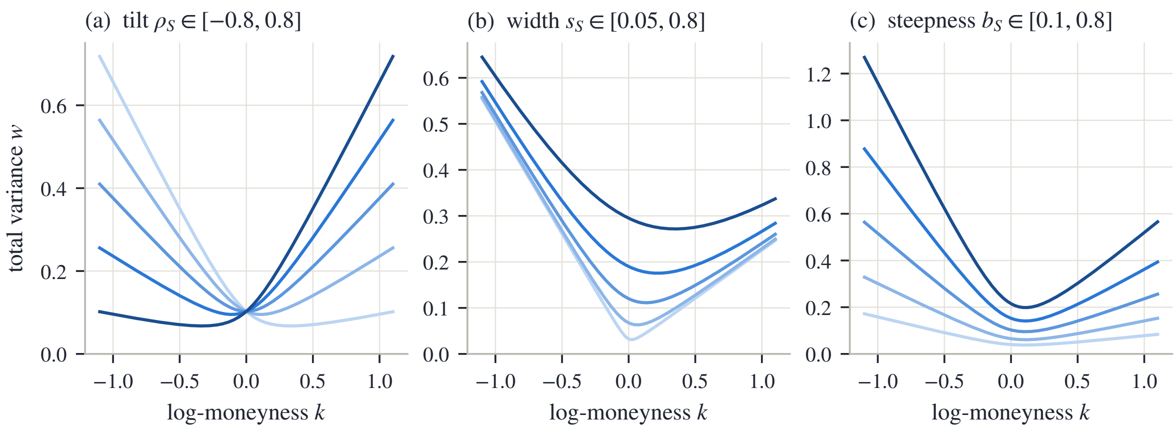

Figure 3.1: The reach of the family (3.1), one parameter at a time around a common slice. (a) The tilt ρ_S redistributes a fixed total steepness between the wings. (b) The width s_S rounds the vertex without moving the rays. (c) The steepness b_S opens or closes both wings together. Every member is one convex curve with one minimum.

We now have the whole geometry. Two consequences of it organize the rest of the chapter.

First, the tails collapse to two numbers. As $|k|$ grows, $w(k)/|k|$ tends to β_L on the put side and β_R on the call side, and no other feature of the slice survives to infinity. Whatever the market's tails are worth, this family records them in (β_L, β_R) alone. That is exactly the pair that Lee's moment bound constrains, and section 3.3 develops the correspondence.

Second, strict convexity is a rigidity. One hyperbola has one belly and one monotone turn of the slope, from $-\beta_L$ up to $+\beta_R$. This excludes a shape that markets do produce: the W, a central trough with a raised shoulder on each side.

Proposition 3.2 (A convex slice cannot carry a W)**.** *A convex function on an interval has no interior strict local maximum. Hence a total-variance curve with a central trough flanked by two shoulders — an interior local maximum on each side — is not representable by any globally convex slice, in particular by* (3.1)*.*

Proof. For a convex function the derivative (or, where it fails to exist, the one-sided derivatives) is non-decreasing. At a strict interior local maximum the derivative would have to pass from positive, on the way up, to negative, on the way down — a decrease, contradicting monotonicity. By proposition 3.1, $w'' > 0$ everywhere for the family (3.1), so the family is convex and the exclusion applies to it. ☐

Whether this rigidity helps or hurts depends on the smile in front of you. On a liquid index chain it helps: it forbids the wiggles that quote noise would otherwise carve into the fitted curve. Ahead of a scheduled event, or in a single stock (a *name*, in desk speech) whose holders are split between two scenarios, it is a ceiling — the market's smile genuinely turns more than once, and the family cannot follow it. Section 3.7 removes the ceiling without giving up the wings.

3.3 Why the wing slopes are capped

We know from section 3.2 that the family's tails collapse to the two slopes (β_L, β_R). Table 3.1 promised a cap on those slopes. This section explains where the cap comes from, why it is set strictly *inside* the theoretical bound rather than at it, and what a fitted slope does and does not say about the return distribution.

The starting point is Lee's moment formula, met in Chapter 2 as (2.23). In plain terms: the steeper a wing of total variance, the fatter the corresponding tail of the return law, and "fatter" is measured by which moments still exist. Recall the actors. Y is the gross forward return of Chapter 2 — the underlying's final level divided by its forward price, the random variable behind every smile in this book. Its moments $\mathbb{E}[Y^r]$ make sense for *any* real exponent r, not just integers, and for a fat-tailed law they stop being finite beyond some critical exponent. Write $1 + r_+^*$ for that critical exponent: $\mathbb{E}[Y^{1+r}]$ is finite for $r < r_+^*$ and infinite for $r > r_+^*$. (The subscript $+$ marks the right tail; a mirror image r_-^* governs the left.) Lee's formula ties the right wing to this critical exponent: $\beta_R = \Psi(r_+^*)$, where

$$\Psi(r) = 2 - 4\big(\sqrt{r^2 + r} - r\big).$$

The function Ψ decreases: more finite moments mean a shallower wing. As the law loses its moments, $r_+^* \downarrow 0$ and $\Psi \uparrow 2$. So the number 2 is a wall: a wing of total variance steeper than 2 is the tail of no probability law at all. For LQD the correspondence between tails and moments was a closed formula in the endpoint speeds (proposition 2.9). For a family whose wings are exactly straight lines it is even simpler — a checkable inequality on two parameters, $\max(\beta_L, \beta_R) \le 2$.

That inequality looks like the whole story. It is not, and the cleanest way to see why is to follow the bound into the function that actually tests for arbitrage. Recall the Durrleman function of (2.5) — up to a positive factor it is the probability density the smile implies, so a valid smile needs $g_{\mathrm{D}} \ge 0$:

$$g_{\mathrm{D}}(k) = \Big(1 - \frac{k\,w'}{2w}\Big)^2 - \frac{(w')^2}{4}\Big(\frac{1}{w} + \frac{1}{4}\Big) + \frac{w''}{2}.$$

It mixes w, w', and w''. Deep in the wings of the family (3.1), however, all three are controlled by the slope alone, and the mixture settles to a limit.

Proposition 3.3 (The wing slope sets the sign of the tail density)**.** *For a slice* (3.1) *with* $w > 0$,

$$\lim_{k\to-\infty} g_{\mathrm{D}}(k) = \frac{4-\beta_L^2}{16}, \qquad \lim_{k\to+\infty} g_{\mathrm{D}}(k) = \frac{4-\beta_R^2}{16}. \tag{3.6}$$

Hence for $\beta < 2$ the corresponding tail of g_{D} is eventually positive and for $\beta > 2$ eventually negative. At $\beta_R = 2$ the limit vanishes and the sign falls to the next order: the right wing is $w(k) = 2k + (a_S - 2m_S) + O(k^{-1})$, and

$$g_{\mathrm{D}}(k) = \frac{(a_S - 2m_S) - 2}{4k} + O(k^{-2}), \tag{3.7}$$

so the boundary slice carries negative tail density whenever $a_S - 2m_S < 2$. The mirror statement holds on the left with the intercept $a_S + 2m_S$.

Proof. Work on the right wing; the left is symmetric. There, by proposition 3.1, $w = \beta k + O(1)$ with $\beta = \beta_R$, $w' \to \beta$, and $w'' \to 0$. Take the three terms of g_{D} in turn:

$$\frac{k\,w'}{2w} \longrightarrow \frac{k\,\beta}{2\,\beta k} = \frac{1}{2}, \qquad \text{so} \qquad \Big(1 - \frac{k\,w'}{2w}\Big)^2 \longrightarrow \frac{1}{4};$$

$$\frac{(w')^2}{4}\Big(\frac{1}{w} + \frac{1}{4}\Big) \longrightarrow \frac{\beta^2}{4}\Big(0 + \frac{1}{4}\Big) = \frac{\beta^2}{16},$$

because $w \to \infty$ kills the $1/w$; and $w''/2 \to 0$. Adding the three limits gives $\frac{1}{4} - \beta^2/16 = (4-\beta^2)/16$, which is (3.6).

For the boundary case set $b_S(1+\rho_S) = 2$ and expand one order deeper. With $x_S = k - m_S$,

$$r_S = x_S\sqrt{1 + s_S^2/x_S^2} = x_S + \frac{s_S^2}{2x_S} + O(x_S^{-3}),$$

so

$$w = a_S + b_S(\rho_S x_S + r_S) = a_S + 2\,(k - m_S) + \frac{b_S s_S^2}{2k} + O(k^{-2}) = 2k + (a_S - 2m_S) + O(k^{-1}),$$

and differentiating, $w' = 2 + O(k^{-2})$. The intercept $a_S - 2m_S$ recurs below; we carry it written out. First term: dividing numerator and denominator by $4k$,

$$\frac{k\,w'}{2w} = \frac{2k + O(k^{-1})}{4k + 2(a_S - 2m_S) + O(k^{-1})} = \frac{1}{2}\Big(1 - \frac{a_S - 2m_S}{2k}\Big) + O(k^{-2}) = \frac{1}{2} - \frac{a_S - 2m_S}{4k} + O(k^{-2}),$$

so

$$\Big(1 - \frac{k\,w'}{2w}\Big)^2 = \Big(\frac{1}{2} + \frac{a_S - 2m_S}{4k}\Big)^2 + O(k^{-2}) = \frac{1}{4} + \frac{a_S - 2m_S}{4k} + O(k^{-2}).$$

Second term: $(w')^2/4 = 1 + O(k^{-2})$ and $1/w = 1/(2k) + O(k^{-2})$, so

$$\frac{(w')^2}{4}\Big(\frac{1}{w} + \frac{1}{4}\Big) = \frac{1}{4} + \frac{1}{2k} + O(k^{-2}).$$

Third term: $w'' = O(k^{-3})$. Subtracting,

$$g_{\mathrm{D}}(k) = \frac{a_S - 2m_S}{4k} - \frac{1}{2k} + O(k^{-2}) = \frac{(a_S - 2m_S) - 2}{4k} + O(k^{-2}),$$

which is (3.7). ∎

The proposition says two things, and the second is the one that is easy to miss. Below the bound, the tail of the implied density is eventually positive: the wing condition genuinely protects the far tail. *At* the bound, it protects nothing. The limit is zero, and the sign is decided by the $1/k$ term — that is, by the intercept, which no condition of the form $\beta \le 2$ even looks at.

A concrete slice makes the boundary failure explicit, and it is worth computing along. Take

$$(a_S, b_S, \rho_S, m_S, s_S) = (0.04,\ 2,\ 0,\ 0,\ 0.2).$$

With $\rho_S = 0$ and $m_S = 0$ the curve is symmetric, and its minimum, at $k = 0$, is $w^* = 0.04 + 2 \cdot 0.2 = 0.44$ — comfortably positive, so the floor condition holds. Both wings have slope $b_S(1 \pm 0) = 2$, so a constraint written as $\beta \le 2$ is satisfied (with equality) and raises no objection. Yet the intercept is $a_S - 2m_S = 0.04$, far below 2, so (3.7) makes the tail density negative at every large strike — at $k = 10$ the predicted leading term is $(0.04 - 2)/40 \approx -0.049$, and the exact value on this slice is $g_{\mathrm{D}}(10) = -0.0485$. A constraint that uses Lee's constant as its right-hand side therefore admits curves that are not the smile of any law.

The reference implementation draws the consequence as a convention, stated once: the enforced cap sits strictly inside the bound,

$$\max(\beta_L, \beta_R) \ \le\ \beta_{\max}, \qquad \beta_{\max} = 1.95 < 2. \tag{3.8}$$

Is the buffer expensive? To answer, invert Lee's formula and ask which laws a slope just under 2 still allows. Set $\beta = \Psi(r)$ and solve for r. Writing $\sqrt{r^2 + r} - r = (2 - \beta)/4$, move r across and square:

$$r^2 + r = \Big(r + \frac{2-\beta}{4}\Big)^2 = r^2 + 2r\,\frac{2-\beta}{4} + \Big(\frac{2-\beta}{4}\Big)^2 \quad\Longrightarrow\quad r\Big(1 - \frac{2-\beta}{2}\Big) = \frac{(2-\beta)^2}{16},$$

and since $1 - (2-\beta)/2 = \beta/2$, this gives the *moment budget* at slope β:

$$r^* = \Psi^{-1}(\beta) = \frac{(2-\beta)^2}{8\beta}, \tag{3.9}$$

the number of moments beyond the first that a wing of slope β still permits. At the cap, $r^* = (0.05)^2/(8 \times 1.95) = 1.6 \times 10^{-4}$: the only laws the buffer excludes are those whose entire moment budget beyond the mean is smaller than about two parts in ten thousand. The cap costs essentially nothing in distributional scope, and it restores the implication the boundary slice broke: wing-admissible now implies that the tail density is eventually positive.

The map (3.9) also tells us when a fitted wing may be read as a moment estimate, because its sensitivity is extreme at both ends of the range. Expand the quotient:

$$r^* = \frac{4 - 4\beta + \beta^2}{8\beta} = \frac{1}{2\beta} - \frac{1}{2} + \frac{\beta}{8}, \qquad \text{so} \qquad \frac{\mathrm{d}r^*}{\mathrm{d}\beta} = -\frac{1}{2\beta^2} + \frac{1}{8} = -\frac{4-\beta^2}{8\beta^2}.$$

For small β the derivative behaves as $-1/(2\beta^2)$: a shallow wing pins essentially no bound on the moments, and a small change in the fitted slope moves the inferred moment violently. As $\beta \to 2$ the budget collapses to zero. A fitted β is a usable moment reading only in the middle of the range — and only for a slice that actually is a law, which is the subject of the next section.

Figure 3.2 draws the three parts of the story. In panel (a), the tail limit (3.6) is plotted against the slope β: a downward parabola that crosses zero exactly at $\beta = 2$ (dotted vertical line), with the working cap $\beta_{\max} = 1.95$ marked dashed just inside it. Left of the crossing the far tail is positive; right of it, negative; at it, undecided. Panel (b) shows the boundary slice of the worked example: its Durrleman function (orange) is negative at every plotted strike, creeping up toward zero from below exactly as the $1/k$ law (3.7) prescribes — while the same slice with its wings brought back to the cap (blue) is positive and settles at the limit $(4 - \beta_{\max}^2)/16$ (dotted

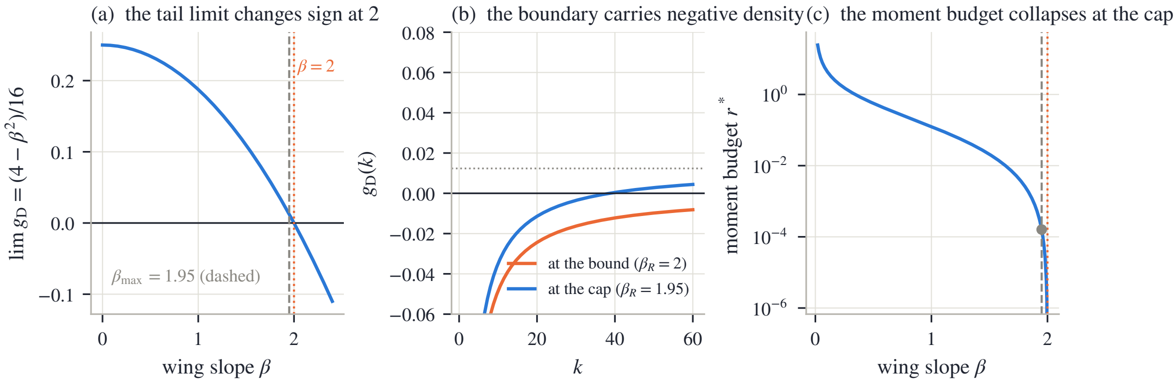

Figure 3.2: Lee's bound inside the density factor. (a) The tail limit (3.6) against the wing slope: a downward parabola crossing zero exactly at $\beta = 2$; the dashed line is the working cap $\beta_{\max} = 1.95$. (b) The boundary slice of the text: at $\beta_R = 2$ (orange) the tail density is negative at every plotted strike, approaching zero from below as (3.7) prescribes; the same slice with its wings at the cap (blue) turns positive and settles at the limit $(4 - \beta^2_{\max})/16$ (dotted). (c) The moment budget (3.9) on a log scale: it explodes as $\beta \to 0$ and collapses at the cap (marked point, $r^* = 1.6 \times 10^{-4}$).

horizontal). Panel (c) plots the moment budget (3.9) on a log scale: it explodes as $\beta \to 0$ — shallow wings say nothing about moments — and collapses at the cap, where the marked point sits at $r^* = 1.6 \times 10^{-4}$.

We now know that the floor guards the vertex and the cap guards the tails. The next question is what guards everything in between — and the answer, for now, is nothing.

3.4 What the inequalities do not check: the belly

The floor condition of table 3.1 guards the vertex, and the cap (3.8) guards the wings. This section shows that the region between them — the belly of the smile — is guarded by neither, on a concrete slice that passes every cheap test and still fails to be a law.

The root of the problem is that the word "convex" hides two different second derivatives. The family (3.1) has $w'' > 0$ by construction: the *variance curve* is convex in log-moneyness. What no-arbitrage requires is different: the *option price* must be convex in the strike. A butterfly spread — buy one option at $K - \delta$, sell two at K, buy one at $K + \delta$ — has a nonnegative payoff, so its price must be nonnegative, and that price is (up to scale) the second difference of the call price in strike. Price convexity is what the sign of g_D tests, and g_D mixes w, w', and w''. In the wings the two notions of convexity agree — proposition 3.3 showed that the sign of g_D out there is decided by the wing slope. In the interior they need not agree.

That the gap is real, and not a defect of some particular fitting code, is shown by a classical example due to Vogt, reproduced as Example 3.1 by Gatheral and Jacquier [13]:

$$(a_S, b_S, \rho_S, m_S, s_S) = (-0.0410,\ 0.1331,\ 0.3060,\ 0.3586,\ 0.4153).$$

Score this slice against table 3.1, row by row. The structural conditions hold: $b_S > 0$, $|\rho_S| < 1$, $s_S > 0$. The floor holds: the minimum total variance, from (3.4), is $w^* = 0.0116$, safely positive. The wings hold with an enormous margin: the larger slope is $\max(\beta_L, \beta_R) = 0.174$, an order of magnitude below the cap, and by (3.6) both tails of g_D settle near 0.25 — clean, positive tail density on both sides. Everywhere the cheap inequalities look, this slice is impeccable.

Yet it is not a law. In the interior, g_D dips to -0.033 near $k = 0.88$ — squarely between the two clean tails. The implied density is negative there, which means the slice prices some butterfly below zero. Figure 3.3 shows both halves of the verdict. In panel (a), the total-variance curve itself: smooth, convex, floor marked, nothing visibly wrong — this is the point; the defect is invisible in the chart everyone looks at. In panel (b), the same slice's Durrleman function: it approaches the positive limit (3.6) on both sides, exactly as the wing analysis promises, and turns negative in between. The wing conditions fence the tails; the belly escapes them.

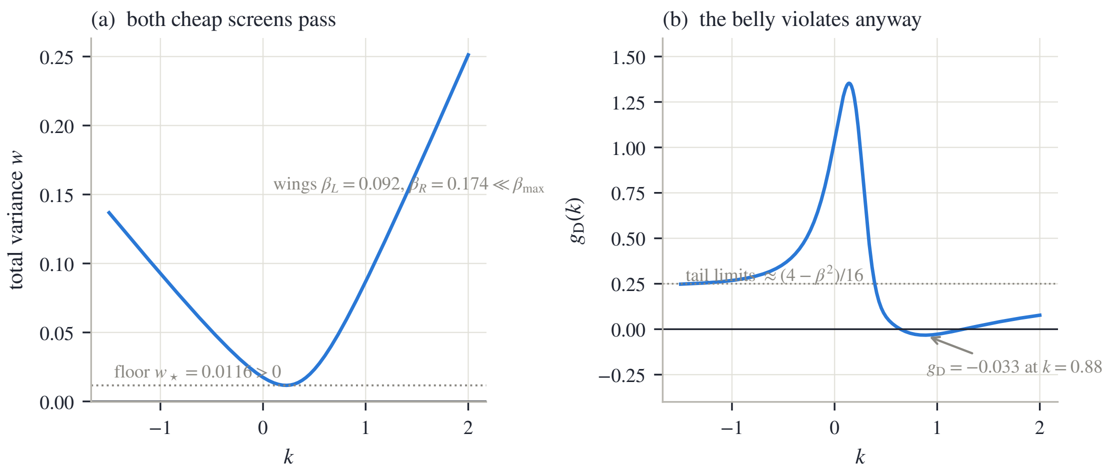

Figure 3.3: The Vogt slice. (a) Total variance: the floor is positive and both wing slopes are far below the cap — every inequality of table 3.1 up to wing-admissibility holds strictly. (b) Its Durrleman function approaches the positive tail limits (3.6) on both sides yet turns negative in the interior. The wing conditions fence the tails; the belly escapes them.

Could a better inequality close the gap? In principle, yes: the exact butterfly-free domain of the five SVI parameters has been characterized completely by Martini and Mingone [21]. But the characterization is not one short inequality — it involves a parameter rescaling, root finding, and a numerical minimization — and it is specific to this five-parameter family, with no analogue for the superposition family of section 3.7. A family-by-family exact domain is not a strategy a multi-family fitter can build on.

The reference implementation adopts a different division of labor for every family in the volatility chart, and it is the chapter's second convention. *During* the fit, the floor and the cap are enforced inside the least-squares residual — the stacked vector of per-quote misfits that the optimizer drives toward zero — by appending two cheap penalty rows to it:

$$W\ \max\big(-(a_S + b_S s_S q_S),\, 0\big), \qquad W\ \max\big(b_S(1+|\rho_S|) - \beta_{\max},\, 0\big), \tag{3.10}$$

with W a multiplier large enough to dominate any violation. Read the two rows: the first is active only when the floor $w^* = a_S + b_S s_S q_S$ of (3.4) goes negative, the second only when the larger wing $b_S(1+|\rho_S|)$ exceeds the cap. A row of this shape is called a *hinge*: it is exactly zero whenever its inequality holds, so a clean fit is identical with or without it, and it only pushes back when the optimizer strays. *After* the fit, the condition the hinges cannot see is checked where it lives: g_{D} is evaluated on a dense grid of strikes across the traded range — always from the model's own derivatives (3.3), never by differencing prices, which would inject curvature noise at exactly the scale being measured — and the slice is accepted only if the sampled minimum clears a small tolerance. This is the same posture as the numerical certificates of section 2.7.2: an analytic guarantee where one exists, and where it does not, a computed check on the finished object, with its scope (the sampled grid on the traded range, not the whole line) stated rather than implied. The minimum of g_{D} over that grid is called the fit's *validity margin*; the comparison of section 3.8 reports it for every fit of every family.

One could try to push the butterfly condition into the optimizer too, as a hinge on sampled values of g_{D} over a grid — and the superposition family of section 3.7, whose localized bump corrections (its *kernels*, built there) can bend the density in ways a hyperbola cannot, carries exactly such a row (appendix 3.A). It helps, but it settles nothing: a finite-weight hinge is traded against the data term like any other residual row, so it discourages violations without excluding them, and section 3.8 shows fits that fail the final check with that hinge active. In the other direction, trusting the cheap fences alone would publish the Vogt slice. So the division stands: fence cheaply during the fit to keep the optimizer in the region where the check usually passes — and let the check decide.

3.5 Trader coordinates: the jump-wing handles

The raw quintuple $(a_S, b_S, \rho_S, m_S, s_S)$ is an excellent computational coordinate system and a poor market language. No single coordinate is "the level" or "the skew," and each of b_S, ρ_S, s_S moves a wing and the interior at once. The jump-wing (JW) coordinates of Gatheral and Jacquier [13] re-read the same curve through five quantities chosen to match the desk's questions. (The name is inherited, not descriptive: the coordinates arose from reading wing behavior in stochastic-volatility models with jumps; nothing in this section requires jumps.) This section defines them, shows how to read them without mixing up their units, and then answers the harder question: given five quoted handles, can the hyperbola be reconstructed — and when it cannot, what information was lost?

A useful way to remember the five handles: three of them describe the *interior* of the smile and two describe its *tails*.

Definition 3.4 (Jump-wing handles)**.** At variance time τ, with $w_0 = w(0)$ the ATM total variance, the jump-wing coordinates of a slice are

$$\underbrace{v_J = \frac{w_0}{\tau}, \qquad \psi_J = \big(\sqrt{w}\big)'(0), \qquad \tilde{v}_J = \frac{\min_k w(k)}{\tau},}_{\text{interior}} \qquad \underbrace{p_J = \frac{\beta_L}{\sqrt{w_0}}, \qquad c_J = \frac{\beta_R}{\sqrt{w_0}}}_{\text{tails}}. \tag{3.11}$$

In words: v_J is the ATM variance per year, ψ_J is the ATM slope (of total volatility $\sqrt{w}$, not of w — this matters below), $\tilde{v}_J$ is the smile's minimum variance per year, and p_J, c_J are the put and call wing slopes, each divided by $\sqrt{w_0}$.

For the family (3.1) all five have closed forms. Four are immediate from proposition 3.1; the ATM slope takes two lines. Since $(\sqrt{w})' = w'/(2\sqrt{w})$, and at $k = 0$ the shorthands (3.2) read $x_S = -m_S$ and $r_S = \sqrt{m_S^2 + s_S^2}$, formula (3.3) gives $w'(0) = b_S(\rho_S - m_S/\sqrt{m_S^2 + s_S^2})$, hence

$$v_J = \frac{w_0}{\tau}, \quad \psi_J = \frac{b_S}{2\sqrt{w_0}}\Big(\rho_S - \frac{m_S}{\sqrt{m_S^2 + s_S^2}}\Big), \quad p_J = \frac{b_S(1-\rho_S)}{\sqrt{w_0}}, \quad c_J = \frac{b_S(1+\rho_S)}{\sqrt{w_0}}, \quad \tilde{v}_J = \frac{w^*}{\tau}, \tag{3.12}$$

with w^* from (3.4).

Reading the handles in their own units

Each handle lives in its own units, and misreading them is the most common JW error, so we spell the dictionary out. Write $I(k) = \sqrt{w(k)/\tau}$ for the implied-volatility curve. Then $I(0) = \sqrt{w_0/\tau} = \sqrt{v_J}$; differentiating, $I'(0) = w'(0)/(2\sqrt{\tau w_0}) = (\sqrt{w})'(0)/\sqrt{\tau} = \psi_J/\sqrt{\tau}$; and inverting the definitions of the wing handles, $\beta_L = p_J\sqrt{w_0} = p_J\sqrt{v_J\tau}$ and likewise for β_R:

$$I(0) = \sqrt{v_J}, \qquad I'(0) = \frac{\psi_J}{\sqrt{\tau}}, \qquad \min_k I(k) = \sqrt{\tilde{v}_J}, \qquad \beta_L = p_J\sqrt{v_J\tau}, \qquad \beta_R = c_J\sqrt{v_J\tau}. \tag{3.13}$$

So: v_J and $\tilde{v}_J$ are variances, and what an implied-volatility chart shows is their square roots. ψ_J is the ATM slope of total *volatility*, so the tangent drawn on the IV chart has slope $\psi_J/\sqrt{\tau}$. And p_J, c_J are *normalized* total-variance wing slopes — they are not slopes of the IV curve at all.

Figure 3.4 annotates a fitted slice in exactly these units. The data is the SPY December 2026 expiry of the frozen snapshot — the fixed market capture, described in appendix 2.C, from which every empirical example in this book is drawn — and this particular (asset, expiry) pair, a *node* in the book's vocabulary, will recur through the chapter. The fit misses the quotes by 7.5 vol bp on average (a *vol bp* is one hundredth of a volatility percentage point, the book's unit of fit error), and carries $\sqrt{v_J} = 14.89\%$, tangent slope -0.430, floor $\sqrt{\tilde{v}_J} = 12.41\%$, and wing handles $(p_J, c_J) = (1.384, 0.724)$. Panel (a) is the implied-volatility chart, where the three *interior* handles are read: the orange dots are the mid quotes, the blue curve is the SVI fit, the two dotted horizontals mark the level $\sqrt{v_J}$ and the floor $\sqrt{\tilde{v}_J}$, and the dashed line through the money is the ATM tangent, of slope $\psi_J/\sqrt{\tau}$ — negative, as an equity skew requires. Panel (b) switches to the total-variance chart, the only place the two *tail* handles can be read honestly: the two dashed rays are the asymptotes, of slopes $\beta_L = p_J\sqrt{v_J\tau}$ and $\beta_R = c_J\sqrt{v_J\tau}$. Plotting all five handles on one IV axis would give the wing handles units they do not have.

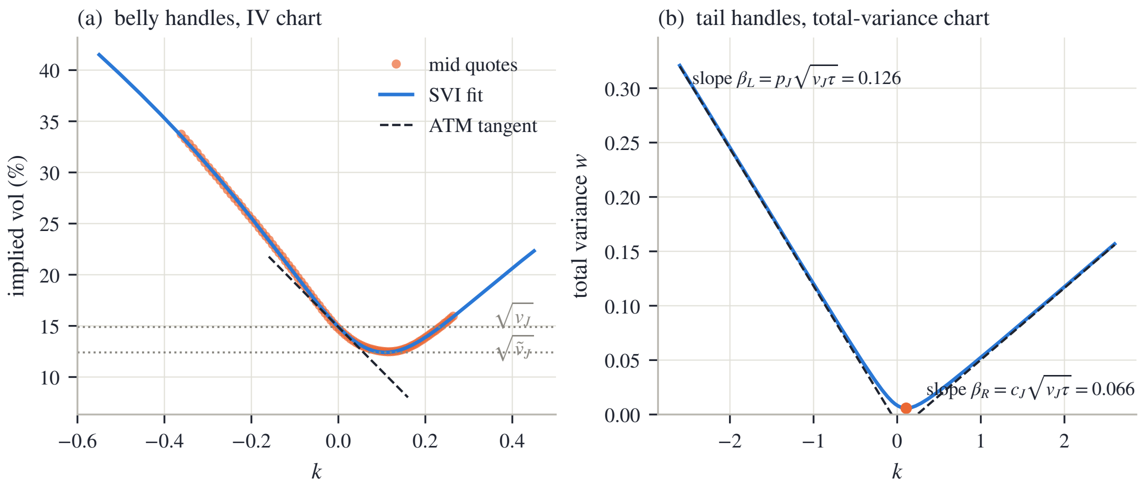

Figure 3.4: The five handles in their honest units, on the SVI fit of the SPY December 2026 quotes (rms 7.5 vol bp). (a) The three interior handles are read on the IV chart: the level $\sqrt{v_J}$, the minimum $\sqrt{\tilde{v}_J}$, and the ATM tangent of slope $\psi_J/\sqrt{\tau}$. (b) The two tail handles are read on the total-variance chart, as the slopes $\beta_L = p_J\sqrt{v_J\tau}$ and $\beta_R = c_J\sqrt{v_J\tau}$ of the asymptotes.

One caution before inverting. The tail handles are *normalized* indicators: by (3.13), the actual asymptotic slopes — the objects Lee's bound and the moment budget (3.9) read — carry a factor $\sqrt{v_J\tau}$ of the supposedly interior level. Raise v_J with (p_J, c_J) held fixed and both actual wings steepen, moving the inferred moments. JW is a quoting convention adapted to the desk, not a decomposition of the slice into independent interior and tail degrees of freedom.

3.5.1 Inverting the handles

Reading (3.12) off a fitted slice is mere evaluation. The interesting direction is the reverse: a desk quotes five handles — can we reconstruct the hyperbola? We build the inverse in three steps and watch where it can break.

Step 1: the tail handles fix the scale and the tilt. From (3.12), $p_J\sqrt{w_0} = b_S(1-\rho_S)$ and $c_J\sqrt{w_0} = b_S(1+\rho_S)$. Adding and subtracting,

$$b_S = \frac{\sqrt{w_0}}{2}(p_J + c_J), \qquad \rho_S = \frac{c_J - p_J}{c_J + p_J}, \qquad w_0 = v_J\tau. \tag{3.14}$$

Step 2: the ATM slope fixes where the vertex sits. The combination of m_S and s_S that appears in ψ_J is the *normalized vertex displacement* $\chi := m_S/\sqrt{m_S^2 + s_S^2} \in (-1, 1)$. Solve (3.12) for it: $\psi_J = \frac{b_S}{2\sqrt{w_0}}(\rho_S - \chi)$ and $b_S/\sqrt{w_0} = (p_J + c_J)/2$ give

$$\chi = \rho_S - \frac{4\psi_J}{p_J + c_J}. \tag{3.15}$$

Step 3: the gap between level and floor sets the width. Evaluate (3.1) at the money and at the vertex:

$$w_0 = a_S + b_S\Big(\sqrt{m_S^2 + s_S^2} - \rho_S m_S\Big), \qquad w^* = a_S + b_S\, s_S\sqrt{1-\rho_S^2}. \tag{3.16}$$

By the definition of χ, $m_S = \chi s_S/\sqrt{1-\chi^2}$ and hence $\sqrt{m_S^2 + s_S^2} = s_S/\sqrt{1-\chi^2}$. Subtract the second identity from the first: the unknown a_S cancels, and

$$w_0 - w^* = b_S\, s_S\Big(\frac{1-\rho_S\chi}{\sqrt{1-\chi^2}} - \sqrt{1-\rho_S^2}\Big) = b_S\, s_S\, \mathcal{D},$$

where we name the bracket

$$\mathcal{D}(\rho_S, \chi) = \frac{1-\rho_S\chi}{\sqrt{1-\chi^2}} - \sqrt{1-\rho_S^2}. \tag{3.17}$$

Since $w_0 - w^* = (v_J - \tilde{v}_J)\tau$, the width and the remaining two parameters follow:

$$s_S = \frac{(v_J - \tilde{v}_J)\,\tau}{b_S\,\mathcal{D}}, \qquad m_S = \frac{\chi\, s_S}{\sqrt{1-\chi^2}}, \qquad a_S = \tilde{v}_J\tau - b_S\, s_S\sqrt{1-\rho_S^2}. \tag{3.18}$$

The whole inverse therefore stands or falls with the sign of the denominator $\mathcal{D}$, and that sign is a disguised Cauchy–Schwarz inequality. Consider the two unit vectors $(\rho_S, \sqrt{1-\rho_S^2})$ and $(\chi, \sqrt{1-\chi^2})$ — two points on the upper half of the unit circle. Their dot product is at most 1:

$$\rho_S\chi + \sqrt{1-\rho_S^2}\sqrt{1-\chi^2} \;\le\; 1, \quad \text{i.e.} \quad \mathcal{D} \;\ge\; 0, \tag{3.19}$$

(multiply the definition (3.17) through by the positive $\sqrt{1-\chi^2}$ to see the two statements are the same), with equality exactly when the two vectors coincide — that is, when $\chi = \rho_S$, which by (3.15) means $\psi_J = 0$. This single inequality gives both the domain of the coordinate system and its one singularity.

Theorem 3.5 (The image of the jump-wing map)**.** *Fix $\tau > 0$ and restrict (3.1) to $b_S > 0$, $|\rho_S| < 1$, $s_S > 0$, $w_0 > 0$.*

(i) A JW point has a unique inverse, given by (3.14)–(3.18), exactly on

$$v_J > 0, \qquad p_J > 0,\ c_J > 0, \qquad -\frac{p_J}{2} < \psi_J < \frac{c_J}{2}, \qquad \psi_J \neq 0, \qquad \tilde{v}_J < v_J. \tag{3.20}$$

(ii) The image also contains the stratum $\psi_J = 0$, $\tilde{v}_J = v_J$ *(with $v_J, p_J, c_J > 0$) — a lower-dimensional sheet of handle points — every point of which is attained by infinitely many slices. There are no other image points.*

Proof. If $p_J, c_J > 0$ then (3.14) gives $b_S > 0$ and $\rho_S \in (-1, 1)$; conversely every admissible slice has $p_J, c_J > 0$ because $b_S > 0$ and $|\rho_S| < 1$ make both $\beta_L, \beta_R > 0$.

Next, the band on ψ_J is exactly the condition $|\chi| < 1$, which keeps m_S finite in (3.18). To see the equivalence, note from (3.14) that

$$\rho_S - 1 = \frac{(c_J - p_J) - (c_J + p_J)}{c_J + p_J} = \frac{-2p_J}{p_J + c_J}, \qquad \rho_S + 1 = \frac{2c_J}{p_J + c_J},$$

so, by (3.15),

$$|\chi| < 1 \iff \rho_S - 1 < \frac{4\psi_J}{p_J + c_J} < \rho_S + 1$$
$$\iff -\frac{2p_J}{p_J + c_J} < \frac{4\psi_J}{p_J + c_J} < \frac{2c_J}{p_J + c_J} \iff -\frac{p_J}{2} < \psi_J < \frac{c_J}{2}.$$

By (3.19), $\mathcal{D} > 0$ exactly when $\chi \neq \rho_S$, i.e. $\psi_J \neq 0$; and then $s_S > 0$ in (3.18) requires $v_J > \tilde{v}_J$. So on the domain (3.20) the three steps produce a unique admissible slice, and each failure of (3.20) breaks a requirement: $p_J \leq 0$ or $c_J \leq 0$ forces $b_S \leq 0$ or $|\rho_S| \geq 1$; ψ_J outside the band sends $|\chi| \geq 1$; $\psi_J = 0$ annihilates $\mathcal{D}$; $\tilde{v}_J \geq v_J$ makes $s_S \leq 0$.

For the stratum: $\psi_J = 0$ gives $\chi = \rho_S$, i.e. $m_S/\sqrt{m_S^2 + s_S^2} = \rho_S$, which solves to $m_S = s_S\rho_S/\sqrt{1-\rho_S^2}$; substituting into (3.4),

$$k^* = m_S - \frac{s_S\rho_S}{q_S} = \frac{s_S\rho_S}{\sqrt{1-\rho_S^2}} - \frac{s_S\rho_S}{\sqrt{1-\rho_S^2}} = 0$$

— the smile bottoms exactly at the money — so the floor is attained there and $\tilde{v}_J = v_J$ automatically. Conversely, fix (v_J, p_J, c_J) on the stratum. Then (3.14) sets (b_S, ρ_S), and for *every* width $s_S > 0$, the slice with $m_S = \rho_S s_S/\sqrt{1-\rho_S^2}$ and $a_S = v_J\tau - b_S s_S\sqrt{1-\rho_S^2}$ reproduces the same five handles: one handle set, a one-parameter family of slices. □

Remark 3.6 (The blind spot is the interior curvature). The free width on the stratum is not an abstract degeneracy; it is a specific piece of missing information. On the stratum the vertex sits at the money, the handles pin the level and both tails — and what they omit is exactly how sharply the curve turns through its bottom. By (3.4) the vertex curvature is $\kappa^* = b_S q_S^3/s_S$, which runs through every positive value as the free width s_S does.

Nor is the failure confined to the stratum itself; it is felt nearby. Differentiate (3.17) in χ (quotient rule, then simplify the numerator):

$$\frac{\partial\mathcal{D}}{\partial\chi} = \frac{-\rho_S(1-\chi^2) + \chi(1-\rho_S\chi)}{(1-\chi^2)^{3/2}} = \frac{\chi - \rho_S}{(1-\chi^2)^{3/2}},$$

which vanishes at $\chi = \rho_S$: the denominator is flat there, not just zero. Differentiate once more by the product rule,

$$\frac{\partial^2\mathcal{D}}{\partial\chi^2} = (1-\chi^2)^{-3/2} + 3\chi\,(\chi - \rho_S)\,(1-\chi^2)^{-5/2},$$

and evaluate at $\chi = \rho_S$, where the second term drops: $\partial^2\mathcal{D}/\partial\chi^2 = (1-\rho_S^2)^{-3/2}$. So by Taylor expansion, and then substituting $\chi - \rho_S = -4\psi_J/(p_J + c_J)$ from (3.15),

$$\mathcal{D} \sim \frac{(\chi-\rho_S)^2}{2\,(1-\rho_S^2)^{3/2}} = \frac{8\,\psi_J^2}{(p_J + c_J)^2(1-\rho_S^2)^{3/2}}, \qquad \psi_J \to 0. \tag{3.21}$$

The denominator closes *quadratically* in ψ_J. Three consequences, in increasing order of practical pain: near the stratum, a finite width forces the gap $v_J - \tilde{v}_J$ to be of order ψ_J^2, so the two handles carry nearly the same number; quote noise in the handles is amplified by $1/\psi_J^2$ in the recovered width; and a direct evaluation of (3.17) subtracts two nearly equal numbers, losing precision exactly where precision is scarce. An implementation can remove that last cancellation by evaluating an algebraically identical form — but no rearrangement can restore a curvature the handles never carried. And the stratum is not exotic: ATM sitting at the smile's minimum is where a desk would expect the coordinates to be simplest, and fitted index slices pass close to it.

Figure 3.5 draws the two halves of the remark. In panel (a), the denominator $\mathcal{D}$ is plotted against ψ_J at fixed (p_J, c_J) (blue): it closes at $\psi_J = 0$, and the dashed curve — the quadratic asymptote (3.21) — hugs it near the origin, confirming that the closing is quadratic, not linear. In panel (b), three slices share one handle quintuple $(v_J, 0, p_J, c_J, v_J)$ on the stratum: same level, same asymptotes, visibly different turns through the bottom, one per width s_S. The script verifies that the three slices agree on all five handles to 1.1×10^{-17} — machine precision — so nothing about the handles distinguishes curves the eye tells apart at a glance.

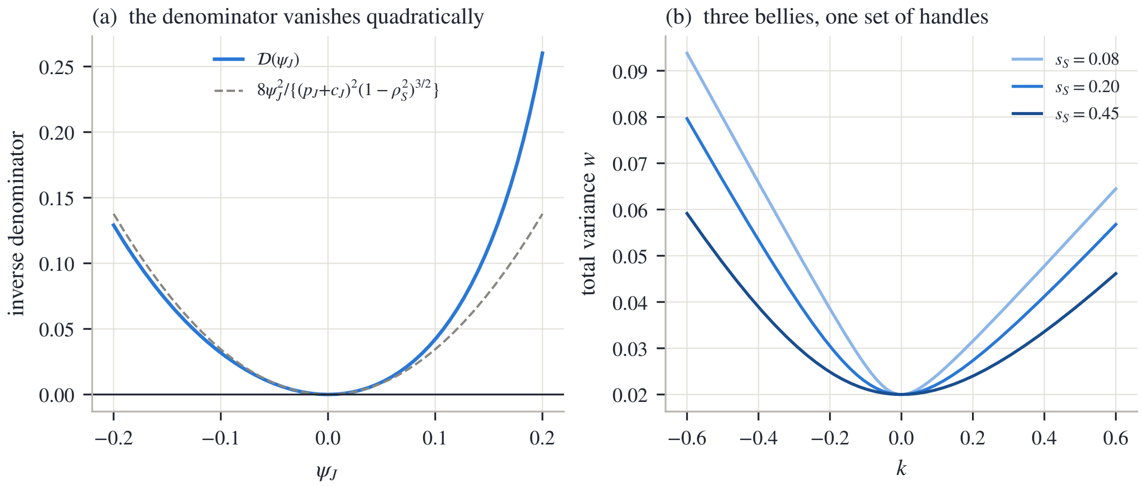

Figure 3.5: Where the jump-wing chart tears. (a) The inverse denominator $\mathcal{D}$ against ψ_J at fixed (p_J, c_J): it closes quadratically at $\psi_J = 0$, following (3.21) (dashed). (b) Three slices with identical handles $(v_J, 0, p_J, c_J, v_J)$ — same level, same asymptotes — differing only in width s_S: the five handles lose the interior curvature (handle agreement across the three slices: 1.1×10^{-17}).

We close the section with a practical dividend. The validity screens of table 3.1 translate directly into handle vocabulary, so a quoted handle set can be sanity-checked before any conversion: the floor condition is $\tilde{v}_J \geq 0$; the domain condition is $-p_J/2 < \psi_J < c_J/2$; and since $\max(\beta_L, \beta_R) = \max(p_J, c_J)\sqrt{w_0}$ by (3.13), the wing cap (3.8) reads

$$\max(p_J, c_J)\,\sqrt{v_J \tau} \;\leq\; \beta_{\max}. \tag{3.22}$$

None of the three certifies the interior, for the reason section 3.4 established: the belly answers to no cheap inequality, in any coordinate system.

3.6 Calibration in the chart

We can now write a slice, read it, and screen it. This section fits one to quotes. Chapter 2 already settled the anatomy of a slice objective — units, band targets, regularization (section 2.6) — and most of it carries over unchanged. Two choices genuinely change when the model lives in the volatility chart, and they are this section's subject: what the objective integrates against, and what coordinates the optimizer moves in.

First, the objective itself. Because the model produces $w(k)$ directly, no price inversion stands between model and data, and the residual can live in implied volatility: with quotes (k_i, σ_i) and weights μ_i, the data term is

$$\sum_i \mu_i \left(I(k_i;\theta) - \sigma_i\right)^2, \qquad I(k;\theta) = \sqrt{w(k;\theta)/\tau}, \tag{3.23}$$

optionally replaced by the band or haircut targets of (2.48) and (2.49) — Chapter 2's variants that fit to the quoted bid–ask spread, or to that spread shrunk by a declared margin, instead of to the mid — evaluated in volatility space.

3.6.1 The measure a fit integrates against

Start with a concrete ladder. A liquid expiry might list thirty strikes within a few percent of the money and three in the far put wing. Fit with equal weights and the wing contributes three rows out of thirty-three: less than a tenth of the objective decides the entire region where the tail lives. Nobody chose that allocation — it happened.

To say precisely what happened, idealize (3.23) as a continuum problem: minimize $\int (I - \sigma)^2 \,\mathrm{d}\mu(k)$ for some measure μ on the strike axis — "measure" in the ordinary sense of a distribution of weight along the axis, saying which strike ranges the fit cares about. A finite quote set turns the integral into a weighted sum, so the weight vector is a *quadrature rule* for μ — a recipe assigning each sample point its share of the integral: the weights *are* the measure. Equal weights make μ the counting measure of the exchange's strike listing — dense at the money, sparse in the wings. That listing was designed by a listings committee for trading convenience, not for inference, and a fit that adopts it inherits an accidental prior. It even fails the most basic sanity check: add more ATM listings — more information — and the fitted wings get *worse*, because the new near-duplicate rows outvote the isolated wing quotes by sheer count.

The repair is to *declare* the measure instead of inheriting it. A natural declaration weights strike ranges by the time value they carry: $\mathrm{d}\mu = \mathrm{TV}(k)\,\mathrm{d}k$, where $\mathrm{TV}(k)$ is the normalized price of the out-of-the-money option at k — the part of the premium that is genuinely optionality rather than intrinsic value, and so a gauge of how much economic content a strike carries. To discretize it on a quote ladder, give each quote the piece of axis nearer to it than to any neighbor: quote i's cell C_i runs half the gap to each adjacent quote (one-sided at the ends), with width $|C_i|$. The scheme

$$\mu_i \;\propto\; \mathrm{TV}_i\,|C_i| \tag{3.24}$$

then weights each quote by time value times the stretch of axis it represents.

Proposition 3.7 (Weights as a quadrature)**.** *For any continuous integrand f,*

$$\sum_i \mathrm{TV}_i\,|C_i|\,f(k_i) \quad \textit{is a Riemann sum of} \quad \int \mathrm{TV}(k)\,f(k)\,\mathrm{d}k$$

over the partition $\{C_i\}$, converging under refinement. In particular the weighted loss with (3.24) discretizes the continuum time-value objective; on a uniform strike grid all $|C_i|$ coincide and the scheme reduces to pure time-value weighting.

Proof. The cells partition the quoted interval and $k_i \in C_i$, so the sum is a Riemann sum over that partition; convergence under refinement is the standard property of Riemann sums. The uniform case is arithmetic. □

Two conventions keep the quadrature usable on real ladders, each in one sentence. The cell of an isolated far-wing quote can be arbitrarily wide, so the width factor is capped (the reference implementation uses ten times the mean spacing): beyond the cap the scheme deliberately under-weights the deep wing rather than let one quote carry a tenth of the objective. And the weights are normalized to unit mean, so switching schemes redistributes the data term without rescaling it against the regularization. Having said all this, the comparisons in this chapter still use equal weights throughout — deliberately. When three families are raced on the same ladder, the simplest shared convention is the fair one, and the accidental measure, once *recognized* as a measure, is no longer accidental. The lesson here is not that one scheme is right; it is that a weight vector *is* a measure, and should be chosen as one rather than inherited from a listings committee.

3.6.2 Two charts

Now the coordinates. Recall from section 2.6.3 what an optimization *chart* is: a smooth map from an unconstrained vector $\theta \in \mathbb{R}^n$ onto the model's parameters, built so that every θ lands in the allowed set — constraints eliminated by construction instead of policed by the solver. (This is the word "chart" in its second sense; the *volatility* chart of section 3.1 is the picture the model draws in, and an optimization chart is a coordinate system on the family.)

The historical chart for SVI maps $\theta \in \mathbb{R}^5$ to

$$a_S = \theta_1, \quad b_S = \mathrm{softplus}(\theta_2), \quad \rho_S = \tanh(\theta_3), \quad m_S = \theta_4, \quad s_S = e^{\theta_5}, \tag{3.25}$$

with $\mathrm{softplus}(\theta) = \log(1 + e^{\theta})$. Each lift does one job: softplus and the exponential keep b_S and s_S positive, tanh keeps ρ_S in $(-1, 1)$, and a_S, m_S need no lift. Every finite θ is therefore a structurally valid hyperbola — the first row of table 3.1, and nothing more. The floor can still go negative and a wing can still exceed the cap; those are fenced by the hinge rows (3.10), which the optimizer can trade against the data term while it runs.

The structural chart pushes the same idea one level up: parameterize the slice by the quantities the inequalities are *about*. Those are the two wing slopes, the vertex location, the floor, and the vertex curvature — five numbers, and the inversion displayed below in (3.27) will show that they determine the hyperbola completely. Lift each so that every finite coordinate is admissible:

$$\beta_L = \beta_{\max}\Lambda(\theta_1), \quad \beta_R = \beta_{\max}\Lambda(\theta_2), \quad k^* = \theta_3, \quad w^* = \mathrm{softplus}(\theta_4), \quad \kappa^* = e^{\theta_5}, \tag{3.26}$$

where Λ is the logistic function of section 2.1, $\Lambda(\theta) = 1/(1 + e^{-\theta})$, which maps the whole line onto $(0, 1)$ — so each wing lands strictly between 0 and $\beta_{\max}$, and the floor is strictly positive, before the optimizer has taken a single step.

The raw parameters are recovered by inverting proposition 3.1, one line per parameter. Adding and subtracting $\beta_L = b_S(1 - \rho_S)$ and $\beta_R = b_S(1 + \rho_S)$ gives b_S and ρ_S; solving $\kappa^* = b_S q_S^3 / s_S$ for s_S gives the width; the vertex location (3.4) then gives m_S, and the floor gives a_S:

$$\begin{gathered} b_S = \frac{\beta_L + \beta_R}{2}, \qquad \rho_S = \frac{\beta_R - \beta_L}{\beta_R + \beta_L}, \qquad s_S = \frac{b_S(1 - \rho_S^2)^{3/2}}{\kappa^*}, \\ m_S = k^* + \frac{s_S \rho_S}{\sqrt{1 - \rho_S^2}}, \qquad a_S = w^* - b_S s_S \sqrt{1 - \rho_S^2}. \end{gathered} \tag{3.27}$$

Under this *structural chart*, every finite iterate has $w(k) \geq w^* > 0$ and both wings strictly inside $(0, \beta_{\max})$: the second and third rows of table 3.1 hold by construction, and the hinge rows (3.10) are identically zero along the entire optimization path. What the chart does *not* deliver is the fourth row. The Vogt slice of section 3.4 satisfies floor and cap, so it corresponds to a perfectly ordinary θ in either chart, and the dense-grid check remains the deciding instance.

The two charts describe the same family, so a well-converged fit should not depend on which one the solver walked through — and it does not: on the SPY December quotes the two charts reach the same slice to 0.00 vol bp (fig. 3.6). What changes is the geometry the solver works in. A hinge row has a kink where the violation starts, exactly where an optimizer that needs derivatives is most likely to stall; the structural chart removes those kinks from the path entirely, because no iterate can reach them.

Figure 3.6 shows both halves. In panel (a), the wing lift $\beta = \beta_{\max}\Lambda(\theta)$: the whole horizontal axis (every finite coordinate) maps into the band between 0 and the dashed cap, itself strictly below the dotted Lee bound at 2 — admissibility is a property of the chart, not a constraint on the step. In panel (b), the five structural coordinates are drawn on the SPY December fit: the marked vertex at $(k^*, w^*) = (0.108, 0.0058)$, the osculating parabola — the parabola matching the curve's value, slope, and curvature $\kappa^* = 0.397$ at the vertex — hugging it at the bottom, and

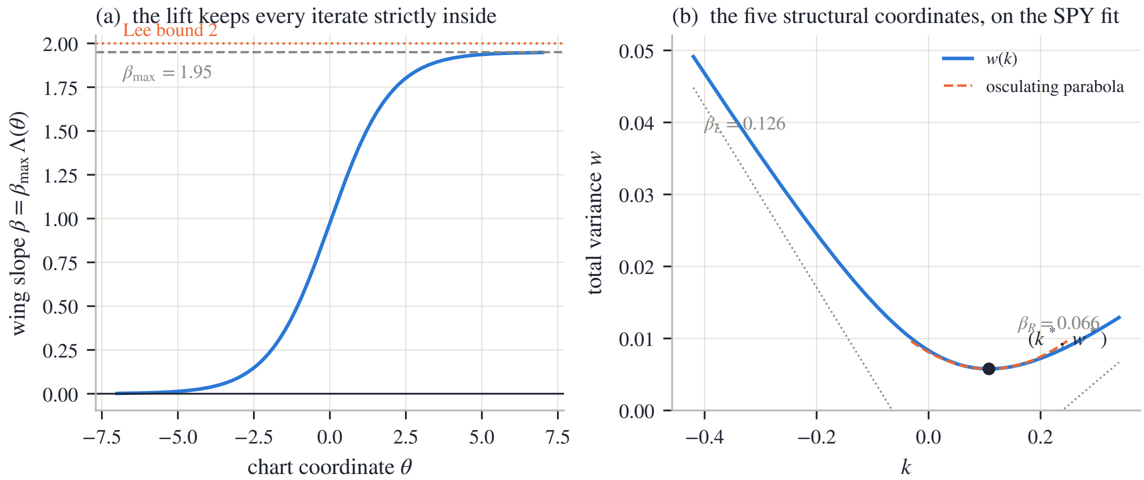

Figure 3.6: The structural chart. (a) The wing lift $\beta = \beta_{\max}\Lambda(\theta)$: every finite coordinate lands strictly between 0 and $\beta_{\max}$, itself strictly below Lee's bound — admissibility is a property of the chart, not a constraint on the step. (b) The five structural coordinates read on the SPY December fit: vertex $(k^*, w^*) = (0.108, 0.0058)$, osculating parabola of curvature $\kappa^* = 0.397$, and the two asymptote slopes.

the two asymptote slopes. The optimizer moves exactly the quantities the inequalities constrain — which is the whole idea.

Both charts are consumed by the same least-squares machinery as Chapter 2's, with analytic derivatives assembled by the chain rule; the assembly is bookkeeping rather than mathematics, and appendix 3.A records it. We now have everything needed to fit one hyperbola well. What remains is the shape it cannot take.

3.7 Superposition: capacity without touching the wings

Proposition 3.2 left a gap: some smiles — ahead of a binary event, or in a name whose holders are split between two scenarios — carry a central trough with a shoulder on each side, and no globally convex slice can follow them. We now fill the gap. But the design problem is sharper than "add flexibility." The wings are the *solved* part of the curve: Lee's bound caps them, the outermost quotes pin them, and section 3.3 showed that the tail density reads them alone. What is missing is detail in the *body*, and whatever supplies it must not move the wings at all.

The Multi-Core Sigmoid (MCS) meets this specification by superposition: a convex base curve that owns the wings, plus localized corrections that are silent — in value, in slope, *and* in curvature — in both tails. The whole construction rests on one elementary mechanism, and we build it first.

3.7.1 The base

The family works in a standardized moneyness

$$\zeta = \frac{k}{\sigma_{\mathrm{ref}}\sqrt{\tau}}, \tag{3.28}$$

where σ_{ref} is a reference volatility fixed before the fit (the quoted volatility nearest the money). The scaling makes ζ count moneyness in standard deviations, so parameters mean roughly the same thing at every expiry: with $\sigma_{\mathrm{ref}} = 20\%$ and $\tau = 0.25$, one ζ-unit is $0.20 \times 0.5 = 0.10$ of log-moneyness; at $\tau = 1$ it is 0.20. Displacements along this axis are written $\bar{\zeta}$: the base below measures its displacement from its center ζ_0, a kernel from its own center ζ_r, and $\bar{\zeta}$ always denotes the distance from the center at hand. (The symbol ζ is new to this chapter's ledger; Chapter 2's z is the log-odds of rank, an unrelated object.)

The convexity primitive of the family is the log-cosh:

$$\mathcal{A}_\kappa(\bar\zeta) = \frac{4}{\kappa^2}\log\cosh\frac{\kappa\bar\zeta}{2}, \qquad \mathcal{A}'_\kappa(\bar\zeta) = \frac{2}{\kappa}\tanh\frac{\kappa\bar\zeta}{2}, \qquad \mathcal{A}''_\kappa(\bar\zeta) = \operatorname{sech}^2\frac{\kappa\bar\zeta}{2}, \tag{3.29}$$

a smooth, even, convex softening of the absolute value, with κ setting how fast the softening straightens out. Two facts about it carry the whole section. Near the origin, $\log\cosh$ of a small argument is half its square, so $\mathcal{A}_\kappa(\bar\zeta) = \bar\zeta^2/2 + O(\bar\zeta^4)$: a rounded vertex, of unit curvature whatever κ. Far out it is *asymptotically affine* — straight line plus a vanishing remainder. To see this, write

$$\cosh x = \frac{e^x + e^{-x}}{2} = \frac{e^{|x|}}{2}\big(1 + e^{-2|x|}\big) \quad\Longrightarrow\quad \log\cosh x = |x| - \log 2 + \log\big(1 + e^{-2|x|}\big),$$

and the last term dies exponentially; scaling by $4/\kappa^2$ at argument $\kappa\bar\zeta/2$,

$$\mathcal{A}_\kappa(\bar\zeta) = \frac{2}{\kappa}|\bar\zeta| - \frac{4}{\kappa^2}\log 2 + o(1). \tag{3.30}$$

The base models annualized variance $V(\zeta) = w/\tau$ as level plus tilt plus one rounded hinge:

$$V_{\text{base}}(\zeta) = V_0 + V_1(\zeta - \zeta_0) + V_2\,\mathcal{A}_\kappa(\zeta - \zeta_0), \qquad \kappa = \begin{cases}\kappa_P, & \zeta < \zeta_0,\\ \kappa_C, & \zeta \ge \zeta_0,\end{cases} \tag{3.31}$$

six parameters: level V_0, slope V_1, and convexity amplitude $V_2 \ge 0$ at the center ζ_0, with separate put-side and call-side steepnesses κ_P, κ_C. Switching the steepness at ζ_0 costs no smoothness: $\mathcal{A}_\kappa$ and $\mathcal{A}'_\kappa$ both vanish at the origin, and $\mathcal{A}''_\kappa(0) = 1$ for *every* κ, so value, slope, and curvature all match across the switch — the curve is C^2, and the asymmetry lies only in how fast each side straightens.

The wings of the base follow from (3.30). As $\zeta \to +\infty$ the hinge contributes slope $2V_2/\kappa_C$, so V_{base} has asymptotic slope $V_1 + 2V_2/\kappa_C$; on the left, $V_1 - 2V_2/\kappa_P$, typically negative — the put wing rises as ζ falls. To convert to the total-variance slopes that Lee's bound reads, two bookkeeping steps: the map $w(k) = \tau V(k/(\sigma_{\text{ref}}\sqrt{\tau}))$ contributes a chain-rule factor $\tau/(\sigma_{\text{ref}}\sqrt{\tau}) = \sqrt{\tau}/\sigma_{\text{ref}}$; and on the put side the convention of (3.5) measures the slope in $|k|$, which flips the sign of the falling left slope:

$$\beta_P = \frac{\sqrt{\tau}}{\sigma_{\text{ref}}}\Big(\frac{2V_2}{\kappa_P} - V_1\Big), \qquad \beta_C = \frac{\sqrt{\tau}}{\sigma_{\text{ref}}}\Big(V_1 + \frac{2V_2}{\kappa_C}\Big). \tag{3.32}$$

These six numbers set the wings. The design goal for everything added next is that nothing moves them.

3.7.2 A kernel that vanishes to second order

Here is the mechanism. A centered second difference — value at the left, minus twice the value at the center, plus value at the right — annihilates affine functions exactly: for any straight line $L(\bar\zeta) = \text{const} + \text{slope}\cdot\bar\zeta$,

$$L(\bar\zeta - \ell) - 2L(\bar\zeta) + L(\bar\zeta + \ell) = \big(-\ell\cdot\text{slope}\big) + \big(+\ell\cdot\text{slope}\big) = 0 \qquad \text{identically.}$$

Now take the second difference of a function that is *almost* affine far out — such as $\mathcal{A}_\kappa$, by (3.30). Far out, the second difference sees an affine function plus an exponentially small remainder, and returns the remainder alone: a bump that dies in both tails. Near the center, where $\mathcal{A}_\kappa$ is genuinely curved, the second difference is not zero — that is the bump itself.

The correction kernel is exactly this, normalized: with $\bar{\zeta} = \zeta - \zeta_r$,

$$\mathcal{B}_{\zeta_r,\ell_r,\kappa_r}(\zeta) = \frac{\mathcal{A}_{\kappa_r}(\bar{\zeta} - \ell_r) - 2\mathcal{A}_{\kappa_r}(\bar{\zeta}) + \mathcal{A}_{\kappa_r}(\bar{\zeta} + \ell_r)}{2\,\mathcal{A}_{\kappa_r}(\ell_r)}, \tag{3.33}$$

centered at ζ_r, of half-width ℓ_r and steepness κ_r. The normalizer makes the peak height exactly one — at $\bar{\zeta} = 0$ the numerator is $2\mathcal{A}_{\kappa_r}(\ell_r)$, because $\mathcal{A}$ is even with $\mathcal{A}(0) = 0$ — so a kernel's amplitude will read directly as a variance displacement. Its derivatives are the same second difference applied to $\mathcal{A}'$ and $\mathcal{A}''$, over the same normalizer.

Lemma 3.8 (Zero wings)**.** *$\mathcal{B} \to 0$, $\mathcal{B}' \to 0$, and $\mathcal{B}'' \to 0$ as $\zeta \to \pm\infty$.*

Proof. By (3.30), $\mathcal{A}_\kappa$ is affine plus $o(1)$ in each tail; a centered second difference annihilates the affine part exactly (the computation displayed above), so what remains tends to zero. The same argument covers the derivatives: $\mathcal{A}'_\kappa$ tends to the constants $\pm 2/\kappa$ in the tails — affine of zero slope — so its second difference also tends to zero, and $\mathcal{A}''_\kappa$ itself decays. □

The lemma is the entire design, and notice that it does not depend on the log-cosh: *any* asymptotically affine primitive yields a tail-silent kernel by the same second difference. The log-cosh is a convenience — the model and its derivatives share one set of formulas (3.29), the unit peak makes amplitudes readable, and the half-width ℓ_r and steepness κ_r are separate dials where a Gaussian bump has only one width. One more reading will be used below: the kernel's curvature at its own center. Apply the second difference to $\mathcal{A}''$ from (3.29) at $\bar{\zeta} = 0$, using that $\mathcal{A}''$ is even with $\mathcal{A}''(0) = 1$:

$$\mathcal{B}''(\zeta_r) = \frac{\mathcal{A}''_{\kappa_r}(-\ell_r) - 2\mathcal{A}''_{\kappa_r}(0) + \mathcal{A}''_{\kappa_r}(\ell_r)}{2\,\mathcal{A}_{\kappa_r}(\ell_r)} = \frac{\operatorname{sech}^2(\kappa_r\ell_r/2) - 1}{\mathcal{A}_{\kappa_r}(\ell_r)} < 0,$$

negative because $\operatorname{sech}^2 < 1$ away from zero. So a *positive* multiple of $\mathcal{B}$ is locally concave — it builds a shoulder — and a negative multiple digs a notch.

3.7.3 The model, and what the superposition preserves

The MCS slice is the base plus M signed kernels:

$$V_M(\zeta) = V_{\text{base}}(\zeta) + \sum_{r=1}^{M} \gamma_r\,\mathcal{B}_{\zeta_r,\ell_r,\kappa_r}(\zeta), \qquad w(k) = \tau\,V_M\Big(\frac{k}{\sigma_{\text{ref}}\sqrt{\tau}}\Big), \tag{3.34}$$

with $6 + 4M$ parameters (6 for the base; center, half-width, steepness, and amplitude γ_r per kernel). By lemma 3.8, every member of the family has the same asymptotic wings as its base, whatever M: expressiveness in the body and control of the tails are decoupled by construction.

Figure 3.7 shows the mechanism end to end, on a synthetic target built to be globally admissible — a hyperbolic total-variance base with two Gaussian shoulders, asymptotically linear wings, and $\min g_{\mathrm{D}} = 0.277$ verified from analytic derivatives on a dense grid out to $|k| = 12$ (appendix 3.A records the construction and why a target must be certified this carefully). Panel (a) shows the primitive $\mathcal{A}_\kappa$ merging with its affine asymptote (3.30): within a couple of units the two are indistinguishable, and only the rounded vertex tells them apart — this exponential merging is what the kernel will inherit as tail silence. Panel (b) shows one kernel (3.33): a unit-height bump (blue) whose slope (orange) and curvature both die visibly in the tails — silent to all three orders, as lemma 3.8 proved. Panel (c) runs the fit on the W-shaped target (orange dots). The convex base alone ($M = 0$) cannot reach the two shoulders — proposition 3.2 forbids it — and misses by 118.2 vol bp, leaving a residual that is not noise but two localized features. Adding two kernels (centers dotted) seats both shoulders and cuts the miss to 0.7 vol bp, while the fitted slice remains butterfly-free: $\min g_{\mathrm{D}} = 0.420$ on the wide diagnostic grid.

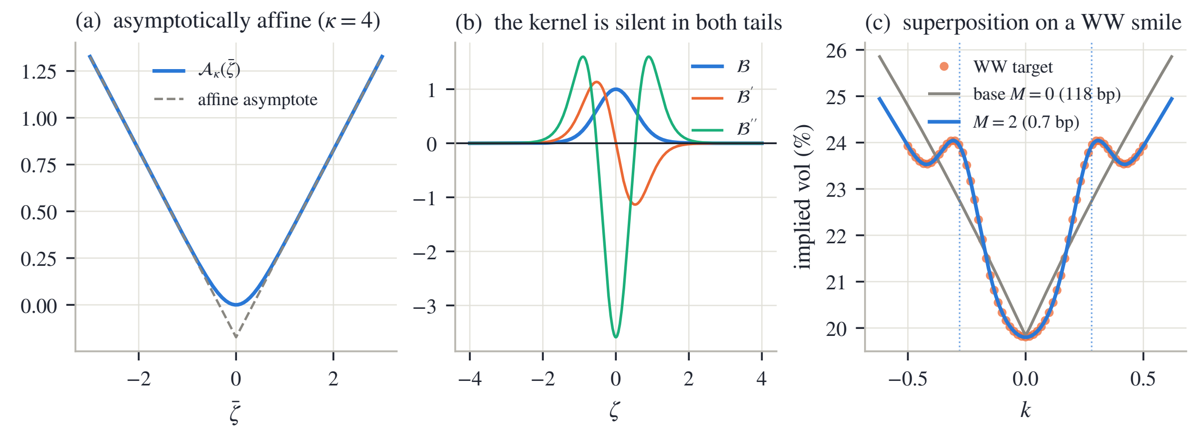

Figure 3.7: Superposition in three steps. (a) The log-cosh primitive merges with its affine asymptote (3.30); only the rounded vertex differs. (b) The centered second difference (3.33): a unit-height bump whose value, slope, and curvature all decay — the correction is silent in the wings to all three orders (lemma 3.8). (c) A W-shaped target (dots): the convex base misses both shoulders by 118.2 vol bp exactly as proposition 3.2 requires; two signed kernels (centers dotted) seat them, reducing the miss to 0.7 vol bp while the wings — and the fitted slice's admissibility — are preserved.

Two boundaries of the construction, each stated once. *Wing-neutral is weaker than wing-admissible*: the kernels preserve the base's slopes (3.32), but nothing in the family forces those slopes under the cap — the base is screened like any other slice. And *the family is not arbitrage-free by construction*: a signed kernel can drive variance negative or break price convexity locally, so the dense-grid check of section 3.4 decides for MCS exactly as it does for SVI — and since two overlapping kernels can trade amplitude without changing the curve, the fitted parameters are means to an end, and only the fitted curve is comparable across fits.

3.7.4 Capacity and its governance

The number of cores M is the family's capacity dial. Counting degrees of freedom already bounds it: $6 + 4M$ parameters against n_q quotes requires

$$M \le \left\lfloor \frac{n_q - 6}{4} \right\rfloor \tag{3.35}$$

for the fit to be determined at all — the analogue, for this family, of the LQD order guard (2.50). A thin chain with fifteen quotes, for instance, admits at most $\lfloor (15-6)/4 \rfloor = 2$ cores.

But identification is the weaker constraint, and the stronger one is worth stating in words. A kernel is a local feature detector. On a smile that has genuine local features — an event trough, a second mode — it finds them. On a smile that has none, it finds quote noise instead: the error against the quotes falls while the error against the underlying smile rises, and the manufactured curvature shows up exactly where section 3.4 taught us to look — as a negative dip of g_{D} in a sparsely quoted region, where no data resists it. Section 3.8 measures this on the frozen snapshot: at $M = 2$ on liquid index chains, MCS fits the mid quotes better than SVI on every node and fails the validity check on most of them; at $M = 0$ the margins are uniformly clean and the fit quality is SVI's. The capacity is real, and so is its price.

The reference implementation therefore treats M as a governed dial — capped by (3.35), ridged in amplitude, and reserved for smiles that are genuinely non-convex — rather than as a default. The calibration is layered to match the model's structure: fit the base first; seed kernels one at a time on the largest remaining residual features; then refine all $6 + 4M$ parameters jointly (appendix 3.A records the settings).

3.8 Three families on the same quotes

The chapter closes by putting the three families side by side on the same data: the frozen snapshot of appendix 2.C, all 16 nodes, under one protocol. Each family is calibrated by the reference implementation to the same mid quotes with equal weights. LQD runs at polynomial order 16 — its capacity dial, cut back by the guard (2.50) on thin chains — in its own constraint-eliminating coordinates (the logistic chart of section 2.6.3); SVI runs in the structural chart of section 3.6.2; MCS runs at $M = 2$. Each fit reports the same three quantities: the rms distance to the quotes, the validity margin $\min g_D$ over a dense grid on the traded range, and its wing slopes. Every Durrleman function is evaluated from the family's own analytic structure; appendix 3.A records the details. The point of the exercise is not a ranking. It is to watch each family's contract — what it guarantees, what it must have checked — show up in numbers.

3.8.1 The December node and the SPY gallery

Figure 3.8 shows the SPY December 2026 slice, the node the chapter has used throughout. In panel (a), the orange dots are the mid quotes with their bid–ask band, and the three fitted curves — LQD in green, SVI in blue, MCS in violet — overlie so closely that at chart scale they read as one curve inside the band. Whatever separates the families, it is not visible here. Panel (b) is where the information lives: it plots each family's *residuals*, quote by quote, against the mids. Now the three separate cleanly. LQD tracks the quotes at 3.4 vol bp rms with a residual that looks like noise — no structure left. SVI carries 7.5 vol bp, and its residual is not noise: it is one slow wave across the strike range, the visible signature of a five-parameter curve absorbing a whole quoted skew — the cost of parsimony made concrete. MCS at $M = 2$ sits between, at 5.6 vol bp: its two kernels have spent themselves on the two largest features of the SVI-shaped wave.

The three fits also agree on the wings to within a few hundredths of slope — $\beta_R = 0.034$, 0.066, and 0.054 respectively — and it is worth pausing on why this is expected rather than reassuring. Section 2.4 made the point for LQD: no finite strip of quotes identifies an asymptotic slope by itself; a fitted wing is the model's convention continuing the last quoted evidence. Here, three different conventions continue similar evidence in similar ways. The agreement measures the evidence, not the truth of any family's tails.

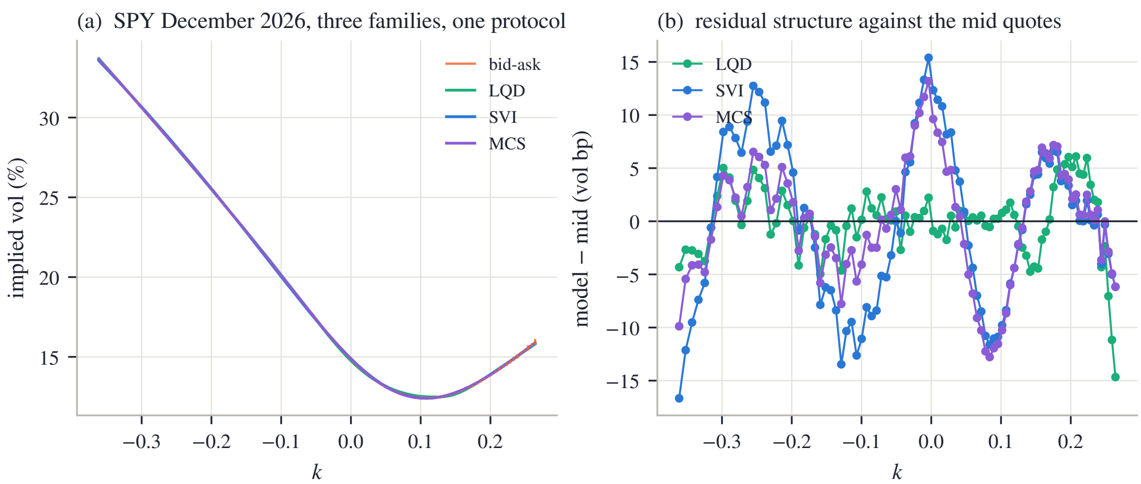

Figure 3.8: Three families, one protocol, on SPY December 2026. (a) The fitted curves overlie at chart scale inside the bid–ask band. (b) The residuals against the mid quotes separate them: LQD (3.4 bp rms, 17 parameters), MCS (5.6 bp, 14), SVI (7.5 bp, 5). More coordinates buy a whiter residual.

Figure 3.9 extends the reading to all eight SPY expiries. Panel (a) plots fit quality by expiry, and the ordering is stable everywhere: LQD, then MCS, then SVI, with SVI's gap widening at the long end, where the skew's shape departs furthest from a hyperbola. More parameters fit better, uniformly — no surprise. Panel (b) is the panel to study: the validity margin $\min g_{\mathrm{D}}$ by expiry, with the zero line marked. The picture inverts. LQD's margin is positive on every node — proposition 2.4 made that structural, so this panel is simply the numerical trace of a theorem. SVI's margin is positive on every node as well, and it cannot be by theorem — the Vogt slice of section 3.4 is proof that the family contains invalid members. It is by rigidity: five parameters fitted to a liquid convex smile have no capacity left over to bend the density negative. MCS at $M = 2$ fails the check on most of the SPY expiries — even though its own optimizer carries a sampled hinge on g_{D} (appendix 3.A), the concrete form of section 3.4's warning that a finite-weight fence discourages violations without excluding them. Its kernels, finding no genuine local features on a liquid index smile, chase quote noise, and the manufactured curvature surfaces as a negative dip of g_{D} — typically in the thinly quoted region, where no data resists it.

Across all 16 nodes (both names), the medians read:

	median rms (bp)	worst rms (bp)	median $\min g_{\mathrm{D}}$	nodes with margin < 0
LQD (order 16, guarded)	7.3	34.0	0.184	0 of 16
SVI (structural chart)	19.0	68.0	0.173	0 of 16
MCS, $M = 2$	11.8	38.5	−0.026	10 of 16
MCS, $M = 0$ (base only)	18.1	—	0.179	0 of 16

Read the two MCS rows against each other: they isolate the dial. Withdraw the kernels and the margins return to uniformly positive while the fit quality falls back to SVI's level. Capacity is neither free nor fatal — it is a choice, and on these liquid chains the correct choice is zero.

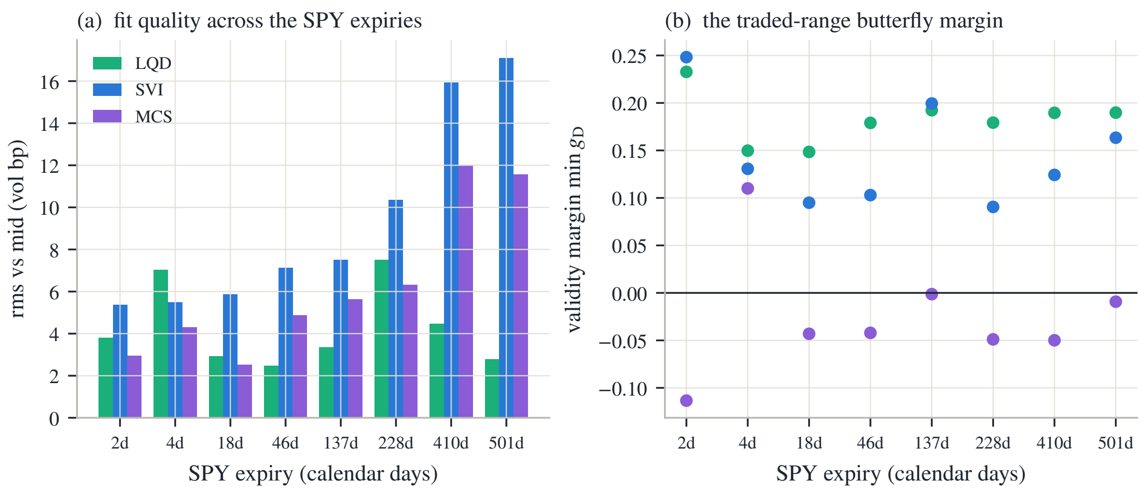

Figure 3.9: The SPY gallery under one protocol. (a) Fit quality by expiry: more parameters fit better, uniformly. (b) The validity margin by expiry: LQD is positive by construction, SVI by rigidity, and MCS at $M = 2$ — its kernels chasing noise on a smile with no genuine local features — fails the dense-grid check on most nodes. The two panels together are the chapter's thesis in one figure: in the volatility chart, fit quality and validity are separate accounts.

3.8.2 Expressiveness measured on a legal target

The gallery shows what excess capacity does where it is not needed. The complementary question — what rigidity costs where capacity *is* needed — has to be asked carefully. It would prove nothing to fit a W-shaped curve that no law can produce; the target must itself be a genuine law, so that an arbitrage-aware family is allowed to reproduce it.

A mixture of two lognormals supplies one. Take component weights 0.65 and 0.35 and component volatilities 14% and 42% — a calm scenario and a stressed one — with a quarter year to expiry. Give the stressed component a forward of 0.85; the martingale identity (the weighted forwards must average to one) then fixes the calm component's forward:

$$0.65\, f_1 + 0.35 \times 0.85 = 1 \quad \Longrightarrow \quad f_1 = \frac{1 - 0.2975}{0.65} = 1.081.$$

This law is genuine by construction — a mixture of laws is a law — and because its density is known exactly, its own Durrleman minimum can be computed rather than estimated: $\min g_{\mathrm{D}} = 0.369$ on the quoted range. Bimodal, but strictly admissible.

Figure 3.10 fits the three families to this smile. In panel (a), the target smile (orange dots) shows the two-mode signature, and the three fitted curves now visibly separate — unlike on the index node, chart scale is enough. LQD (green) follows the two modes at quote scale. SVI (blue) cuts through them with its single belly. MCS (violet) recovers most of the shape. Panel (b) makes the comparison quantitative with the signed errors. LQD fits at 3.6 vol bp with margin 0.371. SVI misses by 166.4 vol bp — a structured, two-lobed error of about two volatility points, which is proposition 3.2 in numbers: one belly cannot seat two modes. Note what SVI's failure is *not*: its margin is 0.431, perfectly valid. Rigidity fails at expressiveness, not at arbitrage. MCS halves SVI's miss to 87.3 vol bp, but overdrives its kernels into a small violation (-0.066): on this target its capacity is genuinely needed — and still needs its check. No family wins all three columns, which is the result.

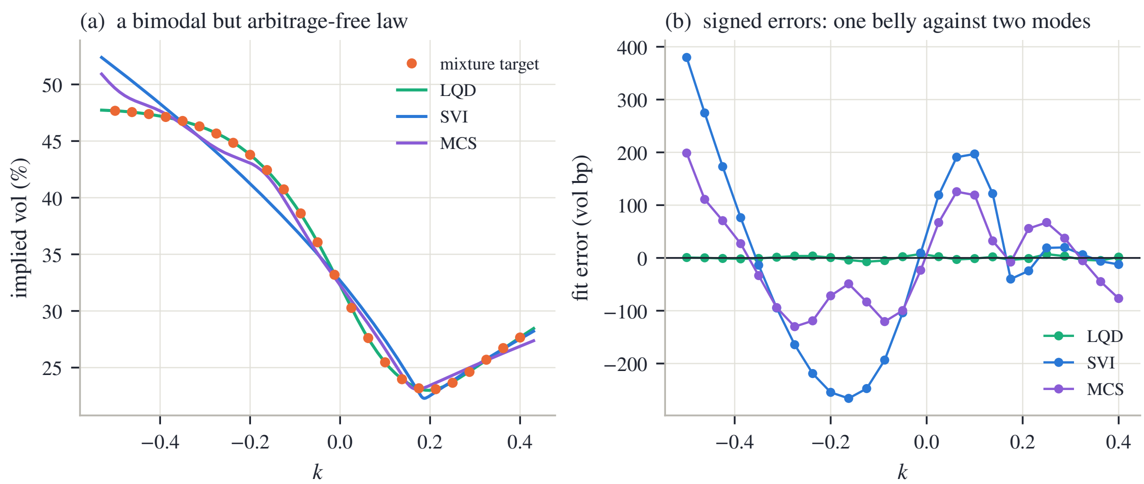

Figure 3.10: The rigidity question asked legally: a bimodal but strictly admissible mixture law ($\min g_{\mathrm{D}} = 0.369$). (a) The smile and the three fits. (b) Signed errors: SVI's single belly leaves a ±two-vol-point structured miss; MCS halves it; LQD follows the modes at quote scale.

3.8.3 The chapter's contract

The node, the gallery, and the mixture say one thing three ways, and it is the chapter's summary. Working in the volatility chart does not remove the constraint that a smile be a probability law. It moves the constraint out of the coordinates and into the process: some of it becomes inequalities a chart can absorb (the floor, the cap), some of it becomes a check no chart absorbs (the butterfly grid), and some of it becomes capacity that must be rationed because nothing structural rations it (the cores). What the chart buys in exchange is legibility: coordinates a desk can quote, wings that are two readable numbers, corrections that are local by construction.

	LQD (Ch. 2)	SVI	MCS
Validity	structural: every admissible vector is a law (proposition 2.4)	policed: floor and cap structural in the chart; butterfly checked on a grid	policed: wings neutral by lemma 3.8; butterfly checked on a grid
Tails	exponential class, slopes from endpoint speeds (proposition 2.9)	exactly linear wings; slopes capped at $\beta_{\max} < 2$	base-owned wings (3.32); admissibility a separate check
Interior capacity	polynomial order, guarded by (2.50)	one belly, one turn (proposition 3.2)	M kernels, guarded by (3.35); governed, not default
Interpretability	endpoint speeds and ledger functionals	JW handles on the domain of theorem 3.5; curvature lost on the stratum	curve-level only: kernels are non-unique by design
What convention supplies	order, ridge, tail chart	$\beta_{\max}$, chart, weights	M, ridge, weights

Table 3.2: What each family guarantees, what must be checked, and what the modeler chooses. The families do not dominate one another; they allocate the same difficulty differently.

Chapter 2 and this chapter are therefore not competitors but two poles of one design axis — validity built into the coordinates at the price of legibility, or coordinates built for the reader at the price of policing. The choice of chart allocates the difficulty; it does not remove it. The next chapter moves to a third position on the same axis: local volatility, where the modeled object is neither a law nor a curve but the field consistent with all of them at once.

3.A Comparison protocol and numerical notes

Protocol

Every number and figure in this chapter is produced by one deterministic script from the frozen snapshot of appendix 2.C and the fixed synthetic constructions below; nothing is typed by hand. The three-family comparison of section 3.8 calibrates each family to the same mid quotes of each node with equal unit-mean weights:

- **LQD**: order 16 reduced by the guard (2.50), logistic tail chart, ridge $\alpha = 10^{-6}$ with $\nu = 1$ — the recipe of section 2.6 with the mid objective.
- **SVI**: the structural chart (3.26) with $\beta_{\max} = 1.95$ and hinge multiplier $W = 10^3$ (the rows are structurally zero in this chart; the multiplier matters only for the historical chart (3.25)).
- **MCS**: $M = 2$ under the cap (3.35), amplitude ridge 10^{-2}, and a one-sided hinge on sampled g_{D} over a grid extending two ζ-units past the quoted range, weighted double on the put side — a soft fence with the same status as (3.10): zero on any admissible fit. Kernel centers may roam half a unit past the quoted range; half-widths are confined to $[0.15, 1.5]$ and steepnesses to $[1, 12]$ in ζ-units. The base is fitted first, kernels are seeded greedily on the largest residuals with a masked neighborhood so two kernels cannot seed one feature, and a joint refinement follows.

The validity margin is $\min g_{\mathrm{D}}$ over 801 equally spaced points spanning exactly the node's quoted log-moneyness range.

Evaluating g_{D} per family

Each family's Durrleman function is computed from its own analytic structure, never by differencing prices — a finite difference of reconstructed prices carries curvature noise at exactly the scale the diagnostic measures. For SVI, (w, w', w'') are the closed forms (3.3). For MCS, the ζ-space derivatives of (3.34) are analytic (second differences of (3.29)) and convert by $w' = \sqrt{\tau}\, V'/\sigma_{\mathrm{ref}}$, $w'' = V''/\sigma_{\mathrm{ref}}^2$. For LQD no derivatives are needed at all: (2.5) states $f_X(k) = g_{\mathrm{D}}(k)\, \varphi(d_-)/\sqrt{w}$, so

$$g_{\mathrm{D}}(k) = \frac{f_X(k)\sqrt{w(k)}}{\varphi\big(d_-(k, w(k))\big)}, \qquad d_- = -\frac{k}{\sqrt{w}} - \frac{\sqrt{w}}{2},$$

with f_X the slice's explicit density — the identity that defined g_{D} is run backwards. For the mixture target of fig. 3.10 the same identity is used with the exact mixture density, which is how the target's own margin is known rather than estimated.

The residual Jacobian, in layers

The least-squares solvers of both charts consume an analytic Jacobian, and it assembles by the chain rule in three layers. The raw-parameter partials of the hyperbola follow from (3.1) with (3.2):

$$\frac{\partial w}{\partial a_S} = 1, \quad \frac{\partial w}{\partial b_S} = \rho_S x_S + r_S, \quad \frac{\partial w}{\partial \rho_S} = b_S x_S, \quad \frac{\partial w}{\partial m_S} = -b_S\Big(\rho_S + \frac{x_S}{r_S}\Big), \quad \frac{\partial w}{\partial s_S} = \frac{b_S s_S}{r_S}. \tag{3.36}$$

The data rows live in volatility, so each is scaled by

$$\frac{\partial I}{\partial w} = \frac{1}{2\sqrt{w\tau}} = \frac{1}{2\tau I}. \tag{3.37}$$

Finally the lift: for (3.25) the three nontrivial factors are

$$\frac{\mathrm{d}b_S}{\mathrm{d}\theta_2} = 1 - e^{-b_S}, \qquad \frac{\mathrm{d}\rho_S}{\mathrm{d}\theta_3} = 1 - \rho_S^2, \qquad \frac{\mathrm{d}s_S}{\mathrm{d}\theta_5} = s_S, \tag{3.38}$$

while the structural chart pushes the same raw-space rows through the closed-form 5×5 matrix $\partial(a_S, b_S, \rho_S, m_S, s_S)/\partial\theta$ obtained by differentiating (3.27) along (3.26). Hinge rows contribute their linear part where the violation is strictly positive and a zero row otherwise. Nothing here is deep; the point of recording the layers is that a chart is a change of variables like any other, and its cost is one matrix product per iterate rather than a constrained solver.

Synthetic constructions

The W-shaped target of fig. 3.7 is built in total variance so that its wings are asymptotically linear and its derivatives are exact:

$$w(k) = 0.008 + 0.012\sqrt{k^2 + 0.15^2} + 0.0025\big(e^{-\{(k-0.28)/0.11\}^2} + e^{-\{(k+0.28)/0.11\}^2}\big), \qquad \tau = 0.25,$$

sampled at 61 equally spaced quotes on $[-0.5, 0.5]$. Gaussians vanish with all derivatives, so both wings carry the asymptotic slope 0.012 — far under Lee's bound — and the tail limit of g_{D} is $(4 - 0.012^2)/16 = 0.250$; the script verifies $\min g_{\mathrm{D}} = 0.277 > 0$ on a dense grid to $|k| = 12$ from the analytic derivatives before any figure is written. A target built carelessly in volatility space with linear volatility tails would violate Lee far outside the plotted window: total variance would grow quadratically. Certify a synthetic target the same way as a fit, globally, or it certifies nothing.

The mixture target is the two-lognormal law of section 3.8: weights $(0.65, 0.35)$, volatilities $(0.14, 0.42)$, second component forward 0.85 and the first fixed by the martingale identity $0.65\, f_1 + 0.35 \cdot 0.85 = 1$ (so $f_1 = 1.0808$), $\tau = 0.25$, quoted at 25 strikes on $k \in [-0.5, 0.4]$ through exact Black inversion of the mixture call.

Floating-point strictness

The structural chart's guarantee — wings strictly inside $(0, \beta_{\max})$, floor strictly positive — is a statement about real numbers, and float64 does not honor it automatically. Two boundary effects deserve record. A logistic lift evaluated at a moderately large coordinate rounds to exactly 1, parking a wing exactly at the cap; the implementation clips the lift one unit in the last place inside 1. And when one wing saturates against an ordinary other, the quotient for ρ_S in (3.27) can round to ± 1, making $1 - \rho_S^2$ an exact zero and m_S undefined; the implementation computes it by the algebraically identical product form

$$1 - \rho_S^2 = \frac{4\,\beta_L \beta_R}{(\beta_L + \beta_R)^2},$$

which never rounds to zero for admissible wings. A chart whose strictness silently fails at the boundary reproduces exactly the failure mode it was built to remove.

Chapter 4

Local Volatility

Chapters 2 and 3 gave each expiry its own probability law and used a calendar condition to keep neighboring expiries consistent. This chapter asks for more: one model that carries every expiry of the surface at once. The Dupire equation provides it. Its coefficient — the local volatility — turns out to be a ratio of the two quantities the earlier chapters already policed: the calendar increment on top, the butterfly factor below. So the field exists exactly where the surface is valid. Computing the field, however, splits into a wrong way and a right way. Read as a formula, the ratio asks for two derivatives of noisy quotes, and fails no matter how good the data gets. Read as a pricing model, it poses a well-behaved fitting problem: a sheet of positive numbers on a grid, priced by a stable marching scheme, differentiated exactly, and fitted to the quotes like every other model in this book. The chapter builds that fit slowly, shows why the quotes alone cannot pin the sheet down, and ends by fitting the book's frozen SPY and NVDA chains with one sheet each.

4.1 One diffusion for the whole surface

Chapters 2 and 3 both ended in the same place. Each expiry of a fitted surface carries its own probability law for the return: Chapter 2 built the law first and read the smile from it, Chapter 3 wrote the smile directly and then checked that it described a law. Between neighboring expiries one further condition held, the calendar order of section 2.8: at the same log-moneyness, the later normalized call is worth at least the earlier one. A surface, so far, is a stack of laws, related two at a time.

This chapter asks for more. Is there one random process — one model of the underlying's path — whose distribution at each expiry is exactly the law that expiry's smile encodes? If there is, the surface stops being a stack and becomes a single object: one dynamics, and every European price of every maturity comes out of it.

Half of the answer is already known. Recall two terms from Chapter 2. Calendar order across all adjacent pairs is *convex order* of the normalized returns — later laws are more spread out. And a *martingale* is a process with no drift: its expected future value, given everything known today, is today's value. A theorem of Kellerer [19] says these two notions meet exactly: if the family of laws is in convex order, then *some* martingale has these laws as its distributions at the given dates. But the theorem is pure existence. It does not name the martingale, it gives it no structure a computation could use, and it does not say what extra information any particular choice smuggles in.

Dupire's observation [8] answers all three objections at once, by restricting attention to one concrete class of processes. Work in the normalized units the book has used since section 2.8: $Y_\tau = S_\tau / F_{0,\tau}$ is the underlying divided by its forward. Dividing by the forward is what strips the drift: the forward is the expected future level under the pricing law (with deterministic carry, as section 2.8 assumed), so the ratio starts at 1 and has no tendency to grow or shrink —

it is a positive martingale worth 1 at time zero. And τ is variance time, the maturity variable the book has quoted variance against since Chapter 2, so that a slice's total implied variance is $w = \sigma^2\tau$. The reader knows the Black–Scholes dynamics $\mathrm{d}Y = \sigma Y\,\mathrm{d}W$, with one constant volatility. Now let the volatility depend on where the process is and on when it is there:

$$\mathrm{d}Y_\tau = \sqrt{v(\tau, Y_\tau)}\, Y_\tau\, \mathrm{d}W_\tau, \qquad Y_0 = 1, \tag{4.1}$$

where $v(\tau, y) \geq 0$ is a fixed, deterministic function of two variables. The function v is called the *local variance*; its square root $\sigma_{\mathrm{loc}} = \sqrt{v}$ is the *local volatility*. It is a field: a whole surface of variances, one for every (time, level) pair, laid out in advance. When the process visits level y at time τ, it diffuses at exactly the rate the field prescribes there. Nothing else is random: the field is the entire model.

Within this class, the vague question above has an exact answer, in both directions. Given the field, one linear equation — derived in the next section — produces the marginal law of every maturity at once. Given the surface, a pointwise formula recovers the field, wherever the implied density is positive. A valid surface is therefore not merely *consistent with* some model. It *is* the coefficient field of one canonical model, written in different coordinates. This chapter is about that change of coordinates: how to compute it, why the obvious way of computing it fails on real data, and what a market's finitely many quotes do and do not say about the field.

Before starting, be clear about what the object is not, on two counts. First, the diffusion (4.1) is a *representative*, not a discovery. A theorem of Gyöngy [16] says that any well-behaved martingale with the same marginal laws — however wild its own volatility, random or path-dependent — is *mimicked* by (4.1): take $v(\tau, y)$ to be the average instantaneous variance of that process over the paths that sit at y at time τ, and the diffusion reproduces every one of its marginals. Many different dynamics share one field. Anything that depends only on the marginal laws — every European price, every quantity of Chapters 2–3 — is therefore safe to compute in the model. Anything that depends on the joint law across dates — above all, how the smile moves when the spot moves — is not determined by the surface at all. That question is postponed to Chapter 9, and no computation in this chapter will touch it.

Second, the class (4.1) contains no jumps. Every path is continuous, so a genuine gap — an earnings night, a scheduled announcement — can only be imitated by a brief burst of very large diffusive variance. The honest treatment of scheduled events is a change of clock, and it waits for Chapter 8. Throughout this chapter, rates, dividends and borrow — the cost of shorting the stock — live inside the forward as usual (Chapter 6 makes that precise), $y = e^k$ is the normalized strike, $c(\tau, y)$ is the normalized call, and the field v is variance per unit of variance time.

The plan follows the two directions of the correspondence. Section 4.2 derives the forward equation, shows that a nonnegative field makes the whole surface arbitrage-free automatically, and turns the equation around into the extraction formula — whose two ingredients will be exactly the two validity conditions of the static chapters. Section 4.3 shows why the extraction formula must never be applied to market quotes. Sections 4.4 and 4.5 build the alternative: a field with finitely many positive parameters, and a pricing scheme that keeps the theory's guarantees on a finite grid. Section 4.6 asks what the quotes actually determine about the field, and sections 4.7 and 4.8 assemble the calibration and run it on the book's frozen data.

4.2 The equation the surface must obey

4.2.1 Forward: from the field to every marginal

Fix a bounded measurable field $v \geq 0$ and let Y solve (4.1). We now derive the equation that the normalized call satisfies as a function of maturity. It is the computational engine of the whole chapter: solve it once, marching left to right in maturity, and every expiry's smile comes out along the way.

Proposition 4.1 (The forward Dupire equation)**.** *Suppose Y_τ has a density $f(\tau,\cdot)$ such that $vy^2 f$ and $y\,\partial_y(vy^2 f)$ are integrable and vanish at $y=0$ and as $y\to\infty$. Then the normalized call $c(\tau,y)=\mathbb{E}[(Y_\tau-y)^+]$ satisfies*

$$\boxed{\partial_\tau c(\tau,y)=\tfrac12\,v(\tau,y)\,y^2\,\partial_{yy}c(\tau,y),\qquad c(0,y)=(1-y)^+.} \tag{4.2}$$

Proof. The proof has three steps: find the equation the density obeys, integrate that equation against the call payoff, and move the derivatives across.

Step 1: the density. Apply Itô's formula to $\Xi(Y_\tau)$, for a smooth function Ξ — recall that Itô adds a second-order term to the ordinary chain rule, because the squared increment $(\mathrm{d}Y)^2$ accumulates at the rate $v\,Y^2\,\mathrm{d}\tau$ here:

$$\mathrm{d}\,\Xi(Y_\tau)=\Xi'(Y_\tau)\,\sqrt{v}\,Y_\tau\,\mathrm{d}W_\tau+\tfrac12\,v(\tau,Y_\tau)\,Y_\tau^2\,\Xi''(Y_\tau)\,\mathrm{d}\tau.$$

Take expectations. The $\mathrm{d}W$ term has mean zero, and writing the remaining expectations against the density f,

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\int_0^\infty \Xi(y)\,f(\tau,y)\,\mathrm{d}y=\tfrac12\int_0^\infty v(\tau,y)\,y^2\,\Xi''(y)\,f(\tau,y)\,\mathrm{d}y.$$

Let Ξ vanish outside a bounded set away from 0, and integrate the right side by parts twice, moving both derivatives off Ξ and onto $vy^2 f$ (no boundary terms, since Ξ vanishes near the ends):

$$\int v\,y^2 f\,\Xi''\,\mathrm{d}y=-\int \partial_y(v\,y^2 f)\,\Xi'\,\mathrm{d}y=\int \partial_{yy}(v\,y^2 f)\,\Xi\,\mathrm{d}y.$$

This holds for every such Ξ; and if the two integrands differed anywhere, choosing a Ξ concentrated near that point would break the equality. So they agree:

$$\partial_\tau f=\tfrac12\,\partial_{yy}(v\,y^2 f).$$

This is the *Fokker–Planck equation* — the equation every diffusion's density obeys; the two displays above are its entire derivation.

Step 2: integrate against the payoff. The call is $c(\tau,y_0)=\int_0^\infty (y-y_0)^+ f(\tau,y)\,\mathrm{d}y$; differentiating under the integral sign (which the decay hypothesis licenses),

$$\partial_\tau c(\tau,y_0)=\int_0^\infty (y-y_0)^+\,\partial_\tau f(\tau,y)\,\mathrm{d}y=\tfrac12\int_0^\infty (y-y_0)^+\,\partial_{yy}(v\,y^2 f)\,\mathrm{d}y.$$

Step 3: move the derivatives onto the payoff. Integrate by parts once. The payoff's derivative in y is the step function $\mathbf{1}_{\{y>y_0\}}$ — slope 0 below the strike, 1 above — and the boundary term $(y-y_0)^+\,\partial_y(vy^2 f)$ vanishes at both ends by the decay hypothesis:

$$\tfrac12\int_0^\infty (y-y_0)^+\,\partial_{yy}(vy^2 f)\,\mathrm{d}y=-\tfrac12\int_{y_0}^\infty \partial_y(v\,y^2 f)\,\mathrm{d}y.$$

The remaining integral is elementary: it integrates an exact derivative, and $vy^2 f$ vanishes at the far end, so

$$-\tfrac12\int_{y_0}^\infty \partial_y(v\,y^2 f)\,\mathrm{d}y=-\tfrac12\Big[v\,y^2 f\Big]_{y_0}^\infty=\tfrac12\,v(\tau,y_0)\,y_0^2\,f(\tau,y_0).$$

Finally, recall the density identity of section 2.1: differentiating the call twice in strike returns the density, $f(\tau,y_0)=\partial_{yy}c(\tau,y_0)$. Substituting gives (4.2). The initial condition is $Y_0=1$, whose call is $(1-y)^+$. □

Read (4.2) as a physics problem. It is a heat equation in the *strike* variable: the call price plays the temperature, the payoff $(1-y)^+$ is the initial temperature profile, maturity plays the role of time, and $\tfrac12 vy^2$ is the conductivity of the material at position y and time τ. One field, one linear march, and every marginal law of the surface comes out at once. The analogy is not decoration; it is the source of every structural property this chapter uses. Heat flow spreads bumps and never sharpens them, and "spreading", translated into option language, is exactly the no-arbitrage direction. The next proposition makes that precise, and it is the reason this chapter can *promise* validity rather than police it. (One name in it deserves its reminder: a *butterfly* is the three-strike spread whose payoff is a tent, and its price is a second difference of call prices — so ruling out negative butterfly prices is exactly the condition $\partial_{yy}c \geq 0$, a nonnegative implied density.)

Proposition 4.2 (Positive local variance propagates no-arbitrage)**.** *Let c solve (4.2) with $v \geq 0$ bounded. Then for every $\tau > 0$: (i) $\partial_{yy}c(\tau, \cdot) \geq 0$ — each slice is butterfly-free; (ii) $\partial_\tau c(\tau, y) \geq 0$ — the normalized call is non-decreasing in maturity at fixed y, which is the calendar order of section 2.8.*

Proof sketch. Differentiate (4.2) twice in y: the function $f = \partial_{yy}c$ — by Step 3's density identity, the same f as in proposition 4.1 — satisfies $\partial_\tau f = \frac{1}{2}\,\partial_{yy}(vy^2 f)$ — the Fokker–Planck equation of Step 1 again — with initial datum $\partial_{yy}(1-y)^+$, the unit point mass at $y = 1$: at $\tau = 0$ all the probability sits at $Y = 1$. Equations of this type preserve nonnegativity. Probabilistically there is nothing to prove: $f(\tau, \cdot)$ *is* the density of the diffusion (4.1) started at 1, and densities are nonnegative. Analytically, the same fact is the parabolic maximum principle, which we quote from PDE theory. With $f \geq 0$ established, part (ii) is the equation (4.2) itself read as a sign statement: $\partial_\tau c = \frac{1}{2}vy^2 f \geq 0$. □

Pause on what this proposition buys, because it is the strategy of the entire chapter. In Chapters 2 and 3, validity was work, done slice by slice: Chapter 2's LQD chart (its log-quantile-density coordinates, in which every parameter choice is automatically a law) made butterfly freedom an identity within one expiry, the volatility chart had to police $g_{\mathrm{D}} \geq 0$, and calendar order across expiries was in both cases a separate condition. Here, one sign condition on one function of two variables delivers both families of conditions, across the whole surface, automatically. The rest of the chapter is organized around protecting that gain. Choose a parametrization in which $v > 0$ is trivial to enforce (section 4.4). Choose a discretization that *inherits* the proposition on a finite grid, rather than merely approximating it (section 4.5). Then measure what little the finite grid leaks, instead of assuming it leaks nothing (section 4.8).

4.2.2 Backward: from the surface to the field

We now know how a field produces a surface. Read the equation the other way. If the surface is given — so that both derivatives of c are known — then (4.2) can simply be solved for the coefficient:

$$v(\tau, y) = \frac{\partial_\tau c(\tau, y)}{\frac{1}{2}\,y^2\,\partial_{yy}c(\tau, y)}, \tag{4.3}$$

wherever the density in the denominator is positive. This is the uniqueness half of the correspondence: within the class (4.1), at most one field is consistent with a given surface, and (4.3) computes it pointwise. The formula becomes most instructive when translated into the coordinates of Chapter 3 — the volatility chart, where a surface is the total-variance function $w(k, \tau)$.

Proposition 4.3 (The field in the volatility chart)**.** *Write the surface in implied form, $c = B(k, w(k, \tau))$ with $w(\cdot, \tau)$ the total-variance slice. Then (4.3) reads*

$$v(\tau, e^k) = \frac{\partial_\tau w(k, \tau)}{g_{\mathrm{D}}(k)}, \tag{4.4}$$

with g_{D} the Durrleman factor of (2.5), evaluated on the slice at maturity τ [12].

Proof. Compute the numerator and the denominator of (4.3) in turn.

Numerator. At fixed k, the call is the Black function of the slice's total variance, so by the chain rule $\partial_\tau c = \partial_w B \cdot \partial_\tau w$. The derivative $\partial_w B$ is the vega of the Black formula in total-variance units — the sensitivity of the price to the variance the formula is fed — and it is worth computing once in full. The normalized call is $B = \Phi(d_+) - e^k \Phi(d_-)$ with $d_\pm = -k/\sqrt{w} \pm \sqrt{w}/2$. The two arguments satisfy

$$d_+ - d_- = \sqrt{w}, \qquad d_+ + d_- = -\frac{2k}{\sqrt{w}}, \qquad \text{so} \qquad d_+^2 - d_-^2 = (d_+ + d_-)(d_+ - d_-) = -2k.$$

Because $\varphi(d) = e^{-d^2/2}/\sqrt{2\pi}$, the ratio of the two normal densities is

$$\frac{\varphi(d_+)}{\varphi(d_-)} = \exp\Big(-\tfrac{d_+^2 - d_-^2}{2}\Big) = e^k, \qquad \text{i.e.} \qquad \varphi(d_+) = e^k \varphi(d_-).$$

Now differentiate B in w and use this identity:

$$\partial_w B = \varphi(d_+)\,\partial_w d_+ - e^k \varphi(d_-)\,\partial_w d_- = \varphi(d_+)\,\partial_w (d_+ - d_-) = \varphi(d_+)\,\partial_w \sqrt{w} = \frac{e^k \varphi(d_-)}{2\sqrt{w}}.$$

Denominator. Two facts convert $\partial_{yy} c$ into chart language. First, the density of Y transforms into the density of the log return $X = \log Y$ by the usual change of variables: with $y = e^k$, $f(\tau, y) = f_X(k)/y$ (the factor $1/y$ is $\mathrm{d}y = y\,\mathrm{d}k$). Second — this is (2.5), quoted from Chapter 2 — the log-return density of a slice with total variance w is

$$f_X(k) = g_{\mathrm{D}}(k)\,\frac{\varphi(d_-)}{\sqrt{w}},$$

where g_{D} is the Durrleman function: up to the positive factor $\varphi(d_-)/\sqrt{w}$, it *is* the probability density the smile implies, so $g_{\mathrm{D}} \geq 0$ is exactly the butterfly condition Chapter 3 policed. The denominator of (4.3) is then

$$\tfrac{1}{2}\, y^2\, \partial_{yy} c = \tfrac{1}{2}\, y^2\, f(\tau, y) = \tfrac{1}{2}\, y\, f_X(k) = \tfrac{1}{2}\, e^k\, g_{\mathrm{D}}(k)\,\frac{\varphi(d_-)}{\sqrt{w}}.$$

Divide numerator by denominator: the factors e^k, $\varphi(d_-)$ and $2\sqrt{w}$ all cancel, leaving $v = \partial_\tau w / g_{\mathrm{D}}$. $\square$

The reader has met both halves of this ratio before, as the two families of static validity. The numerator is the calendar increment: it is nonnegative exactly when the slices around maturity τ are calendar-ordered in the sense of section 2.8. The denominator is the butterfly factor: it is nonnegative exactly when the slice at τ implies a genuine (nonnegative) density. So (4.4) is more than a formula; it is a compatibility statement. *A local-volatility field exists precisely where the surface is valid*, and each mode of static arbitrage has its own place to fail: a calendar violation makes the numerator negative, a butterfly violation the denominator. This chapter therefore takes as its input hypothesis what Chapters 2–3 produced as output: slices that are butterfly-free and calendar-ordered, in the normalized units of section 2.8. Where the hypothesis fails, no field exists, and (4.4) says so by changing sign.

Figure 4.1 evaluates the ratio on the book's frozen SPY surface — the snapshot's eight fitted slices, with total variance interpolated linearly in τ between expiries. Walk its three panels left to right. Panel (a) is the numerator: the calendar increment $\partial_\tau w$, piecewise constant in τ because the interpolation between expiries is linear, and largest toward the short-dated put side, where the smile grows fastest with maturity. Panel (b) is the denominator: each slice's Durrleman factor, the implied-density factor of the chart. Panel (c) is their ratio, displayed as local volatility $\sqrt{v}$: a steep put-side wall, a quiet call side, sharp structure at the short end — the

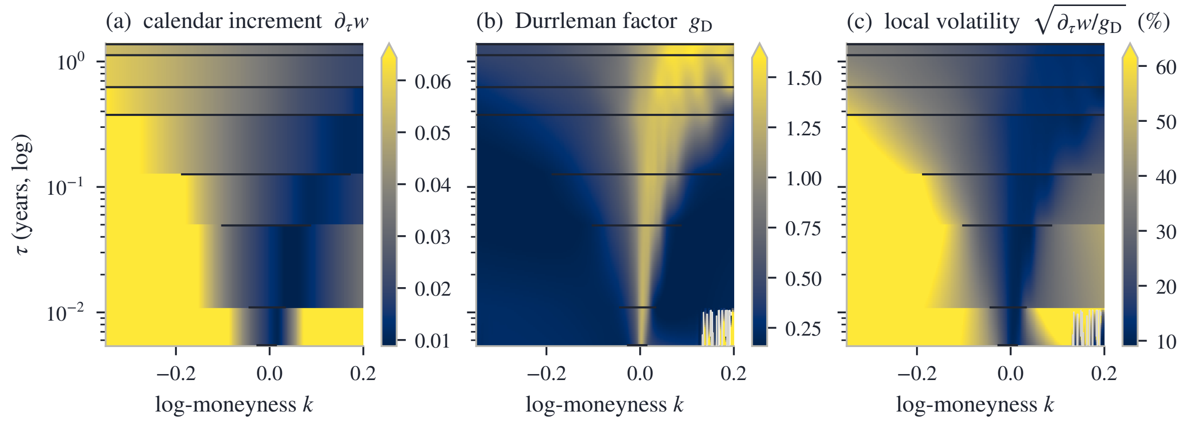

Figure 4.1: The field as a ratio, on the frozen SPY surface (stored per-expiry fits of the snapshot, total variance interpolated linearly in τ between the eight expiries; horizontal bars mark each expiry's quoted span). (a) The calendar increment $\partial_\tau w$. (b) The Durrleman factor of each slice. (c) Their ratio, displayed as local volatility; grey cells have a negative numerator or denominator and are inadmissible.

field inherits the smile's features in recognizable form, and where the data speaks the extracted values range from 9.0% to 61.5%. The grey cells are the diagnostic part of the picture: there one of the two signs fails and no local variance exists. They amount to 0.5% of the displayed mesh, and every one of them sits *outside* the common quoted support of the two expiries that bracket it — restricted to the quoted region the share is 0.00% — in territory owned by extrapolation conventions rather than by prices.

One caution closes the section. The surface in this figure is fitted and smooth, and evaluating the ratio on a fitted surface is a legitimate operation — it is how the figure was drawn. What must never be done is to apply the formula to the market's raw quotes. The next section shows why.

4.3 The wrong direction

Formula (4.3) suggests an obvious procedure: interpolate the quoted smile, differentiate it twice in strike and once in maturity, divide. On a continuum of exact prices, this is a theorem. On a real chain — a few dozen quotes per expiry, each carrying bid–ask noise — it fails. The failure is structural, not bad luck, and it is worth computing exactly how large it is.

Start with how curvature is estimated from data. From three quotes at equally spaced strikes, the standard estimate is the second difference,

$$\partial_{kk} w \;\approx\; \frac{w(k-\Delta k) - 2\,w(k) + w(k+\Delta k)}{\Delta k^2},$$

which recovers the curvature of smooth data as $\Delta k \to 0$. Now let each quote err. Suppose the middle quote reads too low by ε and its two neighbors too high by ε — the alternating pattern a bid–ask bounce produces naturally, one print near the bid, the next near the ask. The three errors pass through the formula as

$$\frac{(+\varepsilon) - 2\,(-\varepsilon) + (+\varepsilon)}{\Delta k^2} = \frac{4\varepsilon}{\Delta k^2},$$

an error in the estimated curvature of 4ε divided by the *square* of the strike spacing. Put numbers in. On a smile at 23% volatility with three months to run, a ripple of 50 volatility basis points (a basis point is 0.01 volatility percent) moves total variance by about $\varepsilon = 2\sigma\tau\,\delta\sigma \approx 5\times10^{-4}$. Sent through strikes $\Delta k = 0.02$ apart, the curvature error is $4\times(5\times10^{-4})/(0.02)^2 = 5$ — several times larger than the true curvature, which for the smile at hand is of order one. And the dependence on the spacing is inverse-square: halving Δk multiplies the error by four. More quotes — a finer, ostensibly better chain — makes the estimate *worse*, without bound. There is no stable limit to converge to. Numerical differentiation of noisy data is the textbook ill-posed operation.

The place this amplified noise lands could not be worse chosen. By (4.4), the curvature sits inside g_{D} — the factor that carries the *sign* of the implied density. Recall its expression from (2.5), with primes denoting strike derivatives of the slice:

$$g_{\mathrm{D}}(k) = \Big(1 - \frac{k\,w'}{2w}\Big)^2 - \frac{(w')^2}{4}\Big(\frac{1}{w} + \frac{1}{4}\Big) + \frac{w''}{2}.$$

The estimated curvature enters additively, through the last term: a curvature error of 5 is a g_{D} error of 2.5, dwarfing the factor's healthy values (panel (b) of fig. 4.1 showed them). The first time the noise pushes the estimated density through zero, the extraction returns a negative variance. That is not a large error; it is a non-object. No diffusion has negative variance, and no pricing equation, simulation or risk system downstream can accept one.

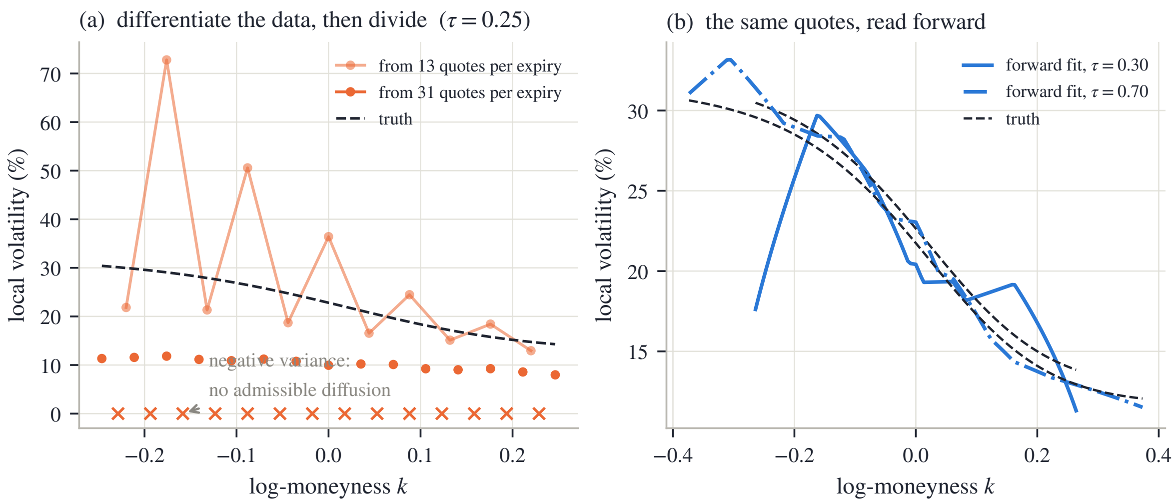

Figure 4.2: The same synthetic market, read both ways. Quotes come from a known smooth field (dashed) plus a deterministic alternating ±50 bp volatility ripple — a stylized bid–ask bounce. (a) The extraction (4.4) from the noisy quotes at $\tau = 0.25$, from 13 and from 31 quotes per expiry; crosses mark strikes where it returns a negative variance or density. (b) The same 31-quote noisy chains handed to this chapter's forward calibration: admissible by construction, with reprice residuals of rms 49 bp — the scale of the ripple itself.

Figure 4.2 performs the experiment on a synthetic market whose answer is known. The smooth field and the quote recipe are recorded in appendix 4.A; the "noise" is a deterministic alternating ripple of ±50 bp of volatility, a stylized bid–ask bounce, so that every number in the figure is reproducible. Panel (a) is the extraction at $\tau = 0.25$. The dashed curve is the true local volatility, whose level at the money is 23%. From 13 quotes per expiry, the extracted curve sawtooths between 13% and 73% — already unusable. From 31 quotes of the *same* market — the same ripple, only finer strike spacing — it does worse than oscillate: at 14 strikes, marked by crosses, it fails outright, reporting a negative variance or a negative density. That is the inverse-square law at work. Refining the grid, which improves every well-posed computation, drives this one to report, at nearly half the strikes, that no diffusion exists.

Panel (b) previews the repair, run on the harder case: the same 31-quote noisy chains are handed to the calibration this chapter builds. The result is admissible by construction — every iterate of the search is a genuine positive field — and it tracks the truth through the quoted region, staying within 12.9 volatility points of it along the plotted cross-sections, while the quotes reprice to 49 bp rms, the scale of the ripple itself. The noise ends up in the residual, where it belongs, instead of in a derivative. One honest caveat is visible in the panel and explained properly in section 4.6: at the edges of the shortest expiry's quoted span, where a single rippled quote is the only witness, the fitted field wanders by as much as 22 volatility points. Noise rejection comes from dense data and declared smoothing, not from magic.

The repair itself, due to Andreasen and Huge [1], is a change of role, not of formula. Stop treating the field as the output of a differentiation. Declare it the unknown. Parametrize it; price it through the forward equation (4.2), which differentiates only the smooth numerical solution and never the data; and ask an optimizer for the positive field whose prices meet the quotes. Positivity is imposed on the *parameters*, so every iterate of the search is an admissible diffusion before any quote is consulted. Readers of the earlier chapters will recognize the design: it is constraint elimination again, the idea that built the LQD chart (section 2.6.3) and the structural SVI chart (section 3.6.2), now one dimension up. Choose the coordinates in which the constraint set is trivial, and let the pricing map do the hard work.

Three questions remain, and they organize the rest of the chapter. What parametrization makes positivity of a two-variable field trivial (section 4.4)? What discretization of (4.2) keeps proposition 4.2 true on a finite grid, rather than approximately true (section 4.5)? And what do the quotes actually determine about the field, once the pointwise formula is off the table (section 4.6)?

4.4 A sheet with finitely many parameters

4.4.1 Positivity as a box

The unknown is now a positive function of two variables — an infinite-dimensional object with a one-sided constraint, which sounds harder than anything fitted so far in this book. The finite version is built from the simplest ingredient available: straight-line interpolation. Picture the smallest example that shows everything. Take two maturities and three strikes — a grid of six points — and write a positive number on each. Split each of the two rectangular cells along a diagonal, giving four triangles. Inside each triangle, define the field as the unique plane through its three corner values. The result is a continuous surface made of flat triangular facets, passing through the six numbers — and those six numbers are the model's only parameters. The general object just replaces six by a few hundred.

Two names before the formal definition. A *tensor grid* is a rectangular grid: pick a list of maturities and a list of strikes, and take every combination as a vertex. The *hat function* of a vertex is the tent that rises to 1 there, slopes to 0 at every neighboring vertex, and stays 0 beyond; on each triangle it is the affine function with those corner values. Its values at a point are also called the point's *barycentric weights* — the three nonnegative numbers, summing to one, that locate the point inside its triangle.

Definition 4.4 (P_1 local-variance sheet)**.** Fix vertices (τ_i, y_j), $i = 1..n_\tau$, $j = 1..n_y$, on a tensor grid, triangulated by splitting each rectangular cell along one diagonal (remark 4.6). The local variance is the continuous piecewise-affine interpolant

$$v_\theta(\tau, y) = \sum_{\ell=1}^{n_\theta} \theta_\ell \, \mathcal{H}_\ell(\tau, y), \qquad n_\theta = n_\tau n_y, \tag{4.5}$$

where $\mathcal{H}_\ell$ is the hat function of vertex ℓ — barycentric on each triangle, equal to 1 at its own vertex and 0 at every other — and the parameters $\theta_\ell = v(\tau_i, y_j) > 0$ are the *nodal local variances*, indexed flat as θ_ℓ or by row and column as $\theta_{i,j}$, whichever is convenient.

With the names in place, read (4.5) as plainly as it deserves: the sum simply interpolates the nodal values. At any point of the grid, only the three hats of the surrounding triangle are nonzero; their values there are the point's barycentric weights; and v_θ at that point is the corresponding weighted average of the three corner parameters.

Why affine pieces, and why triangles? Because a weighted average with nonnegative weights cannot leave the range of the numbers it averages. That one-line fact is the model's entire validity theory:

Proposition 4.5 (Nodal bounds are surface bounds)**.** *If $v_{\mathrm{lo}} \le \theta_\ell \le v_{\mathrm{hi}}$ for all ℓ, then $v_{\mathrm{lo}} \le v_\theta(\tau, y) \le v_{\mathrm{hi}}$ everywhere on the triangulation's hull.*

Proof. At any point of the hull, the only nonzero hat functions are the three belonging to the surrounding triangle, and their values there are nonnegative and sum to one — they are the point's barycentric weights. Hence

$$v_\theta(\tau, y) = \sum_\ell \theta_\ell\, \mathcal{H}_\ell(\tau, y) \;\le\; v_{\mathrm{hi}} \sum_\ell \mathcal{H}_\ell(\tau, y) = v_{\mathrm{hi}},$$

and the lower bound is the same computation with v_{lo}. □

Positivity of a surface has been reduced to positivity of finitely many numbers: keep every θ_ℓ inside the box $[v_{\mathrm{lo}}, v_{\mathrm{hi}}]$, and the whole field stays inside it too, everywhere on the hull — the region the triangles cover. A box is the one constraint that least-squares solvers handle natively — clip each coordinate — with no penalty terms and no feasibility questions. This is the promised chart, one dimension up: as with the LQD chart of Chapter 2 and the structural chart of section 3.6.2, the coordinates are chosen so that every point of the parameter set is a valid object, and the optimizer never learns that a constraint existed. Combined with proposition 4.2, the chain of guarantees is complete before any data arrives:

$$\text{box} \implies \text{positive field} \implies \text{butterfly-free, calendar-ordered surface.}$$

Every iterate of every fit in this chapter is an admissible diffusion.

Figure 4.3 shows the object this chapter will calibrate: the SPY sheet of section 4.8, with 9 maturity rows, 24 strike columns, and therefore 216 vertices, drawn with the very triangulation the pricing uses. Read it as a list of numbers, because that is all it is. Every dot is one calibration parameter; every triangle is a region where local variance is an affine function of (τ, y). The tall put-side wall, the sharp structure at the short end, and the flat call-side plain are nothing but nodal values. Notice also that the strike columns crowd toward the money at short maturities; that is deliberate grid design, and section 4.6 explains it.

Figure 4.4 takes the construction apart on a small demonstration grid. In panel (a), the triangulated vertex grid; the shaded star of triangles around one vertex is the support of that vertex's hat function — the entire region of the surface its parameter can influence. A nodal variance is a local dial: turn it, and only the prices of options whose diffusion passes through that star can move. (The cell diagonals in the panel do not all point the same way; that is real, and remark 4.6 explains it.) In panel (b), a cross-section along one vertex row: each hat rises to 1 at its own vertex, is 0 at every other, and the hats sum to 1 identically — the partition of unity behind the weighted-average argument of proposition 4.5.

Remark 4.6 (Which diagonal, and does it matter)*.* On a tensor grid every cell is a rectangle, and a rectangle's four corners lie on one circle — exactly the tie case of the standard (Delaunay) criterion for well-shaped triangles, which prefers triangulations in which no vertex falls strictly inside another triangle's circumscribed circle. The triangulation routine therefore breaks the tie cell by cell, deterministically but by no designed rule, and the figures draw the resulting patchwork faithfully. The choice is second-order small: the two candidate interpolants of a cell agree at all four corners and along all four edges, and differ inside by at most half the cell's

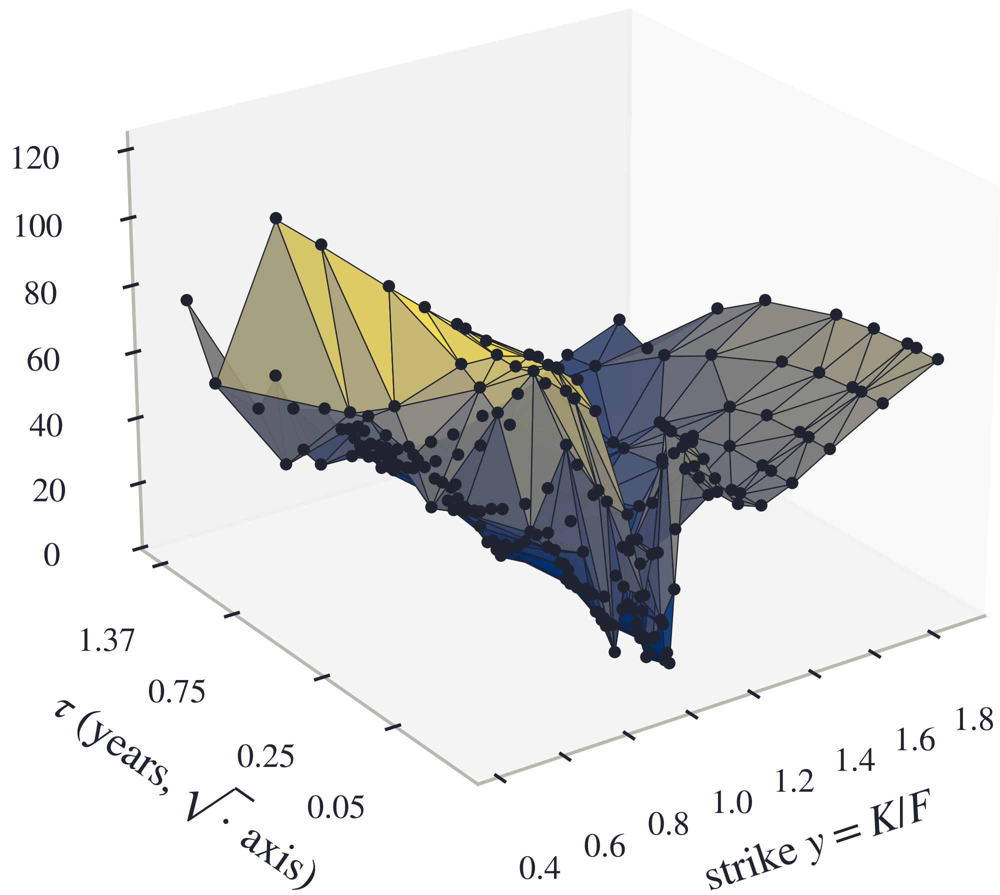

Figure 4.3: The model, drawn: the calibrated SPY local-volatility sheet of section 4.8 (9 maturity rows $\times$ 24 strike columns, 216 vertices), with the triangulation the pricing basis actually uses. Every dot is one calibration parameter; every triangle is a region where local variance is affine in (τ, y).

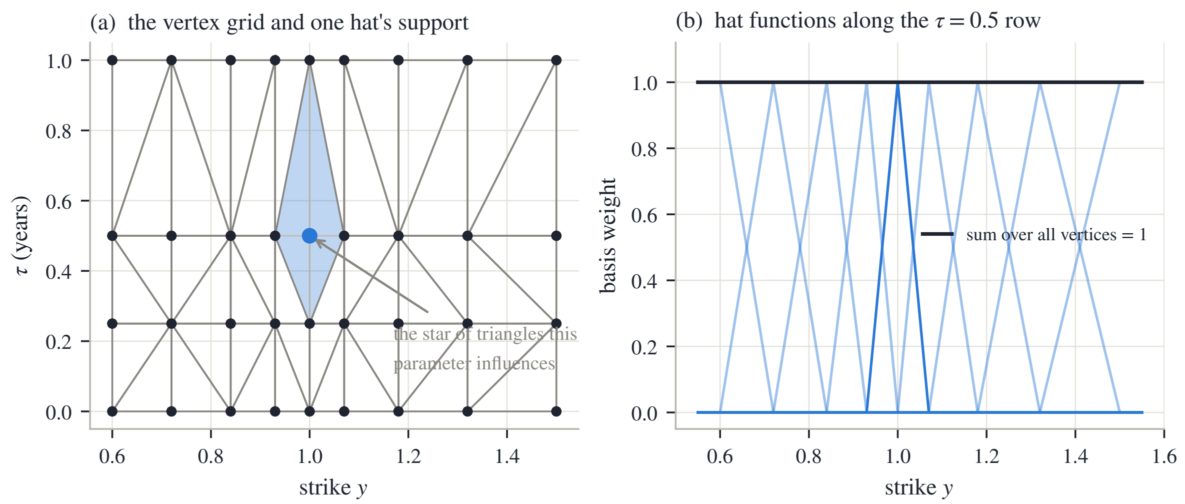

Figure 4.4: The anatomy of the sheet, on a small demonstration grid. (a) The triangulated vertex grid; the shaded star of triangles is the support of one vertex's hat function. The cell diagonals are not uniformly oriented; that is real (remark 4.6). (b) A strike cross-section along a vertex row: each hat is 1 at its own vertex, 0 at every other, affine between, and they sum to 1 identically.

twist, $(\theta_{i,j} + \theta_{i+1,j+1} - \theta_{i,j+1} - \theta_{i+1,j})/2$, at the center — zero whenever the four values are locally bilinear. It is nonetheless a convention, so it is pinned: the triangulation is computed once, cached, and the same triangles price every iterate.

4.4.2 Beyond the hull: the wing contract

The grid must end somewhere, and pricing does not. Below the lowest strike column, the sheet continues local variance *linearly*, with slope a times the slope of the first interior cell; above the highest column it is clamped flat. Three settings of a are meaningful. The default, $a = 0$, clamps the put wing flat as well. A fixed $a > 0$ keeps an unquoted put wing rising. And a is handed to the optimizer as a free parameter in exactly one circumstance: when a variance-swap quote is present. A variance swap pays the realized variance of the underlying to expiry, and its fair price integrates the whole smile, deep wings included — it is the one instrument in the problem that genuinely reads the deep-put tail, as Chapter 5 will show.

Remark 4.7 (The extrapolation sits outside the positivity guarantee). Proposition 4.5 stops at the hull, and for a reason: outside it, interpolation becomes extrapolation, and extrapolation weights are signed — an extrapolated value is no longer an average of nodal values. With $a > 0$ and a first cell whose variance decreases toward the boundary, the linear continuation can reach zero before $y = 0$ (pricing floors it there, so the continued curve kinks rather than turning negative), and a rising continuation can exceed v_{hi} freely. Every guarantee in this chapter that leans on the box is therefore scoped to the hull and to the flat-clamped wings. The wing is a disclosed convention — the same standing as the tail conventions of Chapters 2–3 — and the chapter's contract table records it.

Remark 4.8 (Lee's bound, inherited). A bounded field bounds the smile, and the bound is worth deriving because Chapter 3 had to legislate what comes out here for free. The first step is that the marched call is *monotone in the field*: raising local variance anywhere can only raise call prices. To see it, let c_v and $c_{v+\delta v}$ solve (4.2) for fields v and $v + \delta v$ with $\delta v \geq 0$, and subtract the two equations. Their difference $s = c_{v+\delta v} - c_v$ starts at $s(0, \cdot) = 0$ and satisfies

$$\partial_\tau s = \tfrac{1}{2}\,(v + \delta v)\, y^2\, \partial_{yy} s + \tfrac{1}{2}\,\delta v\, y^2\, \partial_{yy} c_v$$

(add and subtract $\frac{1}{2}(v+\delta v)y^2\partial_{yy}c_v$ to check). The source term is nonnegative — $\delta v \geq 0$ by assumption, $\partial_{yy}c_v \geq 0$ by proposition 4.2 — and a heat equation driven by a nonnegative source from a zero start stays nonnegative (the maximum principle again). So $s \geq 0$. Now compare with the constant field v_{hi}. With flat-clamped wings, $v \leq v_{\mathrm{hi}}$ holds on the entire strike line, so every call price is dominated by its price in the Black–Scholes model with variance rate v_{hi}; and since the Black price increases in total variance (vega is positive), the domination transfers to implied variance:

$$w(k,\tau) \;\leq\; v_{\mathrm{hi}}\,\tau \qquad \text{for every } k.$$

Total implied variance bounded in strike means its wing slope decays to zero: the surface sits strictly inside Lee's cap of 2 automatically, where Chapter 3 had to impose a cap by hand (section 3.3). A steep released continuation (a free) escapes v_{hi}, and with it this bound.

4.5 Pricing on the lattice

4.5.1 The implicit march and its M-matrix

The forward equation must now be solved numerically, and this section chooses the scheme with one criterion above all others: it must inherit proposition 4.2, not merely approximate it.

First the setting. The equation is solved on a strike grid over $[0, y_{\max}]$, with the two end values pinned: $c(\cdot,0)=1$ (a call struck at zero delivers the forward, worth 1 in normalized units) and $c(\cdot,y_{\max})=0$ (far enough out, the call is worthless). Pinned end values are called *Dirichlet* boundaries. The march then steps maturity forward in increments $\Delta\tau$, and at each step it must decide at which time level to evaluate the strike derivative. The *fully implicit* choice — evaluate at the new level — is the one this chapter defends.

On the grid, the second derivative becomes a weighted second difference, and the weights are worth deriving because their signs carry the whole theory. The grid need not be uniform; write $\Delta y_i^- = y_i - y_{i-1}$ and $\Delta y_i^+ = y_{i+1} - y_i$ for the gaps around node i. Taylor-expand the two neighbors of a smooth function c around y_i:

$$c_{i\pm1} = c_i \pm \Delta y_i^\pm\, c'(y_i) + \tfrac{1}{2}\,(\Delta y_i^\pm)^2\, c''(y_i) + O(\Delta y^3).$$

Multiply the "+" expansion by Δy_i^-, the "−" expansion by Δy_i^+, and add: the first-derivative terms cancel, leaving

$$\Delta y_i^-\, c_{i+1} + \Delta y_i^+\, c_{i-1} - (\Delta y_i^- + \Delta y_i^+)\, c_i = \tfrac{1}{2}\,\Delta y_i^+ \Delta y_i^- (\Delta y_i^+ + \Delta y_i^-)\, c''(y_i) + O(\Delta y^4),$$

and solving for c'' gives the three-point weights. Multiplying by $\frac{1}{2}v_i y_i^2$, the discrete generator $\mathcal{L}$ encoding $\frac{1}{2}vy^2\partial_{yy}$ has the stencil

$$\mathcal{L}_{i,i\mp1} = \frac{v_i\, y_i^2}{(\Delta y_i^- + \Delta y_i^+)\,\Delta y_i^\mp} \;\geq 0, \qquad \mathcal{L}_{ii} = -(\mathcal{L}_{i,i-1} + \mathcal{L}_{i,i+1}), \tag{4.6}$$

which reduces to the familiar $(1,-2,1)/\Delta y^2$ pattern on a uniform grid. Note the two signs that will do all the work: the off-diagonal entries are nonnegative, and each diagonal is exactly minus the sum of its row's off-diagonals.

An implicit step then solves a linear system for the vector U^{n+1} of calls at the new maturity, the operator evaluated at the new level:

$$\mathcal{M}^{n+1}U^{n+1} = U^n + (\text{boundary terms}), \qquad \mathcal{M}^{n+1} = I - \Delta\tau\,\mathcal{L}^{n+1}. \tag{4.7}$$

Because the stencil couples only neighbors, $\mathcal{M}$ is tridiagonal, and each step is one tridiagonal solve — linear cost in the number of nodes. The boundary terms carry the pinned values 1 and 0 through the stencil's end columns, and two later proofs lean on them, so write them out once as a vector b^{n+1}: only the row whose lower neighbor is the boundary node $y = 0$ has a nonzero entry,

$$b_i^{n+1} = \Delta\tau\, \mathcal{L}_{i,i-1}^{n+1} \times 1 \quad \text{(the pinned call value at } y = 0\text{)}, \qquad b_i^{n+1} = 0 \ \text{ at every other row}$$

— the far boundary would contribute through its own stencil arm, but its pinned value is 0.

Now the promised discipline. Proposition 4.2 was proved for the continuous equation. Does the discrete march *inherit* it — no invented negative densities, no broken calendar order — or merely approximate it, with errors of unknown sign? The answer runs through one classical notion from linear algebra. A matrix with positive diagonal, nonpositive off-diagonal entries and enough diagonal dominance is called an *M-matrix*, and the property that sign pattern buys is exactly the one needed here: the inverse of an M-matrix has no negative entries.

Proposition 4.9 (The step matrix is an M-matrix)**.** *For any $v \geq 0$ and any $\Delta\tau > 0$, the matrix $\mathcal{M} = I - \Delta\tau\mathcal{L}$ of (4.7) has positive diagonal, non-positive off-diagonals, and each diagonal entry exceeds the absolute sum of its row's off-diagonals by at least 1 (exactly 1 at fully interior rows). Hence $\mathcal{M}$ is a strictly diagonally dominant nonsingular M-matrix, and $\mathcal{M}^{-1} \geq 0$ elementwise.*

Proof. The sign pattern reads off (4.6): the off-diagonals of $\mathcal{M}$ are $-\Delta\tau\mathcal{L}_{i,i\mp1} \leq 0$; the diagonal is $1 + \Delta\tau(\mathcal{L}_{i,i-1} + \mathcal{L}_{i,i+1}) \geq 1$; and the dominance gap is the 1 contributed by the identity at interior rows, more at rows whose stencil arm reaches a boundary column (that neighbor is not among the unknowns). This is the defining pattern of a nonsingular M-matrix, and such matrices are inverse-positive [3]. A self-contained argument fits in four lines. To solve $\mathcal{M}u = b$ with $b \geq 0$, split $\mathcal{M} = D - \mathcal{N}$ into its diagonal $D > 0$ and off-diagonal part $\mathcal{N} \geq 0$, and iterate Jacobi's method from zero:

$$u^{(m+1)} = D^{-1}(\mathcal{N}u^{(m)} + b), \qquad u^{(0)} = 0.$$

Every iterate is nonnegative, being assembled from nonnegative pieces; the iteration converges, because every row of $D^{-1}\mathcal{N}$ sums to less than one (that is diagonal dominance), so each update multiplies the largest error entry by a factor below one; and the limit solves $\mathcal{M}u = b$. So $b \geq 0$ forces $u \geq 0$, for every such b — which is the statement $\mathcal{M}^{-1} \geq 0$, column by column. □

Inverse positivity is the discrete counterpart of heat flow never turning a nonnegative temperature profile negative, and it translates directly into the property numerical analysts call *monotonicity* — order in, order out — and into stability in the strongest everyday sense: the largest absolute value in the solution can never grow.

Corollary 4.10 (Discrete maximum principle, unconditionally)**.** *For any $\Delta\tau > 0$ the scheme (4.7) is* monotone *— $U^n \geq V^n$ with the same boundary data implies $U^{n+1} \geq V^{n+1}$ — and every step keeps the values between the extremes of the previous step and the boundary data: the march is unconditionally ℓ^∞-stable (the maximum absolute value never grows), with no step-size restriction, and cannot create new extrema.*

Proof. Monotonicity is inverse positivity applied to a difference: if $U^n \geq V^n$, the two systems (4.7) subtract to $\mathcal{M}^{n+1}(U^{n+1} - V^{n+1}) = U^n - V^n \geq 0$, and multiplying by $\mathcal{M}^{-1} \geq 0$ preserves the sign. For the upper bound, let U_* be the larger of $\max_i U_i^n$ and the boundary values 1 and 0, and test the constant vector $U_*\mathbf{1}$, where $\mathbf{1}$ is the all-ones vector (a vector here, not the indicator of proposition 4.1's proof). At a fully interior row, the row of $\mathcal{M}$ sums to 1 (because the stencil's row sums to zero), so $(\mathcal{M}\,U_*\mathbf{1})_i = U_* \geq U_i^n$. At a row next to a boundary, the arm that would reach the boundary column is missing from $\mathcal{M}$, leaving the larger value $(\mathcal{M}\,U_*\mathbf{1})_i = U_* + \Delta\tau\,\mathcal{L}_{i,\mathrm{bd}}\,U_*$, with $\mathcal{L}_{i,\mathrm{bd}}$ the missing arm's weight; the right-hand side there is $U_i^n + b_i^{n+1}$ with $b_i^{n+1} \leq \Delta\tau\,\mathcal{L}_{i,\mathrm{bd}} \times 1$. Since U_* dominates both U_i^n and the boundary value 1, the left side dominates the right at every row: $\mathcal{M}(U_*\mathbf{1}) \geq U^n + b^{n+1}$ entrywise. Multiplying by $\mathcal{M}^{-1} \geq 0$ gives $U_*\mathbf{1} \geq U^{n+1}$. The lower bound is symmetric. □

Remark 4.11 (What the M-matrix argument does and does not prove). Proposition 4.9 and corollary 4.10 control the call *values*: order preservation, boundedness, no invented extrema. They do not by themselves give nonnegative second differences — a bounded monotone vector can still be locally concave — so "the scheme cannot manufacture negative densities" would need a separate convexity-preservation proof for the actual stencil and boundary treatment, and this chapter does not claim one. What it does instead is measure: the scheme is monotone and stable by theorem, and the butterfly and calendar cleanliness of the *output* is checked on the price grid with every fit. Section 4.8 reports the measured minima for the chapter's two calibrated sheets; they sit at the scale of solver rounding.

4.5.2 Why not Crank–Nicolson

The implicit march is only first-order accurate in the time step, and the standard second-order upgrade is Crank–Nicolson: evaluate the operator half at the old level and half at the new, replacing (4.7) by

$$(I - \tfrac{\Delta\tau}{2}\mathcal{L})U^{n+1} = (I + \tfrac{\Delta\tau}{2}\mathcal{L})U^n.$$

Look at what the upgrade costs. The implicit factor on the left is still an M-matrix — that part survives. But monotonicity now also needs the *explicit* factor on the right to map nonnegative vectors to nonnegative vectors, and its diagonal entries are $1 - \frac{\Delta\tau}{2}(\mathcal{L}_{i,i-1} + \mathcal{L}_{i,i+1})$: nonnegative only when the step is small enough,

$$\Delta\tau \;\le\; \frac{2}{\mathcal{L}_{i,i-1} + \mathcal{L}_{i,i+1}} \;=\; \frac{2\,\Delta y^2}{v_i\, y_i^2} \quad \text{(uniform grid).} \tag{4.8}$$

This is a step-size restriction of the Courant–Friedrichs–Lewy (CFL) type — the generic fate of schemes with an explicit part — and it tightens with the *square* of the strike step. Put numbers in: at a 40% local volatility on a $\Delta y = 0.005$ lattice — the resolution section 4.6 will show short-dated smiles require — the bound is 0.0003 years, more than thirty times smaller than the standard 0.01 marching step. Monotone Crank–Nicolson on realistic lattices is simply not on offer. When the bound is violated, the scheme remains second-order *accurate* but is no longer *monotone*, and the payoff kink at $y = 1$ seeds the oscillation that fig. 4.5 displays.

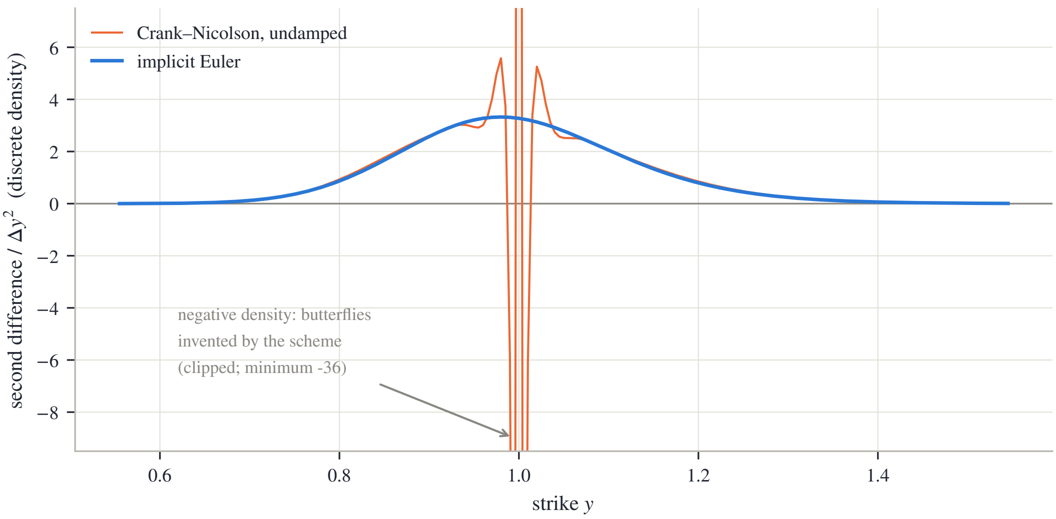

Figure 4.5: The same flat 40% surface, the same $\Delta y = 0.005$ lattice, marched to $\tau = 0.1$ in ten steps by the same solver — only the scheme differs. (a) Implicit Euler: the discrete density is a clean bell, minimum 0.0012 in the plotted window. (b) Undamped Crank–Nicolson: the kink oscillation swings the discrete density to -36.

Figure 4.5 shows what breaking monotonicity does to the object we actually care about: the discrete density, i.e. the second differences of the marched prices. Both panels march the same flat 40% surface on the same $\Delta y = 0.005$ lattice to $\tau = 0.1$ in ten steps; only the scheme differs. In panel (a), implicit Euler: the second differences form a clean bell — a discrete Gaussian, as they should — whose minimum over the plotted window is 0.0012: nonnegative, exactly as corollary 4.10 guarantees. In panel (b), undamped Crank–Nicolson: the oscillation seeded by the payoff kink swings the discrete density down to -36 — a negative density, which is butterfly arbitrage, manufactured by the scheme that is, in the textbook ordering, the more accurate of the two. Damped start-up steps — a few implicit steps before switching to Crank–Nicolson — are the standard repair [15]. The reference implementation declines the trade altogether: it pays one order of time accuracy and keeps corollary 4.10 unconditionally.

4.5.3 Reading out, and the operator's blind spot

The observation operator is plain: the normalized call at a quote's (τ, y) is read off the marched solution by interpolation in strike, and quoted expiries are forced to be lattice time-nodes, so no interpolation in maturity is ever needed.

One principle from inverse problems completes the section, because it dictates part of the chapter's protocol. An optimizer facing a *fixed* discretization will happily bend its parameters to cancel the operator's own discretization error. The residual measured on the fitting operator therefore flatters the fit, and validating a reconstruction with the operator that produced it has a name in the inverse-problems literature: the *inverse crime*. The chapter's protocol answers it by repricing every fit on a *refined* operator — half the strike step, four times the time steps — and reporting both numbers. The gap between them is not noise; it is the measured size of the compensation, and section 4.8 will show it is the dominant error term at the short end.

4.6 What the quotes determine

Before writing an objective function, a chapter should ask whether the problem it is about to pose has one answer. This one does not, in three distinct ways — one about resolution, one about averages, one about entire regions of the sheet — and each failure dictates one piece of the calibration's design. The section works through the three in order, then shows the sharpest of them as an experiment.

The length scale. Everything interesting about a smile at maturity τ happens within a few multiples of $\sigma\sqrt{\tau}$ of the money — the diffusive width, the distance the underlying typically moves by expiry. Work out its size and the design problem appears: a two-day smile at 20% volatility is about 1.5% wide, while a one-year smile at the same volatility is 20% wide — more than ten times wider. Both live on the same grids. Three placement rules follow. First, *vertex placement*: strike columns are spaced by probability, not by strike. Fix a ladder of probabilities (the desk's "deltas"), and place a column at the strike the return ends beyond with each given probability — under a normal approximation, the strike at probability u is $y = \exp\big(\sigma\sqrt{\tau}\,\Phi^{-1}(u)\big)$. Equal steps of probability make strike steps that are tight near the money and wide in the wings, so parameter density automatically follows where optionality lives; the exact ladder is in appendix 4.A. Second, *vertex coverage*: an axis laid out for the longest expiry undersamples the shortest, whose entire smile can fall between two adjacent columns; so every expiry is guaranteed several columns within its own traded range, spread evenly across it — this is what crowds the short-dated money in fig. 4.3. Third, *lattice resolution*: the strike step must resolve the narrowest smile, and it must also be at least as fine as the quote spacing — a lattice coarser than the quotes cannot represent them faithfully, and the fitted smile develops spurious wiggles at the quote spacing (an *alias* of the data, in signal-processing language) until the lattice out-resolves it. The exact placement rules used in this chapter are collected in appendix 4.A.

An averaging fact, and the floor. At the money, implied and local variance are related as average to integrand. If the field near the money depends only on time, $v(\tau, y) \approx v(\tau, 1)$, the model is Black–Scholes with a time-dependent volatility, and the first-course identity applies exactly:

$$w(0,\tau) = \int_0^\tau v(s,1)\,\mathrm{d}s.$$

In general the identity holds to leading order near the money — a fact we quote rather than prove [12]. Its consequence for calibration is simple: an average can only be matched by an integrand that crosses it. (One reading convention for the numbers that follow: volatilities are quoted in percent, and the field v is their *square*.) Whenever the ATM *term structure* — implied variance read across maturities at the money — is *increasing*, early local variance must sit strictly *below* the shortest implied variance: if the one-week implied is 12% and the one-month is 18%, the field's first days must run below $(0.12)^2$ in variance units. A parameter floor set carelessly — say at a round 5% volatility, $v_{\mathrm{lo}} = 0.0025$ — therefore makes some low-volatility short-dated term structures literally unfittable, with the optimizer pinned to the box and the misfit looking like a mystery. The box of proposition 4.5 must be generous and *adaptive*: the cap a fixed multiple of the largest implied volatility in view, the floor a fraction of the smallest *ATM* implied — keyed to ATM quotes deliberately, so that one noisy deep-wing print cannot drag down the floor where it does real stabilizing work.

Rows the quotes cannot see. The averaging fact has a sharper corollary at the front of the grid. The quotes at the first expiry τ_1 constrain — through the integral above — only the *total* of local variance accumulated over $[0, \tau_1]$. A vertex row strictly inside that interval is individually invisible: the optimizer can ring one sub-front row up and the next one down, leave every integral unchanged, and pay nothing in fit. Under noise it will do exactly that — this is how the rippled fit of fig. 4.2 spent its freedom at the shortest expiry, wandering 22 volatility points where a single quote was the only witness. The cure is not more data, because no vanilla — an ordinary call or put — expires inside $[0, \tau_1]$. The cure is a declared prior: tie each sub-front row to the first identified row — the *front tie* of section 4.7 — turning an invisible subspace into a stated constant continuation.

Two sheets, one quote set. The three paragraphs above are instances of one geometry, and it is best seen as an experiment. A fit like this chapter's carries a couple of hundred nodal parameters. The quotes outnumber them — nearly five to one on the frozen SPY chain of section 4.8, between two and three on NVDA — but the counting misleads. The quotes cluster near the money of a handful of expiries, and each vanilla reads the field only through a smoothing integral along the diffusion's paths. Whole regions of the sheet — deep wings, sub-front rows, maturities beyond the last expiry — are therefore weakly determined no matter what the count says.

Figure 4.6 stages the experiment on the synthetic market. In panel (a), a strike cross-section of the calibrated sheet at $\tau = 0.5$ (solid), and the same sheet with one deliberate change: its deep-put column at $y = 0.45$, below every quoted strike, is pushed down by 10 full volatility points (dashed). Inside the shaded band — the quoted region — the two curves are indistinguishable; the change lives entirely in unquoted territory. In panel (b), the cost of the change where the market can see it: the absolute reprice difference at every quote of every expiry. The largest is 22.6 bp, the rms across the quote set is 2.5 bp, and the effect concentrates in the longest expiry's lowest strikes — the only quotes whose diffusion reaches the moved column at all. A thousand basis points of surface, twenty basis points of quotes.

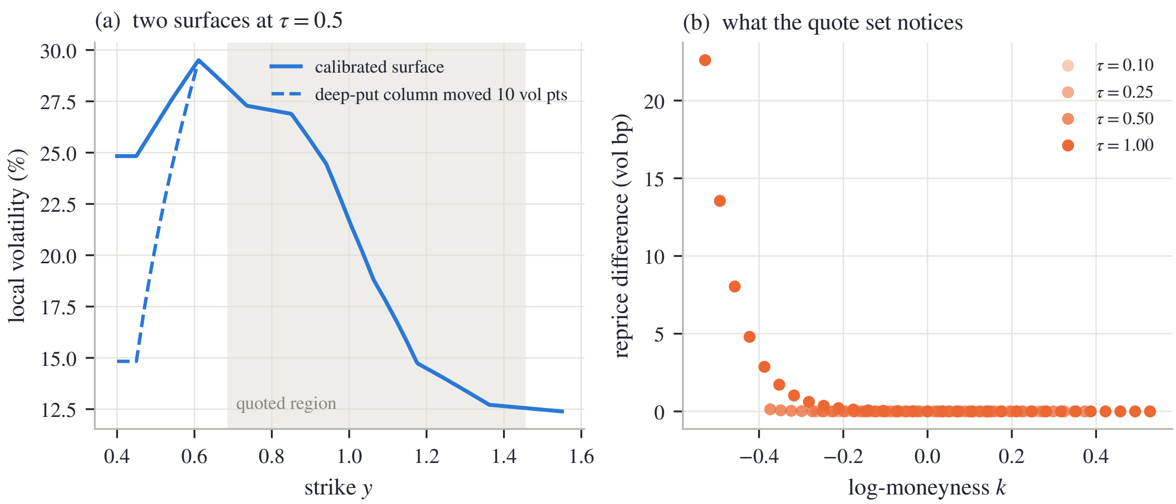

Figure 4.6: What sparse quotes do not see. (a) The calibrated synthetic sheet at $\tau = 0.5$ (solid) and the same sheet with its unquoted deep-put column pushed down by 10 volatility points (dashed); the shaded band is the quoted region. (b) The absolute reprice change at every quote of every expiry: at most 22.6 bp, concentrated where the longest expiry's lowest strikes feel the moved column.

What a calibration returns, then, is one *representative* of an equivalence class of sheets the data cannot tell apart, and every device in the next section's objective is a declared choice of representative. The roughness term selects the smoothest member near a reference. The box clips the unidentified extremes. The front tie pins the rows below the first expiry. The seed decides which basin a non-convex search lands in. None of this is hidden tuning; it is the "convention" column of the book's thesis, operating inside a calibration. And it is why section 4.8 reports quote error and surface error as two separate numbers: the first says how well the representative prices the market; only the second says how far the representative can be trusted *as the field*.

4.7 The objective and its derivatives

4.7.1 The objective

The pieces now assemble into a bound-constrained least-squares problem over the nodal variances:

$$\min_{\theta \in [v_{\mathrm{lo}}, v_{\mathrm{hi}}]^{n_\theta}} \ \tfrac{1}{2}\Big(\sum_q r_q(\theta)^2 \ + \ \lambda \left\|\Gamma(\theta - \theta_{\mathrm{ref}})\right\|^2 \ + \ \lambda_0 \left\|\theta_{1,\cdot} - \theta_{2,\cdot}\right\|^2\Big). \tag{4.9}$$

Its three blocks, in order: a data term, a roughness penalty, and a tie between the first two vertex rows — $\theta_{1,\cdot}$ is the $\tau = 0$ row, $\theta_{2,\cdot}$ the row at the first quoted expiry. Each block was argued for before it was written down; we take them in turn.

The *data block* compares the marched call c_θ at quote q's coordinates with the quoted one, in volatility units: $r_q = (c_\theta(\tau_q, y_q) - c_q)/\eta_q$, where η_q is one volatility percent of the quote's vega, floored. Recall what vega is: the derivative of the Black price with respect to implied volatility — the exchange rate between price units and volatility units. Dividing each price error by it makes every residual read as a volatility error, so a deep wing's tiny price errors and the money's large ones compete on the same scale; the floor (the same device as Chapter 2's vega floor η; this chapter's constant is in appendix 4.A) keeps the scale from vanishing where vega does. This is the residual design of section 2.6 — solve in price space, measure in volatility space — carried over unchanged, and when band edges rather than mids are trusted, the band objective of that section — residuals measured to the nearest bid–ask band edge, zero inside the band — applies verbatim.

The *roughness block* $\lambda\|\Gamma(\theta - \theta_{\mathrm{ref}})\|^2$ is Tikhonov regularization — the standard name for adding a penalty so that, among sheets fitting the data equally well, the optimizer returns the smoothest. The operator Γ takes second divided differences along strike within each maturity row and along time within each strike column — the discrete curvature of the sheet — normalized by the local vertex spacing, so that a tightly spaced money column is not penalized more than a wide wing one (on a uniform grid it reduces to the $(1, -2, 1)$ stencil). The penalty pulls toward a reference sheet θ_{ref} — flat, at the median ATM variance, in this chapter's protocol — and it is the declared answer to section 4.6: of the equivalence class the data allows, return the smoothest member near the reference.

The *front tie*, at weight λ_0, pins the $\tau = 0$ row to the first quoted row, closing the invisible sub-front subspace of section 4.6.

Two further row types plug into the same stack when their instruments are present, and are only named here. A variance-swap quote adds one scalar residual per quoted expiry, constraining the deep-put tail integral that vanillas cannot see — Chapter 5 develops the integral, and it is exactly the instrument that justifies releasing the wing slope a of remark 4.7. And prior structures from a previous fit enter as reference terms in the same Γ-and-θ_{ref} position — Chapter 10 develops when and how much to trust them.

4.7.2 Every derivative from the same factorization

A least-squares solver runs on the Jacobian: the matrix of derivatives of every residual with respect to every parameter. For a model priced by a PDE march, the naive route is finite differences — perturb θ_ℓ, re-march the surface, subtract — which costs n_θ extra marches per iteration, hundreds here. The sheet's structure gives something better: *exact* derivatives, for the price of extra right-hand sides through a factorization already computed. The key is linearity. By (4.5) the field depends linearly on θ, so the discrete generator does too, and its derivative in one parameter is itself a stencil:

$$\frac{\partial \mathcal{L}}{\partial \theta_\ell} = \mathcal{L}[\mathcal{H}_\ell],$$

the stencil (4.6) with the field v replaced by the single hat function $\mathcal{H}_\ell$ — zero outside vertex ℓ's star of triangles.

Proposition 4.12 (Tangent: same factor, extra right-hand sides)**.** *The sensitivities $s_\ell^n = \partial U^n / \partial \theta_\ell$ obey*

$$\mathcal{M}^{n+1} s_\ell^{n+1} = s_\ell^n + \Delta\tau\, \mathcal{L}[\mathcal{H}_\ell]\, U^{n+1}, \qquad s_\ell^0 = 0, \tag{4.10}$$

where $\mathcal{L}[\mathcal{H}_\ell]\, U^{n+1}$ applies the boundary-inclusive stencil to the solution extended by its fixed boundary values. This is the discretization of the continuous sensitivity equation $\partial_\tau s_\ell = \frac{1}{2} v y^2 \partial_{yy} s_\ell + \frac{1}{2} \mathcal{H}_\ell\, y^2 \partial_{yy} c$.

Proof. Write one step over the interior unknowns as $\mathcal{M}^{n+1} U^{n+1} = U^n + b^{n+1}$, with b^{n+1} the boundary vector displayed after (4.7) — it carries the Dirichlet values through the stencil's boundary columns, and therefore depends on θ through the field at boundary-adjacent nodes. Differentiate the whole equation with respect to θ_ℓ, using the product rule on the left:

$$\left(\partial_{\theta_\ell} \mathcal{M}^{n+1}\right) U^{n+1} + \mathcal{M}^{n+1} s_\ell^{n+1} = s_\ell^n + \partial_{\theta_\ell} b^{n+1}.$$

By linearity, $\partial_{\theta_\ell} \mathcal{M}^{n+1} = -\Delta\tau\, \mathcal{L}[\mathcal{H}_\ell]$ exactly — no approximation is involved. Move it to the right-hand side:

$$\mathcal{M}^{n+1} s_\ell^{n+1} = s_\ell^n + \Delta\tau\, \mathcal{L}[\mathcal{H}_\ell]\, U^{n+1} + \partial_{\theta_\ell} b^{n+1},$$

and the last two terms combine into the boundary-inclusive product of the statement, because the boundary *values* themselves (1 and 0) do not move with θ — only the stencil weights through which they enter do. The start is $s_\ell^0 = 0$: the payoff does not depend on the field. $\square$

Read (4.10) twice. As a computation: the sensitivity system has the *same* matrix $\mathcal{M}^{n+1}$ as the value system — only the right-hand side differs — so every sensitivity marches behind the same tridiagonal factorization as the value solve. The entire n_θ-column Jacobian costs one factorization plus n_θ extra back-substitutions per step; the columns form a block of right-hand sides that a solver processes together. As probability: the source term says that vertex ℓ influences a price exactly through the diffusion's visits to ℓ's star of triangles, weighted by the convexity $\partial_{yy}c$ it finds there.

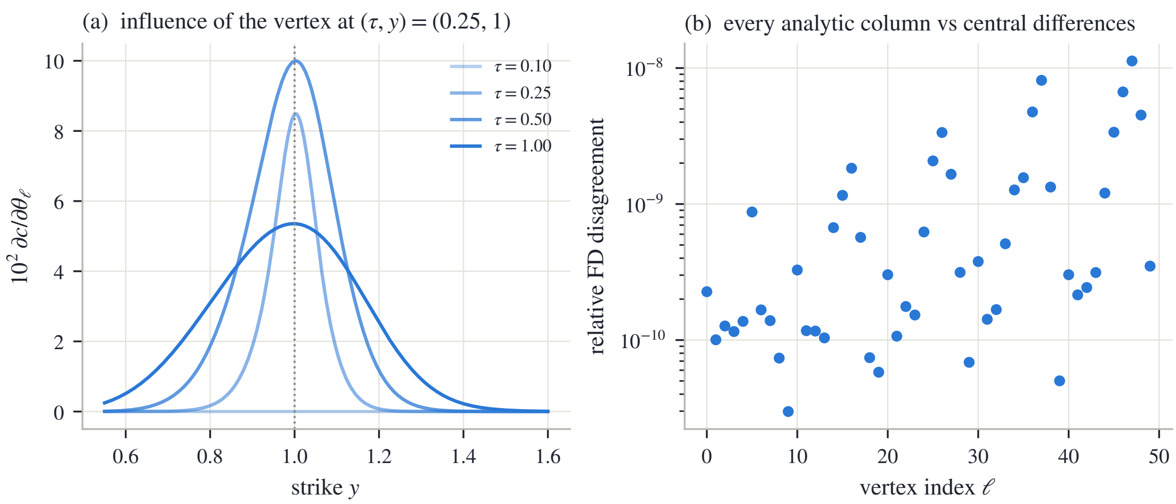

Figure 4.7: The tangent system, seen and audited. (a) Sensitivity of the priced call curve to the single vertex at $(\tau, y) = (0.25, 1)$, across the four quoted expiries. (b) Every analytic column of the Jacobian against a central finite difference of the full march, over all 50 vertices: worst relative disagreement 1.1×10^{-8}.

Figure 4.7 shows both readings on the synthetic market. In panel (a), the sensitivity of the priced call curve to the single vertex at $(\tau, y) = (0.25, 1)$, plotted across the four quoted expiries. At $\tau = 0.10$ the curve is identically zero: an earlier expiry cannot feel a later vertex, because the march has not reached it yet. At $\tau = 0.25$, the vertex's own maturity, the sensitivity is a sharp cone centered on its strike. At the two later expiries the cone spreads and flattens: the diffusion gradually forgets where it collected its variance. In panel (b), the audit: every analytic Jacobian column is compared against a central finite difference of the full march, over all 50 vertices, and the worst relative disagreement is 1.1×10^{-8}. The derivative is exact, not approximate.

The optimization itself is then standard machinery used plainly: a trust-region reflective least-squares iteration on (4.9) — a method of the Gauss–Newton family (linearize the residuals at the current iterate, solve the resulting linear least squares, repeat) that handles box bounds natively, shrinking its steps rather than leaving the box [5] — with the analytic Jacobian of proposition 4.12 supplied at every iterate and a flat seed at the reference sheet. The tangent route's Jacobian cost grows linearly with n_θ; the classical *adjoint* alternative — run the march backward once through its transposed steps, collecting the whole gradient in a single sweep, at a cost independent of the parameter count — is derived in appendix 4.A for the reader who will need it at larger grids.

4.8 Examples and the contract

All fits in this section use one protocol — the grids of section 4.6, the objective of section 4.7, every constant recorded in appendix 4.A — applied unchanged to a synthetic market whose answer is known and to the book's frozen SPY and NVDA chains. No fit is tuned separately.

4.8.1 The round trip, and its two separate numbers

The first test is a market whose answer is known. The synthetic field of appendix 4.A is priced through an *independently implemented* dense scheme — different grids, different code, so the test cannot flatter the chapter's march by validating it against itself — and quoted cleanly at four expiries: 124 quotes in all. The calibration starts from a flat seed. Following section 4.6, the result is reported as two numbers, not one. The *quote* number: the fitted sheet reprices the quotes to 0.6 bp rms, 1.8 bp at worst — in the space the optimizer sees, the fit is essentially exact. The *surface* number: the fitted field agrees with the true field to 0.78 volatility points rms over the quote-covered region, 3.10 points at worst — more than a hundred times the quote error.

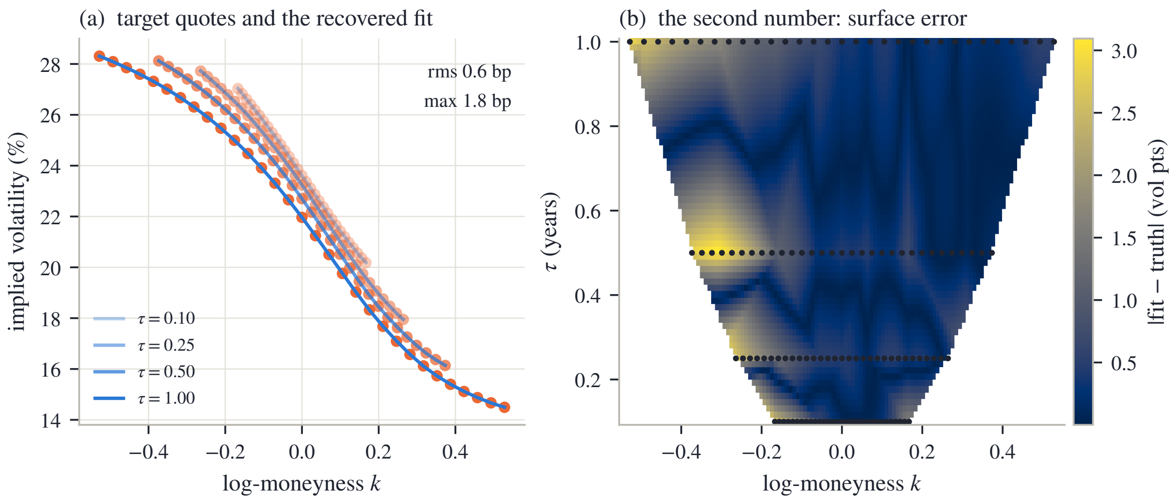

Figure 4.8: The synthetic round trip. (a) Target quotes (dots) and the recovered fit (lines) at the four expiries: rms 0.6 bp, max 1.8 bp. (b) |fitted − true| local volatility over the quote-covered region, quote positions dotted.

Figure 4.8 shows where each number lives. In panel (a), the four quoted smiles (dots) and the recovered fit (lines): the curves pass through the data, and the two numbers printed above — rms 0.6 bp, worst 1.8 bp — are all there is to say. In panel (b), the map of |fitted − true| local volatility over the quote-covered region, with the quote positions dotted. The error is not spread evenly: it concentrates between the expiries and toward the edges of each quoted span — exactly where fig. 4.6 said the quotes do not look, and largest where the nearest quote is farthest. Both numbers are correct at once: the quotes are matched almost perfectly, and the field between them is known only to a fraction of a volatility point.

4.8.2 The frozen chains

The real test fits one sheet per underlier to every mid quote of the snapshot's eight expiries at once: 1026 quotes against 216 nodal variances for SPY, 506 against 207 for NVDA. Figure 4.3 already showed the calibrated SPY sheet itself; fig. 4.9 shows what it prices. Its four panels are four expiries of the same surface, each priced by the *one* calibrated diffusion, against the frozen quotes (dots are mids, whiskers the bid–ask band; each panel's annotation gives that expiry's rms on the fitting operator). Read them as a progression. In the first panel, the two-day smile: narrow, steep, and carrying visibly the most residual of the four — the next figure isolates why. In the last, a year-and-a-half LEAP (market shorthand for a long-dated listed option): wide, gentle, and essentially exact. The two middle panels bridge the regimes, and the entire gallery comes from one list of positive numbers.

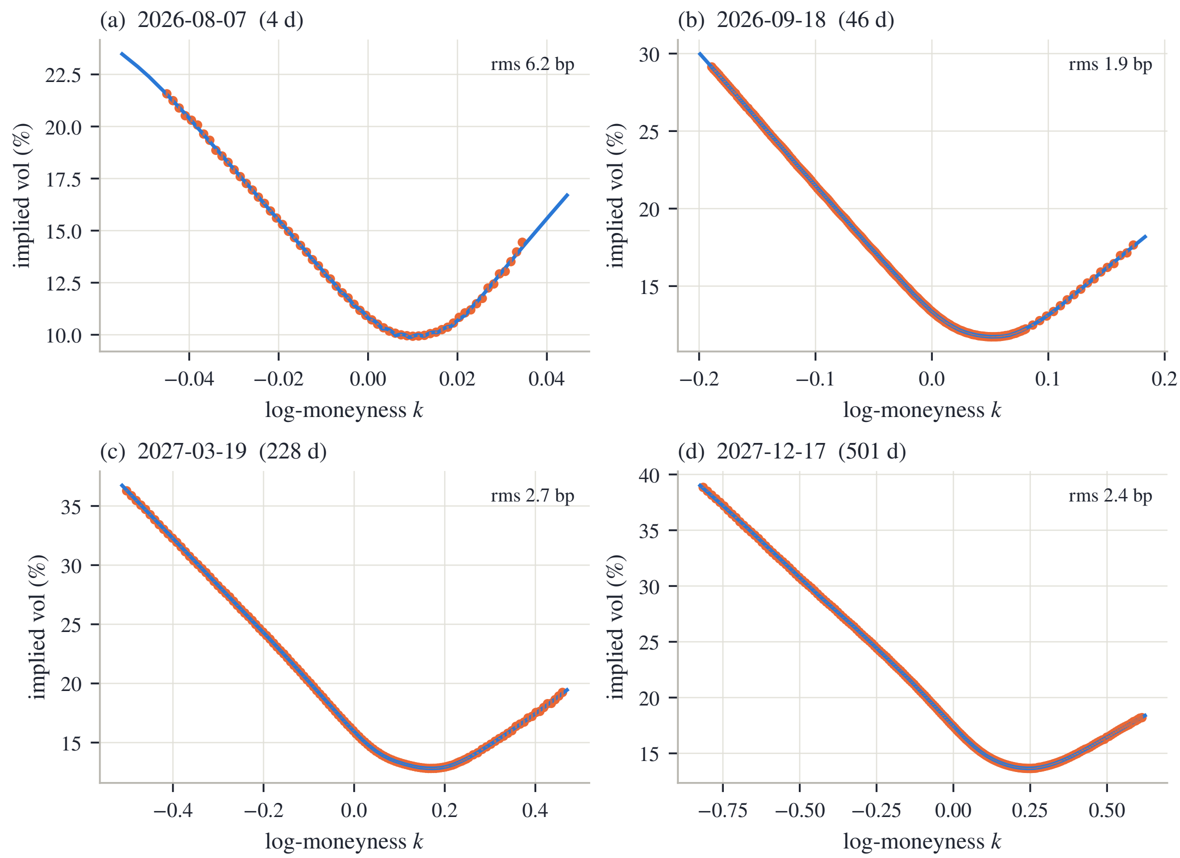

Figure 4.9: Four SPY expiries priced by the single calibrated sheet, against the frozen quotes (dots: mids; whiskers: bid–ask). Panel titles give the expiry and its distance; annotations the per-expiry rms on the fitting operator.

Figure 4.10 reports the chapter's two metrics side by side, per expiry and on a log scale: SPY in panel (a), NVDA in panel (b). The grey bars are the rms on the fitting operator — across all quotes, 2.7 bp for SPY and 16.1 bp for NVDA, whose wider spreads and harder short-dated smiles carry more residual everywhere. The blue bars are the same fitted sheets repriced on the refined operator of section 4.5: 13.6 and 46.1 bp, with the worst single expiry — in both names the one-day front — at 28.2 and 116.9 bp. The gap between the paired bars is the operator's blind spot made quantitative: the optimizer partially cancels the coarse march's discretization error at the expiries it is fitted on, the in-operator residual flatters accordingly, and the refined number is the one to quote. Where the pairs agree — the long end — the fit is genuinely converged. Where they diverge — day-scale maturities — the remaining error is discretization, not disagreement with the quotes.

The measured cleanliness promised by remark 4.11 completes the report. Across every expiry of both calibrated sheets, the minimum divided second difference of the marched prices is -8.5×10^{-13} (SPY) and -2.1×10^{-13} (NVDA), and the minimum adjacent-expiry increment is -8.9×10^{-16} and -7.8×10^{-16}. Both sit at the scale of solver rounding — as proposition 4.2 predicts, and as remark 4.11 declined to assume without measuring.

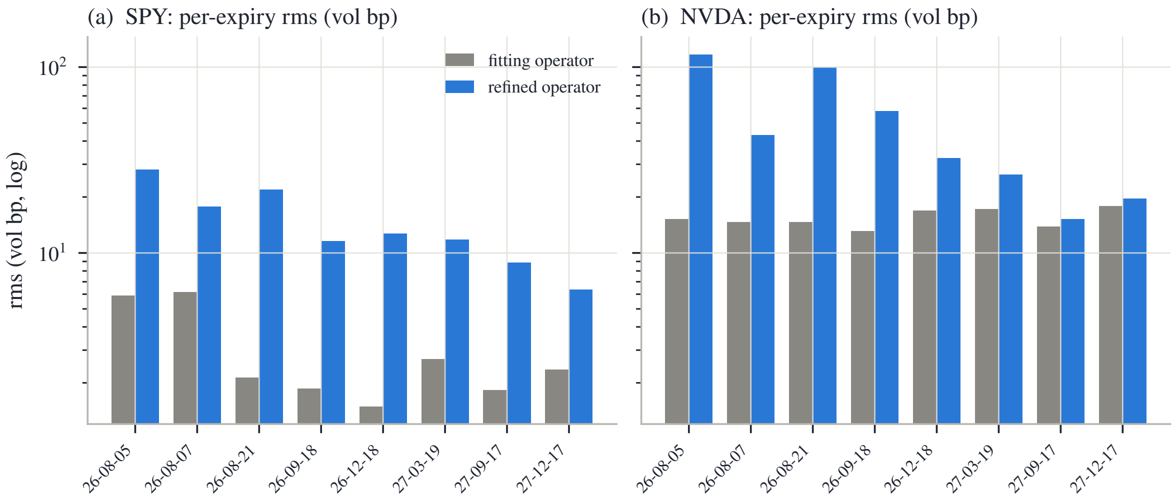

Figure 4.10: Per-expiry rms for SPY (a) and NVDA (b), on the fitting operator (grey) and on the refined operator (blue), log scale.

4.8.3 The contract

The chapter closes the way the book's model chapters do, with its contract. Read the table's three columns as decreasing strength. The first column holds by theorem, and no data can break it. The second column is not guaranteed but is *checked*, on every fit, and reported. The third column is what a user must not read into the model, with the chapter that takes each item up.

Structural	**Numerically controlled**	**Outside the claim**
Box bounds imply a positive field on the hull; a positive field implies butterfly-free, calendar-ordered marginals (propositions 4.2 and 4.5); every optimizer iterate is an admissible diffusion	Butterfly, calendar and bound cleanliness of the marched prices, measured per fit; refined-operator reprice beside the in-operator residual	Convexity preservation of the discrete scheme as a theorem; the extrapolation region beyond the hull, where basis weights are signed
Monotone, unconditionally ℓ^∞-stable march (proposition 4.9 and corollary 4.10); exact tangent Jacobian from the same factorization (proposition 4.12)	Tangent columns audited against finite differences; an independent discretization cross-checks the march	Uniqueness of the calibrated sheet: the data determines an equivalence class, the conventions of section 4.7 a representative
Bounded field $\Rightarrow$ implied variance bounded in strike, strictly inside Lee's cap on the hull (remark 4.8)	Grid placement rules resolving each expiry's $\sigma\sqrt{\tau}$; adaptive variance box keyed to ATM quotes	Jumps and scheduled-event spikes (the clock, Chapter 8); spot–volatility dynamics (Chapter 9); the deep-put tail's shape without a variance-swap quote (Chapter 5)

Table 4.1: The local-volatility model contract: theorem, numerical control, and excluded scope.

The book's separation thesis reads off the table. *Validity* is structural and cheap: a box on nodal values. *Information* is scarcer than the formula (4.3) pretends: the quotes read the field only through smoothing integrals, and only where the quotes exist. *Convention* fills the remainder — roughness, reference, front tie, wing, seed. The calibration does not eliminate those choices. It declares them, measures what the discretization leaks, and reports its error twice: once on the operator the optimizer saw, and once repriced on the refined one.

4.A Numerical facts and the chapter protocol

4.A.1 The protocol

Every fit in this chapter — synthetic and real — uses the following recipe; no constant is tuned per name.

Vertex grid. Maturity rows at $\tau = 0$ and at every quoted expiry. Strike columns delta-spaced: the ladder of put deltas $\{1, 2, 5, 10, 25, 40\}\%$, the same ladder mirrored on the call side, and the money, mapped to strikes through $y = \exp\left(\sigma_A \sqrt{\tau_{\max}}\, \Phi^{-1}(\cdot)\right)$ with σ_A the longest expiry's ATM volatility; the two end columns are extended to cover the pooled quoted range, so no quote falls outside the hull. Coverage rule: every expiry must own at least 6 columns inside its own traded range (section 4.6), enforced as follows: list the columns falling inside the expiry's range together with the range's two endpoints, and while the widest gap between consecutive entries exceeds one fifth of the range, split that gap with a new column; this is what densifies the short-dated money in fig. 4.3.

Lattice. Strike step

$$\Delta y = \min\left(0.01,\ 0.15 \min_e \sigma_e \sqrt{\tau_e}\right),$$

capped at 800 nodes over $[0,\ \max(2.5,\ 1.05\, y_{n_y})]$ with y_{n_y} the highest vertex column, and $y = 1$ a node; every quoted expiry is a time node and each maturity interval receives $\max(8, \lceil \Delta\tau_{\text{int}}/0.01 \rceil)$ implicit steps. The refined (audit) operator halves the strike step and quadruples each interval's step count.

Box and objective. Adaptive box

$$v_{\text{lo}} = \min\left(0.0025,\ (\tfrac{1}{2} \min_e \sigma_{\text{ATM},e})^2\right), \qquad v_{\text{hi}} = \min\left(\max(0.36,\ (3\,\sigma_{\max})^2),\ 16\right)$$

— the floor a fraction of the smallest ATM implied per the averaging fact, the cap a multiple of the largest implied in view. Seed and reference $\theta^0 = \theta_{\text{ref}}$ flat at the median ATM variance. Weights $\lambda = 10^{-2}$, $\lambda_0 = 1$; residual scale η_q equal to one volatility percent of the quote's Black vega, floored at 10^{-3}. The snapshot carries no variance-swap quotes, so the wing multiple stays at its default $a = 0$ (flat clamp) throughout.

Solver. Trust-region reflective least squares [5] with the analytic Jacobian of proposition 4.12, termination tolerances 10^{-8}, at most 200 objective evaluations; the fits of section 4.8 converged in 116 (SPY), 94 (NVDA) and 20 (synthetic) evaluations from their flat seeds.

4.A.2 The synthetic market

The running example's truth is the smooth skewed field

$$\sigma_{\text{loc}}(\tau, y) = 0.21 - 0.10\, \tanh\left(\frac{y-1}{0.22}\right) + 0.03\, e^{-2\tau}, \qquad v = \sigma_{\text{loc}}^2,$$

priced by an *independently implemented* dense march (uniform $\Delta y = 0.0025$ on $[0, 3]$, $\Delta\tau = 0.002$, fully implicit) — a second discretization on different grids, so the round trip of fig. 4.8 cross-checks the chapter's march rather than validating it against itself. Quotes are read at the four expiries $\tau \in \{0.10, 0.25, 0.50, 1.00\}$, at 31 strikes per expiry uniform in the standardized moneyness $\zeta \in [-2.2, 2.2]$ of section 3.7, with $\sigma_{\text{ref}} = 0.24$, i.e. $k = 0.24\sqrt{\tau}\,\zeta$ (the coarse arm of fig. 4.2a uses 13). The "noise" of section 4.3 is a deterministic alternating ± 50 bp volatility ripple — no randomness anywhere in the chapter's figures. The synthetic sheet has rows $\{0\} \cup \{\text{expiries}\}$ and the delta-spaced columns at scale 0.24 plus one deliberate deep-put column at $y = 0.45$ for the experiment of fig. 4.6; its box is $[0.005, 0.36]$.

4.A.3 The adjoint

The tangent route of proposition 4.12 prices the Jacobian at $O(N_\tau N_y\, n_\theta)$ per evaluation, for a march of N_τ steps on N_y lattice nodes. The classical alternative makes the cost of a *gradient* independent of n_θ; deriving it costs a few lines, so the chapter records it for the grids on which it will one day pay. The derivation follows the standard optimal-control pattern — adjoin the marching constraints with multiplier vectors p^n, the *adjoint states*, require stationarity, and read the result as a backward march; a reader meeting adjoints for the first time can treat (4.11)–(4.12) as a recipe and check it numerically against the tangent Jacobian.

Proposition 4.13 (Adjoint: one backward sweep)**.** *Let $\mathcal{J}(\theta)$ be a scalar objective depending on the march through the observed values $\mathcal{O}_n U^n$, with $\mathcal{O}_n$ fixed observation rows. Define adjoint states by the backward recursion*

$$(\mathcal{M}^n)^\top p^n = \mathcal{O}_n^\top \frac{\partial \mathcal{J}}{\partial(\mathcal{O}_n U^n)} + p^{n+1}, \qquad p^{N+1} = 0. \tag{4.11}$$

Then

$$\frac{\partial \mathcal{J}}{\partial \theta_\ell} = \sum_n \Delta\tau\, (p^n)^\top\, \mathcal{L}[\mathcal{H}_\ell]\, U^n, \tag{4.12}$$

with the boundary-inclusive product of proposition 4.12, at the cost of one backward solve, $O(N_\tau N_y)$, plus the $O(n_\theta)$ accumulation of (4.12).

Proof. Form the Lagrangian (fraktur $\mathfrak{L}$, kept visually apart from the stencil operator $\mathcal{L}$) $\mathfrak{L} = \mathcal{J} - \sum_n (p^n)^\top (\mathcal{M}^n U^n - U^{n-1} - b^n)$, with b^n the boundary terms of proposition 4.12. Stationarity in each U^n — which appears in step n's system and in step $n{+}1$'s right-hand side — gives (4.11); the θ-derivative of $\mathfrak{L}$ then retains the explicit dependence of $\mathcal{M}^n$ and b^n, which combine as in proposition 4.12 into the boundary-inclusive (4.12). Structurally this is summation by parts: the transpose of a lower block-bidiagonal system is upper block-bidiagonal, i.e. a backward march with transposed tridiagonal factors. □

4.A.4 Measured cleanliness

The diagnostics quoted in section 4.8 are computed on the fitting operator's price grid: the butterfly proxy is the minimum divided second difference $(U_{i+1} - U_i)/\Delta y_i^+ - (U_i - U_{i-1})/\Delta y_i^-$ over all interior nodes and all expiries; the calendar proxy is the minimum of $U^{(e+1)} - U^{(e)}$ over adjacent expiries at fixed y (the lattice prices are already in the normalized units of section 2.8, so the fixed-y comparison is the right one); and the bounds check verifies $0 \le U \le 1$ to the same tolerance. For both calibrated sheets all three sit at solver rounding (section 4.8); they are measured per fit precisely because remark 4.11 declines to promote them to a theorem.

Chapter 5

Integrals and Wings: Variance Swaps Beyond the Last Quote

The three model chapters each ended with a fitted curve that reprices every quote. This chapter asks a harder question of those curves: what do they say where there are no quotes? The probe is a single traded number, the fair strike of a variance swap. It is an integral of the whole smile, it can be computed exactly in each of Part I's three coordinate systems, and on the book's frozen data — the one market snapshot every example reads — a fifth of it comes from strikes beyond the last quote. Models that agree to basis points at every quote therefore disagree on it. The chapter measures that disagreement, proves the two model-free facts that bound it — a ceiling on how steep any wing can be, and an envelope, fanning out from the last quote, that no arbitrage-free continuation can leave — and shows why everything inside the envelope is a choice the modeler must make and disclose. It closes with the term structure: the calendar condition of Chapter 2 forces the fair strike to grow with maturity, so the integral inherits the order the surface already keeps.

5.1 One number for the whole smile

Chapters 2–4 built three ways of fitting one expiry's smile, and each chapter ended the same way: with a curve that reprices every quote to a few basis points. This section introduces a single number, attached to the whole smile at once, that will spend the rest of the chapter testing those curves in the region the quotes never reach.

The number comes from a real instrument. A *variance swap* is a contract on how much the underlying actually moves. Over the life of the contract, each day's log-return is squared, the squares are summed, and at expiry the buyer receives that realized sum and pays a fixed amount agreed at the start. The fixed amount is called the *strike* of the swap, and the strike at which the contract is worth zero today is the *fair strike*. Quoting desks publish fair strikes the way they publish option prices. So the fair strike is not a modeling construct: it is a market number, one per (underlier, expiry), and a fitted smile implies a value for it.

To compute that implied value, work in the book's standing units. The underlying is measured as $Y = S/F$, the level divided by the forward, so Y starts at 1 and is a positive martingale — a process with no drift, whose expected future value is its current value (section 4.1). Time is variance time τ — the maturity variable the book has quoted variance against since Chapter 2, whose clock Chapter 8 will make precise — so a smile's total implied variance at log-moneyness k is $w(k) = \sigma(k)^2\tau$, with $\sigma(k)$ the implied volatility there. Let the martingale move as a diffusion,

$$\mathrm{d}Y_s = \sqrt{v_s}\, Y_s \,\mathrm{d}W_s,$$

where s runs from 0 to the expiry τ and v_s is the *instantaneous variance* at time s — the squared volatility the process is running at that instant. In the representative model of Chapter 4, v_s

is the field evaluated along the path, $v_s = v(s, Y_s)$; but nothing below needs that: v_s may be any reasonable random process. As the monitoring interval shrinks, the sum of squared returns converges to the accumulated instantaneous variance, so in these units the realized variance the swap pays is the integral $\int_0^\tau v_s \, ds$, and the fair strike is its expected value under the pricing law — the law the smile encodes, the same one behind every expectation in this book. (Two conventions separate this from the listed contract, and it is worth stating them once, here. The contract sums squared *daily* returns while the integral monitors continuously, and the contract's clock is the calendar while ours is variance time; both refinements have standard corrections and Chapter 8 returns to the clock. A third gap — paths that jump — is genuine and reappears in the chapter's closing table of guarantees, section 5.7.1.)

The expected value looks like it should depend on the whole path law of Y. It does not, and the two-line computation that shows this is the heart of the chapter. Apply Itô's formula — the chain rule for diffusions, which adds a second-derivative term — to $\log Y_s$, using the multiplication rule $(dW_s)^2 = ds$, so that $(dY_s)^2 = v_s Y_s^2 \, ds$:

$$d \log Y_s = \frac{dY_s}{Y_s} - \frac{1}{2} \frac{(dY_s)^2}{Y_s^2} = \sqrt{v_s} \, dW_s - \frac{1}{2} v_s \, ds. \tag{5.1}$$

Integrate from 0 to τ and use $\log Y_0 = 0$:

$$\log Y_\tau = \int_0^\tau \sqrt{v_s} \, dW_s - \frac{1}{2} \int_0^\tau v_s \, ds.$$

Now take expectations. The first integral is a stochastic integral against a Brownian motion, and such integrals have expectation zero — they are the running gains of a bet with no edge. Writing $X = \log Y_\tau$ for the terminal log-return, the surviving terms rearrange to

$$w_{\mathrm{vs}} \;=\; \mathbb{E}\left[\int_0^\tau v_s \, ds\right] \;=\; -2 \, \mathbb{E}[X]. \tag{5.2}$$

The expected realized variance equals minus twice the expected log-return. The right-hand side no longer mentions the path: it is a property of the *terminal law alone*, the same law every smile of this book encodes. A path-dependent payoff has been priced by a European one — the payoff $-2 \log y$, called the *log contract*. Chapter 2 already met this number in passing, as (2.34); this chapter is about taking it seriously. We write w_{vs} for the fair strike in total-variance units and $\sigma_{\mathrm{vs}} = \sqrt{w_{\mathrm{vs}}/\tau}$ for the same number quoted as a volatility.

Before asking what the number says about a real smile, compute it once by hand on the simplest smile there is. Suppose the smile is flat: $w(k) = w_0$ at every strike. A flat smile means the law of X is exactly the one Black–Scholes assumes, a normal with variance w_0. Its mean is pinned by the martingale condition $\mathbb{E}[e^X] = \mathbb{E}[Y_\tau] = Y_0 = 1$: for a normal with mean μ_X and variance w_0, the standard identity $\mathbb{E}[e^X] = e^{\mu_X + w_0/2}$ forces

$$\mu_X + \frac{w_0}{2} = 0 \qquad \Longrightarrow \qquad \mathbb{E}[X] = -\frac{w_0}{2},$$

and (5.2) gives

$$w_{\mathrm{vs}} = -2 \, \mathbb{E}[X] = w_0. \tag{5.3}$$

On a flat smile the fair strike *equals* the ATM total variance, exactly. This small computation earns its keep twice over. It is a sanity check — a variance swap on a constant-volatility world is worth that variance. And it is a decomposition: whatever a real fair strike shows *beyond* the ATM level is entirely a statement about the smile's departure from flatness — its curvature and, above all, its wings. Section 5.7 will measure that excess on real data and find it worth hundreds of basis points.

Why the wings, specifically? Look at the shape of the payoff. The function $-2 \log y$ is large where y is far from 1, and it grows without bound as $y \to 0$. At the same time the martingale condition $\mathbb{E}[e^X] = 1$ forces a balance: a law cannot put weight on large upside without compensating weight below, and every rearrangement of tail mass moves $\mathbb{E}[X]$. The fair strike therefore loads exactly on the part of the law the quoted strikes describe least — a suspicion the next two sections make precise and then quantify.

Figure 5.1 shows why the number deserves a chapter. It fits the three families of Part I — LQD from Chapter 2, SVI and MCS from Chapter 3 — to the same 94 mid quotes (bid–ask midpoints) of the book's running node: the (underlier, expiry) pair SPY December 2026 that Chapters 2–4 kept returning to. The fits follow Chapter 3's comparison protocol (section 3.8) — same quotes, equal weights, each family's reference settings. In panel (a), the quoted span: the three fitted curves pass through the same quotes and through each other — the worst fit error of any family is 7.5 bp, and no two of the curves are ever farther than 13.3 bp apart on the span (one vol bp is 10^{-4} of absolute volatility, the chapter's unit for volatility distances). By every measure the data offers, the three models are the same smile. In panel (b), the same three fits drawn wide, in total variance, with the quoted span shaded: outside the shading the curves separate — each family continues the smile according to its own structure — and the legend prints each family's implied fair strike. The three values of σ_{vs} span 58.2 vol bp: LQD says 19.83%, SVI says 19.33%, MCS says 19.25%. Nothing has gone wrong. Three valid laws thread the same quotes and integrate to different numbers.

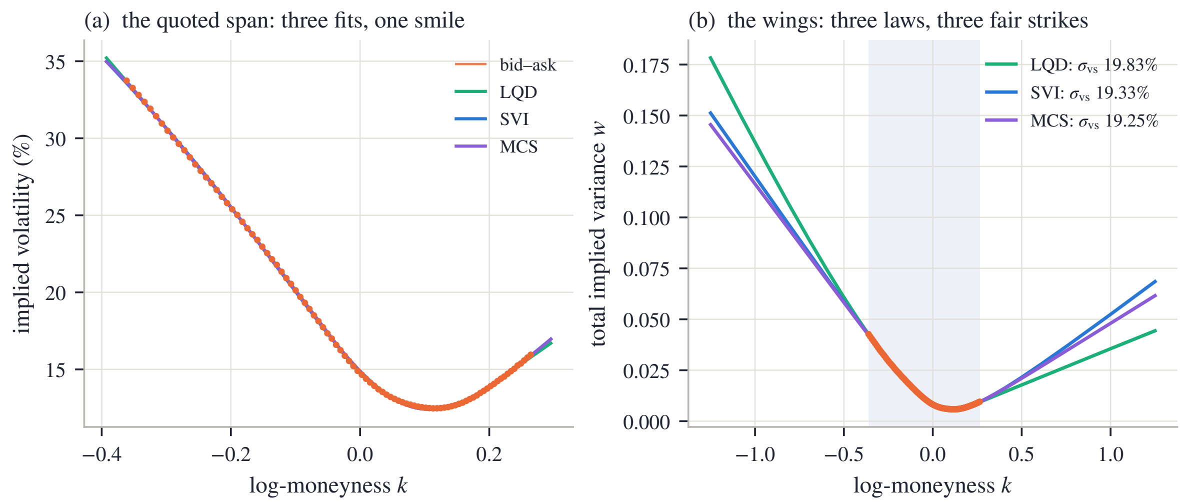

Figure 5.1: Three families, one quote set, three fair strikes (SPY December 2026, 94 quotes). (a) On the quoted span the LQD, SVI and MCS fits agree with the quotes and with each other to 13.3 bp. (b) In total variance, wide: beyond the shaded quoted span the three curves separate, and their fair var-swap strikes differ by up to 58.2 vol bp.

We now have the chapter's protagonist and its puzzle. The fair strike is one market-traded number that the whole smile determines — and that fitted smiles, in perfect agreement wherever quotes exist, do not agree on. The plan follows the thesis of the book. Section 5.2 computes the number exactly, in each model's own coordinates, and checks that all routes agree on one model at a time. Section 5.3 locates the disagreement: it lives precisely in the share of the integral that accrues beyond the last quote. Sections 5.4 and 5.5 then prove what *can* be said with no quotes at all — a ceiling and an envelope — and sections 5.6 and 5.7 treat what remains inside those bounds for what it is: a convention to be stated, policed at finite strikes, and kept consistent across the term structure.

5.2 Three integrals for one number

We have one number, $w_{\mathrm{vs}} = -2\,\mathbb{E}[X]$, defined by the terminal law. But the three model families of Part I hold that law in three different coordinate systems — three *charts*, in the word Chapter 3 introduced for a choice of coordinates: Chapter 3's families state the *smile* $w(k)$, Chapter 2's model states the *law* itself through its quantile, and Chapter 4's model states the *field* $v(\tau, y)$ that generates every law at once. This section derives one exact integral for w_{vs} in each chart. The three integrals look nothing alike, and watching them return the same number — to a tenth of a basis point — is the section's payoff.

5.2.1 Against option prices

The first route rewrites the expectation of *any* smooth payoff as a portfolio of the options themselves.

Proposition 5.1 (Spanning). *Let φ be twice continuously differentiable on $(0,\infty)$ and let $Y > 0$ have $\mathbb{E}[Y] = 1$. Write c and P for the normalized call and put at strike K, i.e. $c = \mathbb{E}[(Y-K)^+]$ and $P = \mathbb{E}[(K-Y)^+]$. Then, whenever the integrals converge,*

$$\mathbb{E}[\varphi(Y)] = \varphi(1) + \int_0^1 \varphi''(K)\,P(K)\,\mathrm{d}K + \int_1^\infty \varphi''(K)\,c(K)\,\mathrm{d}K. \tag{5.4}$$

Proof. The claim reduces to a pointwise identity: for every $y > 0$,

$$\varphi(y) = \varphi(1) + \varphi'(1)(y-1) + \int_0^1 \varphi''(K)(K-y)^+\mathrm{d}K + \int_1^\infty \varphi''(K)(y-K)^+\mathrm{d}K. \tag{5.5}$$

Check it for $y \geq 1$; the case $y < 1$ is the mirror image. For $y \geq 1$ the put-type kernel $(K-y)^+$ vanishes on $K \leq 1$, and the call-type kernel is nonzero only for $K \in [1, y]$, so the right-hand side's integrals reduce to $\int_1^y \varphi''(K)(y-K)\,\mathrm{d}K$. Integrate by parts once, differentiating $(y-K)$:

$$\int_1^y \varphi''(K)(y-K)\,\mathrm{d}K = \Big[\varphi'(K)(y-K)\Big]_1^y + \int_1^y \varphi'(K)\,\mathrm{d}K = -\varphi'(1)(y-1) + \varphi(y) - \varphi(1).$$

Substituting back into (5.5) cancels everything but $\varphi(y)$. Now set $y = Y$ in (5.5) and take expectations. The linear term carries $\mathbb{E}[Y] - 1 = 0$ and dies; the two kernels become put and call prices. $\square$

In words: any smooth payoff is a constant, plus a forward position, plus a strip of puts below the forward and calls above it, held in amounts given by the payoff's second derivative. The identity is completely model-free — it never asks where the prices came from.

Now specialize to the log contract, $\varphi(y) = -2\log y$. Then $\varphi(1) = 0$ and $\varphi''(y) = 2/y^2$, so (5.4) reads

$$w_{\mathrm{vs}} = 2\int_0^1 \frac{P(K)}{K^2}\,\mathrm{d}K + 2\int_1^\infty \frac{c(K)}{K^2}\,\mathrm{d}K.$$

Every strike appears, weighted by $1/K^2$: the variance swap is a strip of out-of-the-money options, densest below the forward. For computation it is tidier in log-moneyness. Substitute $K = e^k$, $\mathrm{d}K = e^k\mathrm{d}k$, and index the same prices by log-strike, writing $c(k)$ for $c(e^k)$ and likewise for the put; each $1/K^2\,\mathrm{d}K$ becomes $e^{-k}\mathrm{d}k$. Now recall put–call parity in normalized units ($P(k) = c(k) + (e^k - 1)$, from eq. (2.2)) and write $(x)^- = \min(x, 0)$ for the negative part. On the call leg ($k > 0$) the quantity $e^k - 1$ is positive, so $(e^k-1)^- = 0$ and the bracket below is just the call; on the put leg ($k < 0$) it is negative, so $(e^k-1)^- = e^k - 1$ and the bracket is $c(k) + (e^k - 1) = P(k)$, exactly the put parity demands. One bracket therefore covers both legs, and the two integrals merge into one integral over the smile:

$$w_{\mathrm{vs}} = 2\int_{-\infty}^{\infty} \Big[B(k, w(k)) + (e^k - 1)^-\Big]\,e^{-k}\,\mathrm{d}k, \tag{5.6}$$

where B is the Black call function of eq. (2.3), so $B(k, w(k)) = c(k)$ is the smile's call price. Equation (5.6) consumes nothing but the total-variance curve $w(k)$. Any model that can state its smile can be integrated — this is the natural route for SVI and MCS, whose $w(k)$ is a formula.

5.2.2 Against the law

The second route never leaves the law. The mean of any function of X can be taken in rank coordinates — this is eq. (2.21), the change of variables under which Chapter 2 built everything: if U is uniform on $(0,1)$ then $Q(U)$ has the law of X, so integrating the quantile function Q over ranks *is* taking the mean. Applying this to $w_{\text{vs}} = -2\,\mathbb{E}[X]$ gives Chapter 2's eq. (2.34):

$$w_{\text{vs}} = -2\int_0^1 Q(u)\,\mathrm{d}u = -2\int_{-\infty}^{\infty} x(z)\,\rho(z)\,\mathrm{d}z,$$

where the second form substitutes the log-odds coordinate z, with $x(z)$ the transport and ρ the logistic density (section 2.2). For an LQD slice — one expiry's fitted law — this is one dot product against arrays the slice has already built on the fixed log-odds grid every slice carries: no Black function, no inversion, no new grid. The integrand $-2\,x(z)\rho(z)$ decays like $|z|e^{-|z|}$ — Chapter 2's exponential tails make the transport asymptotically linear in z, and the logistic weight supplies the $e^{-|z|}$ — so the slice's grid exhausts it.

5.2.3 Along paths

The third route uses neither prices nor the terminal law, but the field. Return to the left-hand side of (5.2), the expected accumulated variance, in the representative model of Chapter 4 where $v_s = v(s, Y_s)$. The expectation of a time integral is the time integral of expectations, and each instant's expectation is an integral against that instant's density $f(s, y)$ — the marginal density of Y_s that section 4.2 evolved with the forward equation:

$$w_{\text{vs}} = \mathbb{E}\Big[\int_0^\tau v(s, Y_s)\,\mathrm{d}s\Big] = \int_0^\tau \mathbb{E}\big[v(s, Y_s)\big]\,\mathrm{d}s = \int_0^\tau\int_0^\infty v(s, y)\,f(s, y)\,\mathrm{d}y\,\mathrm{d}s. \tag{5.7}$$

Read the double integral as a pairing of two surfaces: the field $v(s, y)$, against the sheet of densities $f(s, y)$ that says how much time the paths spend at each level. The density sheet is called the *occupation measure* of the diffusion, and (5.7) says the fair strike is the field averaged over where the paths actually go.

Computing (5.7) does not require assembling f. Define the *expected remaining variance*

$$\mathcal{W}(s, y) = \mathbb{E}\Big[\int_s^\tau v(r, Y_r)\,\mathrm{d}r \;\Big|\; Y_s = y\Big],$$

the fair strike of a swap started at time s from level y. It satisfies a backward equation of exactly the form Chapter 4 marched, with the field itself as a source term:

$$\partial_s\mathcal{W} + \tfrac{1}{2}\,v(s, y)\,y^2\,\partial_{yy}\mathcal{W} + v(s, y) = 0, \qquad \mathcal{W}(\tau, \cdot) = 0, \qquad w_{\text{vs}} = \mathcal{W}(0, 1). \tag{5.8}$$

The derivation is two applications of ideas already in hand — Itô's formula and the martingale property — and is written out in appendix 5.A. Operationally: one backward sweep of the same solver Chapter 4 marched forward, adding $v\,\Delta s$ at each step, and the number is read off at the spot. This is the natural route for the local-volatility model, which owns neither a smile formula nor a terminal quantile — but owns the field.

5.2.4 One number

Collecting the three routes with the definition they all serve:

$$\boxed{w_{\mathrm{vs}} = -2\,\mathbb{E}[X] = 2\int_{\mathbb{R}} \Big[B(k, w(k)) + (e^k - 1)^-\Big] e^{-k}\mathrm{d}k = -2\int_{\mathbb{R}} x(z)\,\rho(z)\,\mathrm{d}z = \int_0^\tau \!\! \int v\, f\,\mathrm{d}y\,\mathrm{d}s.} \tag{5.9}$$

One number; an integral against the smile, an integral against the law, an integral against the field. Each equality is a theorem, so any two routes evaluated on the same model must agree — and checking that they do is a free, end-to-end audit of an implementation. (Route two costs one *quadrature* — one numerical integral — and the audits below are limited only by quadrature accuracy.)

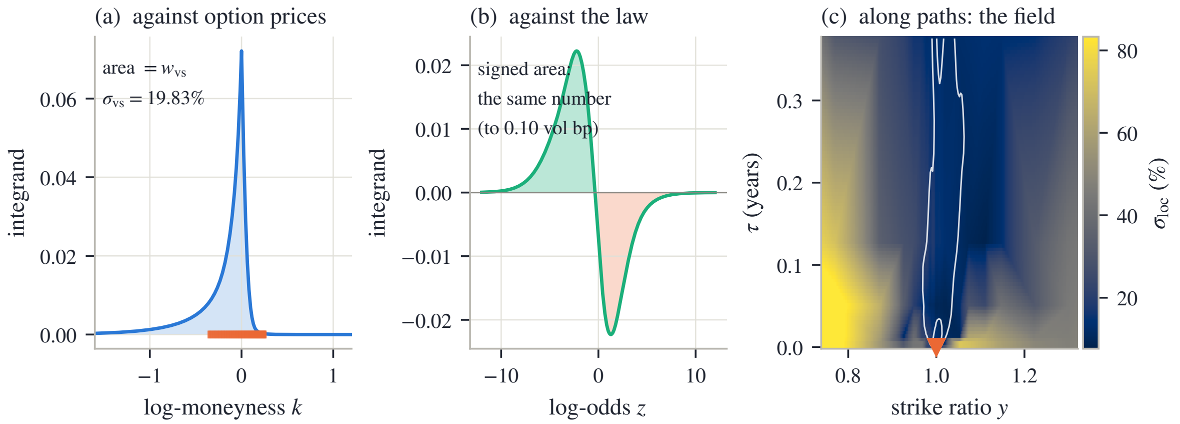

Figure 5.2: One number, three integrands. (a) The strike-side integrand of (5.6) on the running LQD slice; the orange ticks mark the quoted strikes, and the shaded area is w_{vs}. (b) The law-side integrand $-2\,x(z)\rho(z)$ on the same slice's log-odds grid; the signed area is the same number to 0.10 vol bp. (c) The field route on Chapter 4's calibrated SPY sheet: the local volatility as a heatmap, the occupation measure as white density contours funnelling out of the spot, and the marker at $(y, \tau) = (1, 0)$ where the backward solve (5.8) reads off the number.

Figure 5.2 draws the three integrands for real fitted objects, and it repays a slow look. In panel (a), the strike side on the running LQD slice of fig. 5.1: the integrand of (5.6) peaks near the money — the e^{-k} weight decays, but the option prices it multiplies decay much faster — and the orange ticks show how little of the strike axis the quotes occupy. The area under this curve is the fair strike, $\sigma_{\mathrm{vs}} = 19.83\%$. In panel (b), the law side of the *same* slice: the integrand $-2\,x(z)\rho(z)$ against log-odds. It has two lobes, because $-2x(z)$ is positive on the downside ranks ($x < 0$) and negative on the upside: the green lobe is downside outcomes pushing the fair strike up, the orange lobe upside outcomes pulling it down. The lobes are visibly unequal — that asymmetry is the skew — and their *signed* area agrees with panel (a)'s area to 0.10 vol bp: two utterly different integrals, one number, the agreement limited only by quadrature. In panel (c), the field route on a different model of the same market: Chapter 4's SPY sheet, the field fitted to all eight expiries at once. The heatmap is the calibrated local volatility; the white contours are the occupation measure — the densities $f(s, y)$ fanning out of the spot as time runs up the axis — and the pairing (5.7) weighs the heatmap by the contours. The marker at the origin of the fan is where (5.8) reads the answer.

The field route's audit deserves its own sentence, because it teaches the same lesson as Chapter 4's two operators. On the sheet, the strike-side route (5.6) and the field-side route (5.8) are both computed by finite differences, and both carry a first-order bias in the time step. At the march's own step the two reads differ by 23.2 vol bp; refine the step eightfold and the gap collapses to 2.9 vol bp, halving with each halving of the step. The mathematics says the integrals are equal; the computation agrees — in the limit, at the first-order rate the schemes promise, and measuring that convergence is the audit.

One structural observation organizes all of this, and it is worth stating because it predicts which route each model finds cheap. The fair strike is *linear* in the option prices and *linear* in the quantile Q — both appear under plain integral signs in (5.9) — so a model whose own coordinates are one of those objects gets the number, and by the same linearity its parameter derivatives, for the price of one quadrature: LQD holds Q, and the law route is a dot product. It is *not* linear in the smile $w(k)$ — the smile enters (5.6) through the nonlinear Black function — so a model whose coordinates are the smile must pay the Black evaluations: cheap when $w(k)$ is a formula, as for SVI and MCS, and expensive precisely when the smile must be *solved for* point by point. The field route's economy has a third source. Its pairing (5.7) is against an occupation measure the field itself generates, so the number is not linear in v. What saves it is (5.8): $\mathcal{W}$ solves a *linear* equation in which v enters only as a coefficient and a source, so the number still costs one backward sweep. This section's rule, stated once: *evaluate a shared functional in each model's own chart.* It stands behind every route above.

We can now compute the number exactly, in any model, and trust the result. The next question is what part of it the quotes actually pin down.

5.3 Where the number lives

We can compute the fair strike three ways and get one answer per model. Figure 5.1 showed that different models give different answers. This section locates the disagreement on the strike axis, and the location explains everything.

The tool is the accrual function. Write $\mathcal{S}(k)$ for the share of the strike-side integral (5.6) accrued by strikes up to k: run the integral from the far left to k, divide by the whole. Then $\mathcal{S}$ climbs from 0 to 1, and reading it at two strikes tells you what fraction of the fair strike lives between them.

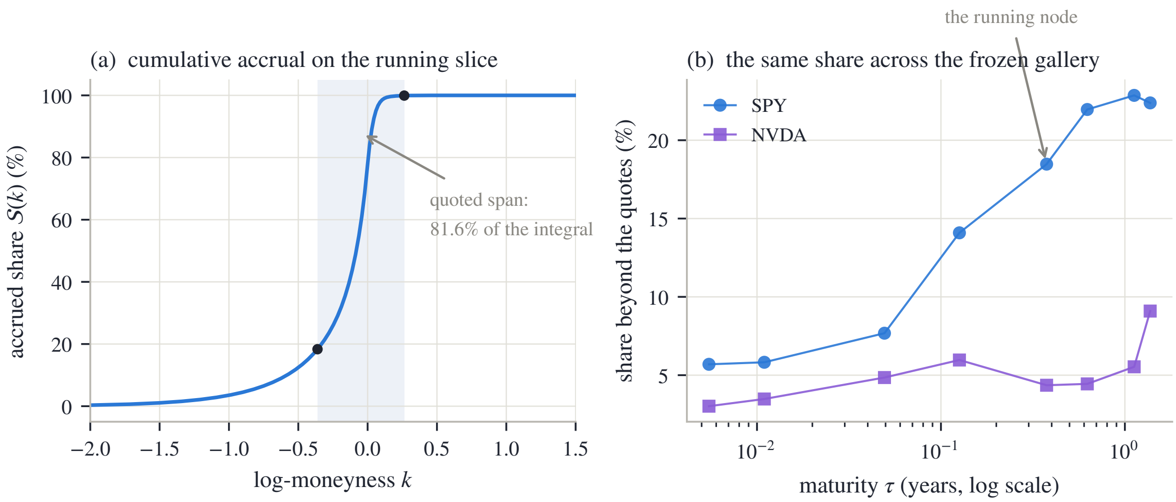

Figure 5.3: Where the fair strike accrues. (a) The accrual share $\mathcal{S}(k)$ on the running slice, with the quoted span shaded and its endpoints dotted: the span carries 81.6% of the integral, so 18.4% accrues where there are no quotes at all. (b) The beyond-quotes share for all sixteen frozen nodes (the snapshot's stored fits), against maturity: from 3.0% to 22.9%, growing with maturity for SPY.

Figure 5.3(a) draws $\mathcal{S}$ on the running slice, with the quoted span shaded. Follow the curve from the left. It is essentially flat until $k \approx -1$: the very deep puts, for all their $1/K^2$ weighting, carry prices too small to matter. It then climbs steeply through the belly — $|k| \leq 0.10$ alone accrues 52.4% of the number — and flattens again on the call side, where prices die quickly. The two dots mark the edges of the quoted span: between them the market's quotes constrain the integrand directly, and that stretch carries 81.6% of the integral. The remaining 18.4% accrues at strikes where no quote exists — where the integrand is whatever the fitted model's continuation says it is. About a fifth of this traded number is, on this node, pure extrapolation.

Panel (b) repeats the measurement across the whole frozen gallery — all sixteen (underlier, expiry) nodes of the book's snapshot, each read from the fit it was published with (the snapshot's stored fits; appendix 5.A records the provenance) and each with its own quoted span. The beyond-quotes share runs from 3.0% to 22.9%. The pattern is worth reading. What protects a node is not how many quotes it has but how many *widths of the smile* its quoted span covers. A strike matters through its standardized distance from the money, $k/(\sigma\sqrt{\tau})$ with σ that expiry's ATM implied volatility: the smile's natural width is $\sigma\sqrt{\tau}$, and a quoted span of roughly fixed width in k covers fewer of those widths as $\sqrt{\tau}$ grows. SPY's long-dated spans cover fewer, and the unquoted share climbs steadily with maturity toward a quarter of the number; the marked running node sits at 18.4%. NVDA's quoted spans are generous relative to its volatility and its share stays under ten percent.

Now put the two exhibits of the chapter together. Figure 5.1(b) showed three models disagreeing on the fair strike by up to 58.2 vol bp while agreeing on the quoted span to 13.3 bp. Figure 5.3 says the disagreement has exactly one place to live: the 18.4% of the integral that accrues beyond the quotes. The same comparison runs across model chapters: Chapter 4's whole-surface sheet, fitted to the same mid quotes, implies $\sigma_{\mathrm{vs}} = 19.70\%$ at this expiry — 13.7 vol bp from the running slice's 19.83%. None of these models is wrong. Each reprices the quotes; the quotes simply do not decide the number. A fit error of a few basis points *per quote* coexists with a fair-strike spread ten times larger, because the fair strike reads a region the fit error never visits.

This is the chapter's instance of a distinction the book keeps returning to. What the quotes pin — the belly, the well-quoted central stretch of strikes — is information. What they leave free — the wings' share — is not noise around zero; it is a quantity the model must supply, and different structural choices supply different amounts. Sections 5.4 and 5.5 will show that the freedom is bounded: no admissible continuation — none that still prices a genuine law — can push the share arbitrarily high. But within those bounds, the number is chosen, not estimated.

One consequence deserves a paragraph before we turn to the bounds, because it changes how a desk should read the instrument. A traded var-swap quote is one *observation* of precisely the combination of unquoted prices the listed strikes leave loose. Chapter 2's calibration objective (section 2.6) accommodates it directly: the model's own fair strike — computed by whichever route of section 5.2 is native — becomes one more residual against the quoted level, in volatility units, with a weight the modeler declares; the derivative of w_{vs} in the model's parameters comes in closed form by the linearity of section 5.2 (Chapter 2 already recorded it as eq. (2.46)). Weighted softly, the quote bargains with the options rather than overruling them, and it spends its influence exactly where fig. 5.3 says its information lives: in the wings. A desk that has such a quote should use it; the sections that follow are addressed to the more common case where it does not.

We now know where the number lives and why fitted models disagree about it. The next two sections establish what no model, and no market, can do in that region.

5.4 The ceiling

A fifth of the fair strike accrues where nothing is quoted. Before deciding what a model *should* draw there, establish what any model *may* draw there. This section proves the one bound that holds with no data at all: a universal ceiling on how steeply total variance can grow with strike.

Recall the objects. The wing slopes of a smile are its asymptotic growth rates in total variance,

$$\beta_R = \limsup_{k\to+\infty} \frac{w(k)}{k}, \qquad \beta_L = \limsup_{k\to-\infty} \frac{w(k)}{|k|},$$

the same β_L, β_R carried since section 2.4 (a limsup is the largest value a ratio keeps returning to; for every smile in this book the ratios simply converge). The claim is that no admissible smile — no smile that prices a positive law with mean one — can have either slope above 2.

The proof rests on one physical fact and one computation. The fact: a call at an absurdly high strike must be worth almost nothing. Formally, $c(k) = \mathbb{E}[(Y - e^k)^+]$, the payoff is dominated by Y itself (which has finite mean) and falls to zero pointwise as $k \to \infty$, so dominated convergence gives

$$c(k) \longrightarrow 0 \qquad (k \to +\infty). \tag{5.10}$$

The computation: what the Black function does when total variance grows along a ray. Fix a slope $\beta > 0$ and evaluate $B(k, \beta k)$ for large k. The Black arguments (from eq. (2.3)) are $d_\pm = -k/\sqrt{w} \pm \sqrt{w}/2$; substituting $w = \beta k$,

$$d_+ = \sqrt{k}\left(\frac{\sqrt{\beta}}{2} - \frac{1}{\sqrt{\beta}}\right), \qquad d_- = -\sqrt{k}\left(\frac{1}{\sqrt{\beta}} + \frac{\sqrt{\beta}}{2}\right). \tag{5.11}$$

Everything hinges on the sign of the parenthesis in d_+: multiplying $\sqrt{\beta}/2 > 1/\sqrt{\beta}$ through by $\sqrt{\beta}$ shows it is positive exactly when $\beta > 2$. So as $k \to \infty$, d_+ runs to $+\infty$ for $\beta > 2$, to $-\infty$ for $\beta < 2$, and sits at exactly 0 for $\beta = 2$.

The second Black term dies in every case. The tool is the normal tail bound $\Phi(d) \le \varphi(d)/|d|$ for $d < 0$, with Φ, φ the standard normal distribution function and density. Its one-line proof: for $t \le d < 0$ the ratio t/d is at least 1, so

$$\Phi(d) = \int_{-\infty}^{d} \varphi(t)\,\mathrm{d}t \le \int_{-\infty}^{d} \frac{t}{d}\,\varphi(t)\,\mathrm{d}t = \frac{\varphi(d)}{|d|},$$

the last step because $t\varphi(t)$ integrates to $-\varphi(d)$. Applying the bound,

$$\log\left(e^k \Phi(d_-)\right) \le k - \frac{d_-^2}{2} - \log\left(|d_-|\sqrt{2\pi}\right), \qquad \frac{d_-^2}{2} = \frac{k}{2}\left(\frac{1}{\beta} + 1 + \frac{\beta}{4}\right),$$

and the coefficient of k collects to

$$1 - \frac{1}{2}\left(\frac{1}{\beta} + 1 + \frac{\beta}{4}\right) = \frac{4\beta - 4 - \beta^2}{8\beta} = -\frac{(\beta-2)^2}{8\beta} \;\le\; 0, \tag{5.12}$$

strictly negative except at $\beta = 2$, where the $\log|d_-|$ term still drives the bound to $-\infty$. Hence $e^k\Phi(d_-) \to 0$ along every ray, and

$$B(k, \beta k) = \Phi(d_+) - e^k \Phi(d_-) \;\longrightarrow\; \begin{cases} 0, & \beta < 2, \\ \frac{1}{2}, & \beta = 2, \\ 1, & \beta > 2. \end{cases} \tag{5.13}$$

Proposition 5.2 (The model-free ceiling)**.** *For any admissible smile, $\beta_L \le 2$ and $\beta_R \le 2$.*

Proof. Right wing. Suppose $\beta_R > 2$ and pick β strictly between 2 and β_R. By the definition of the limsup there are arbitrarily large strikes with $w(k) \ge \beta k$. The Black function increases in its variance argument, so at those strikes $c(k) = B(k, w(k)) \ge B(k, \beta k)$, and by (5.13) the right side approaches 1. A smile that steep prices calls that refuse to become worthless, contradicting (5.10).

Left wing. The mirror argument runs through the put, measured per unit of strike: $P(k)/e^k = \mathbb{E}[(1 - Ye^{-k})^+]$, which by dominated convergence tends to $\mathbb{P}(Y = 0)$ as $k \to -\infty$ — and $\mathbb{P}(Y = 0) = 1$ is impossible for a law with $\mathbb{E}[Y] = 1$. It remains to show a left ray of slope $\beta > 2$ forces that limit to be 1. Parity turns the Black call into the Black put per unit of strike,

$$\frac{P(k)}{e^k} = \frac{B(k,w) - 1 + e^k}{e^k} = \Phi(-d_-) - e^{-k}\,\Phi(-d_+),$$

and along $w = \beta|k|$ (so $k = -|k|$) the arguments (5.11) give

$$-d_- = \sqrt{|k|}\left(\frac{\sqrt{\beta}}{2} - \frac{1}{\sqrt{\beta}}\right), \qquad -d_+ = -\sqrt{|k|}\left(\frac{1}{\sqrt{\beta}} + \frac{\sqrt{\beta}}{2}\right):$$

exactly the two expressions of the right-wing computation with the roles of the arguments exchanged. So for $\beta > 2$ the first term $\Phi(-d_-) \to 1$, and the second, $e^{|k|}\Phi(-d_+)$, dies by the same exponent $-(\beta - 2)^2/(8\beta)$ as in (5.12). Hence $P(k)/e^k \to 1$, the contradiction. $\square$

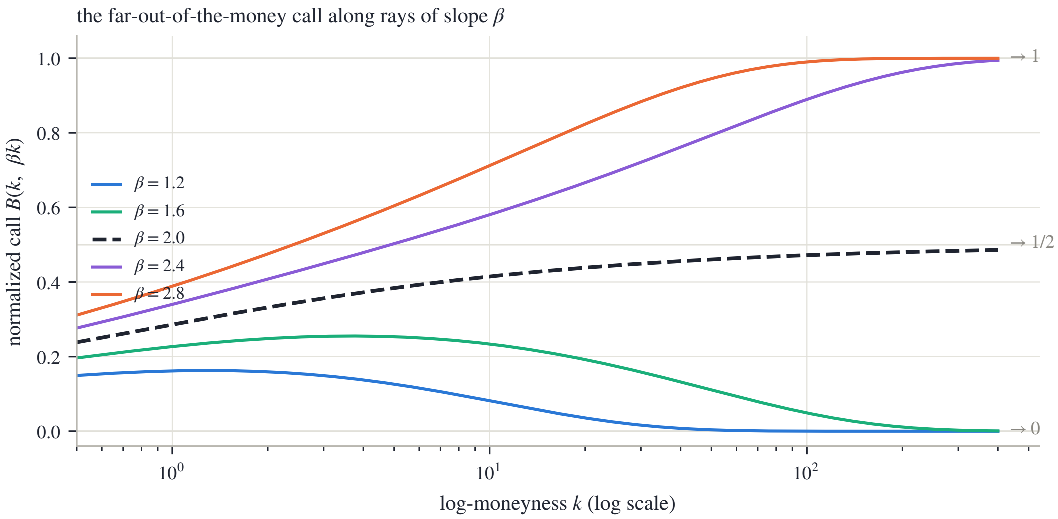

Figure 5.4: The ceiling, computed. The normalized Black call along total-variance rays $w = \beta k$, against log-strike on a log axis. Below the ceiling the call dies (at $k = 400$ the $\beta = 1.6$ ray reads 0.001); above it the call saturates at forward value (0.995 for $\beta = 2.4$); on the boundary ray it creeps up to one half (0.486 at the right edge). Only the dying curves are smiles of any law.

Figure 5.4 draws the computation. Each curve is $B(k, \beta k)$ for one slope, out to $k = 400$ on a logarithmic strike axis — far past anything tradeable, because the convergence is slow and honesty requires showing it. The three regimes of (5.13) separate cleanly: the $\beta < 2$ curves sink to zero (the $\beta = 1.6$ ray reads 0.001 at the right edge), the $\beta > 2$ curves climb toward one (0.995 for $\beta = 2.4$), and the dashed boundary ray $\beta = 2$ creeps up to one half — there $\Phi(d_+)$ equals $\frac{1}{2}$ exactly at every strike, and what remains is the second term dying at its slow boundary rate, leaving 0.486 at the right edge. Notice that the boundary ray also violates (5.10): a call permanently worth half the forward is no better than one worth the whole forward. The ceiling is therefore a *supremum, not a member*: an admissible smile can approach slope 2 from below, but no admissible smile lies on the ray itself. Chapter 3 met the same phenomenon inside the density factor: at $\beta = 2$ the leading tail term of g_{D} vanishes, and whether the far density is positive is decided by the next, smaller term — one the slope condition never examines — which is why the working cap of eq. (3.8) sits strictly inside the bound. Section 5.5 will meet the same phenomenon a third time, as the top edge of the envelope.

The ceiling says how steep a wing *cannot* be. Lee's moment formula — stated in Chapter 2 as eq. (2.23) — says how steep it *is*: $\beta = \Psi(r^*)$, with $\Psi(r) = 2 - 4(\sqrt{r^2 + r} - r)$, where r^* is the tail's critical moment order and Ψ decreases from $\Psi(0) = 2$. The two results are faces of one mechanism, and the computation above quietly contained the link. For $\beta < 2$ the surviving Black term is $\Phi(d_+)$ with $d_+ \to -\infty$, and the same tail bound gives its decay exponent:

$$\frac{d_+^2}{2} = \frac{k}{2}\left(\frac{1}{\sqrt{\beta}} - \frac{\sqrt{\beta}}{2}\right)^2 = \frac{k\,(2-\beta)^2}{8\beta} \implies \Phi(d_+) \approx e^{-k\,(2-\beta)^2/(8\beta)} = K^{-r}, \quad r = \frac{(2-\beta)^2}{8\beta},$$

up to algebraic factors — and this r is exactly the moment budget $\Psi^{-1}(\beta)$ that Chapter 3 computed in eq. (3.9). The bridge between call decay and moments is elementary: whenever $Y \geq K$ one has $(Y-K)^+ \leq Y \leq Y\,(Y/K)^r$, so a finite moment caps the call, $\mathbb{E}[(Y-K)^+] \leq \mathbb{E}[Y^{1+r}]\,K^{-r}$ — more finite moments force faster call decay, and (this is Lee's harder half) the converse holds too. A wing of slope β prices calls that decay with power r; call decay and finite moments carry the same information; so slope pins moments and moments pin slope. Steeper wings, fatter tails, fewer moments — one dial.

What mathematics gives, then, is a cone: every admissible wing lies between slope 0 and slope 2, and its slope names its tail's moment order. What mathematics does not give is a selection. The next section sharpens the cone into a finite-strike envelope anchored at the last quote — and shows that it, too, refuses to select.

5.5 The envelope of admissible completions

The ceiling constrains slopes at infinity. A desk's question is more concrete: the quotes end at some strike — what prices may an honest model print at the strikes just past it? This section answers with an envelope: two curves, computable from the last two quotes, between which every admissible continuation must run.

Fix the setting on the call side (the put side mirrors it). A slice has been fitted through the quotes; call everything up to the last quoted strike *the belly* and regard it as settled. Bars mark quantities frozen at the last quote: $\bar{k}$ is the last quoted log-strike, $\bar{y} = e^{\bar{k}}$ the strike itself, and $\bar{c} = c(\bar{y})$ the normalized call there. Write y_1 for the second-to-last quoted strike; the *last secant slope* is the price change per unit of strike across the final quoted interval,

$$\bar{s} = \frac{c(\bar{y}) - c(y_1)}{\bar{y} - y_1} \;<\; 0.$$

A *completion* is any way of continuing the call curve to strikes beyond $\bar{y}$ so that the whole curve still prices a law. Chapter 2 established what that requires (section 2.1 and eq. (2.4)): as a function of strike, the call curve of a law is nonincreasing, and it is convex — its second derivative *is* the density, so convexity is positivity of probability. And by (5.10) it must eventually reach zero.

Those three properties already imply the envelope.

Proposition 5.3 (The admissible cone)**.** *Every admissible completion satisfies, for all $y \geq \bar{y}$,*

$$\max(0,\; \bar{c} + \bar{s}\,(y - \bar{y})) \;\leq\; c(y) \;\leq\; \bar{c}. \tag{5.14}$$

The lower edge is itself an admissible completion; the upper edge is approached by admissible completions but attained by none.

Proof. Write the completion through its slope: $c(y) = \bar{c} + \int_{\bar{y}}^{y} c'(t)\,\mathrm{d}t$. Convexity of the whole curve says c' is nondecreasing; in particular, for a convex function a chord's slope never exceeds the slope just past its right endpoint, so applying this to the chord over $[y_1, \bar{y}]$ gives $c'(\bar{y}^+) \geq \bar{s}$, hence $c'(t) \geq \bar{s}$ for all $t \geq \bar{y}$. Monotonicity says $c' \leq 0$. Integrating the two slope bounds gives $\bar{c} + \bar{s}(y - \bar{y}) \leq c(y) \leq \bar{c}$, and prices are nonnegative, which is (5.14).

For the lower edge, let the slope stay exactly at $\bar{s}$ until the line reaches zero, at the strike

$$y_0 = \bar{y} - \frac{\bar{c}}{\bar{s}}, \tag{5.15}$$

and stay at zero after. Slopes step from $\bar{s}$ up to 0, so the curve is convex and nonincreasing, and it reaches zero: admissible. Its kink at y_0 is a point mass — the law parks all remaining upside probability at the single strike y_0 and owns nothing beyond. For the upper edge, a concrete family does the work. For small $\alpha > 0$ consider the completion $c(y) = \bar{c}\,(y/\bar{y})^{-\alpha}$: it is decreasing and convex, it tends to zero, and it joins the belly with a convex kink at $\bar{y}$ (its opening slope $-\alpha\bar{c}/\bar{y}$ is closer to zero than $\bar{s}$, and slopes may jump upward at a kink). Being decreasing, convex and vanishing at infinity, it is the call curve of a genuine law (section 2.1), one that pushes its remaining mass to ever higher strikes as α shrinks. As $\alpha \to 0$ these curves approach the constant $\bar{c}$ at every strike — but the constant itself never reaches zero, so it is not admissible. ☐

Pause on what the proposition says. At any strike beyond the last quote, the admissible call prices fill an interval: from a linear descent that hits zero — at y_0, which (5.15) computes from two quotes — up to (but excluding) the last quoted price itself. On the running node the numbers are stark: the last call-side quote sits at $\bar{k} = 0.264$, worth $\bar{c} = 1.17 \times 10^{-4}$ of the forward, and the lower edge reaches zero at $k = 0.350$. The data permits the smile to *end* — price zero, no law beyond — less than a tenth of a log-unit past the last quote. It also permits a wing that is still worth nearly the full $\bar{c}$ a log-unit out. Everything between is allowed.

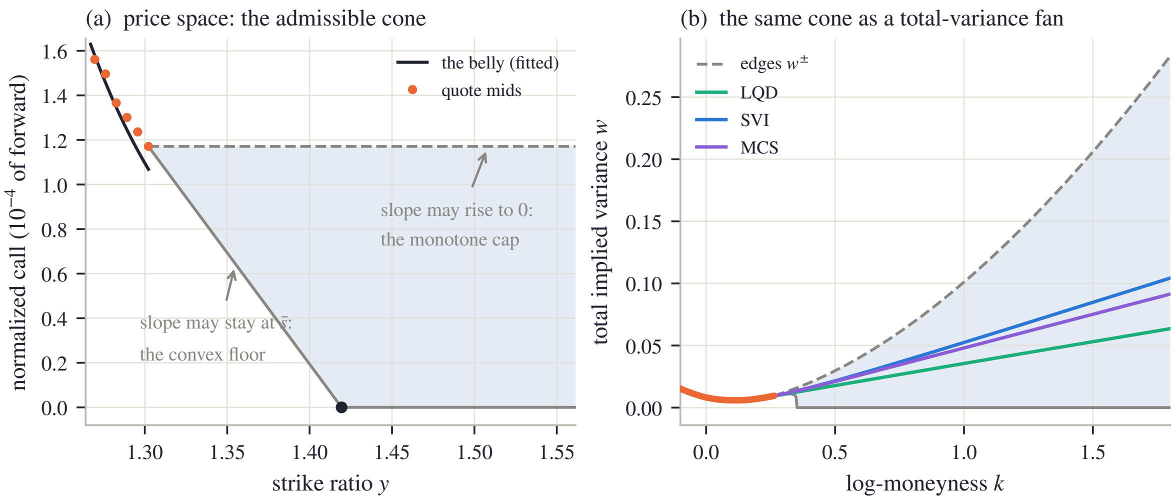

Figure 5.5: The envelope of admissible completions (running node, call side). (a) Price space: the fitted belly (black) through the last quote mids (orange), and the cone (5.14) beyond the last quote: the dashed monotone cap at $\bar{c}$, the convex floor descending at the secant slope $\bar{s}$, reaching zero at y_0 (dot). (b) The same cone mapped through the Black chart: the shaded fan $[w^-, w^+]$ of admissible total-variance completions, with the three families' fitted wings threading it.

Figure 5.5(a) draws the cone in price space on the running node. The black curve is the fitted belly, ending at the last quote; prices here are a few 10^{-4} of the forward, so the axis is in those units. From the endpoint two edges open: the dashed horizontal — the *monotone cap*, prices may not rise — and the descending line — the *convex floor*: prices may never fall faster than the last secant slope, so the straight continuation at $\bar{s}$, dying into the axis at y_0 (dot), is the steepest descent any completion is allowed. The shaded region between them is every call price any law consistent with the belly can print.

Panel (b) shows the same cone where this book draws smiles: in total variance. The Black function is increasing in w, so the price interval (5.14) maps strike by strike onto an interval

$[w^-(k),\, w^+(k)]$, the *fan*: w^- is the implied variance of the descending line (zero once the line dies), w^+ that of the constant $\bar{c}$. The fan is vast. By $k = 1$ the admissible total variance runs from 0 to several times the ATM level; the three families' wings — the same three fits of fig. 5.1 — thread it in a narrow bundle near the bottom. Numerically, each family's wing clears the floor and cap of its *own* cone — the cone rebuilt from that family's fitted call values at the same two strikes, since proposition 5.3 constrains a completion relative to its own belly (appendix 5.A) — with room to spare: worst clearances 0.0016/0.0010/0.0010 below and 0.0016/0.0012/0.0013 above, in total variance. That is proposition 5.3 confirmed on real fits. The bundle's narrowness against the fan's width restates section 5.3: the models' structural choices are far more selective than the data's constraints.

The fan's top edge has a closed form, and deriving it closes the loop with section 5.4. The upper edge holds the call at the constant $\bar{c}$, so $w^+(k)$ solves $B(k, w) = \bar{c}$. Suppose for a moment the second Black term is negligible; then $\Phi(d_+) = \bar{c}$, i.e. $d_+ = q$ with $q = \Phi^{-1}(\bar{c})$ (a negative number — on the running node $\bar{c} = 1.17 \times 10^{-4}$, far below one half). Unpack d_+:

$$\frac{\sqrt{w}}{2} - \frac{k}{\sqrt{w}} = q \qquad \Longrightarrow \qquad w - 2q\sqrt{w} - 2k = 0 \qquad \Longrightarrow \qquad \sqrt{w^+(k)} = q + \sqrt{q^2 + 2k},$$

taking the positive root of the quadratic in $\sqrt{w}$. The neglected term justifies itself with unexpected grace. The two Black arguments always differ by $\sqrt{w}$, so along this solution $d_- = d_+ - \sqrt{w} = q - \sqrt{w}$, and

$$\frac{d_-^2}{2} = \frac{q^2 - 2q\sqrt{w} + w}{2} = \frac{q^2 + 2k}{2} \qquad \Longrightarrow \qquad \log\left(e^k \Phi(d_-)\right) \le k - \frac{q^2 + 2k}{2} = -\frac{k}{2} - \frac{q^2}{2},$$

using the quadratic itself in the middle step, and the tail bound of section 5.4 (whose $-\log|d_-|$ term only strengthens the inequality) in the last: the correction decays like $e^{-k/2}$, exponentially small. (At $k = 30$ the closed form agrees with a full Black inversion by the reference implementation — the working code behind every number in this book — to 1.1×10^{-10} in total variance.) Differentiating the closed form gives the edge's slope,

$$\frac{\mathrm{d}w^+}{\mathrm{d}k} = 2\left(q + \sqrt{q^2 + 2k}\right) \cdot \frac{1}{\sqrt{q^2 + 2k}} = 2 + \frac{2q}{\sqrt{q^2 + 2k}} \longrightarrow 2 \qquad (k \to \infty), \tag{5.16}$$

rising monotonically (recall $q < 0$, and the negative term shrinks) — it reads 1.14 at $k = 30$ — and approaching 2 from below, never reaching it. The top edge of the finite-strike envelope is the ceiling of proposition 5.2, met for the third time and in its sharpest form: the steepest admissible completion climbs at slope $2 - 2|q|/\sqrt{q^2 + 2k}$, a curve that grazes the ceiling asymptotically while staying strictly under it — the supremum, not a member, made into a formula.

The section's summary is uncomfortable and important. Between the floor and the cap, mathematics is silent: every curve in the fan prices a legitimate law consistent with everything quoted. The data cannot choose. The model must — and the next section is about doing that in the open.

5.6 The wing as a stated choice

Inside the envelope the data has no further opinion, yet every fitted model draws exactly one curve there. What selects the curve is the model's structure — decisions made when the family was designed, long before this node's quotes arrived. This section reads those decisions off the three families, states them as contracts, and then demolishes a comfortable inference about them.

Recall the three families' wings; each was derived in its own chapter, and here they only need to be set side by side. The LQD slice of Chapter 2 has exponential log-return tails *by* *construction*, and its wing slopes are closed functions of its two endpoint speeds (proposition 2.9): admissibility is structural, no cap or check required. The SVI slice of Chapter 3 has exactly straight total-variance wings, $\beta_{L,R} = b_S(1 \mp \rho_S)$ — the S-subscripts mark Chapter 3's raw SVI parameters (proposition 3.1) — and its calibration fences them with a cap set strictly inside the ceiling (eq. (3.8)): strictly, because section 5.4 showed the boundary ray itself is inadmissible. The MCS slice of Chapter 3 splits the job: its base owns the wings (eq. (3.32)), and its hats are built to vanish at large strikes, so adding hats cannot move the slopes — but nothing in the family *forces* the base's slopes into the admissible cone; their admissibility is diagnosed after the fit, not enforced during it. On the running node the three contracts print the slopes of table 5.1: every wing far below the ceiling, and the put slopes several times the call slopes. That asymmetry is the equity skew — the standing pattern that downside strikes carry more implied variance than upside ones — read here in its asymptotic form.

Table 5.1: The three wing contracts on the running node. Slopes are asymptotic total-variance wing slopes; all sit far inside the ceiling of proposition 5.2.

Family	β_L (put)	β_R (call)	how the wing is governed
LQD	0.178	0.034	structural: exponential tails, slopes from proposition 2.9
SVI	0.126	0.066	capped: straight wings under eq. (3.8)
MCS	0.117	0.054	inherited: base wings, hats neutral; diagnosed, not enforced

Three families, three slopes per side, all admissible — and all different, as fig. 5.5(b) drew. It bears saying plainly: past the last quote, the divergence between the families is not an error to be averaged away. Choosing a family *is* choosing an extrapolation law. What a careful fitter owes its user is not agreement in the wings — nothing can supply that — but a stated contract: which curve will be drawn, governed by what mechanism, with what guarantee.

There is, however, a gap in those guarantees, and it is the section's second lesson. All three contracts of table 5.1 speak about *slopes* — asymptotic statements, properties of the limit $k \to \infty$. The comfortable inference is: admissible slopes, therefore admissible wing. It is false, and one deliberate construction shows why.

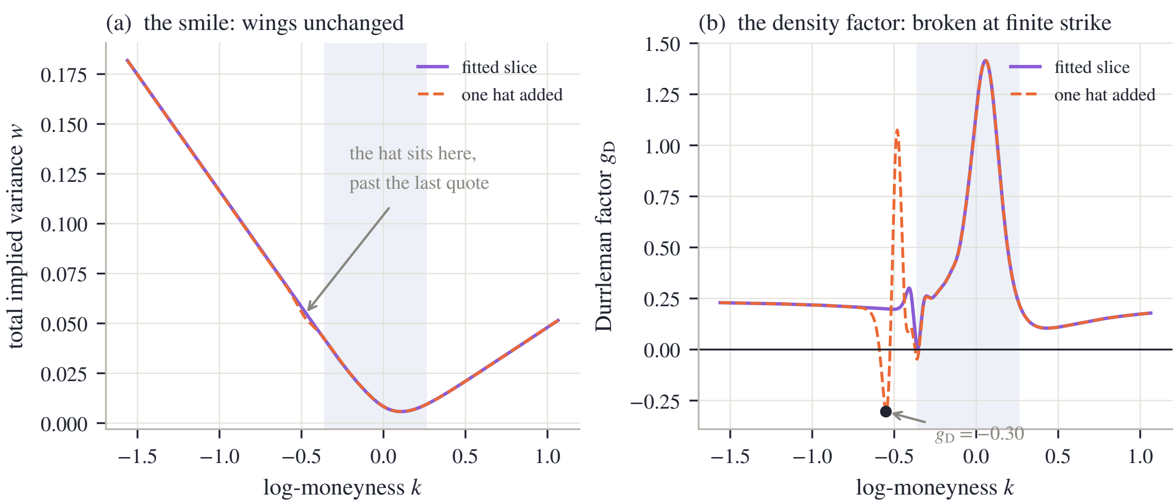

Figure 5.6: The limit is not the wing. One zero-wing hat is added to the fitted MCS slice, centered beyond the last put-side quote. (a) Total variance: the two curves are indistinguishable except for the small bump at the hat; the measured far-field wing slopes change by 0 — an exact zero in double precision. (b) The density factor g_D: the hat drives it to -0.30 at $k = -0.55$, negative density at a finite strike, while the fitted slice (solid) stays nonnegative to rounding (worst -0.001).

Take the fitted MCS slice of the running node and add one more of its own hat kernels (eq. (3.33)) — amplitude negative, centered just beyond the last put-side quote, where quotes no longer object. Figure 5.6(a) shows the smile before and after: the two curves coincide to the eye everywhere except a small dent near $k = -0.55$, and the wings are untouched — measuring the realized slope far out, the change is 0, an exact zero at double precision. The inheritance contract has done its job perfectly. Panel (b) shows what the contract never promised. Recall the Durrleman factor g_D from eq. (2.5): up to a positive factor it is the probability density the smile implies, so a valid smile needs $g_D \geq 0$ everywhere. The fitted slice (solid) keeps it nonnegative across the range — its worst value is -0.001, zero at rounding scale. The hatted slice (dashed) plunges to $g_D = -0.30$ at $k = -0.55$: negative probability, a butterfly arbitrage, parked at a finite strike in the extrapolated region — with both asymptotic slopes exactly as admissible as before. The dent sits in the quiet stretch just outside the quoted span, precisely where no quote penalizes it and no slope condition sees it. Asymptotic admissibility and finite-strike admissibility are independent, and the second needs its own police.

The demonstration settles how the extrapolated region must be policed, and the rule is worth stating because it cuts both ways. Conditions that are properties of a *single* curve — density positivity above, call convexity — must be checked *into* the wings: one law is being asserted there, the condition needs no second curve to compare against, and abandoning it past the last quote is exactly how the hat's dent would go unnoticed. Conditions that *compare two* curves are different. Chapter 2's calendar control (eq. (2.59)) already made the choice: it enforces the order between neighboring expiries only on the span where quotes pin both curves. Beyond that span, both curves are chosen conventions — two entries of table 5.1, possibly from different rows. A calendar inversion there — the later expiry's curve dipping below the earlier one's at some unquoted strike — is a disagreement between two fictions, not evidence of mispricing; enforcing the order there would bend the fits where the data *does* live in order to reconcile them where it does not. One curve: extend the check. Two curves: confine it to common data.

The chapter's statics are now complete for a single expiry: the number, its three computations, its unquoted share, the proved envelope, and the disclosed choice within it. One dimension remains — maturity.

5.7 The ordered term structure, and the contract

Everything so far concerned one expiry. A surface carries a fair strike per expiry, and the last question is how the family of numbers $w_{\mathrm{vs}}(\tau)$ behaves across maturities. The answer is that the calendar condition of Chapter 2 orders it — and the proof is a three-line reuse of the spanning identity.

Recall the condition. A surface is calendar-ordered when, at every log-moneyness, the later expiry's normalized call is worth at least the earlier one's: $c_1(k) \leq c_2(k)$ for all k (eq. (2.56)). Chapter 2 showed this is the same as *convex order* of the two laws — every convex payoff is priced at least as high under the later law — and, by Kellerer's theorem, exactly the condition for some martingale to have these two laws as its distributions at the two expiries (theorem 2.16 and section 2.8).

Proposition 5.4 (Calendar order integrates)**.** *If two admissible mean-one slices satisfy $c_1(k) \leq c_2(k)$ for all k, then $w_{\mathrm{vs},1} \leq w_{\mathrm{vs},2}$.*

Proof. Puts inherit the order from calls: at a common strike K, parity gives $P = c + K - 1$ for both slices, so $c_1 \leq c_2$ implies $P_1 \leq P_2$. Now write each fair strike through proposition 5.1 with $\varphi(y) = -2\log y$, whose second derivative $\varphi''(K) = 2/K^2$ is positive:

$$w_{\mathrm{vs}} = 2\int_0^1 \frac{P(K)}{K^2}\,\mathrm{d}K + 2\int_1^\infty \frac{c(K)}{K^2}\,\mathrm{d}K.$$

Every integrand of slice 2 dominates the corresponding integrand of slice 1, so the integrals are ordered. □

The same argument proves more: *any* payoff with $\varphi'' \geq 0$ — any convex payoff — has its price ordered by the calendar, which is just convex order saying its own name. The variance swap is the special case a desk trades directly, and for it the order has a market reading. The difference $w_{\text{vs}}(\tau_2) - w_{\text{vs}}(\tau_1)$ is the *forward variance* between the two dates — the variance a client buys by holding the long swap against the short one. Proposition 5.4 says a calendar-ordered surface never prices forward variance below zero; a violation would let a desk lock in variance at a negative price, which is the calendar arbitrage of Chapter 2 restated for the integral.

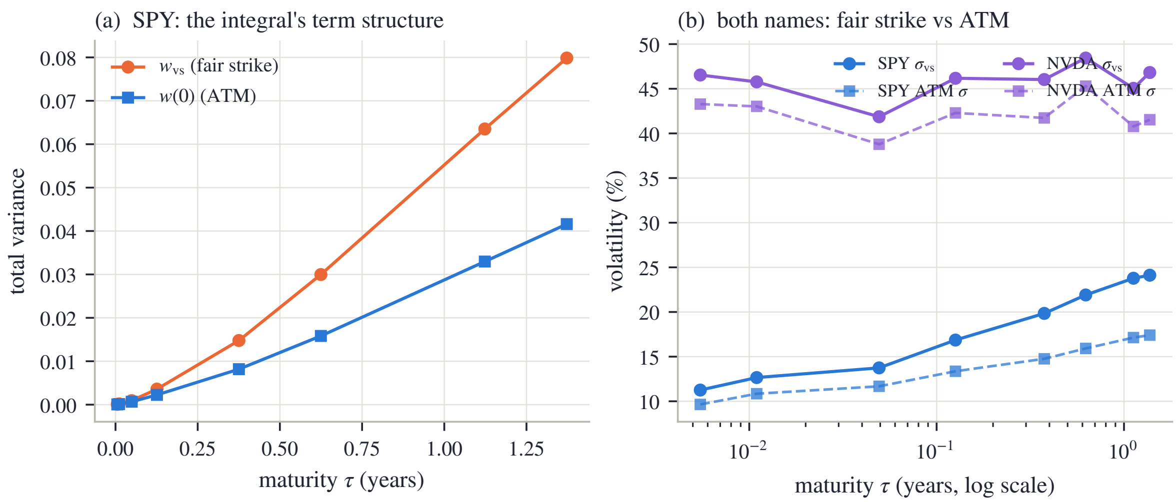

Figure 5.7: The integral's term structure on the frozen gallery (stored surface fits). (a) SPY in total variance: the fair strike w_{vs} and the ATM total variance $w(0)$ both rise with maturity, and the gap between them widens. (b) Both names in volatility units: the fair-strike vol rides above the ATM vol everywhere, by up to 671 vol bp for SPY and 529 for NVDA. NVDA's vol *levels* are not monotone in maturity — the order lives in total variance, panel (a)'s units, not in vol.

Figure 5.7 reads the term structure off the frozen gallery — the snapshot's stored surface fits, the same objects as fig. 5.3(b). In panel (a), SPY in total variance: the orange curve is $w_{\text{vs}}(\tau)$, the blue is the ATM total variance $w(0)(\tau)$, both increasing. Across every adjacent pair of expiries, in both names, the fair strike's increment is positive — the smallest measured increment is +1.1 variance bp (a variance bp is 10^{-4} of total variance, the sibling of the vol bp of section 5.1), between the two front expiries a day apart, and the smallest forward variance, expressed as a forward volatility, is 13.9% — so the published surface keeps the order proposition 5.4 demands, with its thinnest margin exactly where maturities crowd together. Panel (a) shows something else: the *gap* between the curves widens with maturity. By the flat-smile computation (5.3), that gap is precisely the part of the fair strike owed to the smile's departure from flatness, and it compounds with maturity just as the unquoted share of fig. 5.3(b) did — the two figures are the same fact in two currencies.

Panel (b) restates the term structure in the units desks quote, volatility, for both names. The fair-strike vol σ_{vs} rides above the ATM vol at every expiry — by up to 671 vol bp at SPY's long end and 529 for NVDA. On a flat smile the two curves would coincide; the spread *is* the priced skew and wing mass, and a desk can read it directly as what convexity in the smile costs. One trap is visible and worth flagging: NVDA's vol curves are not monotone in maturity, and nothing is wrong — the calendar orders *total variance*, panel (a)'s units, and a falling vol term structure is perfectly consistent with rising total variance. Ordering vols is not a no-arbitrage condition; ordering their $\sigma^2\tau$ is.

5.7.1 The contract

The chapter closes the way the model chapters do, with its contract — three columns of decreasing strength. The first holds by theorem. The second is measured on the book's frozen data, and the chapter quoted every number. The third is convention: chosen, stated, and owed to the user, because no data selects it.

Table 5.2: The chapter's contract.

Proved	**Checked (frozen data)**	**Convention**
The identity chain (5.9): one fair strike, three exact integrals (smile, law, field).	Law vs strike route on one slice: 0.10 vol bp; field vs strike route on one sheet: 23.2 vol bp at the march's step, 2.9 at an eightfold refinement.	Which route a model evaluates (its native chart), and the reading of the fair strike: diffusion paths, continuous monitoring, variance time. Jumps sit outside the identity (5.2).
The ceiling (proposition 5.2): no admissible wing steeper than 2; the boundary ray itself inadmissible.	Fitted slopes (table 5.1): all wings far inside the ceiling.	The wing drawn *within* the cone: each family's stated mechanism — structural, capped, or inherited-and-diagnosed.
The envelope (proposition 5.3): every completion between the convex floor and the monotone cap; top edge climbs at the ceiling (5.16).	All three families' wings clear their cones (clearances of section 5.5); the hatted counterexample caught by the extended g_{D} check, slopes unmoved (fig. 5.6).	Where conditions are enforced: one-curve checks extended into the wings; two-curve (calendar) checks confined to common quote support (eq. (2.59)).
Calendar order integrates (proposition 5.4): the fair strike's term structure is nondecreasing; forward variance is nonnegative.	Smallest measured increment across both names: +1.1 variance bp; smallest forward vol 13.9%.	The weight given a traded varswap quote when one exists: one more residual in the objective of section 2.6, at a declared strength.

Step back, once, to the book's thesis. The fair strike is a single number in which the three ingredients the thesis separates are unusually easy to see. Validity supplied the theorems: the identity chain, the ceiling, the envelope, the order. Information is the quoted belly's 81.6%. Convention is the rest — the wing contract that completes the integral, disclosed in a table rather than hidden in an extrapolation routine. Part I is now complete: laws, smiles, fields, and the integrals that test them. Everything so far took the forward and the discount for granted at every expiry. Part II starts by earning them.

5.A Proofs and numerical protocol

5.A.1 The backward equation for the expected remaining variance

Section 5.2 stated that $\mathcal{W}(s,y) = \mathbb{E}[\int_s^\tau v(r,Y_r)\,\mathrm{d}r \mid Y_s = y]$ solves the backward problem (5.8). Here is the derivation, in two steps the chapter has already rehearsed.

First, a martingale. Consider the running quantity

$$M_s = \int_0^s v(r,Y_r)\,\mathrm{d}r + \mathcal{W}(s,Y_s),$$

the variance accumulated so far plus the expectation of what remains. Write $\mathcal{F}_s$ for the information available at time s. Conditioning the total $\int_0^\tau v\,\mathrm{d}r$ on $\mathcal{F}_s$ splits it into exactly these two pieces — the first is known at time s, and the second is its conditional expectation by the Markov property of the diffusion. So $M_s = \mathbb{E}[\int_0^\tau v\,\mathrm{d}r \mid \mathcal{F}_s]$: every M_s is a conditional expectation of one fixed random variable, and such a family is a martingale by the tower property — conditioning M_t on the smaller information set $\mathcal{F}_s$ $(s < t)$ collapses it back to M_s, which is exactly the martingale identity $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$.

Second, Itô. Assume $\mathcal{W}$ is smooth and apply Itô's formula to $\mathcal{W}(s, Y_s)$ with $\mathrm{d}Y = \sqrt{v}\, Y \mathrm{d}W$, using $(\mathrm{d}W)^2 = \mathrm{d}s$ as in section 5.1:

$$\mathrm{d}\mathcal{W}(s, Y_s) = \Big[\partial_s \mathcal{W} + \tfrac{1}{2}\, v(s, Y_s)\, Y_s^2\, \partial_{yy}\mathcal{W}\Big]\mathrm{d}s + \partial_y \mathcal{W}\, \sqrt{v}\, Y_s\, \mathrm{d}W_s,$$

so that

$$\mathrm{d}M_s = \Big[v + \partial_s \mathcal{W} + \tfrac{1}{2}\, v\, Y_s^2\, \partial_{yy}\mathcal{W}\Big]\mathrm{d}s + \partial_y \mathcal{W}\, \sqrt{v}\, Y_s\, \mathrm{d}W_s.$$

A martingale has no drift, so the bracket must vanish at every state the process can reach — which is the PDE of (5.8). The terminal condition $\mathcal{W}(\tau, \cdot) = 0$ holds because nothing remains at expiry, and evaluating the definition at $(s, y) = (0, 1)$ gives $\mathcal{W}(0, 1) = \mathbb{E}[\int_0^\tau v\, \mathrm{d}r] = w_{\mathrm{vs}}$.

5.A.2 Numerical protocol

Every empirical number in the chapter comes from one deterministic generator reading the book's frozen snapshot; none is typed by hand. The conventions, in the order the chapter uses them.

Fits. The running node is SPY December 2026. Its three family fits are Chapter 3's comparison protocol applied verbatim (section 3.8): the same mid quotes, equal weights, each family's reference settings. The gallery and term-structure figures (fig. 5.3b and 5.7) instead read the snapshot's *stored* surface fits — the calibration the surface was published with, under the fit recipe of section 2.6 — because those are calendar-coupled across expiries, which is what a term-structure exhibit should show.

The strike-side quadrature. Equation (5.6) is evaluated by a trapezoid on $k \in [-6, 6]$ with 4001 points — a finer copy of the 801-point grid the reference implementation uses inside its optimizer loop; the total variance inside the Black call is floored at 10^{-12}. At these settings the quadrature error is far below every number quoted. The law-side integral rides each slice's own fixed log-odds grid.

The field-side audit. The sheet is Chapter 4's whole-surface SPY calibration, unchanged. Both of its var-swap routes run on an audit lattice with the calibration lattice's strike step, widened to strike ratio 12 so the static replication sees the entire wing the extrapolated field implies; the put leg is cut off below strike ratio 0.01 (moving the cutoff to 0.001 changes nothing at the quoted precision). Both routes are first-order in the time step, so the audit reports the pair of reads at the march's own step and at an eightfold refinement; the gap halves with the step, as first order promises.

The envelope. The drawn cone is built from the Black prices of the last two call-side quote *mids*. The per-family clearances are measured against each family's *own* cone — the same construction fed that family's fitted call values at the same two strikes — because proposition 5.3 constrains a completion relative to its own belly; nonnegativity of all six clearances is the proposition confirmed numerically. Clearances are measured on $\bar{k} + 0.05 \leq k \leq 1.8$; beyond 1.8 the flattest wing's call price falls under the double-precision floor and its Black inversion is noise.

The hat. The counterexample of fig. 5.6 adds one kernel of eq. (3.33) with half-width 0.55 and steepness 4 in standardized moneyness, centered 1.4 standardized units past the last put-side quote. Its amplitude is the smallest entry of a fixed ladder (5, 10, 20, 35, 55 percent of the base variance at the center) that drives the minimum of g_{D} below -0.05 while total variance stays positive. Realized wing slopes are measured by finite differences of w at $|k| = 8$.

Part II

The Observation

Chapter 6

Forwards, Dividends, and Carry

Part I fitted smiles in normalized coordinates: every strike was measured against a forward F, every price against a discount D, and both were treated as known. Neither is quoted anywhere. This chapter earns them. Put–call parity makes the pair the two coefficients of one straight line through the option quotes themselves, so estimating (F, D) is a regression — and the chapter reads the whole problem as a statistician would. The line determines its level superbly and its slope poorly: on a stated test board the forward is pinned to a fraction of a basis point while the implied rate wanders by an order of magnitude more, and every safeguard the chapter builds is matched to that asymmetry — a trim for outliers, a prior band for runaway rates, and a re-derivation of the forward from the part of the data that stays trustworthy. Dividends then give the carry *— the growth rate linking spot to forward — its content, and the choice among dividend conventions turns out to be a risk decision, not a cosmetic one. The stakes are measured at the end: a forward error read back through the Black formula imprints a put–call gap at the money and a fake skew — on the frozen node, tens of volatility basis points for every ten basis points of forward error — and the borrow — the fee for shorting a stock, the carry's last component — is identifiable only where a closed-form noise floor says the board can resolve it.*

6.1 Two numbers under every smile

Part I built three families of smile models, and all three lived in the same coordinates. A strike K entered as its log-moneyness $k = \log(K/F)$; a dollar price entered divided by DF (section 2.1). Two numbers sat under every computation: the *forward* F, the price agreed today for delivery of the underlying at expiry, and the *discount factor* D, the value today of one dollar paid at expiry. Part I treated both as given. This chapter asks where they come from.

The honest answer is that, for a single stock or an ETF, neither is quoted. There is no screen showing "the forward of SPY for December 2026". There is a spot price, and there is the option chain itself: for each strike, a call and a put. The forward and the discount must be *estimated* from those option prices — which makes them fitted quantities, with error bars, exactly like the smile parameters they will later support. That change of status is the subject of the chapter, and it opens Part II's theme: before a model can be fitted, the observation itself has to be built, and every step of that construction is either measured from the data or chosen and disclosed.

The estimate rests on one identity. Fix an expiry and a strike K, and write $\mathsf{C}(K)$ and $\mathsf{P}(K)$ for the dollar mid prices of the call and the put there — sans-serif letters, here and for the rest of the book, mark a raw market quote, as opposed to the normalized model prices c and P of Chapter 2. Assume for now that both options are *European* — exercisable only at expiry, in the standard vocabulary; section 6.2 meets the American reality. Consider the portfolio that is long the call and short the put, and follow it to expiry. If the underlying finishes at S_T above K, the call pays $S_T - K$ and the put pays nothing; if it finishes below, the call pays nothing and the short put costs $K - S_T$. In both cases the portfolio delivers

$$(S_T - K)^+ - (K - S_T)^+ = S_T - K,$$

the payoff of a forward contract struck at K. What is that worth today? Price the two pieces separately. The constant $-K$ is easy: a sure payment of K at expiry is worth DK now, so the piece contributes $-DK$. For the S_T piece, use the definition of the forward price: a forward contract struck at F costs *nothing* to enter — that is what makes F the forward price — and it pays $S_T - F$. Prices of payoffs add, so the value of receiving S_T minus the value of receiving the constant F equals the value of receiving $S_T - F$, which is zero. Receiving S_T is therefore worth exactly what receiving F is worth: DF. The portfolio's price today must therefore be

$$\Pi(K) \;\equiv\; \mathsf{C}(K) - \mathsf{P}(K) \;=\; D\,(F - K). \tag{6.1}$$

This is *put–call parity* [17, 26]. Chapter 2 already carried its normalized form — the rightmost identity of eq. 2.2, $c(k) - P(k) = 1 - e^k$; multiplying it by DF and writing $K = Fe^k$ recovers (6.1) exactly. Note what the derivation did *not* use: no model of how the underlying moves, no volatility, no distribution. Parity is a validity statement in the sense of the book's thesis — any European call and put prices that violate it admit an arbitrage, whatever the model.

Now read (6.1) the other way. The right-hand side is a straight line in K, with slope $-D$ and root F. The left-hand side is observable: one number per strike, computed from quotes. So the chain itself draws a line, and the two numbers Part I assumed are that line's two coefficients. Estimating (F, D) means fitting a straight line — nothing more exotic than that.

Before any general machinery, solve the smallest possible case by hand. Suppose an expiry half a year away quotes two usable strikes, and the portfolio prices come out as

$$\Pi(90) = 11.27, \qquad \Pi(110) = -8.33$$

(constructed numbers, chosen to come out exactly; appendix 6.A). Two points determine the line. The slope is

$$\frac{\Pi(110) - \Pi(90)}{110 - 90} = \frac{-8.33 - 11.27}{20} = \frac{-19.60}{20} = -0.98,$$

so the discount is $D = 0.98$. The intercept — the line's value at $K = 0$ — is $\Pi(90) + D \cdot 90 = 11.27 + 88.20 = 99.47$, and by (6.1) the intercept equals DF, so

$$F = \frac{99.47}{0.98} = 101.50.$$

Finally, the discount converts to an interest rate. Writing t for the *calendar* year fraction to expiry ($t = 0.5$ here) and $D = e^{-rt}$,

$$r = -\frac{\log D}{t} = -\frac{\log 0.98}{0.5} \approx 4.04\%.$$

Two portfolio prices — four option quotes — have produced a forward, a discount, and an interest rate, out of prices alone. This is worth pausing on: the option market carries its own funding information, and the fitter never needs an external rate feed to normalize a smile.

One clock deserves a sentence before we go on. The book has used τ, variance time, since Chapter 2, and this chapter introduces t, the calendar year fraction. They are different objects with different jobs: variance accrues on the market's clock (Chapter 8 makes that precise), but discounting and *carry* — the growth rate that links spot to forward, given its full content in section 6.6 — accrue on the calendar: a dollar earns interest on weekends too. In everything below, $D = e^{-rt}$ and the carry exponents use t; total variance $w = \sigma^2\tau$ keeps τ. In this chapter's data the two clocks happen to agree numerically, so nothing subtle hides in the figures; the distinction starts to matter in Chapter 8.

We now know what the two numbers are and that the chain determines them. The next question is what happens with a real chain: many strikes, quoted with noise, where the line must be fitted rather than solved.

6.2 One straight line, drawn by a real chain

We know the chain draws a line. This section fits it on the book's data and meets the two facts that shape the whole chapter: the line is astonishingly straight, and its residuals are not noise.

With more than two strikes the line is overdetermined, and the natural fit is least squares: choose the slope and intercept that minimize the sum of squared residuals across strikes. Write the fitted intercept as $\hat{a}$ and the fitted slope as $\hat{b}$; then, reading (6.1) backwards as in the hand example,

$$\hat{D} = -\hat{b}, \qquad \hat{F} = \frac{\hat{a}}{\hat{D}}, \qquad \hat{r} = -\frac{\log \hat{D}}{t}. \tag{6.2}$$

Hats, here and from now on, mark estimates — numbers computed from quotes, as opposed to the true (F, D) the market is carrying. The whole estimator is four lines of arithmetic; there is nothing else inside it.

Figure 6.1 runs it on the running node — SPY December 2026, the same (underlier, expiry) pair every chapter has used — taking mid prices from the frozen snapshot's raw chain and keeping the 158 strikes where both the call and the put quote a live bid. Panel (a) shows the observable: $\Pi(K)$ against K, falling from +\$609 deep in the left wing to −\$242 at the top of the board, a range of \$850. Through that range the points hug the fitted line so tightly that the eye cannot separate them from it: the rms residual is \$2.97, which is 0.35% of the observable's range. The open circle marks the fitted root $\hat{F} = 762.71$: the strike at which a call and a put cost the same, which is by (6.1) the forward estimate. Parity, a theorem about prices that nobody enforces centrally, is visibly enforced by the market to 0.35% of the observable's own range.

Panel (b) is the same fit with the line subtracted, and it carries the section's real lesson. If the residuals were quote noise they would look like the cent-scale jitter of the hand example — a fuzz around zero with no shape. Instead they *bow*: a smooth arch, dollars deep at both wings, cresting at +\$4.8 near the money — hundreds of times the *tick*, the one-cent minimum increment a quote can move by — with every point sitting on the arch rather than scattered around it. A systematic shape in residuals means a systematic error in the model of the observation — and the model here was equation (6.1), which prices *European* options. SPY's listed options are American: each one carries the extra right to exercise early, that right has value, and the value is not the same for the call and the put at a given strike. The mids on the line are therefore each shifted by a smooth, strike-dependent premium the European identity knows nothing about. Chapter 7 is devoted to measuring and removing that premium; this chapter only needs to know it is there.

The bow has a consequence the figure also shows. Because the contamination is asymmetric — deep in-the-money puts carry much more early-exercise value than calls do — it does not average away; it tilts the fitted line and drags the root. The reference implementation — the working software whose conventions the book's figures audit — deals with this by de-Americanizing the quotes before regressing (Chapter 7's machinery); its resolved forward for this node, stored with the snapshot, is $F = 767.92$, sitting 68 basis points above the naive root $\hat{F} = 762.71$. (A *basis point*, bp, is 10^{-4} throughout: of the forward when we compare forwards, of the rate when we compare rates — that one is absolute — and later a *vol* bp, 10^{-4} of implied volatility.) Both vertical lines are drawn in panel (b); at the scale of the strike axis they almost coincide, and that is worth noticing too — 68 bp is \$5.2 on a spot of \$757.67, invisible in price space. Section 6.7 will show the same five dollars is anything but invisible in volatility space. Until then the chapter works on European boards, where (6.1) is exact, and returns to American data only when the estimation theory is in hand.

For that theory we need a controlled instrument: a board whose true (F, D) we *know*, so that estimator errors can be measured rather than argued about. The chapter's *running board* is synthetic and European: spot $S = 100$, rate $r = 4\%$, a 1% continuous dividend yield (dividends enter properly in section 6.6), half a year to expiry, and 25 strikes from 70 to 130 every 2.5, each

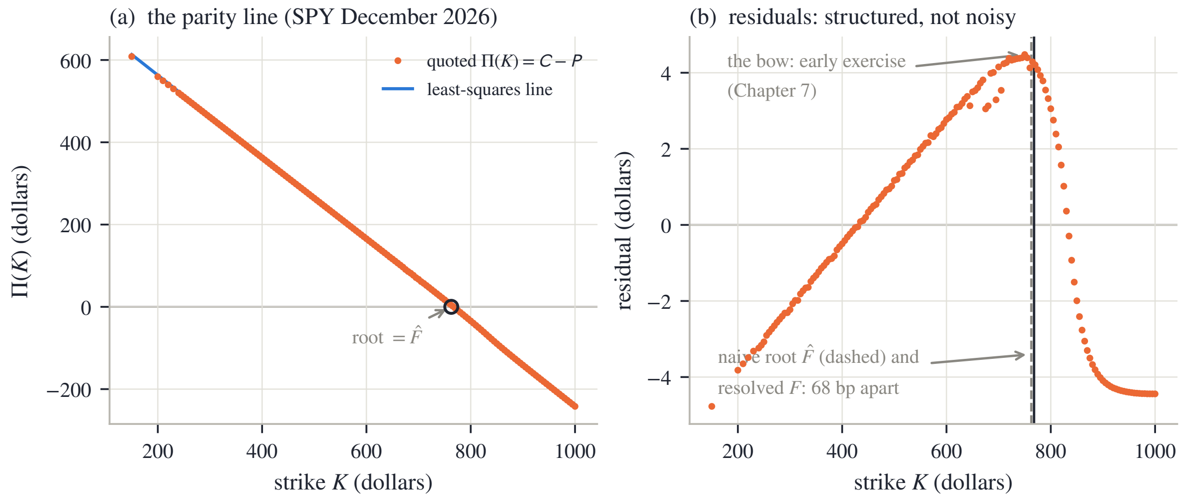

Figure 6.1: The parity line on the running node (SPY December 2026, 158 paired strikes from the frozen chain). (a) $\Pi(K) = \mathsf{C} - \mathsf{P}$ against strike: a straight line to 0.35% of its \$850 range, with the fitted root $\hat{F}$ circled. (b) The residuals bow smoothly — dollars, not cents, and all of one shape: the early-exercise premium of American quotes, the subject of Chapter 7. The naive root and the reference implementation's resolved forward differ by 68 bp.

mid priced by the Black formula at a flat 22% volatility and rounded to the cent (all constants in appendix 6.A). Its true forward is $F = Se^{(r-q_d)t} = 101.51$, where q_d is the dividend yield — take the yield-in-the-exponent form on faith for two more sections; section 6.6 derives it. Its true discount is $D = e^{-rt} = 0.9802$. On this board the least-squares fit recovers the forward to 0.15 bp and the rate to 1.0 bp — the cent rounding is the only noise. The interesting question is not whether it recovers them but *how unequally well* it recovers the two, and that question gets its own section.

We now have the estimator (6.2), a real chain where the line holds to 0.35% of its range, and a clean board with known truth. Next: the error bars.

6.3 What the line identifies well, and what poorly

The estimator (6.2) recovers a clean board's truth. This section asks the sharper question: with quote noise of a stated size, how precise is each of the two fitted numbers? The answer is lopsided — and the lopsidedness explains every safeguard built later in the chapter, so we derive it slowly and then verify it by simulation.

First the noise model. Let each quoted mid — each $\mathsf{C}(K_i)$ and each $\mathsf{P}(K_i)$ — carry an independent error of standard deviation ε, a couple of cents on a liquid board. The observable is the difference of two mids, and independent errors add in variance, so each Π_i carries error of standard deviation $\sqrt{\varepsilon^2 + \varepsilon^2} = \varepsilon\sqrt{2}$. Write n for the number of paired strikes, $\bar{K} = \frac{1}{n}\sum_i K_i$ for their mean, and

$$S_{KK} = \sum_{i=1}^{n} (K_i - \bar{K})^2$$

for the *strike dispersion* — the sum of squared distances of the strikes from their center, the number that measures how wide a lever the board gives the slope.

Proposition 6.1 (error bars of the parity line)**.** *Fit $\Pi_i = a + bK_i$ by least squares under the noise model above. Then the fitted slope and the fitted* level at the center, $\widehat{\Pi(\bar{K})}$, *are uncorrelated,* *with standard deviations*

$$\operatorname{sd}(\hat{D}) = \frac{\varepsilon\sqrt{2}}{\sqrt{S_{KK}}}, \qquad \operatorname{sd}(\widehat{\Pi(\bar{K})}) = \frac{\varepsilon\sqrt{2}}{\sqrt{n}}. \tag{6.3}$$

The induced first-order errors on the market parameters are

$$\delta\hat{r} = -\frac{\delta\hat{D}}{D\,t}, \qquad \delta\hat{F} = \underbrace{\frac{\delta\widehat{\Pi(\bar{K})}}{D}}_{\textit{level noise}} - \underbrace{(F - \bar{K})\,\frac{\delta\hat{D}}{D}}_{\textit{slope noise}\times\textit{lever}}. \tag{6.4}$$

Proof. Center the regressor: put $x_i = K_i - \bar{K}$, so that $\sum_i x_i = 0$ and the model reads $\Pi_i = \alpha + b x_i$ with $\alpha = a + b\bar{K}$ — the line's value at the center, so $\hat{\alpha}$ is exactly the $\widehat{\Pi(\bar{K})}$ of the statement. Least squares on orthogonal columns separates: minimizing $\sum_i(\Pi_i - \alpha - b x_i)^2$ over (α, b) gives

$$\hat{\alpha} = \frac{1}{n}\sum_i \Pi_i, \qquad \hat{b} = \frac{\sum_i x_i \Pi_i}{\sum_i x_i^2} = \frac{\sum_i x_i \Pi_i}{S_{KK}}.$$

Each is a fixed linear combination of the noisy Π_i, so their variances follow from the noise model directly:

$$\operatorname{var}(\hat{\alpha}) = \frac{2\varepsilon^2}{n}, \qquad \operatorname{var}(\hat{b}) = \frac{\sum_i x_i^2\,(2\varepsilon^2)}{S_{KK}^2} = \frac{2\varepsilon^2}{S_{KK}},$$

and their covariance is $\sum_i \frac{1}{n}\cdot\frac{x_i}{S_{KK}}\,(2\varepsilon^2) = 0$ because $\sum_i x_i = 0$. Since $\hat{D} = -\hat{b}$, the first half of (6.3) follows. For (6.4): the rate map is $\hat{r} = -\log\hat{D}/t$, and differentiating gives $\delta\hat{r} = -\delta\hat{D}/(Dt)$. The root satisfies $\hat{F} = \bar{K} + \hat{\alpha}/\hat{D}$ exactly — the fitted line passes through $(\bar{K}, \hat{\alpha})$ and falls with slope $-\hat{D}$, so it crosses zero $\hat{\alpha}/\hat{D}$ to the right of $\bar{K}$. Differentiate in the two independent errors, using $\alpha = D(F - \bar{K})$ for the true values:

$$\delta\hat{F} = \frac{\delta\hat{\alpha}}{D} - \frac{\alpha}{D^2}\,\delta\hat{D} = \frac{\delta\hat{\alpha}}{D} - (F - \bar{K})\,\frac{\delta\hat{D}}{D}. \qquad \square$$

Now put the running board's numbers into (6.3)–(6.4), because the asymmetry *is* the story. The board has $n = 25$ strikes and $\sqrt{S_{KK}} \approx 90$ strike-dollars, against $\sqrt{n} = 5$. With ε of 2 cents:

$$\operatorname{sd}(\hat{D}) = \frac{0.0283}{90} \approx 3.1\times10^{-4}, \qquad \operatorname{sd}(\hat{r}) = \frac{3.1\times10^{-4}}{0.9802\times0.5} \approx 6.4\times10^{-4},$$

more than six basis points of rate scatter from two cents of quote noise. The forward, by contrast, is dominated by the level term:

$$\operatorname{sd}(\hat{F}) \approx \frac{\varepsilon\sqrt{2}}{\sqrt{n}\,D} = \frac{0.0283}{5\times0.9802} \approx 0.6 \text{ cents},$$

about half a basis point of a \$100 forward — and the lever term adds almost nothing, because on this board $F - \bar{K} = \$1.51$: the slope's error reaches the forward only through that short lever. So the same two cents of noise leave the forward 11 times better determined than the rate — rate basis points absolute, forward basis points relative, each parameter judged against its own scale, as the section 6.2 convention set out. Why is the slope the delicate direction? Compare the two errors as *fractions of the numbers they perturb.* The slope's absolute error is tiny, but the slope itself is a discount near one, so relatively it is $3.1\times10^{-4}/0.98 \approx 3.2$ bp; the level's error sits on a level of $DF \approx \$100$, so relatively it is $0.006/100 \approx 0.6$ bp. The design already favors the level by a factor $F\sqrt{n}/\sqrt{S_{KK}} \approx 5.6$. Then the maps out of the line finish the job in opposite directions: the level passes to the forward almost unchanged, while the slope's error is divided by $D\,t$ on its way to the rate — doubling the gap at $t = 0.5$, and worse the shorter the expiry, since $\delta\hat{r}$ carries $1/t$.

Figure 6.2 is the audit. It runs 2000 noisy copies of the running board — fresh two-cent noise on every mid, fitted by (6.2) each time — and histograms the two errors. In panel (a), the implied rate: the measured scatter is 6.3 bp against the predicted 6.4 bp, the dashed lines marking the prediction of (6.3). In panel (b), the forward: measured 0.57 bp of F against predicted 0.57 bp. The formulas are not asymptotic decoration; they are the histograms' widths. And the $1/t$ amplification is real: rerunning the same experiment with the expiry moved from half a year to $t = 0.05$ — less than three weeks — leaves the forward essentially unchanged but blows the rate scatter up to 63 bp. A short-dated parity slope is close to meaningless as a rate estimate, and nothing about that is fixable by fitting harder: it is the design of the data, not the algorithm.

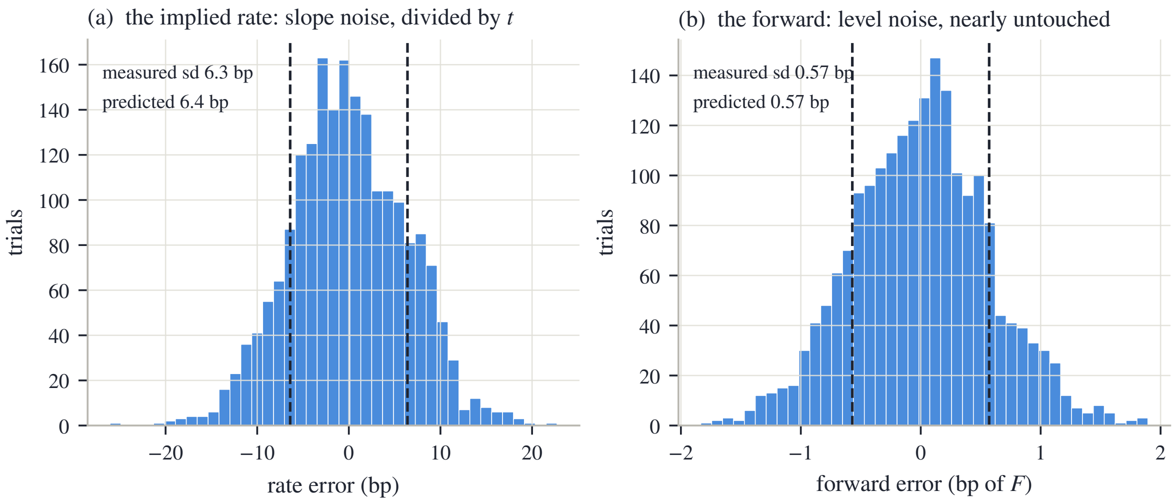

Figure 6.2: The identifiability asymmetry, measured (2000 trials of 2-cent noise on the running board, prediction of proposition 6.1 in dashed lines). (a) The implied rate scatters by 6.3 bp (predicted 6.4 bp). (b) The forward scatters by 0.57 bp (predicted 0.57 bp). Same line, same noise: the level is measured by every quote, the slope is teased out of small differences and then divided by t.

This section's fact is worth restating in one plain sentence, because the next two sections are both consequences of it: *the parity line's level — and with it the forward — is nearly noiseless, while its slope — and with it the discount and the rate — is the worst-determined object in the problem.* A defense built for real chains should therefore spend its suspicion on the slope and its trust on the level. The next section starts with the failures that hit both: quotes that are not merely noisy but wrong.

6.4 Estimating on a dirty chain

The noise model of section 6.3 was polite: every quote wrong by a little, none wrong by a lot. Real chains are not polite. A quote can be *stale* — last updated minutes ago, while the market moved — or *crossed* — its bid above its ask, a pair no live market would let stand — or simply wrong, and one such point is not two cents off the line but a dollar off. This section adds the classical defense, watches it work, and then — more important — finds its blind spot.

The defense is an *iterated trim*, and it is standard robust statistics rather than anything of ours [25]. Fit the line. Compute the residuals. Estimate their typical size robustly — take the median of the absolute residuals and multiply by 1.4826, the constant that makes this estimator agree with the standard deviation on clean Gaussian data, so one huge outlier cannot inflate the scale used to judge it; floor the estimate at one basis point of spot so a suspiciously perfect board does not set an impossibly strict bar. Call the result $\hat{\sigma}$. Drop every pair whose residual exceeds $4\hat{\sigma}$, refit, and repeat — at most three rounds, and never below three surviving pairs, since two points can always draw a perfect line. On a clean board the loop finds nothing, drops nothing, and returns the plain fit bit for bit; the defense costs nothing when it is not needed.

Figure 6.3 stages the intended failure on the running board. One deep put, at strike 75, is stale: its quoted mid is too high by \$1.20, so $\Pi(75)$ sits \$1.20 below the line. In panel (a) the corrupted point is ringed; the fitted line shown is the trimmed one, and the eye confirms there was never any doubt about where the line should go. Panel (b) shows the residuals against that trimmed fit, with the $\pm 4\hat{\sigma}$ band shaded: every clean point lives inside a band a few cents tall, and the stale put stands roughly 120 robust sigmas outside it. It is dropped in the first round (1 point trimmed), and the refit recovers the true forward to 0.1 bp.

The raw fit's damage, before the trim, is itself a lesson in section 6.3's asymmetry. The untrimmed line's forward is off by only 4 bp — one gross point among 25 barely moves the level — but its implied rate comes out at 4.8% against a true 4%. Even vandalism obeys the proposition: the damage concentrates in the slope.

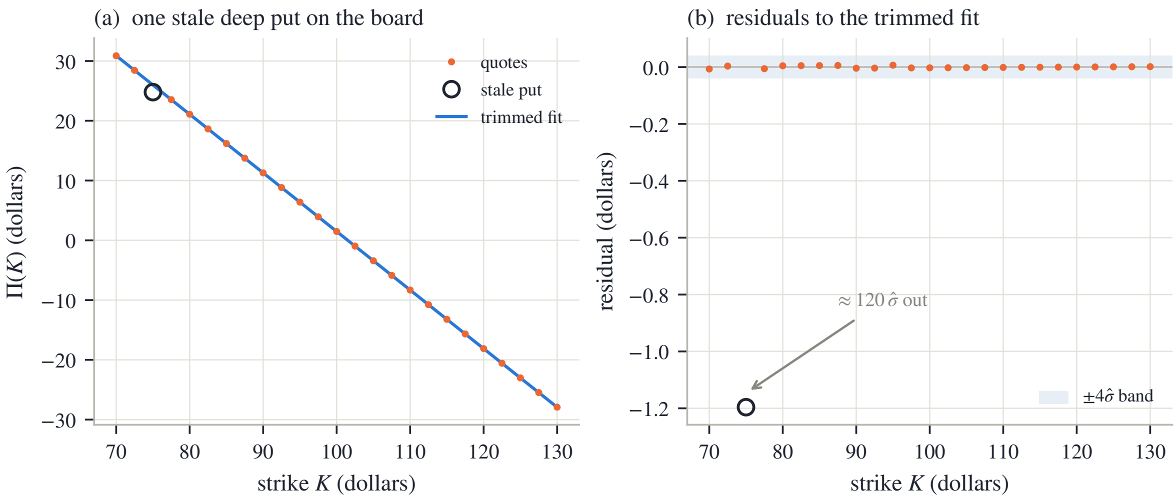

Figure 6.3: The trim doing its job (running board, one staged stale put). (a) The corrupted point (ringed) sits visibly off the line; the trimmed fit ignores it. (b) Residuals against the trimmed fit: the clean points live in the shaded $\pm 4\hat{\sigma}$ band, cents tall; the stale quote stands $\approx 120\,\hat{\sigma}$ out and is dropped in one round. The refit is exact to 0.1 bp.

Remark 6.2 (what the trim cannot see: coherent staleness). A residual trim detects points that disagree with the consensus line. Stale points that agree *with each other* are a different animal. Rerun the experiment with a coherent failure: instead of one random error, let the 5 lowest strikes — the deep put wing, exactly the quotes that update least often — all go stale *together*, their prices recomputed at a financing rate 2% away from the truth, as if that whole wing last traded under yesterday's funding. Each stale quote is now consistent with its stale neighbors, the blended least-squares line splits the difference between the two regimes, and the residuals spread out enough that *nothing* exceeds $4\hat{\sigma}$: the trim flags 0 points. This is the *masking* phenomenon of robust statistics [25] — contamination that moves the consensus toward itself hides from any test built on distance to the consensus. The fitted rate lands at 1.7% against a true 4%: badly wrong, yet entirely plausible — no test on the rate's *value* would blink at it. What survives is, once again, the level: the forward moves only 13 bp, because 5 stale quotes on one wing tilt the line without lifting its center much. Coherent staleness looks, from inside the chain, exactly like a change in the financing rate; it cannot be detected there, only bounded — and the bound is proposition 6.1's asymmetry itself.

Where we stand: the trim removes quotes that betray themselves; coherent staleness does not betray itself, corrupts the slope, and spares the level. So the slope needs protection that does not come from the chain at all: prior knowledge of where a rate can plausibly live. The next section builds that defense — and is honest about its reach. It stops slopes that leave the plausible range; the masking experiment's slope, plausible by construction, is beyond any within-chain repair, and stays bounded rather than fixed.

6.5 A prior on the rate, and the lever-arm identity

Here is where the chapter stands. The slope is the fragile parameter (section 6.3); some failures corrupt it without tripping any alarm inside the chain (remark 6.2). On a slow or stale data feed the corruption can go further than the staged examples: the fitted discount drifts above one, which means a *negative* implied rate on an ordinary US equity chain. The question this section answers is what to do when the data's least reliable parameter contradicts something we know from outside the data.

The answer is a prior — prior knowledge, imposed as a hard constraint. A financing rate for a US equity has a physical range: not below zero by percents, not above by tens. Fix a band of plausible rates (its endpoints are a disclosed convention, reviewed like any other; the reference implementation ships a wide one) and apply one rule: *if the fitted rate $\hat{r}$ falls inside the band, keep the regression's answer untouched; if it falls outside, stop trusting the slope* — set the discount to the band's edge, the nearest value prior knowledge allows. Clean chains sit strictly inside the band and pass bit for bit, like the trim before: the device only exists on demonstrably bad inputs.

But rejecting the slope creates an obligation. The forward estimate $\hat{F} = \hat{a}/\hat{D}$ has the rejected slope in its denominator; if $\hat{D}$ was untrustworthy, so is $\hat{F}$ computed from it. The forward must be re-derived from what is still trustworthy — the level — and *how* to do that well is a small piece of mathematics worth isolating, because it explains a design choice that looks arbitrary otherwise.

Solve (6.1) for F one strike at a time: on a noiseless board, $F = K_i + \Pi_i/D$ for every i. So given any discount D' — possibly the clamped one, possibly wrong — a whole family of *level estimators* is available: pick weights $\mu_i \geq 0$ summing to one (μ_i in its Chapter 3 role, a weight per quote) and average,

$$\hat{F}_\mu(D') = \sum_i \mu_i \Big(K_i + \frac{\Pi_i}{D'} \Big).$$

The question is how much a wrong D' hurts, and the answer is exact and short.

Proposition 6.3 (the lever arm of a level estimator)**.** *On a noiseless board obeying* (6.1),

$$\hat{F}_\mu(D') - F = \Big(1 - \frac{D}{D'}\Big)(\bar{K}_\mu - F), \qquad \bar{K}_\mu = \sum_i \mu_i K_i. \tag{6.5}$$

Proof. Substitute the truth $\Pi_i = D(F - K_i)$ into one term of the average:

$$K_i + \frac{\Pi_i}{D'} = K_i + \frac{D}{D'}(F - K_i) = K_i\Big(1 - \frac{D}{D'}\Big) + \frac{D}{D'}\,F.$$

Averaging with the weights μ_i replaces K_i by $\bar{K}_\mu$:

$$\hat{F}_\mu(D') = \bar{K}_\mu\Big(1 - \frac{D}{D'}\Big) + \frac{D}{D'}\,F = F + \Big(1 - \frac{D}{D'}\Big)(\bar{K}_\mu - F). \qquad \square$$

Read (6.5) in words: the transmitted error is the relative discount error times $\bar{K}_\mu - F$, the distance from the estimator's center of weight to the forward. That distance is a *lever arm.*

Where an estimator centers its strikes decides how much a wrong discount can hurt it — and the modeler chooses the center.

Figure 6.4 measures the identity on a deliberately asymmetric board (strikes from 85 to 140, so that averages are pulled away from the money; constants in appendix 6.A), imposing rate errors of up to two percent and re-deriving the forward three ways. In panel (a), the three curves. The steep one is the regression reading $\hat{a}/D'$. It belongs to the family too, in a degenerate way: the intercept $\hat{a}$ is the fitted line's value extrapolated to strike *zero* — a strike no board quotes — so reading $\hat{a}/D'$ amounts to placing all the weight at that phantom point, $\bar{K}_\mu = 0$. Its lever arm is then F itself, the worst available, and it transmits 50.0 bp of forward error per percent of rate error. The middle curve is the uniform average over the board: its lever is $\bar{K} - F$ — mean strike \$112.5 against forward \$101.5, a \$11.0 arm — transmitting 5.4 bp per percent: better by an order of magnitude, but only by the accident of where the board happens to center. The flat curve is the estimator the reference implementation uses when its band bites: weights concentrated *at the money* by a Gaussian kernel in log-moneyness (centered at spot — the one anchor that needs no estimate), whose center of weight lands at \$102.1: a lever of \$0.6, transmitting 0.31 bp per percent. The open circles confirm the middle curve against (6.5) evaluated directly — identity, not approximation. Panel (b) simply draws the three lever arms on a log scale: \$101.5 versus \$11.0 versus \$0.6. The design lesson is compact: *the clamp can afford to be crude about the rate because the level estimator it hands the problem to barely feels the rate at all.* A last-resort guard remains behind it — if even the level answer lands absurdly far from spot, the estimate falls back to spot itself and says so — and on such a chain no smile should be published at all.

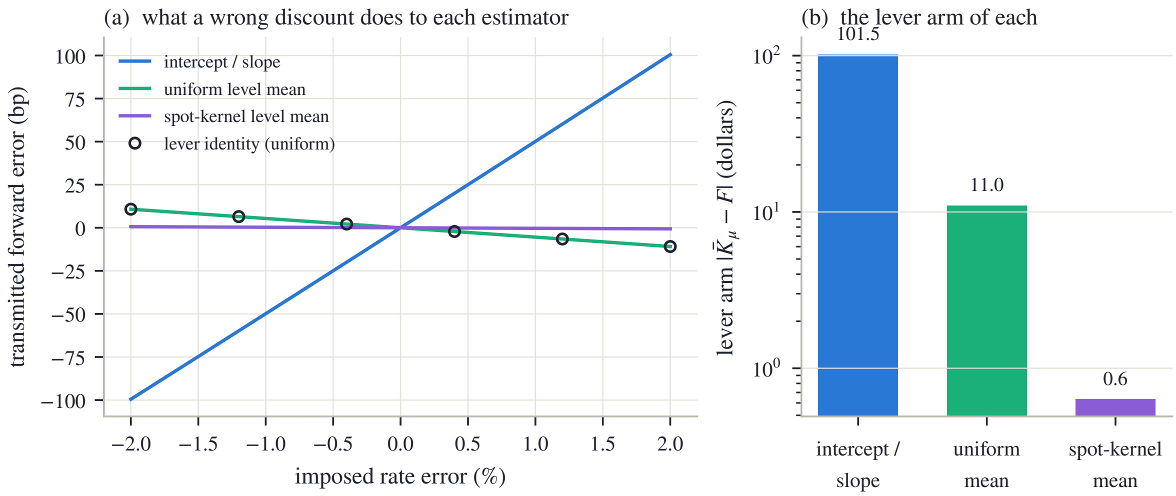

Figure 6.4: The lever-arm identity (6.5) on the asymmetric board. (a) Forward error transmitted by an imposed rate error, for the three level estimators; circles are the identity evaluated directly. (b) Their lever arms $|\bar{K}_\mu - F|$: the regression reading levers on F itself (50.0 bp per percent), the uniform mean on the board's accidental centering (5.4), the spot-centered kernel on under a dollar (0.31).

One honest caveat closes the section. The band is a prior, and a prior can be wrong: a regime of genuinely negative dollar rates would sit outside it, and the device would then clamp away information rather than noise. That is the standing trade of this chapter — stated once here, collected in the contract table of section 6.8. What we now have is a defense in two layers, each matched to a failure mode and each inert on clean data: the trim for quotes that betray themselves, the band plus the short-lever re-derivation for slopes that go absurd. Between them sits the failure neither catches — coherent, plausible staleness — bounded by the level's robustness and nothing else.

6.6 Dividends: what the carry contains

So far the pair (F, D) has been estimated without asking what makes F differ from spot in the first place. This section opens that box, because two later needs depend on it: the borrow of section 6.8 is defined as what remains of the carry after dividends are accounted for, and Chapter 7's early-exercise machinery runs on a dividend model as its fuel.

Call *carry* the growth rate the market charges between spot and forward: the number $\log(F/S)/t$. For a stock that pays nothing, carry is just the rate — $F = Se^{rt}$, because the alternative to agreeing on delivery is to buy the stock today with borrowed money, and the borrowing costs r. Dividends change the arithmetic: whoever holds the stock until expiry *collects* them, and the forward buyer does not, so their value is subtracted from what delivery is worth. How exactly it is subtracted depends on how the dividends are modeled, and there are three standard conventions.

Discrete cash. The stock will pay fixed dollar amounts d_i at *ex-dates* $t_i \le t$ — the dates from which a buyer of the stock no longer receives the next dividend, so the price drops by roughly the payment as each arrives. (Both symbols are always subscripted; the upright d of an integral is unrelated.) Replicate delivery: buy the stock today for S, borrow the money, hold to expiry. Along the way each dividend d_i arrives and can be lent out from its ex-date onward, growing to $d_i e^{r(t-t_i)}$ by expiry. The all-in cost of delivering the stock at expiry is therefore the financed purchase minus the collected, reinvested dividends:

$$F = Se^{rt} - \sum_i d_i\, e^{r(t-t_i)} = \Big(S - \sum_i d_i\, e^{-rt_i}\Big)e^{rt}. \tag{6.6}$$

The second form is the useful one: spot minus the *present value* of the dividend schedule, carried forward at the rate. This is called the escrowed-dividend forward — as if the dividends were set aside in escrow at time zero.

Discrete proportional. Each ex-date the price drops not by a fixed dollar amount but by a fixed *fraction* f_i of whatever the stock is worth that day. The replication changes in one place: reinvest each dividend into the stock itself rather than lending it. Follow one ex-date to see the bookkeeping. Just before it, a share is worth some price S^-; the holder receives $f_i S^-$ in cash, and the price drops to $(1 - f_i)S^-$. Spending the cash on shares at the dropped price buys

$$\frac{f_i S^-}{(1-f_i)S^-} = \frac{f_i}{1-f_i} \quad \text{new shares per share held,}$$

multiplying the holding by $1 + \frac{f_i}{1-f_i} = \frac{1}{1-f_i}$. Each ex-date therefore cancels one factor $(1 - f_i)$: start with $\prod_i(1 - f_i)$ of a share today and exactly one share exists at expiry. The cost today is $S\prod_i(1 - f_i)$, and

$$F = S \prod_i (1 - f_i)\, e^{rt}. \tag{6.7}$$

Continuous yield. Smear the proportional drops into a constant leak q_d (subscripted — bare q is Chapter 2's quantile density): the limit of (6.7) with many small events is

$$F = S\, e^{(r-q_d)\, t}. \tag{6.8}$$

This is the cheapest convention, and the one the running board has been using since section 6.2.

Which convention should a modeler pick? At first sight it barely matters. Figure 6.5(a) shows the chapter's running schedule — \$0.65 per quarter on a \$100 stock, first ex-date 30 days out — through the lens of the *implied dividend yield*: the constant q_d that would produce the same forward at each maturity, $q_d(T) = r - \log(F(T)/S)/T$. The cash schedule draws a sawtooth. Each vertical jump is one ex-date entering the horizon: just after the first one enters, a single \$0.65 payment thirty days away annualizes to 7.9% — violent at the short end, where dividing by a small T amplifies everything (the same $1/t$ the rate suffered in section 6.3). By a year the teeth have shrunk toward the flat equivalent yield of 2.59% drawn dashed, and the two conventions price long-dated forwards almost alike. If the term structure of F were the whole story, the choice would be cosmetic.

It is not, and panel (b) shows the honest discriminator: what happens to the forward when *spot moves*. Differentiate each convention in S at a fixed schedule. For the yield and proportional forms, (6.8) and (6.7), F is proportional to S, so

$$\frac{\partial \log F}{\partial \log S} = 1 :$$

the forward rides spot one for one. For the cash form, write $\mathrm{PV} = \sum_i d_i e^{-rt_i}$ for the schedule's present value; then $F = (S - \mathrm{PV})\, e^{rt}$ with PV *fixed in dollars*, and

$$\frac{\partial \log F}{\partial \log S} = S\, \frac{\partial}{\partial S} \log\left(S - \mathrm{PV}\right) = \frac{S}{S - \mathrm{PV}} \;>\; 1. \tag{6.9}$$

A cash schedule does not scale with the stock, so on a rally the forward outruns spot: the dividends claim a fixed slice, and the slice matters relatively less as the pie grows. Panel (b) plots this *spot elasticity* against maturity. The cash staircase climbs — each step one more ex-date inside the horizon — reaching $S/(S - \mathrm{PV}) = 1.053$ at two years, where the escrowed value is $\mathrm{PV} = \$5.01$; the dotted line is the closed form (6.9) and the solid one a direct bump measurement, in agreement. The proportional and yield conventions sit at exactly one, at every maturity.

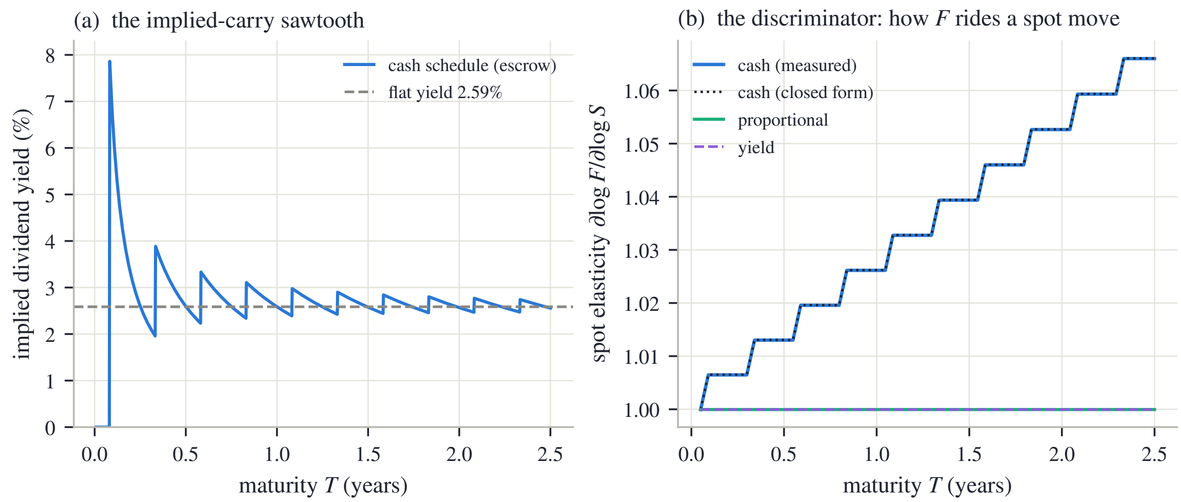

Figure 6.5: Dividend conventions on the running schedule (\$0.65 quarterly, first ex-date 30 days out). (a) Implied dividend yield of the cash schedule against maturity: a sawtooth, each tooth one ex-date entering the horizon, spiking to 7.9% at the short end; dashed, the flat equivalent yield 2.59%. (b) The spot elasticity (6.9): cash climbs to 1.053 by two years (dotted: closed form; solid: measured by bumping spot); proportional and yield sit at exactly one.

Why does an elasticity deserve this much attention in a chapter about statics? Because the smile is drawn against F. When spot moves and the surface is transported to the new spot — the scenario logic Chapter 9 develops — every strike's log-moneyness $k = \log(K/F)$ moves by *minus* however much $\log F$ moves, and (6.9) says the dividend convention decides that amount. Two modelers with identical fits today, one carrying cash dividends and one a yield, will disagree about which strike is at the money *after* a two-percent rally — a risk disagreement, not a pricing one. The dividend model is thus a convention in the full sense this book gives the word: not inferable from today's chain (panel (a) showed the term structures nearly agree), consequential for what the desk does next (panel (b)), and therefore to be stated, not assumed silently.

The carry now has two named parts — rate and dividends — and the chapter can ask its last estimation question: whether what is left over, the borrow, can be measured at all. One tool is still missing: how a carry error shows up *in the smile.* That tool is the next section, and it doubles as the payoff of the whole chapter.

6.7 What a wrong forward does to the smile

The chapter opened by claiming that a small forward error is a large smile error. Every tool needed to prove that claim is now on the table, and the proof is a short computation with a memorable shape.

Set the scene precisely. The market's quotes are what they are; the error lives in the modeler's head. The true forward is F, but the smile is built at a misread forward $F' = Fe^{\delta}$ — δ is the relative error, a few basis points. Every dollar quote is then pushed through the Black formula at the wrong normalization: the strike's log-moneyness is computed as $k' = \log(K/F') = k - \delta$, and the price is divided by DF' instead of DF. The discount is read correctly throughout — the forward error is the subject. The question: how far does the re-implied volatility move, per unit of δ?

Take a call first. Its dollar price, held fixed, is $DF\,B(k,w)$ with the normalized Black call of Chapter 2 (eq. 2.3),

$$B(k,w) = \Phi(d_+) - e^k\Phi(d_-), \qquad d_\pm = -\frac{k}{\sqrt{w}} \pm \frac{\sqrt{w}}{2},$$

where Φ and φ are, as throughout the book, the standard normal distribution function and density. The call's two partial derivatives were computed once in Chapter 2's appendix, and this chapter reuses them:

$$\partial_k B = -e^k\,\Phi(d_-), \qquad \partial_w B = \frac{\varphi(d_+)}{2\sqrt{w}}.$$

Differentiate the fixed dollar price along the misreading. Three things move at once: the prefactor F (its log by $\mathrm{d}\delta$), the moneyness k (by $\mathrm{d}k = -\mathrm{d}\delta$), and the implied total variance w (by whatever it must, to keep the price fixed). To first order,

$$0 = \frac{\mathrm{d}\big[F\,B(k,w)\big]}{F} = B\,\mathrm{d}\delta + \partial_k B\,\mathrm{d}k + \partial_w B\,\mathrm{d}w = \big(B - \partial_k B\big)\,\mathrm{d}\delta + \partial_w B\,\mathrm{d}w,$$

so

$$\mathrm{d}w = -\frac{B - \partial_k B}{\partial_w B}\,\mathrm{d}\delta.$$

The numerator collapses — this is the pleasant step:

$$B - \partial_k B = \Phi(d_+) - e^k\Phi(d_-) + e^k\Phi(d_-) = \Phi(d_+),$$

and inserting $\partial_w B$ — the price's sensitivity to total variance, the object desks call the (Black) *vega* when quoted per unit of volatility —

$$\mathrm{d}w = -\frac{2\sqrt{w}\,\Phi(d_+)}{\varphi(d_+)}\,\mathrm{d}\delta.$$

Convert to volatility: $\sigma = \sqrt{w/\tau}$, so $\mathrm{d}\sigma = \mathrm{d}w/(2\sqrt{w\tau})$, and the $\sqrt{w}$ cancels:

$$\left.\frac{\mathrm{d}\sigma}{\mathrm{d}\delta}\right|_{\text{call}} = -\frac{\Phi(d_+)}{\varphi(d_+)\sqrt{\tau}}, \qquad \left.\frac{\mathrm{d}\sigma}{\mathrm{d}\delta}\right|_{\text{put}} = +\frac{\Phi(-d_+)}{\varphi(d_+)\sqrt{\tau}}. \tag{6.10}$$

The put case is the same computation run on the normalized put $P = B + e^k - 1$ (parity again): its numerator is $P - \partial_k P = B - \partial_k B - 1 = \Phi(d_+) - 1 = -\Phi(-d_+)$, which flips the sign and swaps $\Phi(d_+)$ for $\Phi(-d_+)$.

Before the figure, read (6.10) at the money, where $k = 0$ and $d_+ = \sqrt{w}/2$ is small: both $\Phi(d_+)$ and $\Phi(-d_+)$ are close to $\frac{1}{2}$, and $\varphi(d_+) \approx \varphi(0) = 0.399$. A forward misread *up* by δ therefore moves the two sides in opposite directions — call vols down, put vols up, each by about $\frac{1}{2}\delta/(0.4\sqrt{\tau})$. Now recall how a smile is assembled from quotes: at each strike the market's liquid side is the *out-of-the-money* (OTM) option — the put below the money, the call above — so the drawn curve switches sides at $k = 0$, and the opposite moves become a *jump* there, of size

$$\text{ATM gap} \approx \frac{\delta}{\varphi(0)\sqrt{\tau}} \approx \frac{2.5\,\delta}{\sqrt{\tau}}. \tag{6.11}$$

The intuition for the two directions is worth one sentence each. Thinking the forward is higher makes the call look closer to the money than it is, and a closer-to-the-money option carrying the same price must carry *less* volatility; the same misreading pushes the put deeper out of the money, and a deeper option at an unchanged price must carry *more*.

Figure 6.6 draws both halves of the story on the running node's fitted smile — the frozen SPY December 2026 fit of the earlier chapters, δ set to 10 bp. Panel (a) is the closed form (6.10) across the quoted span, scaled to vol basis points per 10 bp of forward error: the put side (positive curve) and call side (negative curve) approach the money from opposite signs, and both fade toward the wings — far from the money an option's price responds weakly to the forward relative to how strongly it responds to volatility, so the damage concentrates exactly where the book is most liquid. Panel (b) is the measurement: every out-of-the-money quote on the node re-implied at the bumped forward, minus its true volatility. The orange curve is the measurement; it lands on the dotted linearization so exactly that the two read as one curve — and it shows the two symptoms a desk would actually see. At the money, the advertised jump: 41 vol basis points of put–call gap for 10 basis points of forward error — (6.11) with the node's numbers. Away from the money, a tilt: the put wing lifted by 11 bp at $k = -0.10$, the call wing dropped by 9 bp at $k = +0.10$. Lifted left wing, lowered right wing — that is precisely what a steepening of the skew looks like. A desk that trusts its forward would read this picture as the market repricing crash risk; the correct reading is a data error \$5.2 wide.

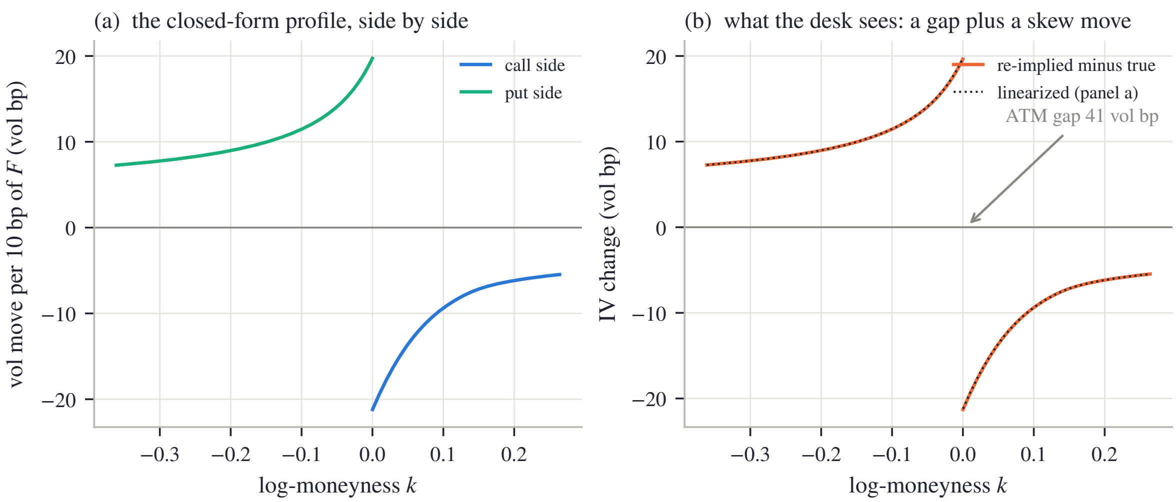

Figure 6.6: A forward error of 10 bp on the running node's smile. (a) The closed form (6.10): put side up, call side down, both largest near the money. (b) Every OTM quote re-implied at the bumped forward (orange; dotted: the linearization): an ATM put–call gap of 41 vol bp plus a wing tilt that reads exactly as a skew move.

Now close the loop the chapter opened in section 6.2. There, the naive whole-chain regression on the real (American) node missed the resolved forward by 68 bp — five dollars, invisible in price space. Push that error through (6.11): it imprints an ATM put–call gap of about 278 vol basis points — nearly three volatility *points* of artificial structure at the most liquid point of the surface, dwarfing the few-basis-point fit errors Part I worked to achieve. This is why the forward cannot be an afterthought: every error in it is repaid, amplified, at the money. And it is why the early-exercise bow of fig. 6.1(b) must be removed rather than tolerated — the task Chapter 7 takes up.

One tool from this section travels forward. Equation (6.10) converts *any* carry-side error into smile units: a rate error, a dividend error, or — next section — a borrow error, each reaching the smile through the forward it misplaces. The chapter's last question is whether that final component of carry can be measured well enough to be worth carrying at all.

6.8 The residual rate: borrow, at the edge of identifiability

The carry now has two named parts, rate and dividends. On most names, that is the whole story. On some it is not, and the leftover has a name and a market of its own.

To sell a stock short, one must first *borrow* the shares, and the lender charges a fee. For most large names the fee is a few basis points and nobody prices it. But when a stock is crowded on the short side — *hard to borrow*, in desk language — the fee can run to hundreds or thousands of basis points a year, and it enters the forward through the same replication logic as everything else in this chapter — from both sides. The arbitrage that keeps the forward from sitting too *low* requires selling the stock and buying it forward, and that arbitrageur pays the fee: the forward can sink below its frictionless level by the fee before pushing it back turns a profit. And the fee pulls from above too: whoever *holds* the stock can lend it out and collect the same fee, so holding shares unlent while the forward trades higher leaves money on the table, and competition among lenders presses the forward down to the fee-adjusted level. (In practice lendable supply and fee splits leave a band rather than a line; the equation below reads as its middle.) Writing b for the borrow rate, the carried forward becomes

$$F = S\,e^{(r-q_d-b)\,t} \qquad \Longleftrightarrow \qquad b = r - q_d - \frac{1}{t}\log\frac{F}{S}. \tag{6.12}$$

Read the second form as a measurement recipe: given the rate, the dividend model, and the parity-fitted forward, the borrow is the carry *residual* — the rate left over when the named parts are subtracted. Nothing new has to be fitted; the chapter's line already contains the number. The question, in this chapter's spirit, is not whether b can be computed — it always can — but whether the computation *means* anything. Two closed forms answer that, one for the stakes and one for the precision, and the chapter has already built both.

Materiality. A borrow error is a carry error, and a carry error reaches the smile through the forward it misplaces: by (6.12), a shift δb moves $\log F$ by $-t\,\delta b$ — the sign says a costlier borrow lowers the forward; the size is what matters here — and section 6.7's conversion (6.10) turns that into implied volatility. At the money,

$$\left|\frac{\mathrm{d}\sigma}{\mathrm{d}b}\right| = t\cdot\frac{\Phi(d_+)}{\varphi(d_+)\sqrt{\tau}} \;\approx\; 1.25\sqrt{t} \qquad \text{(clocks agreeing, } d_+ \text{ small)}, \tag{6.13}$$

about $125\sqrt{t}$ volatility basis points per hundred basis points of borrow. Figure 6.7(a) draws the exact curve at a flat 20% volatility: 65 vol bp at three months, 136 at a year. On a genuinely hard-to-borrow name, the borrow is not a financing footnote; it is a first-order input to the smile's *level*, on par with the volatility quotes themselves.

Precision. How well does the board pin the residual? By proposition 6.1, the parity forward is known to about $\varepsilon\sqrt{2}/\sqrt{n}$ in dollars — the level line — hence to $\varepsilon\sqrt{2}/(\sqrt{n}\,S)$ as a relative error, taking $DF \approx S$ for this back-of-envelope. Inverting (6.12), a borrow perturbs the log-forward through the factor t, so the forward's noise divides by t on its way to b: the smallest borrow the board can distinguish from zero is

$$b_{\min} \approx \frac{\varepsilon\sqrt{2}}{\sqrt{n}\,S\,t}. \tag{6.14}$$

This is the borrow's *identifiability floor* — one standard error of the residual, so "resolve" here means the one-sigma convention — and its shape is the now-familiar $1/t$: precisely the amplification that made short-dated parity rates meaningless in section 6.3 makes short-dated borrow reads meaningless too. Figure 6.7(b) draws (6.14) for a board sized like a typical single name rather than the running board — 20 pairs, quote noise stated in basis points of spot; constants in appendix 6.A — at two noise levels, against the shaded range of borrow fees typical of genuinely hard names. Read it maturity by maturity. At one week, even a clean board resolves nothing finer than 164 bp — larger than most hard-to-borrow fees themselves, so a week-dated "borrow read" is noise wearing a suit. At three months and ten basis points of quote noise the floor is 13 bp; at a year it is 3.2 bp on a clean board and 16 bp on a noisy one — comfortably below the shaded band. A borrow read is a measurement exactly where its floor sits under the fee it claims to measure; elsewhere it is a guess dressed in the estimator's output format.

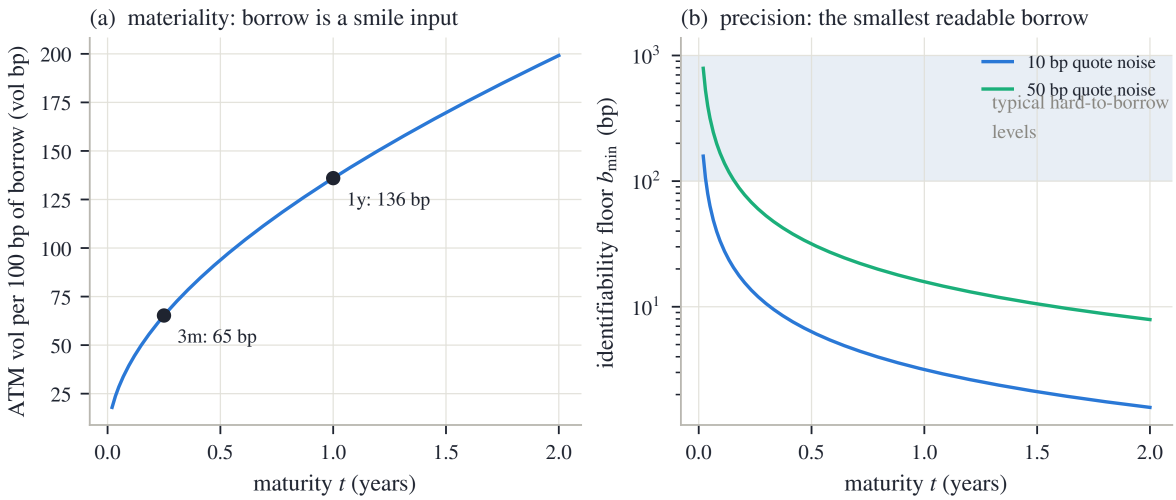

Figure 6.7: The borrow's two closed forms (flat 20% volatility, 20 paired strikes). (a) Materiality (6.13): ATM implied vol moved per 100 bp of borrow — 65 vol bp at three months, 136 at a year. (b) Precision (6.14): the identifiability floor at two quote-noise levels, against typical hard-to-borrow fee levels (shaded). A read is publishable where the floor clears the band.

The tension between the two panels is the section's conclusion, and it is a design principle rather than a formula. Borrow matters most at the short end of a stressed name — exactly where (6.14) says it cannot be measured from one chain. The honest output of an estimator in that position is not a number but a declared state: *unidentified*, reported as such, with the floor that justifies it — never a silent zero, which is also a borrow estimate, just an undeclared one. The reference implementation carries its carry as a per-expiry decomposition in which every leg is tagged with its source — measured, supplied, or unidentified — and lets a fit consume a borrow only where the read clears both its floor and a materiality gate. Two caveats bound the whole enterprise, each one sentence. A flat rate error is indistinguishable from a borrow error at a single expiry — both are carry — so splitting them honestly needs an external rate curve, and (6.12) inherits any rate bias one for one. And on American chains the early-exercise premium moves with the very carry being solved for, so the borrow and the de-Americanization must close jointly — one more reason this chapter hands its output to Chapter 7 rather than declaring victory.

6.8.1 The contract

The chapter closes the way the earlier ones do, with its contract — three columns of decreasing strength. The first holds by theorem. The second is measured, on the frozen chain or on the stated boards, and the chapter quoted every number. The third is convention: chosen, stated, and owed to the user, because no data selects it.

Table 6.1: The chapter's contract.

Proved	**Checked**	**Convention**
Parity (6.1): European call and put prices determine (F, D) as one line's coefficients, model-free.	The frozen node's line is straight to 0.35% of its range over 158 strikes (fig. 6.1a).	Mids as the regression target, and the both-sides-live pairing rule; fit-to-bid/ask variants change the observable, not the identity.
Error bars (6.3)–(6.4): level $\sim \varepsilon/\sqrt{n}$, slope $\sim \varepsilon/\sqrt{S_{KK}}$ amplified by $1/t$; the two uncorrelated.	Predicted vs measured on 2000 trials: 6.4 bp vs 6.3 bp (rate), 0.57 bp vs 0.57 bp (forward) (fig. 6.2).	The noise model itself (independent, homoskedastic mids) — adequate for design decisions, not a law.
The lever identity (6.5): a level estimator transmits a discount error through $\bar{K}_\mu - F$ only.	50.0 / 5.4 / 0.31 bp per percent for the three estimators (fig. 6.4).	The trim's thresholds, the rate band's endpoints, the kernel's width and spot centering; all inert on clean chains. Coherent staleness (remark 6.2) is bounded, not detected.
Escrow (6.6)–(6.8) and the elasticity split (6.9): cash > 1, proportional and yield $= 1$.	Sawtooth vs flat yield 2.59%; elasticity 1.053 at two years, closed form on top of the bump measurement (fig. 6.5).	The dividend convention per name — a risk decision (section 6.6), disclosed, never inferred from one chain.
The conversion (6.10) and its corollaries: ATM gap (6.11), borrow materiality (6.13), floor (6.14).	41 vol bp of gap per 10 bp of forward error on the frozen node; the naive American-mid forward would imprint 278 vol bp (fig. 6.6).	Below its floor a borrow is reported *unidentified*, never zero; a read enters fits only past a declared materiality gate.

Step back to the book's thesis, one last time for this chapter. Validity was parity itself — an arbitrage statement no model can opt out of. Information was distributed exactly as proposition 6.1 said it would be: abundant in the line's level, scarce in its slope, absent past the borrow's floor. Convention was everything the data could not select — the rate band, the dividend model, the declared treatment of what cannot be measured. One contamination ran through the chapter unresolved: the American bow of fig. 6.1(b), tolerated here, priced in section 6.7, and entangled with the borrow just above. The next chapter removes it.

6.A Numerical protocol

Every measured number in the chapter — every fit result, scatter, gap, share, or error — comes from one deterministic generator and enters the text as a generated macro. Three kinds of numbers are typed instead, deliberately: the constants of the constructions themselves (the boards, schedules, and noise levels below, which *define* the experiments); arithmetic the reader is meant to reproduce by hand (the two-strike solve of section 6.1, whose portfolio prices were placed exactly on the line with $D = 0.98$ and $F = 101.50$, and the plug-in evaluations of sections 6.3, 6.7 and 6.8); and stated approximation constants such as $\varphi(0) = 0.399$. Everything else derives from the frozen snapshot or from the boards, in the order the chapter uses them.

The real chain (figs. 6.1 and 6.6). The raw SPY chain embedded in the book's frozen snapshot, December 2026 expiry. A strike enters the parity fit when both its call and its put quote a live (positive) bid; the observable is the difference of the two mids, $\Pi = \mathsf{C} - \mathsf{P}$ with each mid the bid–ask midpoint. The naive fit is plain least squares over all surviving pairs. The *resolved* forward quoted beside it is the one stored with the snapshot — the reference implementation's resolution for that node, which de-Americanizes quotes before regressing (Chapter 7) — and the smile used in fig. 6.6 is the node's stored fit from the same snapshot. The re-implied volatilities of fig. 6.6(b) solve the one-dimensional Black inversion by bracketing to machine tolerance, put side for $k < 0$ and call side for $k > 0$; in this data the calendar and variance clocks coincide, as section 6.1 noted.

The running board (figs. 6.2 and 6.3). European and synthetic: spot $S = 100$, rate $r = 4\%$, continuous dividend yield $q_d = 1\%$, flat volatility 22%, half a year to expiry; strikes from 70 to 130 every 2.5 (25 pairs); every mid priced by the Black formula and rounded to the cent. The identifiability audit adds independent Gaussian noise of standard deviation two cents to every call and put mid separately, refits (6.2), and repeats for 2,000 trials from a fixed-seed generator — the chapter's only use of randomness; the short-dated variant repeats the experiment at $t = 0.05$. (The noise rides on top of the board's cent rounding, which adds ≈ 0.3 cents in quadrature; the predictions ignore it, and the measured–predicted agreement shows how little it matters.) Predictions come from (6.3)–(6.4) with the board's n, S_{KK}, and truth.

Trim and masking (fig. 6.3, remark 6.2). The trim is the loop of section 6.4 verbatim: robust scale 1.4826× the median absolute residual, floored at one basis point of spot; threshold four robust sigmas; at most three rounds; never below three surviving pairs. The staged failure marks the strike-75 put mid \$1.20 high. The masking variant re-prices the five lowest strikes at a financing rate two percent above the truth: their portfolio differences Π are recomputed exactly at the stale (F', D') and rounded to the cent directly (the difference, not the two mids — immaterial at this size, noted for exactness), and the identical trim runs.

The lever board (fig. 6.4). Same construction, strikes from 85 to 140, prices kept exact (no rounding) so the figure isolates the identity (6.5). Imposed rate errors span ±2%. The kernel estimator weights each strike by a Gaussian in $\log(K/S)$ of width 0.10, centered at spot.

Dividends (fig. 6.5). The schedule pays \$0.65 each quarter, first ex-date 30 days out, horizon 2.5 years, at $r = 4\%$; the proportional variant converts each payment to the fraction 0.65/100 of the prevailing level; the flat yield is the schedule's one-year equivalent. Elasticities are measured by recomputing each convention's forward under a one-basis-point spot bump, against the closed form (6.9).

Borrow (fig. 6.7). Closed forms only: (6.13) at a flat 20% volatility, and (6.14) with $n = 20$ paired strikes at quote noise of 10 and 50 basis points of spot. The shaded hard-to-borrow range, 100–1,000 bp per year, is an illustrative convention of the figure, not an estimate.

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